

5G RAN

Energy Conservation and Emission Reduction Feature Parameter Description

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Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

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Contents

1 Change History..... 1

2 About This Document..... 21

2.1 General Statements..... 21

2.2 Features in This Document..... 21

2.3 Differences..... 24

3 Overview..... 29

4 Device-Level Energy Saving (Time Domain)..... 32

4.1 Symbol Power Saving..... 32

4.1.1 Principles..... 32

4.1.2 Network Analysis..... 32

4.1.2.1 Benefits..... 32

4.1.2.2 Impacts..... 33

4.1.3 Requirements..... 33

4.1.3.1 Licenses..... 33

4.1.3.2 Functions..... 33

4.1.3.2.1 Prerequisite Functions..... 34

4.1.3.2.2 Mutually Exclusive Functions..... 34

4.1.3.2.3 Function Impacts..... 35

4.1.3.3 Hardware..... 44

4.1.3.4 Networking..... 45

4.1.3.5 Cells..... 45

4.1.3.6 Others..... 45

4.1.4 Operation and Maintenance..... 45

4.1.4.1 Data Configuration..... 45

4.1.4.1.1 Data Preparation..... 45

4.1.4.1.2 Using MML Commands..... 46

4.1.4.1.3 Using the MAE-Deployment..... 46

4.1.4.2 Activation Verification..... 47

4.1.4.3 Network Monitoring..... 47

4.2 Intelligent Scheduling for Power Saving (Low-Frequency TDD)..... 48

4.2.1 Principles..... 48

4.2.1.1 Power Saving Scheduling..... 48

4.2.1.2 Deep Symbol Power Saving.....	56
4.2.2 Network Analysis.....	56
4.2.2.1 Benefits.....	57
4.2.2.2 Impacts.....	57
4.2.3 Requirements.....	58
4.2.3.1 Licenses.....	58
4.2.3.2 Functions.....	58
4.2.3.2.1 Prerequisite Functions.....	59
4.2.3.2.2 Mutually Exclusive Functions.....	59
4.2.3.2.3 Function Impacts.....	63
4.2.3.3 Hardware.....	79
4.2.3.4 Cells.....	80
4.2.3.5 Others.....	80
4.2.4 Operation and Maintenance.....	80
4.2.4.1 Data Configuration.....	80
4.2.4.1.1 Data Preparation.....	80
4.2.4.1.2 Using MML Commands.....	87
4.2.4.1.3 Using the MAE-Deployment.....	88
4.2.4.2 Activation Verification.....	88
4.2.4.3 Network Monitoring.....	91
4.3 Intelligent Scheduling for Power Saving (High-frequency TDD).....	91
4.3.1 Principles.....	92
4.3.2 Network Analysis.....	95
4.3.2.1 Benefits.....	95
4.3.2.2 Impacts.....	96
4.3.3 Requirements.....	96
4.3.3.1 Licenses.....	96
4.3.3.2 Functions.....	96
4.3.3.2.1 Prerequisite Functions.....	97
4.3.3.2.2 Mutually Exclusive Functions.....	97
4.3.3.2.3 Function Impacts.....	98
4.3.3.3 Hardware.....	100
4.3.3.4 Cells.....	100
4.3.3.5 Others.....	100
4.3.4 Operation and Maintenance.....	100
4.3.4.1 Data Configuration.....	100
4.3.4.1.1 Data Preparation.....	100
4.3.4.1.2 Using MML Commands.....	105
4.3.4.1.3 Using the MAE-Deployment.....	105
4.3.4.2 Activation Verification.....	105
4.3.4.3 Network Monitoring.....	107
4.4 MetaAAU Beam Power Saving (Low-Frequency TDD).....	108

4.4.1 Principles..... 108

4.4.2 Network Analysis..... 108

4.4.2.1 Benefits..... 108

4.4.2.2 Impacts..... 108

4.4.3 Requirements..... 108

4.4.3.1 Licenses..... 108

4.4.3.2 Functions..... 108

4.4.3.2.1 Prerequisite Functions..... 109

4.4.3.2.2 Mutually Exclusive Functions..... 109

4.4.3.2.3 Function Impacts..... 109

4.4.3.3 Hardware..... 109

4.4.3.4 Networking..... 109

4.4.3.5 Others..... 109

4.4.4 Operation and Maintenance..... 109

4.4.4.1 Data Configuration..... 109

4.4.4.1.1 Data Preparation..... 109

4.4.4.1.2 Using MML Commands..... 110

4.4.4.1.3 Using the MAE-Deployment..... 110

4.4.4.2 Activation Verification..... 110

4.4.4.3 Network Monitoring..... 111

4.5 Uplink Symbol Power Saving (Low-Frequency TDD)..... 111

4.5.1 Principles..... 111

4.5.2 Network Analysis..... 114

4.5.2.1 Benefits..... 114

4.5.2.2 Impacts..... 114

4.5.3 Requirements..... 115

4.5.3.1 Licenses..... 115

4.5.3.2 Functions..... 115

4.5.3.2.1 Prerequisite Functions..... 115

4.5.3.2.2 Mutually Exclusive Functions..... 116

4.5.3.2.3 Function Impacts..... 119

4.5.3.3 Hardware..... 124

4.5.3.4 Networking..... 125

4.5.3.5 Cells..... 125

4.5.3.6 Others..... 125

4.5.4 Operation and Maintenance..... 125

4.5.4.1 Data Configuration..... 125

4.5.4.1.1 Data Preparation..... 125

4.5.4.1.2 Using MML Commands..... 128

4.5.4.1.3 Using the MAE-Deployment..... 129

4.5.4.2 Activation Verification..... 129

4.5.4.3 Network Monitoring..... 132

4.6 pRRU-Level Precise Symbol Power Saving (Low-Frequency TDD)	132
4.6.1 Principles	132
4.6.2 Network Analysis	133
4.6.2.1 Benefits	133
4.6.2.2 Impacts	133
4.6.3 Requirements	133
4.6.3.1 Licenses	134
4.6.3.2 Functions	134
4.6.3.2.1 Prerequisite Functions	134
4.6.3.2.2 Mutually Exclusive Functions	134
4.6.3.2.3 Function Impacts	136
4.6.3.3 Hardware	137
4.6.3.4 Networking	138
4.6.3.5 Cells	138
4.6.3.6 Others	138
4.6.4 Operation and Maintenance	138
4.6.4.1 Data Configuration	138
4.6.4.1.1 Data Preparation	138
4.6.4.1.2 Using MML Commands	139
4.6.4.1.3 Using the MAE-Deployment	139
4.6.4.2 Activation Verification	139
4.6.4.3 Network Monitoring	140
5 Device-Level Energy Saving (Space Domain)	141
5.1 RF Channel Intelligent Shutdown	141
5.1.1 Principles	141
5.1.2 Network Analysis	148
5.1.2.1 Benefits	148
5.1.2.2 Impacts	149
5.1.3 Requirements	154
5.1.3.1 Licenses	154
5.1.3.2 Functions	155
5.1.3.2.1 Prerequisite Functions	155
5.1.3.2.2 Mutually Exclusive Functions	156
5.1.3.2.3 Function Impacts	164
5.1.3.3 Hardware	195
5.1.3.4 Cells	198
5.1.3.5 Others	199
5.1.4 Operation and Maintenance	200
5.1.4.1 Data Configuration	200
5.1.4.1.1 Data Preparation	200
5.1.4.1.2 Using MML Commands	208
5.1.4.1.3 Using the MAE-Deployment	211

5.1.4.2 Activation Verification.....	211
5.1.4.3 Network Monitoring.....	218
5.2 MetaAAU Deep Energy Saving (Low-Frequency TDD).....	218
5.2.1 Principles.....	218
5.2.2 Network Analysis.....	219
5.2.2.1 Benefits.....	219
5.2.2.2 Impacts.....	219
5.2.3 Requirements.....	219
5.2.3.1 Licenses.....	219
5.2.3.2 Functions.....	220
5.2.3.2.1 Prerequisite Functions.....	220
5.2.3.2.2 Mutually Exclusive Functions.....	220
5.2.3.2.3 Function Impacts.....	220
5.2.3.3 Hardware.....	224
5.2.3.4 Cells.....	224
5.2.3.5 Others.....	225
5.2.4 Operation and Maintenance.....	225
5.2.4.1 Data Configuration.....	225
5.2.4.1.1 Data Preparation.....	225
5.2.4.1.2 Using MML Commands.....	225
5.2.4.1.3 Using the MAE-Deployment.....	226
5.2.4.2 Activation Verification.....	226
5.2.4.3 Network Monitoring.....	226
5.3 Low-Power RF Channel Dynamic Shutdown (Low-Frequency TDD).....	227
5.3.1 Principles.....	227
5.3.2 Network Analysis.....	231
5.3.2.1 Benefits.....	232
5.3.2.2 Impacts.....	232
5.3.3 Requirements.....	232
5.3.3.1 Licenses.....	232
5.3.3.2 Functions.....	233
5.3.3.2.1 Prerequisite Functions.....	233
5.3.3.2.2 Mutually Exclusive Functions.....	233
5.3.3.2.3 Function Impacts.....	234
5.3.3.3 Hardware.....	239
5.3.3.4 Cells.....	239
5.3.3.5 Others.....	240
5.3.4 Operation and Maintenance.....	240
5.3.4.1 Data Configuration.....	240
5.3.4.1.1 Data Preparation.....	240
5.3.4.1.2 Using MML Commands.....	243
5.3.4.1.3 Using the MAE-Deployment.....	244

5.3.4.2 Activation Verification..... 244

5.3.4.3 Network Monitoring..... 246

5.4 TTI-Level Channel Shutdown..... 246

5.4.1 Principles..... 246

5.4.2 Network Analysis..... 254

5.4.2.1 Benefits..... 255

5.4.2.2 Impacts..... 255

5.4.3 Requirements..... 256

5.4.3.1 Licenses..... 256

5.4.3.2 Functions..... 256

5.4.3.2.1 Prerequisite Functions..... 256

5.4.3.2.2 Mutually Exclusive Functions..... 257

5.4.3.2.3 Function Impacts..... 260

5.4.3.3 Hardware..... 274

5.4.3.4 Cells..... 275

5.4.3.5 Others..... 276

5.4.4 Operation and Maintenance..... 276

5.4.4.1 Data Configuration..... 276

5.4.4.1.1 Data Preparation..... 276

5.4.4.1.2 Using MML Commands..... 281

5.4.4.1.3 Using the MAE-Deployment..... 282

5.4.4.2 Activation Verification..... 282

5.4.4.3 Network Monitoring..... 286

5.5 Optimal Channel Shutdown (Low-Frequency TDD)..... 286

5.5.1 Principles..... 287

5.5.2 Network Analysis..... 287

5.5.2.1 Benefits..... 288

5.5.2.2 Impacts..... 288

5.5.3 Requirements..... 288

5.5.3.1 Licenses..... 288

5.5.3.2 Functions..... 288

5.5.3.2.1 Prerequisite Functions..... 289

5.5.3.2.2 Mutually Exclusive Functions..... 289

5.5.3.2.3 Function Impacts..... 289

5.5.3.3 Hardware..... 289

5.5.3.4 Cells..... 290

5.5.3.5 Others..... 290

5.5.4 Operation and Maintenance..... 290

5.5.4.1 Data Configuration..... 290

5.5.4.1.1 Data Preparation..... 290

5.5.4.1.2 Using MML Commands..... 290

5.5.4.1.3 Using the MAE-Deployment..... 291

5.5.4.2 Activation Verification.....	291
5.5.4.3 Network Monitoring.....	291
5.6 RF Channel Power Adaptation (Low-Frequency TDD).....	291
5.6.1 Principles.....	292
5.6.2 Network Analysis.....	292
5.6.2.1 Benefits.....	292
5.6.2.2 Impacts.....	293
5.6.3 Requirements.....	293
5.6.3.1 Licenses.....	293
5.6.3.2 Functions.....	293
5.6.3.2.1 Prerequisite Functions.....	293
5.6.3.2.2 Mutually Exclusive Functions.....	294
5.6.3.2.3 Function Impacts.....	295
5.6.3.3 Hardware.....	295
5.6.3.4 Networking.....	295
5.6.3.5 Cells.....	295
5.6.3.6 Others.....	295
5.6.4 Operation and Maintenance.....	295
5.6.4.1 Data Configuration.....	296
5.6.4.1.1 Data Preparation.....	296
5.6.4.1.2 Using MML Commands.....	296
5.6.4.1.3 Using the MAE-Deployment.....	296
5.6.4.2 Activation Verification.....	297
5.6.4.3 Network Monitoring.....	297
5.7 Dynamic RF Channel Shutdown.....	297
5.7.1 Principles.....	297
5.7.2 Network Analysis.....	305
5.7.2.1 Benefits.....	305
5.7.2.2 Impacts.....	306
5.7.3 Requirements.....	307
5.7.3.1 Licenses.....	307
5.7.3.2 Functions.....	307
5.7.3.2.1 Prerequisite Functions.....	308
5.7.3.2.2 Mutually Exclusive Functions.....	308
5.7.3.2.3 Function Impacts.....	311
5.7.3.3 Hardware.....	320
5.7.3.4 Cells.....	321
5.7.3.5 Others.....	322
5.7.4 Operation and Maintenance.....	322
5.7.4.1 Data Configuration.....	322
5.7.4.1.1 Data Preparation.....	322
5.7.4.1.2 Using MML Commands.....	328

5.7.4.1.3 Using the MAE-Deployment.....	330
5.7.4.2 Activation Verification.....	330
5.7.4.3 Network Monitoring.....	334
6 Device-Level Energy Saving (Frequency Domain).....	336
6.1 Timing Carrier Shutdown.....	336
6.1.1 Principles.....	336
6.1.2 Energy Saving Procedure.....	339
6.1.3 Network Analysis.....	341
6.1.3.1 Benefits.....	341
6.1.3.2 Impacts.....	341
6.1.4 Requirements.....	344
6.1.4.1 Licenses.....	344
6.1.4.2 Functions.....	344
6.1.4.2.1 Prerequisite Functions.....	344
6.1.4.2.2 Mutually Exclusive Functions.....	345
6.1.4.2.3 Function Impacts.....	354
6.1.4.3 Hardware.....	373
6.1.4.4 Cells.....	374
6.1.4.5 Others.....	374
6.1.5 Operation and Maintenance.....	374
6.1.5.1 Data Configuration.....	374
6.1.5.1.1 Data Preparation.....	374
6.1.5.1.2 Using MML Commands.....	380
6.1.5.1.3 Using the MAE-Deployment.....	381
6.1.5.2 Activation Verification.....	382
6.1.5.3 Network Monitoring.....	385
6.2 Intelligent Carrier Shutdown.....	386
6.2.1 Principles.....	386
6.2.1.1 Overview.....	386
6.2.1.2 Energy Saving Procedure.....	391
6.2.2 Network Analysis.....	402
6.2.2.1 Benefits.....	402
6.2.2.2 Impacts.....	402
6.2.3 Requirements.....	405
6.2.3.1 Licenses.....	405
6.2.3.2 Functions.....	406
6.2.3.2.1 Prerequisite Functions.....	406
6.2.3.2.2 Mutually Exclusive Functions.....	407
6.2.3.2.3 Function Impacts.....	410
6.2.3.3 Hardware.....	439
6.2.3.4 Cells.....	440
6.2.3.5 Others.....	440

6.2.4 Operation and Maintenance..... 440

6.2.4.1 Data Configuration..... 440

6.2.4.1.1 Data Preparation..... 440

6.2.4.1.2 Using MML Commands..... 451

6.2.4.1.3 Using the MAE-Deployment..... 456

6.2.4.2 Activation Verification..... 456

6.2.4.3 Network Monitoring..... 460

6.3 Fast Carrier Shutdown..... 461

6.3.1 Principles..... 461

6.3.2 Energy Saving Procedure..... 461

6.3.3 Network Analysis..... 471

6.3.3.1 Benefits..... 471

6.3.3.2 Impacts..... 471

6.3.4 Requirements..... 473

6.3.4.1 Licenses..... 473

6.3.4.2 Functions..... 474

6.3.4.2.1 Prerequisite Functions..... 474

6.3.4.2.2 Mutually Exclusive Functions..... 475

6.3.4.2.3 Function Impacts..... 477

6.3.4.3 Hardware..... 501

6.3.4.4 Cells..... 502

6.3.4.5 Others..... 502

6.3.5 Operation and Maintenance..... 502

6.3.5.1 Data Configuration..... 502

6.3.5.1.1 Data Preparation..... 502

6.3.5.1.2 Using MML Commands..... 512

6.3.5.1.3 Using the MAE-Deployment..... 516

6.3.5.2 Activation Verification..... 517

6.3.5.3 Network Monitoring..... 521

6.4 Multi-Carrier Joint Carrier Shutdown..... 522

6.4.1 Principles..... 522

6.4.2 Network Analysis..... 523

6.4.2.1 Benefits..... 523

6.4.2.2 Impacts..... 523

6.4.3 Requirements..... 524

6.4.3.1 Licenses..... 524

6.4.3.2 Functions..... 524

6.4.3.2.1 Prerequisite Functions..... 524

6.4.3.2.2 Mutually Exclusive Functions..... 525

6.4.3.2.3 Function Impacts..... 528

6.4.3.3 Hardware..... 528

6.4.3.4 Others..... 528

6.4.4 Operation and Maintenance.....	529
6.4.4.1 Data Configuration.....	529
6.4.4.1.1 Data Preparation.....	529
6.4.4.1.2 Using MML Commands.....	529
6.4.4.1.3 Using the MAE-Deployment.....	529
6.4.4.2 Activation Verification.....	530
6.4.4.3 Network Monitoring.....	530
6.5 Multi-Frequency Deep Carrier Shutdown (Low-Frequency TDD).....	530
6.5.1 Principles.....	530
6.5.1.1 Overview.....	531
6.5.1.2 Energy Saving Procedure.....	531
6.5.2 Network Analysis.....	536
6.5.2.1 Benefits.....	537
6.5.2.2 Impacts.....	537
6.5.3 Requirements.....	537
6.5.3.1 Licenses.....	537
6.5.3.2 Functions.....	537
6.5.3.2.1 Prerequisite Functions.....	538
6.5.3.2.2 Mutually Exclusive Functions.....	539
6.5.3.2.3 Function Impacts.....	539
6.5.3.3 Hardware.....	539
6.5.3.4 Others.....	540
6.5.4 Operation and Maintenance.....	540
6.5.4.1 Data Configuration.....	540
6.5.4.1.1 Data Preparation.....	540
6.5.4.1.2 Using MML Commands.....	543
6.5.4.1.3 Using the MAE-Deployment.....	544
6.5.4.2 Activation Verification.....	545
6.5.4.3 Network Monitoring.....	547
7 Device-Level Energy Saving (Power Domain).....	548
7.1 Energy Saving Based on Flexible Frequency-Domain Scheduling.....	548
7.1.1 Principles.....	548
7.1.2 Network Analysis.....	554
7.1.2.1 Benefits.....	554
7.1.2.2 Impacts.....	555
7.1.3 Requirements.....	555
7.1.3.1 Licenses.....	556
7.1.3.2 Functions.....	556
7.1.3.2.1 Prerequisite Functions.....	556
7.1.3.2.2 Mutually Exclusive Functions.....	557
7.1.3.2.3 Function Impacts.....	558
7.1.3.3 Hardware.....	570

7.1.3.4 Cells.....	571
7.1.3.5 Others.....	571
7.1.4 Operation and Maintenance.....	571
7.1.4.1 Data Configuration.....	571
7.1.4.1.1 Data Preparation.....	571
7.1.4.1.2 Using MML Commands.....	575
7.1.4.1.3 Using the MAE-Deployment.....	576
7.1.4.2 Activation Verification.....	576
7.1.4.3 Network Monitoring.....	578
7.2 MetaAAU Power Reduction for Cell-Center UEs (Low-Frequency TDD).....	579
7.2.1 Principles.....	579
7.2.2 Network Analysis.....	579
7.2.2.1 Benefits.....	579
7.2.2.2 Impacts.....	580
7.2.3 Requirements.....	580
7.2.3.1 Licenses.....	580
7.2.3.2 Functions.....	580
7.2.3.2.1 Prerequisite Functions.....	580
7.2.3.2.2 Mutually Exclusive Functions.....	580
7.2.3.2.3 Function Impacts.....	581
7.2.3.3 Hardware.....	582
7.2.3.4 Cells.....	582
7.2.3.5 Others.....	582
7.2.4 Operation and Maintenance.....	582
7.2.4.1 Data Configuration.....	583
7.2.4.1.1 Data Preparation.....	583
7.2.4.1.2 Using MML Commands.....	583
7.2.4.1.3 Using the MAE-Deployment.....	584
7.2.4.2 Activation Verification.....	584
7.2.4.3 Network Monitoring.....	584
7.3 AMC-based Precise Power Control Energy Saving.....	584
7.3.1 Principles.....	584
7.3.2 Network Analysis.....	588
7.3.2.1 Benefits.....	588
7.3.2.2 Impacts.....	588
7.3.3 Requirements.....	589
7.3.3.1 Licenses.....	589
7.3.3.2 Functions.....	589
7.3.3.2.1 Prerequisite Functions.....	590
7.3.3.2.2 Mutually Exclusive Functions.....	590
7.3.3.2.3 Function Impacts.....	591
7.3.3.3 Hardware.....	597

7.3.3.4 Cells.....	598
7.3.3.5 Others.....	598
7.3.4 Operation and Maintenance.....	598
7.3.4.1 Data Configuration.....	598
7.3.4.1.1 Data Preparation.....	598
7.3.4.1.2 Using MML Commands.....	601
7.3.4.1.3 Using the MAE-Deployment.....	602
7.3.4.2 Activation Verification.....	602
7.3.4.3 Network Monitoring.....	604
8 Site Level Energy Saving.....	606
8.1 Baseband Processor Shutdown.....	606
8.1.1 Principles.....	606
8.1.2 Network Analysis.....	608
8.1.2.1 Benefits.....	608
8.1.2.2 Impacts.....	608
8.1.3 Requirements.....	608
8.1.3.1 Licenses.....	608
8.1.3.2 Functions.....	608
8.1.3.2.1 Prerequisite Functions.....	608
8.1.3.2.2 Mutually Exclusive Functions.....	609
8.1.3.2.3 Function Impacts.....	611
8.1.3.3 Hardware.....	613
8.1.3.4 Others.....	614
8.1.4 Operation and Maintenance.....	614
8.1.4.1 Data Configuration.....	614
8.1.4.1.1 Data Preparation.....	614
8.1.4.1.2 Using MML Commands.....	615
8.1.4.1.3 Using the MAE-Deployment.....	616
8.1.4.2 Activation Verification.....	616
8.1.4.3 Network Monitoring.....	617
9 Network Level Energy Saving.....	618
9.1 Multi-RF-Module Coordinated Energy Saving.....	618
9.1.1 Principles.....	618
9.1.1.1 Downlink Spectral Efficiency and Module Power Consumption Model Building.....	619
9.1.1.2 Energy-Efficiency-based Inter-Frequency Handover.....	621
9.1.1.2.1 Measurement Configuration Delivery.....	622
9.1.1.2.2 Measurement Reporting.....	623
9.1.1.2.3 Energy Efficiency and User Experience Evaluation.....	623
9.1.1.2.4 Optimal Cell Selection.....	624
9.1.1.2.5 Energy-Efficiency-based Inter-Frequency Handover Execution.....	625
9.1.1.3 Energy-Efficiency-based Coordinated Scheduling.....	625
9.1.2 Network Analysis.....	625

9.1.2.1 Benefits.....	626
9.1.2.2 Impacts.....	626
9.1.3 Requirements.....	627
9.1.3.1 Licenses.....	627
9.1.3.2 Functions.....	627
9.1.3.2.1 Prerequisite Functions.....	628
9.1.3.2.2 Mutually Exclusive Functions.....	631
9.1.3.2.3 Function Impacts.....	633
9.1.3.3 Hardware.....	639
9.1.3.4 Networking.....	640
9.1.3.5 Others.....	640
9.1.4 Operation and Maintenance.....	640
9.1.4.1 Data Configuration.....	640
9.1.4.1.1 Data Preparation.....	640
9.1.4.1.2 Using MML Commands.....	642
9.1.4.1.3 Using the MAE-Deployment.....	643
9.1.4.2 Activation Verification.....	643
9.1.4.3 Network Monitoring.....	644
9.2 Intelligent Power Control Energy Saving.....	644
9.2.1 Principles.....	644
9.2.2 Network Analysis.....	647
9.2.2.1 Benefits.....	647
9.2.2.2 Impacts.....	648
9.2.3 Requirements.....	648
9.2.3.1 Licenses.....	648
9.2.3.2 Functions.....	648
9.2.3.2.1 Prerequisite Functions.....	649
9.2.3.2.2 Mutually Exclusive Functions.....	649
9.2.3.2.3 Function Impacts.....	650
9.2.3.3 Hardware.....	658
9.2.3.4 Networking.....	658
9.2.3.5 Cells.....	658
9.2.3.6 Others.....	658
9.2.4 Operation and Maintenance.....	658
9.2.4.1 Data Configuration.....	658
9.2.4.1.1 Data Preparation.....	658
9.2.4.1.2 Using MML Commands.....	662
9.2.4.1.3 Using the MAE-Deployment.....	663
9.2.4.2 Activation Verification.....	663
9.2.4.3 Network Monitoring.....	664
10 Energy Consumption Monitoring.....	666
10.1 Principles.....	666

10.2 Network Analysis.....	670
10.2.1 Benefits.....	670
10.2.2 Impacts.....	670
10.3 Requirements.....	670
10.3.1 Licenses.....	670
10.3.2 Functions.....	670
10.3.2.1 Prerequisite Functions.....	670
10.3.2.2 Mutually Exclusive Functions.....	671
10.3.2.3 Function Impacts.....	671
10.3.3 Hardware.....	671
10.3.4 Others.....	671
10.4 Operation and Maintenance.....	672
10.4.1 Data Configuration.....	672
10.4.2 Activation Verification.....	673
10.4.3 Network Monitoring.....	673
11 Parameters.....	674
12 Counters.....	676
13 Glossary.....	677
14 Reference Documents.....	678

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for AAU-level baseband processor shutdown to not take effect on the AAU5685md. For details, see 8.1.3.3 Hardware .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between uplink symbol power saving and GP symbol shutdown in special slots. For details, see 4.5.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes

Revised descriptions of "Neighboring Cell Information Unreachable", a possible reason why intelligent power control energy saving did not take effect. For details, see [9.2.4.2 Activation Verification](#).

Revised descriptions of the impact relationship between timing carrier shutdown and inter-RAT redirection back to the HPLMN based on roaming UE identification. For details, see [6.1.4.2.3 Function Impacts](#).

Revised descriptions of activation verification for intelligent power control energy saving. For details, see [9.2.4.2 Activation Verification](#).

Revised descriptions of **Power Saving Cell Flag** for intelligent carrier shutdown. For details, see [6.2.1.1 Overview](#).

Revised descriptions of the impacts of intelligent power control energy saving. For details, see [9.2.2.2 Impacts](#).

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 06 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for dynamic RF channel shutdown in SSB slots with UE scheduling. For details, see: • 5.7.1 Principles • 5.7.2.1 Benefits • 5.7.2.2 Impacts • 5.7.4.1.1 Data Preparation • 5.7.4.1.2 Using MML Commands	Modified parameter: Added the RF_CHN_DYN_S HUTDOWN_SSB_SW option to the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added support for dynamic RF channel shutdown in single-carrier scenarios. For details, see: • 5.7.1 Principles • 5.7.2.1 Benefits • 5.7.2.2 Impacts • 5.7.3.4 Cells • 5.7.4.1.1 Data Preparation • 5.7.4.1.2 Using MML Commands • 5.7.4.2 Activation Verification	Modified parameter: Added the RF_CHN_DYN_S HDN_SGL_CARR_SW option to the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added support for load-based evaluation for entering timing carrier shutdown. For details, see: 6.1.1 Principles 6.1.2 Energy Saving Procedure 6.1.3.1 Benefits 6.1.5.1.1 Data Preparation 6.1.5.1.2 Using MML Commands 6.1.5.2 Activation Verification	Added the NRDUCellPowerSaving.CarrierShutdownJudgeMode parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the SUL UE number check to the conditions for entering and exiting intelligent carrier shutdown/fast carrier shutdown/timing carrier shutdown. For details, see: • 6.2.1.2 Energy Saving Procedure • 6.2.4.1.1 Data Preparation • 6.2.4.1.2 Using MML Commands • 6.3.2 Energy Saving Procedure • 6.3.5.1.1 Data Preparation • 6.3.5.1.2 Using MML Commands • 6.1.1 Principles • 6.1.2 Energy Saving Procedure • 6.1.5.1.1 Data Preparation • 6.1.5.1.2 Using MML Commands • 6.1.5.2 Activation Verification	Added the NRDUCellPowerSaving.SulUserNumThld and NRDUCellPowerSaving.SulUserNumOffset parameters.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for uplink symbol power saving in multi-carrier scenarios. For details, see: • 4.5.1 Principles • 4.5.3.3 Hardware • 4.5.3.4 Networking • 4.5.4.1.1 Data Preparation • 4.5.4.1.2 Using MML Commands • 4.5.4.2 Activation Verification	Modified parameter: Added the UL_SYM_SHUTDOWN_MULTI_CARRIER_SW option to the gNB PwrSavingAlgo.PwrSavingAlgorithmSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added uplink symbol power saving optimization in remote interference avoidance scenarios. For details, see: • 4.5.1 Principles • 4.5.2.1 Benefits • 4.5.3.2.1 Prerequisite Functions • 4.5.3.2.3 Function Impacts • 4.5.4.1.1 Data Preparation • 4.5.4.1.2 Using MML Commands	Modified parameter: Added the RIM_UL_SYMBOL_SHUTDOWN_OPT_SW option to the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Optimized baseband processor shutdown. For details, see: • 8.1.1 Principles • 8.1.2.2 Impacts • 8.1.3.2.1 Prerequisite Functions • 8.1.3.2.2 Mutually Exclusive Functions • 8.1.3.2.3 Function Impacts • 8.1.3.3 Hardware • 8.1.4.1.1 Data Preparation • 8.1.4.1.2 Using MML Commands • 8.1.4.2 Activation Verification	Modified parameter: Added the value ADAPT_BBP_SHUTDOWN to the BBP.CELLDEPLOY parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for RF channel intelligent shutdown in pseudo-multimode scenarios. For details, see: • 5.1.3.5 Others • 5.1.4.1.1 Data Preparation • 5.1.4.1.2 Using MML Commands • 5.1.4.2 Activation Verification	Modified parameter: Added the PSE_MULTI_MODE_RF_SHUTDOWN_SW option to the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Removed the mutually exclusive relationship between RF channel intelligent shutdown and cell combination in low-speed 8T8R scenarios. For details, see 5.1.3.2.2 Mutually Exclusive Functions.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added a parameter to control paging signaling shutdown in common-channel dynamic shutdown. For details, see: • 5.4.1 Principles • 5.4.2.2 Impacts • 5.4.4.1.1 Data Preparation • 5.4.4.1.2 Using MML Commands • 5.7.1 Principles • 5.7.2.2 Impacts • 5.7.4.1.1 Data Preparation • 5.7.4.1.2 Using MML Commands	Modified parameter: Added the PAGING_DYN_SHUTDOWN_SW option to the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the AMC-based Precise Power Control Energy Saving feature. For details, see: • 2.2 Features in This Document • 2.3 Differences • 3 Overview • 7.3 AMC-based Precise Power Control Energy Saving	Modified parameters: • Added the INIT_TRANS_POWER_REDUCE_SW and RETRANS_POWER_REDUCE_SW options to the NRDUCellPwrSaveAlgo.PowerSavingAlgorithmSwitch parameter. • Added the values INIT_TRANS_POWER_REDUCE and RETRANS_POWER_REDUCE to the NRDUCellPowerSaving.PowerSavingType parameter. • Added the INIT_TRANS_PWR_RD_EXIT_SW and RETRANS_PWR_RD_EXIT_SW options to the NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw parameter. Added the NRDUCellPowerSaving.PowerProtectionMargin parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added counter-based activation verification for fast carrier shutdown. For details, see 6.3.5.2 Activation Verification and 6.3.5.3 Network Monitoring .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added the impact of "downlink PDSCH coverage enhancement" on the benefits of "RF channel intelligent shutdown". For details, see 5.1.2.1 Benefits .	None	High-frequency TDD	5900 series base stations (macro base stations)
Modified the impact relationship between RF channel intelligent shutdown and Virtual Grid-based Multi-Frequency Coordination due to optimized policy of cooperation between them. For details, see 5.1.3.2.3 Function Impacts .	Modified parameter: Added the VIRTUAL_GRID_CH_SD_COOP_OPT_SW option to the gNodeBAlgo.NrSmartMultiCarrSw parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for checking the number of UEs in an energy saving policy group for entering or exiting intelligent carrier shutdown/fast carrier shutdown. For details, see: • 6.2.1.2 Energy Saving Procedure • 6.2.4.1.1 Data Preparation • 6.2.4.1.2 Using MML Commands • 6.2.4.2 Activation Verification • 6.3.2 Energy Saving Procedure • 6.3.5.1.1 Data Preparation • 6.3.5.1.2 Using MML Commands • 6.3.5.2 Activation Verification	Modified parameter: Added the USER_NUM_CSD_EXIT_SW option to the NRCellPsPolicyGrp.ServTypeDynExitPwrSavingSw parameter. Added parameters: • NRCellPsPolicyGrp.GrpUse rNumThld • NRCellPsPolicyGrp.GrpUse rNumOffset	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added compatibility of multi-RF-module coordinated energy saving with UMPThb. For details, see 9.1.3.3 Hardware .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added compatibility of intelligent power control energy saving with UMPThb. For details, see 9.2.3.3 Hardware .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between data convergence in slots and time-domain power saving BWP. For details, see 4.2.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added the impact relationship between RF channel intelligent shutdown and intelligent beam selection. For details, see 5.1.3.2.3 Function Impacts .	None	High-frequency TDD	5900 series base stations (macro base stations)
Added the impact relationship between power saving scheduling and L/NR Inter-band Deep Coordination. For details, see 4.2.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between RF channel intelligent shutdown and L/NR Inter-band Deep Coordination. For details, see 5.1.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between TTI-level channel shutdown and L/NR Inter-band Deep Coordination. For details, see 5.4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between timing carrier shutdown and L/NR Inter-band Deep Coordination. For details, see 6.1.4.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between intelligent carrier shutdown and L/NR Inter-band Deep Coordination. For details, see 6.2.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between fast carrier shutdown and L/NR Inter-band Deep Coordination. For details, see 6.3.4.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between timing carrier shutdown and UL and DL Decoupling. For details, see 6.1.4.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between intelligent carrier shutdown and UL and DL Decoupling. For details, see 6.2.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationship between fast carrier shutdown and UL and DL Decoupling. For details, see 6.3.4.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the mutually exclusive relationship between timing carrier shutdown and the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios. For details, see 6.1.4.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the mutually exclusive relationship between power saving scheduling and the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios. For details, see 4.2.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	5900 series base stations (macro base stations)
Added the mutually exclusive relationship between deep symbol power saving and the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios. For details, see 4.2.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	5900 series base stations (macro base stations)
Added the mutually exclusive relationship between energy saving based on flexible frequency-domain scheduling and the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios. For details, see 7.1.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	5900 series base stations (macro base stations)
Added the mutually exclusive relationship between RF channel intelligent shutdown and the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios. For details, see 5.1.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the mutually exclusive relationship between intelligent carrier shutdown and the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios. For details, see 6.2.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	5900 series base stations (macro base stations)
Added the mutually exclusive relationship between fast carrier shutdown and the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios. For details, see 6.3.4.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	5900 series base stations (macro base stations)
Added the mutually exclusive relationship between baseband shutdown for energy saving and the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios. For details, see 8.1.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	5900 series base stations (macro base stations)
Added the mutually exclusive relationship between TTI-level channel shutdown and the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios. For details, see 5.4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the mutually exclusive relationship between dynamic RF channel shutdown and the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios. For details, see 5.7.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	5900 series base stations (macro base stations)
Added the impact relationship between symbol power saving and RF module fast deep dormancy. For details, see 4.1.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between power saving scheduling and RF module fast deep dormancy. For details, see 4.2.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between deep symbol power saving and RF module fast deep dormancy. For details, see 4.2.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between uplink symbol power saving and RF module fast deep dormancy. For details, see 4.5.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between RF channel intelligent shutdown and RF module fast deep dormancy. For details, see 5.1.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between low-power RF channel dynamic shutdown and RF module fast deep dormancy. For details, see 5.3.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between TTI-level channel shutdown and RF module fast deep dormancy. For details, see 5.4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between dynamic RF channel shutdown and RF module fast deep dormancy. For details, see 5.7.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the mutually exclusive relationship between timing carrier shutdown and RF module fast deep dormancy. For details, see 6.1.4.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between intelligent carrier shutdown and RF module fast deep dormancy. For details, see 6.2.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between energy saving based on flexible frequency-domain scheduling and RF module fast deep dormancy. For details, see 7.1.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between initial transmission power reduction and RF module fast deep dormancy. For details, see 7.3.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between retransmission power reduction and RF module fast deep dormancy. For details, see 7.3.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between intelligent power control energy saving and RF module fast deep dormancy. For details, see 9.2.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the mutually exclusive relationship between timing carrier shutdown and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy. For details, see 6.1.4.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD High-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the mutually exclusive relationship between timing carrier shutdown and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy. For details, see 6.1.4.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD High-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationships between timing carrier shutdown and initial transmission power reduction and retransmission power reduction. For details, see 6.1.4.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationships between intelligent carrier shutdown and initial transmission power reduction and retransmission power reduction. For details, see 6.2.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationships between fast carrier shutdown and initial transmission power reduction and retransmission power reduction. For details, see 6.3.4.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Replaced the reserved parameter with a formal parameter for the cooperation between TTI-level channel shutdown and Virtual Grid-based Multi-Frequency Coordination, and modified the impact relationship between them. For details, see 5.4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Replaced the reserved parameter with a formal parameter for the cooperation between dynamic RF channel shutdown and Virtual Grid-based Multi-Frequency Coordination, and modified the impact relationship between them. For details, see 5.7.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes

Revised descriptions of baseband processor shutdown and changed the function name controlled by the **BB_SHUTDOWN_POWERSAVING_SW** option of the **gNodeBParam.BaseStationDeepDormancySw** parameter to baseband shutdown for energy saving. For details, see [8.1 Baseband Processor Shutdown](#).

Revised descriptions of the impact relationship between fast carrier shutdown and clock out-of-synchronization detection. For details, see [6.3.4.2.3 Function Impacts](#).

Revised descriptions of "Multi-Carrier Not Coordinated", a possible reason why TTI-level channel shutdown did not take effect. For details, see [5.4.4.2 Activation Verification](#).

Revised descriptions of "Multi-Carrier Not Coordinated", a possible reason why dynamic RF channel shutdown did not take effect. For details, see [5.7.4.2 Activation Verification](#).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-021203	gNodeB Power Saving	4.1 Symbol Power Saving 4.6 pRRU-Level Precise Symbol Power Saving (Low-Frequency TDD) 6.1 Timing Carrier Shutdown
FOFD-031204	Intelligent Scheduling for Power Saving	4.2 Intelligent Scheduling for Power Saving (Low-Frequency TDD) 4.3 Intelligent Scheduling for Power Saving (High-frequency TDD)
FOFD-031205	RF Channel Intelligent Shutdown	5.1 RF Channel Intelligent Shutdown
FOFD-081203	MetaAAU Deep Energy Saving	4.5 Uplink Symbol Power Saving (Low-Frequency TDD) 5.2 MetaAAU Deep Energy Saving (Low-Frequency TDD)

Feature ID	Feature Name	Chapter/Section
FOFD-061203	Intelligent Channel Dynamic Shutdown	5.3 Low-Power RF Channel Dynamic Shutdown (Low-Frequency TDD) 5.4 TTI-Level Channel Shutdown 5.5 Optimal Channel Shutdown (Low-Frequency TDD) 5.6 RF Channel Power Adaptation (Low-Frequency TDD) 5.7 Dynamic RF Channel Shutdown
FOFD-050203	Intelligent Carrier Shutdown	6.2 Intelligent Carrier Shutdown
FOFD-081250	Fast Carrier Shutdown	6.3 Fast Carrier Shutdown
FOFD-060203	Energy Saving Based on Flexible Frequency-Domain Scheduling	7.1 Energy Saving Based on Flexible Frequency-Domain Scheduling
FOFD-100252	AMC-based Precise Power Control Energy Saving	7.3 AMC-based Precise Power Control Energy Saving
FOFD-071904	Intelligent Multi-RF-Module Coordinated Energy Saving	9.1 Multi-RF-Module Coordinated Energy Saving
FOFD-071903	Intelligent Power Control Energy Saving	9.2 Intelligent Power Control Energy Saving
FBFD-010025	Basic O&M Package	10 Energy Consumption Monitoring
Note: For details about FOFD-030216 pRRU Deep Dormancy, see <i>Multi-RAT Coordinated Energy Saving</i> .		

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Symbol power saving	None	4.1 Symbol Power Saving
Intelligent scheduling for power saving	None	4.2 Intelligent Scheduling for Power Saving (Low-Frequency TDD) 4.3 Intelligent Scheduling for Power Saving (High-frequency TDD)
MetaAAU beam power saving	None	4.4 MetaAAU Beam Power Saving (Low-Frequency TDD)
Uplink symbol power saving	None	4.5 Uplink Symbol Power Saving (Low-Frequency TDD)
pRRU-level precise symbol power saving	Supported only in SA networking	4.6 pRRU-Level Precise Symbol Power Saving (Low-Frequency TDD)
RF channel intelligent shutdown	None	5.1 RF Channel Intelligent Shutdown
MetaAAU deep energy saving	None	5.2 MetaAAU Deep Energy Saving (Low-Frequency TDD)
Low-power RF channel dynamic shutdown	None	5.3 Low-Power RF Channel Dynamic Shutdown (Low-Frequency TDD)
TTI-level channel shutdown	None	5.4 TTI-Level Channel Shutdown

Function Name	Difference	Chapter/Section
Optimal channel shutdown	None	5.5 Optimal Channel Shutdown (Low-Frequency TDD)
RF channel power adaptation	None	5.6 RF Channel Power Adaptation (Low-Frequency TDD)
Dynamic RF channel shutdown	None	5.7 Dynamic RF Channel Shutdown
Timing carrier shutdown	None	6.1 Timing Carrier Shutdown
Intelligent carrier shutdown	Supported in SA networking and hybrid SA and NSA networking. In NSA networking, this function works in intra-base-station cases but not inter-base-station cases.	6.2 Intelligent Carrier Shutdown
Fast carrier shutdown	Supported in SA networking and hybrid SA and NSA networking. In NSA networking, this function works in intra-base-station cases but not inter-base-station cases.	6.3 Fast Carrier Shutdown
Multi-carrier joint carrier shutdown	None	6.4 Multi-Carrier Joint Carrier Shutdown
Multi-frequency deep carrier shutdown	None	6.5 Multi-Frequency Deep Carrier Shutdown (Low-Frequency TDD)
Energy saving based on flexible frequency-domain scheduling	None	7.1 Energy Saving Based on Flexible Frequency-Domain Scheduling
MetaAAU power reduction for cell-center UEs	None	7.2 MetaAAU Power Reduction for Cell-Center UEs (Low-Frequency TDD)
AMC-based Precise Power Control Energy Saving	None	7.3 AMC-based Precise Power Control Energy Saving

Function Name	Difference	Chapter/Section
Baseband processor shutdown	None	8.1 Baseband Processor Shutdown
Multi-RF-module coordinated energy saving	None	9.1 Multi-RF-Module Coordinated Energy Saving
Intelligent power control energy saving	None	9.2 Intelligent Power Control Energy Saving
Energy consumption monitoring	None	10 Energy Consumption Monitoring

NOTE

More common messages are involved on the gNodeB side in SA networking than in NSA networking. Therefore, when the network architecture is changed from NSA to SA, the basic energy consumption of RF modules may increase.

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Symbol power saving	Supported in both low and high frequency bands	4.1 Symbol Power Saving
Intelligent scheduling for power saving	- Power saving scheduling: Supported in both low and high frequency bands. - Deep symbol power saving: Supported only in low frequency bands.	4.2 Intelligent Scheduling for Power Saving (Low-Frequency TDD) 4.3 Intelligent Scheduling for Power Saving (High-frequency TDD)
MetaAAU beam power saving	Supported only in low frequency bands	4.4 MetaAAU Beam Power Saving (Low-Frequency TDD)

Function Name	Difference	Chapter/Section
Uplink symbol power saving	Supported only in low frequency bands	4.5 Uplink Symbol Power Saving (Low-Frequency TDD)
pRRU-level precise symbol power saving	Supported only in low frequency bands	4.6 pRRU-Level Precise Symbol Power Saving (Low-Frequency TDD)
RF channel intelligent shutdown	Supported in both low and high frequency bands	5.1 RF Channel Intelligent Shutdown
MetaAAU deep energy saving	Supported only in low frequency bands	5.2 MetaAAU Deep Energy Saving (Low-Frequency TDD)
Low-power RF channel dynamic shutdown	Supported only in low frequency bands	5.3 Low-Power RF Channel Dynamic Shutdown (Low-Frequency TDD)
TTI-level channel shutdown	Supported only in low frequency bands	5.4 TTI-Level Channel Shutdown
Optimal channel shutdown	Supported only in low frequency bands	5.5 Optimal Channel Shutdown (Low-Frequency TDD)
RF channel power adaptation	Supported only in low frequency bands	5.6 RF Channel Power Adaptation (Low-Frequency TDD)
Dynamic RF channel shutdown	Supported only in low frequency bands	5.7 Dynamic RF Channel Shutdown
Timing carrier shutdown	Supported in both low and high frequency bands (However, IF digital processor shutdown for RF modules is supported only in low frequency bands.)	6.1 Timing Carrier Shutdown
Intelligent carrier shutdown	- Precise shutdown mode: Supported only in low frequency bands. - Non-precise shutdown mode: Supported in both low and high frequency bands.	6.2 Intelligent Carrier Shutdown

Function Name	Difference	Chapter/Section
Fast carrier shutdown	Supported only in low frequency bands	6.3 Fast Carrier Shutdown
Multi-carrier joint carrier shutdown	Supported only in low frequency bands	6.4 Multi-Carrier Joint Carrier Shutdown
Multi-frequency deep carrier shutdown	Supported only in low frequency bands	6.5 Multi-Frequency Deep Carrier Shutdown (Low-Frequency TDD)
Energy saving based on flexible frequency-domain scheduling	Supported only in low frequency bands	7.1 Energy Saving Based on Flexible Frequency-Domain Scheduling
MetaAAU power reduction for cell-center UEs	Supported only in low frequency bands	7.2 MetaAAU Power Reduction for Cell-Center UEs (Low-Frequency TDD)
AMC-based Precise Power Control Energy Saving	Supported only in low frequency bands	7.3 AMC-based Precise Power Control Energy Saving
Baseband processor shutdown	Supported only in low frequency bands	8.1 Baseband Processor Shutdown
Multi-RF-module coordinated energy saving	Supported only in low frequency bands	9.1 Multi-RF-Module Coordinated Energy Saving
Intelligent power control energy saving	Supported only in low frequency bands	9.2 Intelligent Power Control Energy Saving
Energy consumption monitoring	Supported in both low and high frequency bands	10 Energy Consumption Monitoring

3 Overview

The expanding wireless services add to network scale and increase the overall equipment energy consumption. The resulting growth of base station energy consumption is a major contributor to the increasing operating expense (OPEX) of telecom operators. Huawei's energy saving solutions are launched against this background. They help operators improve equipment operating efficiency, reduce equipment energy consumption and OPEX, and enhance product competitiveness. The solutions can be implemented at the device, site, and network levels.

- Device level

The energy saving solutions at this level are implemented on RF modules from the following dimensions and involve individual cells:

- Time domain

When there is no data to be transmitted in symbols, power amplifiers (PAs) and radio-on-a-chip (RoC) systems are shut down to reduce the energy consumption of RF modules.

- Space domain

In light-load scenarios, certain RF channels of RF modules are shut down to reduce RF module energy consumption.

- Frequency domain

In light-load scenarios, certain carriers of RF modules or the entire RF modules are shut down to reduce their energy consumption.

- Power domain

In light-load scenarios, the power over the air interface is decreased to reduce the PA power consumption.

- Site level

The energy saving solutions at this level are implemented on power supply systems, baseband equipment, and others.

- Network level

The energy saving solutions at this level are implemented on RF modules and require inter-cell coordination.

Energy saving functions fall into different categories, as listed in [Table 3-1](#) and [Table 3-2](#).

Table 3-1 Energy saving classification (low-frequency)

Category	Sub-Category	Energy Saving Function
Device level	Time domain	4.1 Symbol Power Saving 4.2 Intelligent Scheduling for Power Saving (Low-Frequency TDD) 4.4 MetaAAU Beam Power Saving (Low-Frequency TDD) 4.5 Uplink Symbol Power Saving (Low-Frequency TDD) 4.6 pRRU-Level Precise Symbol Power Saving (Low-Frequency TDD)
	Space domain	5.1 RF Channel Intelligent Shutdown 5.3 Low-Power RF Channel Dynamic Shutdown (Low-Frequency TDD) 5.4 TTI-Level Channel Shutdown 5.5 Optimal Channel Shutdown (Low-Frequency TDD) 5.6 RF Channel Power Adaptation (Low-Frequency TDD) 5.7 Dynamic RF Channel Shutdown
	Frequency domain	6.1 Timing Carrier Shutdown 6.2 Intelligent Carrier Shutdown 6.3 Fast Carrier Shutdown 6.4 Multi-Carrier Joint Carrier Shutdown 6.5 Multi-Frequency Deep Carrier Shutdown (Low-Frequency TDD) For details about pRRU deep dormancy, see <i>Multi-RAT Coordinated Energy Saving</i>.
	Power domain	7.1 Energy Saving Based on Flexible Frequency-Domain Scheduling 7.2 MetaAAU Power Reduction for Cell-Center UEs (Low-Frequency TDD) 7.3 AMC-based Precise Power Control Energy Saving
Site level	N/A	8.1 Baseband Processor Shutdown For the energy saving solutions applicable to power supply systems, see <i>Green Site</i>.
Network level	N/A	9.1 Multi-RF-Module Coordinated Energy Saving 9.2 Intelligent Power Control Energy Saving

Table 3-2 Energy saving classification (high-frequency TDD)

Category	Sub-Category	Energy Saving Function
Device level	Time domain	4.1 Symbol Power Saving 4.3 Intelligent Scheduling for Power Saving (High-frequency TDD)
	Space domain	5.1 RF Channel Intelligent Shutdown
	Frequency domain	6.1 Timing Carrier Shutdown 6.2 Intelligent Carrier Shutdown
	Power domain	None
Site level	N/A	None
Network level	N/A	None

 NOTE

- One energy saving function may fall into more than one category. The categories listed in [Table 3-1](#) and [Table 3-2](#) are only the major categories of energy saving functions.
- When multi-RAT carriers share a PA, energy saving gains by shutting down PAs can be obtained only after carriers of all RATs on the RF channels enter the energy saving state.

Operators can use the MAE-Access to query base stations' or base station equipment's energy consumption distribution and changes. For details, see [10 Energy Consumption Monitoring](#).

To prevent possible hardware faults caused by an excessively large temperature differential, this function may not take effect or may be terminated when the temperature differential of RF modules is excessively large. Energy saving gains may decrease in this case.

4.1.2.2 Impacts

The impacts between this function and other functions are described in [4.1.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.1.3 Requirements

4.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021203	gNodeB Power Saving	NR0S0GNBPS00	per Cell

NOTE

- When NR and LTE (including NB-IoT) share a PA, the licenses for Multi-RAT Coordinated Symbol Power Saving (NR), Multi-RAT Coordinated Symbol Power Saving (LTE), Symbol Power Saving (NR), and Symbol Power Saving (LTE) are required for symbol power saving to take effect.
- For multimode license requirements, see section "LTE and NR Coordinated Symbol Power Saving" in *Multi-RAT Coordinated Energy Saving*. For single-mode license requirements, see section "Symbol Power Saving" in *Energy Conservation and Emission Reduction* and *NB-IoT Enhancements (FDD)* in the eRAN feature documentation.
- The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR coordinated symbol power saving	- On the LTE side: SymbolPwrSaving.MultiRatSymbolShutdownSw - On the NR side: MULTI_RAT_SYMBOL_SHUTDOWN_SW option of NRDUCellAlgoSwitch.PowerSavingSwitch	<i>Multi-RAT Coordinated Energy Saving</i>	In multimode scenarios, as there are both NR and LTE carriers, PAs can be shut down only during downlink symbol periods where neither LTE symbols nor NR symbols carry data. Therefore, in these scenarios, basic symbol power saving requires LTE and NR coordinated symbol power saving to be enabled.

4.1.3.2.2 Mutually Exclusive Functions

None

4.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option (inter-base-station multi-carrier scenarios) of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected - Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitc	<i>Energy Conservation and Emission Reduction</i>	When fast carrier shutdown takes effect: - If the RF module is not serving any other NR carrier, symbol power saving does not take effect. - If the RF module is serving other NR carriers, the status of symbol power saving is not affected.

RAT	Function Name	Function Switch	Reference	Description
		<i>h</i> parameter selected		
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTTI_CARR_SD_SW or INTER_GNB_MULTTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of symbol power saving is not affected.
High-frequency TDD	Intelligent carrier shutdown in non-precise mode	• INTRA_GNB_MULTTI_CARR_SD_SW or INTER_GNB_MULTTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected • MULTI_CARR_PRECISE_SHDN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch <i>h</i> parameter deselected	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of symbol power saving is not affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When timing carrier shutdown takes effect: - If the RF module is not serving any other NR carrier, symbol power saving does not take effect. - If the RF module is serving other NR carriers, the status of symbol power saving is not affected.
Low-frequency TDD High-frequency TDD	RF module deep dormancy	RRU.DORMANCY_SW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Symbol power saving does not take effect when RF module deep dormancy is in effect.
Low-frequency TDD High-frequency TDD	RF module super dormancy	RRU.DORMANCY_SW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Symbol power saving does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR intelligent carrier shutdown	NR: LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter LTE: CellAlgoExtSwitch.LnrSmartCarrierShutdownSw	<i>Multi-RAT Coordinated Energy Saving</i>	When LTE and NR intelligent carrier shutdown takes effect: - If the RF module is not serving any other NR carrier, symbol power saving does not take effect. - If the RF module is serving other NR carriers, the status of symbol power saving is not affected.
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS W (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over symbol power saving when both functions are enabled. Symbol power saving does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH resource allocation enhancement	PDCCH_RES_ALL OC_ENH_SW option of the NRDUCellPdcch.PowerSavingExtSwitch parameter	<i>Channel Management</i>	When 2T, 4T, or 8T RF modules are used, enabling PDCCH resource allocation enhancement will decrease the PDCCH power for some UEs and may cause a decrease in the cell downlink rate. Therefore, you are not advised to enable symbol power saving together with PDCCH resource allocation enhancement when 2T, 4T, or 8T RF modules are used. If both symbol power saving and PDCCH resource allocation enhancement are enabled, selecting the SYM_SHUTDOWN_UE_WEIGHT_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter can reduce the impact on cell throughput.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Maximum Common DCI Power Offset	NRDUCellChnPower.MaxCommonDCIPwrOffset	<i>Power Control</i>	If the common channel power offset (any of the three parameters) is set to a negative value, the common channel power will decrease, affecting cell throughput in symbol power saving scenarios. Therefore, you are not advised to set any of these parameters to a negative value when symbol power saving is enabled in scenarios where 2T, 4T, or 8T RF modules are used. If the common channel power offset is a negative value and symbol power saving is enabled, selecting the SYM_SHUTDOWN_CELL_WEIGHT_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter can reduce the impact on cell throughput.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Max Msg2 Power Offset	NRDUCellChnPwr r.Msg2MaxPwrOf fset	None	If the common channel power offset (any of the three parameters) is set to a negative value, the common channel power will decrease, affecting cell throughput in symbol power saving scenarios. Therefore, you are not advised to set any of these parameters to a negative value when symbol power saving is enabled in scenarios where 2T, 4T, or 8T RF modules are used. If the common channel power offset is a negative value and symbol power saving is enabled, selecting the SYM_SHUTDOWN_CELL_WEIGHT_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter can reduce the impact on cell throughput.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Max Msg4 Power Offset	NRDUCellChnPwr r.Msg4MaxPwrOf fset	None	If the common channel power offset (any of the three parameters) is set to a negative value, the common channel power will decrease, affecting cell throughput in symbol power saving scenarios. Therefore, you are not advised to set any of these parameters to a negative value when symbol power saving is enabled in scenarios where 2T, 4T, or 8T RF modules are used. If the common channel power offset is a negative value and symbol power saving is enabled, selecting the SYM_SHUTDOWN_CELL_WEIGHT_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter can reduce the impact on cell throughput.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	When both RedCap Introduction and symbol power saving take effect, the value of the N.PowerSaving.SymbolShutdown.Dur counter (indicating the symbol power saving duration) may decrease.
Low-frequency TDD	Merged scheduling for downlink RLC status reports	DL_RLC_STAT_REPORT_MERGE_SWITCH option of the NRDUCellDISch.DISchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When both merged scheduling for downlink RLC status reports and symbol power saving take effect, the value of the N.PowerSaving.SymbolShutdown.Dur counter (indicating the symbol power saving duration) may increase.
Low-frequency TDD	Service-priority-based paging message retransmission	NRDUCellPagingConfig.PriBasedPagingRetransPol	<i>VoNR</i>	After service-priority-based paging message retransmission takes effect, symbol power saving yields fewer energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic PUCCH codebook specification improvement for CA UEs	PUCCH_DYN_CB_SPECS_IMP_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Carrier Aggregation</i>	After dynamic PUCCH codebook specification improvement takes effect, the number of scheduling times for CA UEs increases, decreasing the value of N.PowerSaving.SymbolShutdown.Dur.

4.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

Low-frequency bands:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

High-frequency TDD bands:

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Low-frequency bands:

- All NR-capable AAUs/RRUs, except those for which the **Description** parameter value includes "V3" in the command output of **DSP BRDMFRINFO**, support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

- All NR TDD-capable pRRUs support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

High-frequency TDD bands:

Only HAAU5233w and HAAU5235w support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.1.3.4 Networking

When multiple RATs share an RF module, the NR cells enter the symbol power saving state only when this function takes effect for activated cells of all RATs sharing the same channel. The NR cells exit the symbol power saving state if any cell of any RAT exits this state.

4.1.3.5 Cells

This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**. This function can be configured but does not take effect in cells with the **NRDUCell.DuplexMode** parameter set to **CELL_SUL**.

4.1.3.6 Others

None

4.1.4 Operation and Maintenance

4.1.4.1 Data Configuration

4.1.4.1.1 Data Preparation

Table 4-1 describes the parameters used for function activation. Function optimization configurations are not involved. This section does not describe parameters related to cell establishment.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Symbol Shutdown Switch	NRDUCellAlgoSwitch.SymbolShutdownSwitch	Turn on this switch when symbol power saving is required. If an RF module serves multiple NR cells, it is recommended that the same setting for this switch be used across all NR cells.

Parameter Name	Parameter ID	Setting Notes
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	It is recommended that the SYM_SHUTDOWN_CSIRS_PROTECT_SW option of this parameter be selected to enable protection for CSI-RS symbols against symbol power saving. This prevents possible CSI-RS transmission failures. This option is selected by default.

4.1.4.1.2 Using MML Commands

Before using MML commands, refer to [4.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling symbol power saving  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SymbolShutdownSwitch=ON;  
//(Optional) Enabling protection for CSI-RS symbols against symbol power saving  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingSwitch=SYM_SHUTDOWN_CSIRS_PROTECT_SW-1;
```

Deactivation Command Examples

```
//Disabling symbol power saving  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SymbolShutdownSwitch=OFF;  
//(Optional) Disabling protection for CSI-RS symbols against symbol power saving  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingSwitch=SYM_SHUTDOWN_CSIRS_PROTECT_SW-0;
```

4.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.1.4.2 Activation Verification

1. Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Symbol Shutdown Switch** parameter is **On** in the command output, symbol power saving has been enabled.
2. Run the **DSP NRDUCELL** command to check the symbol power saving state of the cell.
 - If the value of the **Symbol Shutdown State** parameter is **Symbol Power Saving**, symbol power saving has taken effect in at least one transmission reception point (TRP) of the cell.
 - If the value of the **Symbol Shutdown State** parameter is **Not Started**, symbol power saving has not taken effect in the cell. The possible reasons are as follows:
 - The RF module does not support symbol power saving. For details about the requirements on RF modules, see [RF Modules](#).
 - The license item "gNodeB Power Saving" has not been acquired.
 - The **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**.
3. (Optional) For a hyper cell or combined cell, run the **DSP NRDUCELLTRP** command to query the symbol power saving state of each TRP in the cell.
 - If the value of the **Power Saving State** parameter is **Symbol Power Saving:Enabled** for a TRP, symbol power saving has taken effect in the TRP.
 - If the value of the **Power Saving State** parameter is **Symbol Power Saving:Disabled** for a TRP, symbol power saving has not taken effect in the TRP. The possible reasons are as follows:
 - The RF module does not support symbol power saving. For details about the requirements on RF modules, see [RF Modules](#).
 - The license items "gNodeB Power Saving" are insufficient.

NOTE

For details about the consumption principles of feature license items in hyper cells or combined cells, see *Hyper Cell* or *Cell Combination*.

4.1.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after symbol power saving is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.
- The duration of symbol power saving can be observed using the counter **N.PowerSaving.SymbolShutdown.Dur** (for a cell) on the MAE-Access. The policy of counter measurement upon carrier shutdown is determined by the

function of energy saving counter optimization upon carrier shutdown, which is controlled by the **COUNTER_OPT_IN_CARROFF_SW** option of the **gNB PwrSavingAlgo.PwrSavingAlgoSwitch** parameter. If this option is deselected in low-frequency scenarios and a carrier has been shut down, the **N.PowerSaving.SymbolShutdown.Dur** counter counts in the measured symbol power saving duration of the carrier. If this option is selected in low-frequency scenarios and a carrier has been shut down, the **N.PowerSaving.SymbolShutdown.Dur** counter does not count in the symbol power saving duration of the carrier. In high-frequency scenarios, if a carrier has been shut down, the **N.PowerSaving.SymbolShutdown.Dur** counter counts in the measured symbol power saving duration of the carrier, regardless of whether this option is selected or deselected.

4.2 Intelligent Scheduling for Power Saving (Low-Frequency TDD)

4.2.1 Principles

In symbols without data transmission, symbol power saving enables the base station to shut down the PAs of RF modules in real time, reducing power consumption. **On the basis of symbol power saving, intelligent scheduling for power saving is introduced to further reduce power consumption. This function increases the number of symbols without data transmission or adds shutdown depth to such symbols.** Hence, there are two sub-functions:

- Power saving scheduling: This type of intelligent time-domain scheduling frees more symbols from data transmission, allowing symbol power saving to take effect for a longer period and ultimately saving more power.
- Deep symbol power saving: The components shut down for symbol power saving are intelligently adjusted based on the capabilities of RF modules, allowing symbol power saving to take effect at a deeper level, again, saving more power.

4.2.1.1 Power Saving Scheduling

Power saving scheduling uses the following four methods to free more symbols from data transmission: **Dynamic Adjustment of the RMSI Broadcast Interval**, **Dynamic Adjustment of the Number of PFs**, **Data Convergence in Symbols**, and **Data Convergence in Slots**.

Power saving scheduling can be enabled by selecting the **POWER_SAVING_SCHEDULE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter. **Table 4-2** describes the parameters in the **NRDUCellPowerSaving** MO that specify the effective period, load threshold, and other information under an energy saving policy of power saving scheduling.

Table 4-2 Parameters related to the power saving scheduling policy

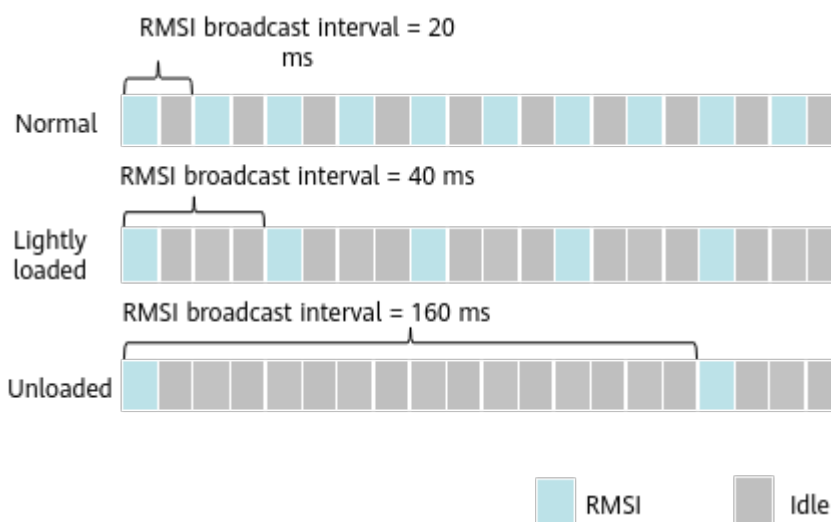
Parameter Name	Parameter ID	Description
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	This parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	This parameter specifies the energy saving type indicated by an energy saving policy index. Power saving scheduling requires that this parameter be set to POWER_SAVING_SCHEDULE .

Parameter Name	Parameter ID	Description
Start Time	NRDUCellPowerSaving.StartTime	These parameters specify the time period in which power saving scheduling takes effect, as well as the evaluation period and conditions for a cell to enter and exit the power saving scheduling state. Within the time period specified by NRDUCellPowerSaving.StartTime and NRDUCellPowerSaving.StopTime for power saving scheduling to take effect for a cell, the base station periodically determines whether the cell enters the power saving scheduling state. The period is specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter. • If the downlink PRB usage in a cell is less than or equal to the value of NRDUCellPowerSaving.DlPrbThld, the cell automatically enters the power saving scheduling state. • If the downlink PRB usage in a cell is greater than the sum of the values of NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset or the current time is not within the period during which power saving scheduling can take effect, the cell automatically exits the power saving scheduling state.
Stop Time	NRDUCellPowerSaving.StopTime	
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	
Note: The following formulas are used to calculate the downlink PRB usage of a cell. • Downlink PRB usage of a common cell = (Average number of PRBs used by PDSCH DRB services – Average number of extended PRBs used by the PDSCH)/Average number of available downlink PRBs • Downlink PRB usage of a TRP of a hyper cell or combined cell = (Average number of PRBs used by PDSCH DRB services in the TRP – Average number of downlink extended PRBs used in the TRP)/Average number of available downlink PRBs in the TRP		

Dynamic Adjustment of the RMSI Broadcast Interval

When the base station detects that the network is under light or no load, it prolongs the remaining minimum system information (RMSI) broadcast interval to free more symbols from data transmission. When the base station detects that the network load is normal, it restores the RMSI broadcast interval. As shown in [Figure 4-2](#), the RMSI broadcast interval can be extended to a maximum of 160 ms.

Figure 4-2 Dynamic adjustment of the RMSI broadcast interval (taking a configured interval of 20 ms as an example)



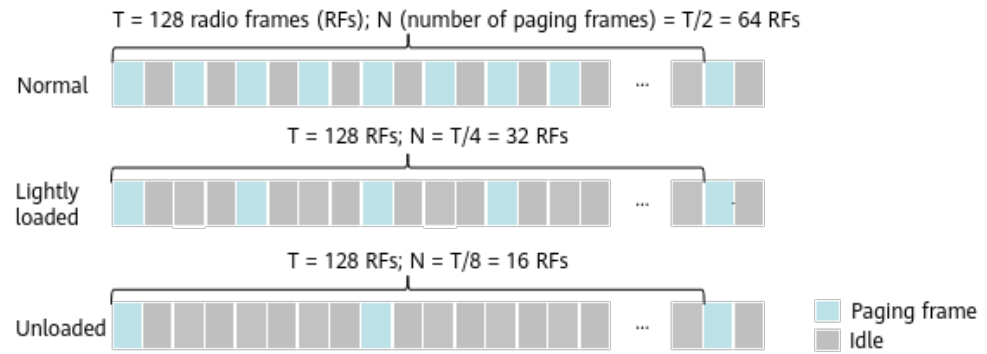
The RMSI broadcast interval can be dynamically adjusted at the cell or beam level, depending on the setting of the **NRDUCellPowerSaving.RmsiAdjustMode** parameter.

- If this parameter is set to **CELL**, the RMSI broadcast interval is dynamically adjusted at the cell level. The base station dynamically adjusts the RMSI broadcast interval for all synchronization signal and PBCH block (SSB) beams when the number of RRC_CONNECTED UEs in the cell (indicated by **N.User.RRConn.Avg**) is less than or equal to the **NRDUCellPowerSaving.UserNumThld** parameter value.
- If this parameter is set to **BEAM**, the RMSI broadcast interval is dynamically adjusted at the beam level. The base station dynamically adjusts the RMSI broadcast interval for an individual SSB beam when the number of RRC_CONNECTED UEs in the beam (indicated by **N.User.RRConn.Avg**) is less than or equal to the **NRDUCellPowerSaving.UserNumThld** parameter value.

Dynamic Adjustment of the Number of PFs

When the base station detects that the network is under light or no load, it reduces the number of paging frames (PFs) sent within a paging cycle (referred to as T) to increase the number of symbols with no data transmission. When the base station detects that the network load is normal, it restores the number of PFs sent within T . [Figure 4-3](#) shows how this function works.

Figure 4-3 Dynamic adjustment of the number of PFs



Dynamic adjustment of the number of PFs takes effect only when power saving scheduling is in effect and the **PAGING_FRAME_NUM_DYN_ADJ_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter is selected.

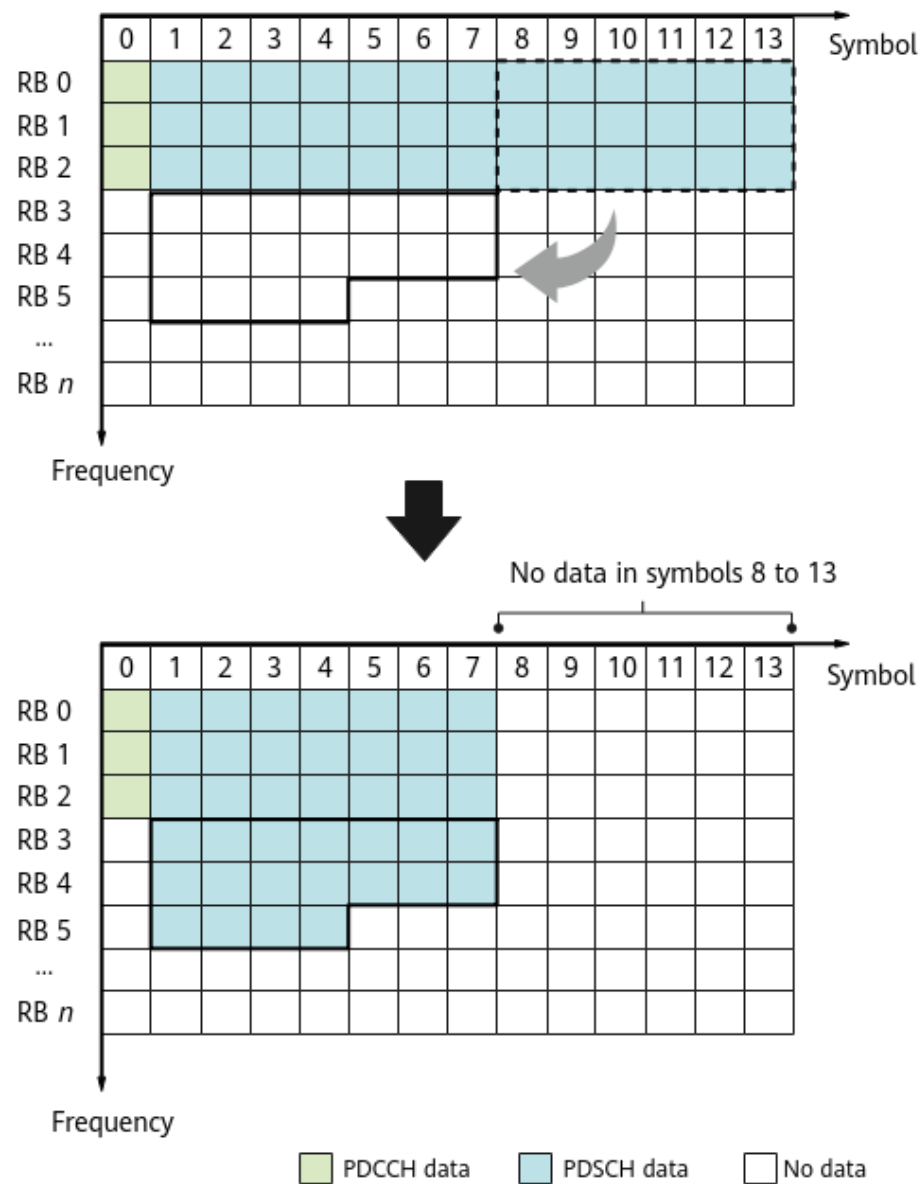
After the **PAGING_FRAME_NUM_DYN_ADJ_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter is deselected, dynamic adjustment of the number of PFs stops up to 5 minutes later.

Data Convergence in Symbols

Based on the size of the physical downlink shared channel (PDSCH) data to be transmitted, the base station converges the PDSCH data on certain symbols within a slot so that PDSCH data occupies fewer time-domain resources but more frequency-domain resources, increasing the number of symbols with no data transmission. The number of occupied symbols after convergence is specified by the **NRDUCellPowerSaving.SymbolCompressLength** parameter.

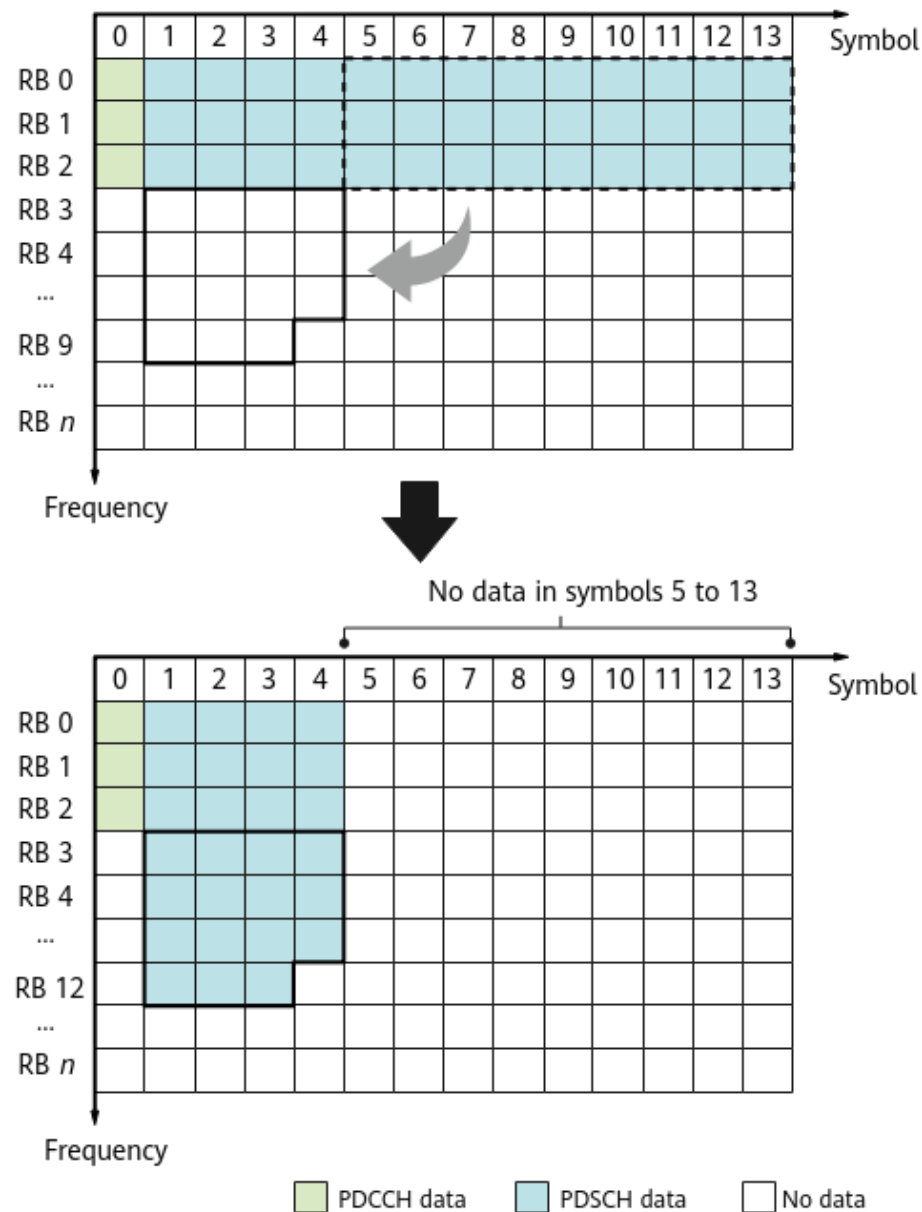
- Parameter set to **7SYMBOL**
 - If seven symbols can carry all PDSCH data, the PDSCH data is converged on seven symbols, as shown in [Figure 4-4](#).
 - Otherwise, the PDSCH data is converged on more than seven symbols.

Figure 4-4 Data convergence on seven symbols



- Parameter set to **4SYMBOL**
 - If four symbols can carry all PDSCH data, the PDSCH data is converged on four symbols, as shown in [Figure 4-5](#).
 - Otherwise, the PDSCH data is converged on more than four symbols.

Figure 4-5 Data convergence on four symbols



Data Convergence in Slots

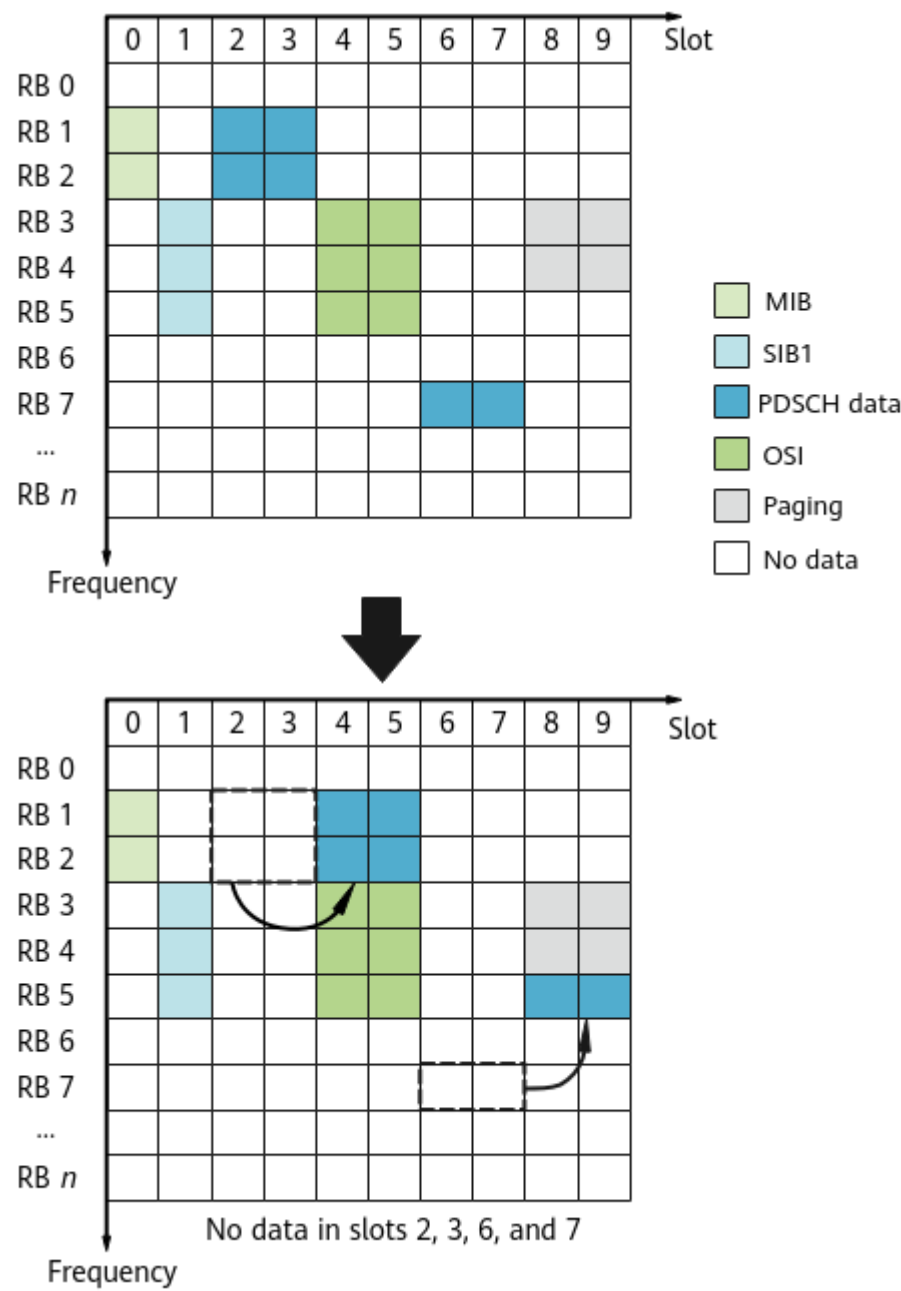
As shown in [Figure 4-6](#), data convergence in slots indicates that the base station converges and sends PDSCH data in the slots corresponding to the master information block (MIB), system information block 1 (SIB1), other system information (OSI), or paging. This concentrates PDSCH data scheduling in the time domain and frees more symbols from data transmission. However, this process also increases the scheduling delay to a certain extent.

The **NRDUCellPowerSaving.RlcFirstPktDelayThld** parameter specifies the first packet delay threshold for data convergence in slots.

- If the first packet delay of a UE is less than or equal to this threshold, data to be scheduled is converged in slots and then transmitted.

- Otherwise, data is directly scheduled for the UE.

Figure 4-6 Data convergence in slots



In NR multi-carrier scenarios, services are randomly distributed on carriers. There are few slots in which there is no data transmission on any of the carriers (referred to as common no-data slots). In the case that the carriers share an RF channel, coordination for data convergence in slots can be used to align scheduling slots between the carriers in order to spare more common no-data slots. This function further lifts the gains offered by data convergence in slots and reduces the energy consumption of the base station.

Coordination for data convergence in slots is enabled by selecting the **SLOT_DATA_CONVERGENCE_COORD_SW** option of the

NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter on condition that power saving scheduling has been enabled. When multiple carriers share an RF channel, this option must be selected for all the NR cells that share this RF channel. Coordination for data convergence in slots works with a maximum of four TRPs. It does not take effect when there are more than four TRPs.

For specific service types, cells can dynamically exit data convergence in slots. This is known as service-type-based dynamic exit from data convergence in slots. When any UE with a specific type of service is to be scheduled in a slot in a cell, the cell exits the data convergence in slots state in this slot. That means the data is scheduled without being postponed. This type of exit ensures the performance of UEs performing that type of service. The service types involved are specified by the **gNBQciBearer.ServiceType** parameter.

Service-type-based dynamic exit from data convergence in slots takes effect when all of the following conditions are met:

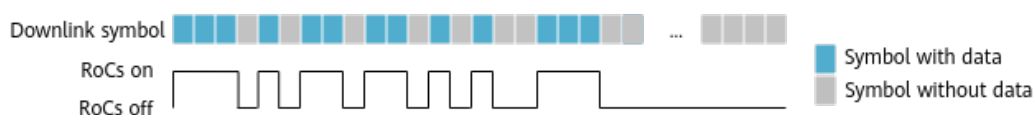
- Power saving scheduling is enabled.
- The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.
- The **SLOT_DATA_CONVERGENCE_EXIT_SW** option of the **NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter is selected in the group identified by the **NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter that is set to the same value as the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter for a specific service type.

4.2.1.2 Deep Symbol Power Saving

Among base station devices, RoC systems of RF modules consume much power. They consume power even when there is no signal output. Deep symbol power saving is introduced to reduce energy consumption while ensuring data transmission integrity, as shown in [Figure 4-7](#).

- When the base station detects that no data needs to be transmitted in a downlink symbol, it shuts down the RoC systems of RF modules in real time to reduce energy consumption.
- When the base station detects that data needs to be transmitted in a downlink symbol, it starts the RoC systems in real time to ensure data transmission integrity.

Figure 4-7 Working principles of deep symbol power saving



Deep symbol power saving can be enabled by selecting the **DEEP_SYMBOL_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter.

4.2.2 Network Analysis

4.2.2.1 Benefits

When the network is under light or no load, enabling intelligent scheduling for power saving can further increase energy saving gains and reduce power consumption on the basis of **symbol power saving**.

Dynamic adjustment of the RMSI broadcast interval provides no energy saving gains if no RMSI is broadcast on SSB beams or the RMSI broadcast interval has been set to 160 ms.

Service-type-based dynamic exit from data convergence in slots relieves the impact of data convergence in slots on specific service types but reduces energy saving gains.

4.2.2.2 Impacts

The impacts between this function and other functions are described in [4.2.3.2.3 Function Impacts](#).

Deep symbol power saving has no impact on the network. Power saving scheduling has the following impacts on the network:

- The access delay of RRC_IDLE UEs camping on the cell increases slightly. For RRC_CONNECTED UEs that have accessed the cell, the maximum waiting delay reaches 10% of the packet delay budget (PDB) corresponding to the QoS class identifier (QCI) of the bearer carrying the data packets.
- If the **gNBPDcpParamGroup.DlDataPdcpSplitMode** parameter is set to **SCG_AND_MCG** in NSA networking, data convergence in slots increases the UE delay, which may increase the proportion of data distributed to LTE and the traffic volume on the LTE side.
- As common messages may be sent at longer intervals, the following changes may occur:
 - The number of random accesses decreases.
 - The average CCE aggregation level decreases.
 - The CCE usage decreases.
 - The PRB usage indicated by **Downlink Resource Block Utilizing Rate (DU)** decreases.
- When dynamic adjustment of the number of PFs takes effect, UEs are informed of the changes in the number of PFs through updated system information. If the UEs do not receive the updated system information due to poor signal quality, paging failures may occur.
- With data convergence in symbols in effect, the downlink PRB usage of the cell is higher when the **NRDUCellPowerSaving.SymbolCompressLength** parameter is set to **4SYMBOL** than when this parameter is set to **7SYMBOL**.
- When data convergence in slots takes effect:
 - The transmission delay for the first downlink packet of a UE may be prolonged.
 - The user-perceived rate may change as follows:
 - Small packets are aggregated into large packets after downlink bundled scheduling, and therefore UEs are more likely to be paired for MU-MIMO. If user data in the local cell and that in a neighboring

cell are scheduled using the same time-domain resources, the downlink interference between the local cell and the neighboring cell may increase. As a result, the block error rate (BLER) and number of retransmissions increase, and the user-perceived rate may decrease.

- The number of PDSCH scheduling times decreases after downlink bundled scheduling, and therefore the interference to the channel state information reference signals (CSI-RSs) may decrease. This may increase the values of certain counters related to channel quality indicators (CQIs) (**N.ChMeas.CQI.SingleCW.0** to **N.ChMeas.CQI.SingleCW.15**), possibly increasing the MCS index and the user-perceived rate and decreasing the PRB usage.
- When coordination for data convergence in slots takes effect:
 - The CPU usage increases because scheduling information needs to be exchanged in real time between the cells that share an RF channel.
 - Due to scheduling slot alignment between carriers, scheduling needs to be performed on all the carriers when scheduling is required on any of these carriers. As a result, the number of scheduling times on each carrier may increase, decreasing the value of the **N.PowerSaving.SymbolShutdown.Dur** counter.
 - As the number of scheduling times on each carrier may increase, small packets are less likely to be aggregated into large packets after downlink bundled scheduling. As a result, the user-perceived rate may decrease.

4.2.3 Requirements

4.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-031204	Intelligent Scheduling for Power Saving	NR0S000PSS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShut downSwitch	<i>Energy Conservation and Emission Reduction</i>	Power saving scheduling and deep symbol power saving depend on symbol power saving.
Low-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Service-type-based dynamic exit from data convergence in slots takes effect only if power saving scheduling has been enabled.

4.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The period specified for power saving scheduling in a cell cannot overlap that specified for timing carrier shutdown in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	When distributed massive MIMO is enabled, the NRDUCellPowerSaving.RmsiAdjust Mode parameter cannot be set to BEAM .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Customized Beam Weight Scenario	NRDUCellTrpBeam.CoverageScenario set to CUSTOMIZED_BEAMWEIGHT_SCENARIO	None	If the NRDUCellPowerSaving.RmsiAdjustMode parameter is set to BEAM , the NRDUCellTrpBeam.CoverageScenario parameter cannot be set to CUSTOMIZED_BEAMWEIGHT_SCENARIO .
Low-frequency TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	- Power saving scheduling is incompatible with the primary/backup function in intra-frequency dual-network redundancy networking. - Deep symbol power saving is incompatible with the primary/backup function in intra-frequency dual-network redundancy networking.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	• Power saving scheduling is incompatible with dual-BBU backup in dual-network redundancy networking. • Deep symbol power saving is incompatible with dual-BBU backup in dual-network redundancy networking.
Low-frequency TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	• Power saving scheduling is incompatible with the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios. • Deep symbol power saving is incompatible with the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	eDRX in idle mode	IDLE_MODE_EDRX_SW option of the NRDUCellUePwrSaving.IdleUePwrSavingSwitch parameter	<i>RedCap</i>	If the IDLE_MODE_EDRX_SW option of the NRDUCellUePwrSaving.IdleUePwrSavingSwitch parameter is selected, dynamic adjustment of the number of PFs cannot be enabled.

4.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option (inter-base-station multi-carrier scenarios) of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected - Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch	<i>Energy Conservation and Emission Reduction</i>	When fast carrier shutdown takes effect: - If the RF module is not serving any other NR carrier, deep symbol power saving does not take effect. - If the RF module is serving other NR carriers, the status of deep symbol power saving is not affected. - If the RF module is not serving any other NR carrier, power saving scheduling does not take effect. - If the RF module is serving other NR carriers, the status of power saving scheduling is not affected.

RAT	Function Name	Function Switch	Reference	Description
		<i>h</i> parameter selected		
Low-frequency TDD	LTE and NR multi-band power adaptive scheduling	- LTE: LNR_MULTI_B AND_PWR_ADPT_SCH_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter - NR: LNR_MULTI_B AND_PWR_ADPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When LTE and NR multi-band power adaptive scheduling and power saving scheduling are both enabled, the effects of power saving scheduling are affected. It is recommended that LTE and NR multi-band power adaptive scheduling not be enabled with this function when the traffic of LTE and NR cells is light.
Low-frequency TDD	L/NR Inter-band Deep Coordination	- LTE: LNR_INTER_B AND_DEEP_COORD_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter - NR: LNR_INTER_B AND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPowerShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	Enabling L/NR Inter-band Deep Coordination together with power saving scheduling affects the performance of power saving scheduling, and is therefore not recommended when the traffic of LTE and NR cells is light.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, power saving scheduling and deep symbol power saving do not take effect. • If the RF module is serving other NR carriers, the status of power saving scheduling and that of deep symbol power saving are not affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	If power saving scheduling is in effect and RF channel intelligent shutdown is enabled in the scenarios listed in Table 5-6, power saving scheduling may stop. It is recommended that a larger downlink PRB usage threshold be configured for power saving scheduling in the period intended for both RF channel intelligent shutdown and power saving scheduling to take effect. For 8T8R RF modules, when RF channel intelligent shutdown takes effect, beam-level adjustment of the RMSI broadcast interval in power saving scheduling does not take effect. Therefore, cell-level adjustment of the RMSI broadcast interval is recommended in such cases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both power saving scheduling and energy saving based on flexible frequency-domain scheduling take effect, the gains of energy saving based on flexible frequency-domain scheduling decrease.
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When timing carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, deep symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of deep symbol power saving is not affected.
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both power saving scheduling of intelligent scheduling for power saving and TTI-level channel shutdown take effect, the energy saving gains of TTI-level channel shutdown decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When power saving scheduling (a sub-function of intelligent scheduling for power saving) and dynamic RF channel shutdown are both enabled, the latter function takes effect preferentially.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCY_SW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Power saving scheduling and deep symbol power saving do not take effect when RF module deep dormancy is in effect.
Low-frequency TDD	RF module super dormancy	RRU.DORMANCY_SW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Power saving scheduling and deep symbol power saving do not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR bundled scheduling	LNR_BUNDLED_SCHEDULING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When both power saving scheduling and LTE and NR bundled scheduling are enabled, it is recommended that the NRDUCellPowerSaving.DlPrbThld or NRDUCellPowerSaving.UserNumThld parameter be set to a larger value with a higher load or larger number of UEs. Otherwise, the gains of LTE and NR bundled scheduling may be affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS W (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over power saving scheduling when both functions are enabled. Power saving scheduling does not take effect when RF module fast deep dormancy is in effect. RF module fast deep dormancy takes precedence over deep symbol power saving when both functions are enabled. Deep symbol power saving does not take effect when RF module fast deep dormancy is in effect.
Low-frequency TDD	Downlink RB estimation optimization	RB_ESTIMATE_OPT_SW option of the NRDUCellPdsch.DLPdschAlgoSwitch parameter	<i>Downlink Scheduling</i>	If data convergence in symbols takes effect but downlink RB estimation optimization does not take effect, the user-perceived rate may decrease. It is recommended that downlink RB estimation optimization be enabled when data convergence in symbols is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RB allocation for downlink experience optimization	NRDUCellDLSchRes.DLExpOptRblncreasePct	<i>Downlink Scheduling</i>	When both RB allocation for downlink experience optimization and power saving scheduling take effect, an exit from power saving scheduling may be performed earlier than expected.
Low-frequency TDD	Common channel interference avoidance	SSB_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnlInterfAvoidSwitch parameter	<i>Interference Avoidance</i>	Assume that cell B is a neighboring cell of cell A. If the RMSI broadcast interval of cell A is adjusted, interference may occur between the PDSCH of cell A and common messages of cell B, decreasing the downlink throughput of cell A. It is recommended that common channel interference avoidance be enabled in both cell A and cell B.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced intra-frequency SSB interference avoidance	INTRA_FREQ_SSB_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.CommChnlInterfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	In a hyper cell, the RMSI interval may differ between TRPs because of dynamic adjustment of the RMSI broadcast interval. In this case, the PDSCH of the serving cell involved in this interference avoidance function gives way to the RMSI of all TRPs in the hyper cell.
Low-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgorithm parameter	<i>Carrier Aggregation</i>	When inter-FR CA is enabled, data convergence in symbols does not take effect in the slots where RAR messages are sent.
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When data convergence in slots is enabled for power saving scheduling, a cell will share less power with others, decreasing the gains of power sharing.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SC_H_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When data convergence in slots is enabled for power saving scheduling in a cell, the cell can allocate less power to others, affecting the gains provided by NR multi-band power adaptive scheduling.
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	When data convergence in slots is enabled for power saving scheduling, a larger value of the NRDUCellPowerSaving.RlcFirstPktDelayThld parameter results in a higher degree of data convergence with a longer time. As a result, the number of scheduling failures of BWP2 UEs in a cell increases. When the scheduling failure rate of BWP2 UEs in a cell exceeds the value of the NRDUCellUePwrSaving.Bwp2SchFailRateThld parameter, UEs are more likely to switch from BWP2 back to BWP1.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Time-domain power saving BWP	TDM_BWP2_SWI TCH option of the NRDUCellUePwr Saving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	A larger value of the NRDUCellPowerSaving.RlcFirstPktDelayThld parameter results in a higher degree of data convergence with a longer time. As a result, the number of scheduling failures of BWP2 UEs in a cell increases. When the scheduling failure rate of BWP2 UEs in a cell exceeds the value of the NRDUCellUePwr Saving.Bwp2SchFailRateThld parameter, UEs are more likely to switch from BWP2 back to BWP1.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CMAS	gNBsibConfig.SibType set to SIB_TYPE_8	<i>CMAS</i>	When power saving scheduling takes effect, the interval at which system information is sent changes, occasionally resulting in some incoming CMAS warning notifications being missed by UEs. However, CMAS warning notifications are usually sent on the CBC side multiple times, increasing the probability for UEs to receive them in power saving scheduling scenarios.
Low-frequency TDD	PEI	UE_PWR_SAVING_ENHANCE_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter and PEI_SW option of the NRDUCellUePwrSaving.IdleUePwrSavingSwitch parameter	<i>UE Power Saving</i>	When dynamic adjustment of the number of PFs is enabled, paging early indications (PEIs) may be inaccurate during the adjustment of the number of PFs. As a result, PEI-based paging fails.
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When Co-MM and power saving scheduling are both enabled, the energy saving gains may be relatively low.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Only low power consumption mode takes effect when both low power consumption mode and power saving scheduling are enabled and a customized backup power saving policy is used in low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUeIdentSw	None	Intelligent scheduling for power saving fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
Low-frequency TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.AppIpv4 • gNBAsAppParam.AppIpv6	None	• Any option of the gNodeBParam.StUeIdentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.AppIpv4 parameter is set. • The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue: • Select the SPEED_TURBO_PWR_SAVING

RAT	Function Name	Function Switch	Reference	Description
				_SW option of the gNB PwrSavin gAlgo.PwrSavin gAlgoSwitch parameter. Intelligent scheduling for power saving can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNB PwrSavin gAlgo.SpeedTurboPwrSavStartTime and gNB PwrSavin gAlgo.SpeedTurboPwrSavEndTime parameters, increasing the energy saving gains. • Select the STATIC_ST_PWR_SAVING_SW option of the gNB PwrSavin gAlgo.PwrSavin gAlgoSwitch parameter. Intelligent scheduling for power saving can take effect for cells served by the base station in some of the preceding

RAT	Function Name	Function Switch	Reference	Description
				scenarios within the period specified by the gNB PwrSavin gAlgo.StaticST PwrSavStartTime and gNB PwrSavin gAlgo.StaticST PwrSavEndTime parameters, increasing the energy saving gains.

4.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

NOTE

LampSite base stations do not support beam-level adjustment of the RMSI broadcast interval, service-type-based dynamic exit from data convergence in slots, or deep symbol power saving.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

For beam-level dynamic adjustment of the RMSI broadcast interval, all NR-capable RF modules that work in 8T8R or a higher specification and in low frequency bands, except those for which the **Description** parameter value includes "V3" in the command output of **DSP BRDMFRINFO**, support this function. For details

about such RF modules, see technical specifications of the related RF modules in *3900 & 5900 Series Base Station Product Documentation*.

For the deep symbol power saving function and the subfunctions of power saving scheduling except beam-level dynamic adjustment of the RMSI broadcast interval, all NR-capable RF modules that work in low frequency bands, except those for which the **Description** parameter value includes "V3" in the command output of **DSP BRDMFRINFO**, support these functions. For details about such RF modules, see technical specifications of the related RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.2.3.4 Cells

This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**.
In TDD, the **NRDUCell.DeployPosition** parameter cannot be set to **DEPLOY_IAB**.

4.2.3.5 Others

None

4.2.4 Operation and Maintenance

4.2.4.1 Data Configuration

4.2.4.1.1 Data Preparation

[Table 4-3](#) and [Table 4-4](#) describe the parameters used for function activation and optimization, respectively. This section does not describe parameters related to cell establishment.

Table 4-3 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellAlgoSwitch.NrDuCellId	Set this parameter based on the network plan.
NR DU Cell ID	NRDUCellPowerSaving.NrDuCellId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	Select the POWER_SAVING_SCHEDULE_SW option of this parameter to enable power saving scheduling: • Dynamic adjustment of the RMSI broadcast interval takes effect by default. • Dynamic adjustment of the number of PFs does not take effect by default. If more energy-saving gains are required, select the PAGING_FRAME_NUM_DYN_ADJ_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter to enable dynamic adjustment of the number of PFs. • Data convergence in symbols takes effect by default. • Data convergence in slots takes effect by default. Select the DEEP_SYMBOL_SHUTDOWN_SW option of this parameter to enable deep symbol power saving.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	When power saving scheduling is used in NR multi-carrier scenarios, select the SLOT_DATA_CONVERGENCE_COORD_SW option of this parameter to enable coordination for data convergence in slots. When multiple carriers share an RF channel, this option must be selected for all the NR cells that share this RF channel.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: - For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. - For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. - The period specified for power saving scheduling in a cell cannot overlap that specified for the function with the NRDUCellPowerSaving.PowerSavingType parameter set to TIMING_CARRIER_SHUTDOWN in the same cell.

Parameter Name	Parameter ID	Setting Notes
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan and the following requirements: - For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. - For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. - The period specified for power saving scheduling in a cell cannot overlap that specified for the function with the NRDUCellPowerSaving.PowerSavingType parameter set to TIMING_CARRIER_SHUTDOWN in the same cell.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to POWER_SAVING_SCHEDULE .

Parameter Name	Parameter ID	Setting Notes
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	You are advised to use the recommended value. If RF channel intelligent shutdown is enabled in the scenarios listed in Table 5-6 and a power saving scheduling policy is configured in the same period, it is recommended that this parameter be set to a large value. This setting prevents power saving scheduling from being stopped when RF channel intelligent shutdown takes effect.
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	You are advised to use the recommended value.
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	You are advised to use the recommended value.
RMSI Adjust Mode	NRDUCellPowerSaving.RmsiAdjustMode	- For cells served by RF modules with eight or more transmit channels: TDD: If the DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter is deselected, the value BEAM is recommended. Otherwise, this parameter can only be set to CELL. - For cells served by RF modules with four or fewer transmit channels, the value CELL is recommended.
Symbol Compress Length	NRDUCellPowerSaving.SymbolCompressLength	The value 4SYMBOL is recommended.

Parameter Name	Parameter ID	Setting Notes
RLC First Packet Delay Thld	NRDUCellPowerSaving.RlcFirstPktDelayThld	- In scenarios with both energy saving and performance considered, the value 2 is recommended. - In energy-saving-preferred scenarios, the value 20 is recommended. A larger value of this parameter results in a higher degree of data convergence, a higher energy saving gain, and a larger decrease in the user-perceived rate. A smaller value of this parameter results in a lower degree of data convergence, a lower energy saving gain, and a smaller decrease in the user-perceived rate. Setting this parameter to 0 equals disabling data convergence in slots.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	For power saving scheduling, this parameter can only be set to its default value.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	You are advised to use the recommended value.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	It is recommended that the SERV_TYPE_DYN_EXIT_PWR_SAVE_SW option be selected when there are specific types of services and user experience of these services needs to be ensured.
Power Saving Policy Group Index	gNBQciBearer.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Power Saving Policy Group Index	NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan. The value of this parameter must be the same as that of the gNBQciBearer.PowerSavingPolicyGrpIndex parameter for services to be guaranteed. Before setting this parameter, ensure that the QCI for which the gNBQciBearer.PowerSavingPolicyGrpIndex parameter is set has been configured.
User Identification Method	NRDUCellPsPolicyGrp.UserIdentMethod	Set this parameter to SERVICE_TYPE .
Service Type Dynamic Exit Power Saving Switch	NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw	To enable service-type-based dynamic exit from data convergence in slots, select the SLOT_DATA_CONVERGENCE_EXIT_SW option.

Table 4-4 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Downlink PDSCH Algorithm Switch	NRDUCellPdsch.DlPdschAlgorithmSwitch	When intelligent scheduling for power saving is enabled, select the RB_ESTIMATE_OPT_SW option of this parameter to enable downlink RB estimation optimization. For details about the impact of downlink RB estimation optimization on network performance, see <i>Downlink Scheduling</i> .

Parameter Name	Parameter ID	Setting Notes
Common Channel Interference Avoid Switch	NRDUCellAlgoSwitch.CommChannelIntrfAvoidSwitch	Select the SSB_INTRF_AVOID_SW option of this parameter to enable SSB interference avoidance. However, if the NR_CELL_SIB1_SW option of the NRDUCellAlgoSwitch.AlgoCompatibilitySwitchExt parameter is deselected, it is recommended that SSB interference avoidance not be enabled.

NOTE

A larger value of the **NRDUCellPowerSaving.UserNumThld** parameter results in a higher probability of triggering dynamic adjustment of the RMSI broadcast interval after a cell enters the power saving scheduling state but a possible decrease in the **N.User.RRCCConn.Avg** counter.

4.2.4.1.2 Using MML Commands

Before using MML commands, refer to [4.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling power saving scheduling
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=POWER_SAVING_SCHEDULE_SW-1;
//(Optional) Enabling dynamic adjustment of the number of PFs (If this function is enabled, UE paging may fail.)
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=PAGING_FRAME_NUM_DYN_ADJ_SW-1;
//Setting an energy saving policy of power saving scheduling
ADD NRDUCCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=01&00&00, StopTime=03&00&00, PowerSavingType=POWER_SAVING_SCHEDULE, DlPrbThld=10, DlPrbOffset=5, RmsiAdjustMode=BEAM, UserNumThld=20, SymbolCompressLength=4SYMBOL, RlcFirstPktDelayThld=2, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5;
//(Optional) Setting another energy saving policy of power saving scheduling
ADD NRDUCCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1, StartTime=04&00&00, StopTime=05&00&00, PowerSavingType=POWER_SAVING_SCHEDULE, DlPrbThld=10, DlPrbOffset=5, RmsiAdjustMode=BEAM, UserNumThld=20, SymbolCompressLength=4SYMBOL, RlcFirstPktDelayThld=2, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5;
//Enabling deep symbol power saving
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=DEEP_SYMBOL_SHUTDOWN_SW-1;
//Enabling coordination for data convergence in slots in NR multi-carrier scenarios
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0,
```



```
PowerSavingExtSwitch=SLOT_DATA_CONVERGENCE_COORD_SW-1;  
//(Optional) Enabling service-type-based dynamic exit from data convergence in slots  
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;  
ADD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,  
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=SLOT_DATA_CONVERGENCE_EXIT_SW-1;
```

Optimization Command Examples

```
//Enabling downlink RB estimation optimization  
MOD NRDUCELLPDSCH: NrDuCellId=0, DiPdschAlgoSwitch=RB_ESTIMATE_OPT_SW-1;  
//Enabling SSB interference avoidance  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CommChnIntrfAvoidSwitch=SSB_INTRF_AVOID_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//(Optional) Disabling service-type-based dynamic exit from data convergence in slots  
MOD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,  
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=SLOT_DATA_CONVERGENCE_EXIT_SW-0;  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;  
//Disabling power saving scheduling  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=POWER_SAVING_SCHEDULE_SW-0;  
//Disabling dynamic adjustment of the number of PFs  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=PAGING_FRAME_NUM_DYN_ADJ_SW-0;  
//Removing an energy saving policy of power saving scheduling  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;  
//(Optional) Removing the other energy saving policy of power saving scheduling  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1;  
//Disabling deep symbol power saving  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=DEEP_SYMBOL_SHUTDOWN_SW-0;  
//Disabling coordination for data convergence in slots in NR multi-carrier scenarios  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=SLOT_DATA_CONVERGENCE_COORD_SW-0;
```

4.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.2.4.2 Activation Verification

[Activation Verification for Power Saving Scheduling](#) and [Activation Verification for Deep Symbol Power Saving](#) describe how to check whether power saving scheduling and deep symbol power saving have taken effect.

Activation Verification for Power Saving Scheduling

After the power saving scheduling function is activated, use either of the following methods to check whether the function has taken effect:

- Method 1:

- a. Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Switch** parameter is **Power Saving Scheduling Switch:On** in the command output, power saving scheduling has been enabled.
- b. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **Power Saving Scheduling**.

 NOTE

If there are two or more records with **Power Saving Type** being **Power Saving Scheduling**, power saving scheduling has been enabled in two or more time periods.

- c. Run the **DSP TIME** command. Check whether the system time is within a power saving scheduling period.
 - If it is, go to **d**.
 - If it is not, wait until the start time of power saving scheduling arrives.
 - d. Check whether the downlink PRB usage of the cell is less than or equal to the downlink PRB usage threshold for starting power saving scheduling (specified by the **NRDUCellPowerSaving.DlPrbThld** parameter).
 - If it is, go to **e**.
 - If it is not, wait until the downlink PRB usage of the local cell is less than or equal to the downlink PRB usage threshold for starting power saving scheduling.
 - e. Wait 1–10 minutes. Then, run the **DSP NRDUCELL** command. If the **Power Saving State** parameter indicates **Power Saving Scheduling:Enabled**, power saving scheduling has taken effect. If the **Power Saving State** parameter indicates **Data Convergence in Slots Coordination:Enabled**, coordination for data convergence in slots has taken effect.
- Method 2:
Run the **DSP NRDUCELLPOWERSAVING** command to query the status of power saving scheduling.
 - If the value of **Power Saving Type** is **Power Saving Schedule** and the value of **Current State** is **Enabled**, power saving scheduling has taken effect.
 - If the value of **Power Saving Type** is **Power Saving Schedule** and the value of **Current State** is **Disabled**, check the **Current Disabled Cause** parameter value to determine the reason why power saving scheduling did not take effect. [Table 4-5](#) lists possible reasons.

Table 4-5 Possible reasons why power saving scheduling did not take effect

Reason	Description
Insufficient License	No license for this function is available.

Reason	Description
Switch Turned Off	None
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable for other reasons.
Triggering Time Not Reached	The current time is not within the range specified for power saving scheduling.
Prerequisite Feature Not Taken Effect	Symbol power saving has not taken effect.
Downlink PRB Threshold Not Reached	None
In Penalty State	A penalty is being imposed upon an exit from the power saving scheduling state.
Impacted Feature Taken Effect	None
Others	- The current time is within the time range specified for power saving scheduling, but this function is in its detection period, which is specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter. - None of the counters is subscribed to. - Internal software or other reasons cause the function to fail.

Activation Verification for Deep Symbol Power Saving

- Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Switch** parameter is **Deep Symbol Power Saving Switch:On** in the command output, deep symbol power saving has been enabled.
- Run the **DSP NRDUCELL** command to check the deep symbol power saving state of the cell.
 - If the value of the **Power Saving State** parameter is **Deep Symbol Power Saving:Enabled**, deep symbol power saving has taken effect in at least one TRP of the cell.
 - If the value of the **Power Saving State** parameter is **Deep Symbol Power Saving:Disabled**, deep symbol power saving has not taken effect in the cell. The possible reasons are as follows:

- The license item "Intelligent Scheduling for Power Saving" has not been acquired.
 - Symbol power saving has not taken effect.
3. (Optional) For a hyper cell or combined cell, run the **DSP NRDUCELLTRP** command to query the deep symbol power saving state of each TRP in the cell.
- If the value of the **Power Saving State** parameter is **Deep Symbol Power Saving:Enabled** for a TRP, deep symbol power saving has taken effect in the TRP.
 - If the value of the **Power Saving State** parameter is **Deep Symbol Power Saving:Disabled** for a TRP, deep symbol power saving has not taken effect in the TRP. The possible reasons are as follows:
 - The license items "Intelligent Scheduling for Power Saving" are insufficient.
 - Symbol power saving has not taken effect.

 NOTE

For details about the consumption principles of feature license items in hyper cells or combined cells, see *Hyper Cell* or *Cell Combination*.

4.2.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after intelligent scheduling for power saving is enabled can be compared to assess the energy saving gains of this feature. It is expected that the base station energy consumption decreases significantly after this feature is enabled.
- The duration of symbol power saving can be observed using the counter **N.PowerSaving.SymbolShutdown.Dur** (for a cell) on the MAE-Access. It is expected that the duration increases after intelligent scheduling for power saving is enabled. The policy of counter measurement upon carrier shutdown is determined by the function of energy saving counter optimization upon carrier shutdown, which is controlled by the **COUNTER_OPT_IN_CARROFF_SW** option of the **gNB PwrSavingAlgo.PwrSavingAlgoSwitch** parameter. If this option is deselected and a carrier has been shut down, the **N.PowerSaving.SymbolShutdown.Dur** counter counts in the measured symbol power saving duration of the carrier. If this option is selected and a carrier has been shut down, the **N.PowerSaving.SymbolShutdown.Dur** counter does not count in the symbol power saving duration of the carrier.

4.3 Intelligent Scheduling for Power Saving (High-frequency TDD)

4.3.1 Principles

In symbols without data transmission, symbol power saving enables the base station to shut down the PAs of RF modules in real time, reducing power consumption. **On the basis of symbol power saving, intelligent scheduling for power saving is introduced to further reduce power consumption. It provides the sub-function power saving scheduling, which increases the number of symbols without data transmission.**

Power saving scheduling uses three methods to free more symbols from data transmission: **Dynamic Adjustment of the RMSI Broadcast Interval**, **Data Convergence in Symbols**, and **Data Convergence in Slots**.

Power saving scheduling can be enabled by selecting the **POWER_SAVING_SCHEDULE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter. **Table 4-6** describes the parameters in the **NRDUCellPowerSaving** MO that specify the effective period, load threshold, and other information under an energy saving policy of power saving scheduling.

Table 4-6 Parameters related to the power saving scheduling policy

Parameter Name	Parameter ID	Description
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	This parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local sector. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local sector.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	This parameter specifies the energy saving type indicated by an energy saving policy index. Power saving scheduling requires that this parameter be set to POWER_SAVING_SCHEDULE .

Parameter Name	Parameter ID	Description
Start Time	NRDUCellPowerSaving.StartTime	These parameters specify the time period in which power saving scheduling takes effect, as well as the evaluation period and conditions for a sector to enter and exit the power saving scheduling state. Within the time period specified by NRDUCellPowerSaving.StartTime and NRDUCellPowerSaving.StopTime for power saving scheduling to take effect for a sector, the base station periodically evaluates whether the sector enters the power saving scheduling state. The evaluation period is specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter. • If the downlink PRB usage of a sector^a is less than or equal to the value of NRDUCellPowerSaving.DlPrbThld, the sector automatically enters the power saving scheduling state. • If the downlink PRB usage of a sector is greater than the sum of the values of NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset or the current time is not within the period during which power saving scheduling can take effect, the sector automatically exits the power saving scheduling state.
Stop Time	NRDUCellPowerSaving.StopTime	
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	For power saving scheduling, this parameter can only be set to MINUTE_LEVEL .
a: Downlink PRB usage of a sector = Total number of PRBs used by PDSCH DRBs of all cells in the sector/Total number of available downlink PRBs in the sector		

The parameters related to the power saving scheduling policy must be set to identical values for all cells in a sector. Otherwise, power saving scheduling does not take effect for the sector.

Dynamic Adjustment of the RMSI Broadcast Interval

When the base station detects that the network is under light or no load, it prolongs the remaining minimum system information (RMSI) broadcast interval to free more symbols from data transmission. When the base station detects that the network load is normal, it restores the RMSI broadcast interval.

With power saving scheduling enabled for cells in a sector, each of the cells periodically checks the number of RRC_CONNECTED UEs. The evaluation period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter. If the number of RRC_CONNECTED UEs in the sector is less than or equal to the UE number threshold specified by the **NRDUCellPowerSaving.UserNumThld** parameter, dynamic adjustment of the RMSI broadcast interval takes effect.

NOTE

The number of RRC_CONNECTED UEs in a sector is the sum of the values of the **N.User.RRConn.Avg** counter for all cells in the sector.

The RMSI broadcast interval is dynamically adjusted based on the ratio of the number of RRC_CONNECTED UEs in the sector to the UE number threshold.

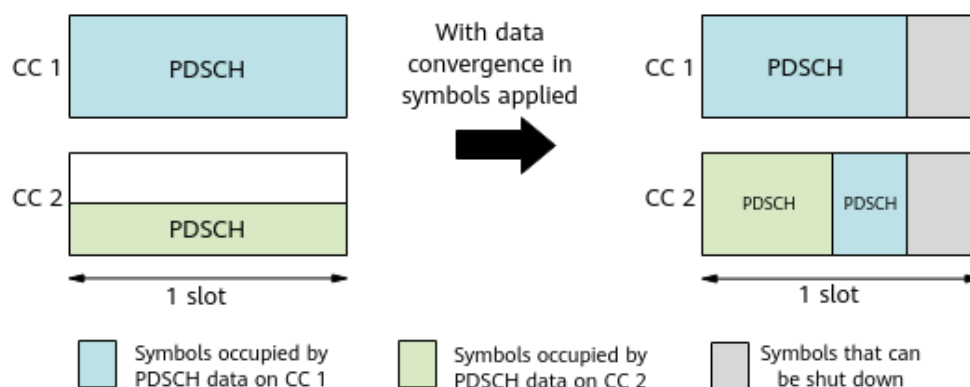
Data Convergence in Symbols

Based on the size of the physical downlink shared channel (PDSCH) data to be transmitted, the base station converges the PDSCH data on certain symbols within a slot so that PDSCH data occupies fewer time-domain resources but more frequency-domain resources, increasing the number of symbols with no data transmission. Data convergence in symbols takes effect if power saving scheduling is enabled.

Data convergence in symbols is classified into two types:

- Common control information convergence in symbols
In slots where common control information is scheduled, the base station converges data in a minimum of three symbols in the time domain based on the TBS of the information. Common control information includes RMSI, other system information (OSI), paging, random access response (RAR), and initial transmission of Msg4.
- Data channel convergence in symbols
The base station converges PDSCH data in 10, 7, or 3 symbols based on the ratio of the number of RBs to be scheduled to the number of available RBs, as shown in [Figure 4-8](#).

Figure 4-8 Data channel convergence in symbols (with 2CC as an example)



Data Convergence in Slots

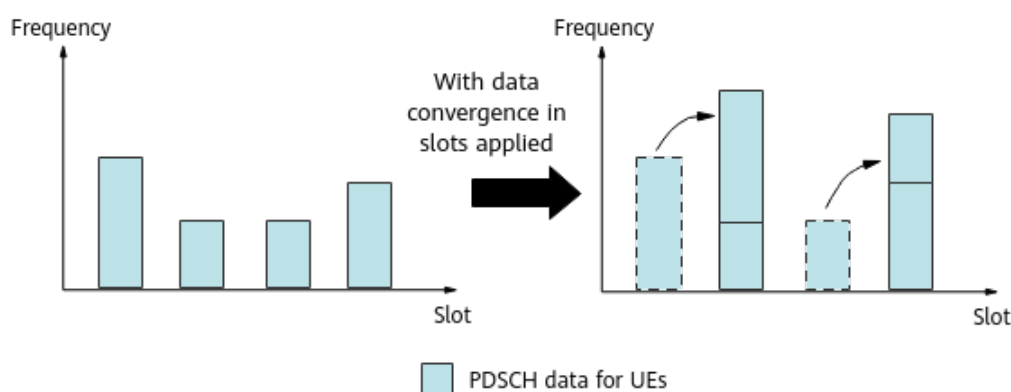
Data convergence in slots combines multiple small packets to be scheduled before scheduling. It reduces the number of slots occupied by small packets, increases the proportion of idle slots, and therefore lifts the symbol shutdown rate.

This function applies only to first packets of PDSCH data, reducing the impact on the user-perceived throughput. [Figure 4-9](#) shows how it works.

The **NRDUCellPowerSaving.RlcFirstPktDelayThld** parameter specifies the first packet delay threshold for data convergence in slots.

- If the first packet delay of a UE is less than or equal to this threshold, first-packet data to be scheduled is converged in slots and then transmitted.
- Otherwise, the data is directly scheduled and transmitted for the UE.

Figure 4-9 Data convergence in slots



4.3.2 Network Analysis

4.3.2.1 Benefits

When the network is under light or no load, enabling power saving scheduling can further increase energy saving gains and reduce base station power consumption on the basis of [symbol power saving](#).

Dynamic adjustment of the RMSI broadcast interval provides no energy saving gains if no RMSI is broadcast on SSB beams.

4.3.2.2 Impacts

The impacts between this function and other functions are described in [4.3.3.2.3 Function Impacts](#).

When power saving scheduling takes effect, common control information is sent less frequently, which may cause the following changes:

- The number of random accesses decreases.
- The average CCE aggregation level decreases.
- The CCE usage decreases.
- The PRB usage indicated by **Downlink Resource Block Utilizing Rate (DU)** decreases.

Data convergence in symbols increases the downlink PRB usage of cells.

Data convergence in slots extends the downlink first packet delay and may increase the downlink initial block error rate (IBLER) and the proportion of downlink retransmissions.

4.3.3 Requirements

4.3.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-031204	Intelligent Scheduling for Power Saving	NR0S000PSS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	Power saving scheduling requires symbol power saving.

4.3.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The period specified for power saving scheduling in a cell cannot overlap that specified for timing carrier shutdown in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.

4.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both power saving scheduling and RF channel intelligent shutdown take effect, more symbols are required for transmitting a given amount of data. In this case, power saving scheduling delivers a lower symbol shutdown rate.
High-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected MULTI_CARR_PRECISE_SHDN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter deselected	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown is working in non-precise mode: - If the RF module is not serving any other NR carrier, power saving scheduling does not take effect. - If the RF module is serving other NR carriers, the status of power saving scheduling is not affected.
High-frequency TDD	RF module deep dormancy	RRU.DORMANCY_SW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Only RF module deep dormancy takes effect when it is enabled together with power saving scheduling.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	RF module super dormancy	RRU.DORMANCY SW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Only RF module super dormancy takes effect when it is enabled together with power saving scheduling.
High-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	Rate-matching-pattern-configuration-based PDCCH rate matching takes effect only for large-packet UEs. However, data convergence in symbols does not take effect for large-packet UEs.
High-frequency TDD	Rate-matching-pattern-configuration-free PDCCH rate matching	PDCCH_NO_PATTERN_RATE_MATCHING_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Downlink Scheduling</i>	Rate-matching-pattern-configuration-free PDCCH rate matching takes effect only for large-packet UEs. However, data convergence in symbols does not take effect for large-packet UEs.
High-frequency TDD	Sparsified RMSI scheduling policy	NRDUCellDISch.RmsiSparsifiedSchedulingPolicy set to a value other than OFF	<i>Downlink Scheduling</i>	If both dynamic adjustment of the RMSI broadcast interval and the sparsified RMSI scheduling policy take effect, the broadcast interval takes the greater one of the interval values involved in the two functions.

4.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function is compatible only with HAAU5233w and HAAU5235w.

4.3.3.4 Cells

- This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**.
- This function requires that the **NRDUCell.FrequencyBand** parameter be set to **N257**, **N258**, **N260**, or **N261** for the cells.
- This function requires that the **NRDUCell.NrDuCellNetworkingMode** parameter be set to **NORMAL_CELL** for the cells.
- This function requires that the **NRDUCell.SlotAssignment** parameter be set to **4_1_DDSU** for the cells.

4.3.3.5 Others

None

4.3.4 Operation and Maintenance

4.3.4.1 Data Configuration

4.3.4.1.1 Data Preparation

Table 4-7 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-7 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellAlgoSwitch.NrDuCellId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	Select the POWER_SAVING_SCHEDULE_SW option of this parameter to enable power saving scheduling: • Dynamic adjustment of the RMSI broadcast interval takes effect by default. • Data convergence in symbols takes effect by default. • Data convergence in slots takes effect by default.
NR DU Cell ID	NRDUCellPowerSaving.NrDuCellId	Set this parameter based on the network plan.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: - For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. - For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. - The period specified for power saving scheduling in a cell cannot overlap that specified for the function with the NRDUCellPowerSaving.PowerSavingType parameter set to TIMING_CARRIER_SHUTDOWN in the same cell.

Parameter Name	Parameter ID	Setting Notes
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan and the following requirements: - For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. - For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. - The period specified for power saving scheduling in a cell cannot overlap that specified for the function with the NRDUCellPowerSaving.PowerSavingType parameter set to TIMING_CARRIER_SHUTDOWN in the same cell.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to POWER_SAVING_SCHEDULE .

Parameter Name	Parameter ID	Setting Notes
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	For power saving scheduling, this parameter can only be set to MINUTE_LEVEL .
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	Use the recommended value.
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	The value 30 is recommended for power saving scheduling in high frequency bands.
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	The value 20 is recommended for power saving scheduling in high frequency bands.
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	Use the recommended value. The larger the value of this parameter, the more likely dynamic adjustment of the RMSI broadcast interval is triggered after power saving scheduling starts. However, the average number of RRC_CONNECTED UEs in the sector may decrease, as indicated by the N.User.RRCCConn.Avg counter.
RLC First Packet Delay Thld	NRDUCellPowerSaving.RlcFirstPktDelayThld	• In scenarios with both energy saving and performance considered, the value 2 is recommended. • In energy-saving-preferred scenarios, the value 20 is recommended. A larger value of this parameter results in a higher degree of data convergence, a higher energy saving gain, and a larger decrease in the user-perceived rate. A smaller value of this parameter results in a lower degree of data convergence, a lower energy saving gain, and a smaller decrease in the user-perceived rate. Setting this parameter to 0 equals disabling data convergence in slots.

4.3.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling power saving scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=POWER_SAVING_SCHEDULE_SW-1;
//Setting an energy saving policy of power saving scheduling
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=01&00&00,
StopTime=03&00&00, PowerSavingType=POWER_SAVING_SCHEDULE, DIPrbThld=30, DIPrbOffset=20,
UserNumThld=20, RlcFirstPktDelayThld=2, EvalPeriodGranularity=MINUTE_LEVEL,
ShutdownTrigJudgePeriod=5;
//(Optional) Setting another energy saving policy of power saving scheduling
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1, StartTime=04&00&00,
StopTime=05&00&00, PowerSavingType=POWER_SAVING_SCHEDULE, DIPrbThld=30, DIPrbOffset=20,
UserNumThld=20, RlcFirstPktDelayThld=2, EvalPeriodGranularity=MINUTE_LEVEL,
ShutdownTrigJudgePeriod=5;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling power saving scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=POWER_SAVING_SCHEDULE_SW-0;
//Removing an energy saving policy of power saving scheduling
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
//(Optional) Removing the other energy saving policy of power saving scheduling
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1;
```

4.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.3.4.2 Activation Verification

After the power saving scheduling function is activated, use either of the following methods to check whether the function has taken effect:

- Method 1:

- a. Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Switch** parameter is **Power Saving Scheduling Switch:On** in the command output, power saving scheduling has been enabled.
- b. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **Power Saving Scheduling**.

 NOTE

If there are two or more records with **Power Saving Type** being **Power Saving Scheduling** for a cell, power saving scheduling has been enabled in two or more time periods.

- c. Run the **DSP TIME** command. Check whether the system time is within a power saving scheduling period.
 - If it is, go to **d**.
 - If it is not, wait until the start time of power saving scheduling arrives.
 - d. Check whether the downlink PRB usage of the cell is less than or equal to the downlink PRB usage threshold for starting power saving scheduling (specified by the **NRDUCellPowerSaving.DlPrbThld** parameter).
 - If it is, go to **e**.
 - If it is not, wait until the downlink PRB usage of the local cell is less than or equal to the downlink PRB usage threshold for starting power saving scheduling.
 - e. Wait 1–10 minutes. Then, run the **DSP NRDUCELL** command. If the **Power Saving State** parameter indicates **Power Saving Scheduling:Enabled**, power saving scheduling has taken effect.
- Method 2:
Run the **DSP NRDUCELLPOWERSAVING** command to query the status of power saving scheduling.
 - If the value of **Power Saving Type** is **Power Saving Schedule** and the value of **Current State** is **Enabled**, power saving scheduling has taken effect.
 - If the value of **Power Saving Type** is **Power Saving Schedule** and the value of **Current State** is **Disabled**, check the **Current Disabled Cause** parameter value to determine the reason why power saving scheduling did not take effect. **Table 4-8** lists possible reasons.

Table 4-8 Possible reasons why power saving scheduling did not take effect

Reason	Description
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable for other reasons.
Switch Turned Off	None
Insufficient License	No license for this function is available.

Reason	Description
Parameter Inconsistency Between Cells	Parameters related to power saving scheduling are set inconsistently between the cells in the same sector.
Triggering Time Not Reached	The current time is not within the range specified for power saving scheduling.
Downlink PRB Threshold Not Reached	None
Prerequisite Feature Not Taken Effect	Symbol power saving has not taken effect.
Impacted Feature Taken Effect	None
In Penalty State	A penalty is being imposed upon an exit from the power saving scheduling state.
Others	- The current time is within the time range specified for power saving scheduling, but this function is in its detection period, which is specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter. - None of the counters is subscribed to. - Internal software or other reasons cause the function to fail.

4.3.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after power saving scheduling is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.

- The duration of symbol power saving can be observed using the counter **N.PowerSaving.SymbolShutdown.Dur** (for a cell) or **VS.RADIOEQM.SymbolShutdown.Dur** (for an RF module) on the MAE-Access. The counter **N.PowerSaving.SymbolShutdown.Dur** (for a cell) is recommended in single-carrier scenarios, and the counter **VS.RADIOEQM.SymbolShutdown.Dur** (for an RF module) is recommended in multi-carrier scenarios. Power saving scheduling is expected to extend the duration of symbol power saving.

4.4 MetaAAU Beam Power Saving (Low-Frequency TDD)

4.4.1 Principles

The energy consumption varies with the analog beam angle of the MetaAAU. MetaAAU beam power saving can adjust the beam angles of the MetaAAU when there are no common signals (such as SSB and RMSI) or data transmission in the downlink, reducing energy consumption.

MetaAAU beam power saving can be enabled by selecting the **BEAM_POWER_SAVING_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter. When there are no common signals (such as SSB and RMSI) or data transmission in the downlink:

- If this option is selected, the beam angle with optimal energy saving gains is used for the MetaAAU, thereby reducing energy consumption.
- If this option is deselected, the SSB beam angle is used by default.

4.4.2 Network Analysis

4.4.2.1 Benefits

MetaAAU beam power saving reduces RF energy consumption in light-load scenarios.

4.4.2.2 Impacts

The impacts between this function and other functions are described in [4.4.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.4.3 Requirements

4.4.3.1 Licenses

None

4.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.4.3.2.1 Prerequisite Functions

None

4.4.3.2.2 Mutually Exclusive Functions

None

4.4.3.2.3 Function Impacts

None

4.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Only MetaAAUs support this function.

4.4.3.4 Networking

This function is supported only in NR single-carrier scenarios. This function is not supported when multiple NR carriers or both LTE and NR carriers are deployed on the MetaAAU.

4.4.3.5 Others

None

4.4.4 Operation and Maintenance

4.4.4.1 Data Configuration

4.4.4.1.1 Data Preparation

[Table 4-9](#) describes the parameters used for function activation.

Table 4-9 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	The BEAM_POWER_SAVING_SW option of this parameter is selected by default, which is also the recommended setting.

4.4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling MetaAAU beam power saving
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=BEAM_POWER_SAVING_SW-1;
```

Deactivation Command Examples

```
//Disabling MetaAAU beam power saving
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=BEAM_POWER_SAVING_SW-0;
```

4.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.4.2 Activation Verification

To check whether MetaAAU beam power saving has taken effect, perform the following steps:

1. Run the **DSP NRDUCCELL** command on the gNodeB side.
2. Query the cell status. If the value of **Power Saving State** is **BEAM_POWER_SAVING:Enabled**, MetaAAU beam power saving has taken effect.

4.4.4.3 Network Monitoring

Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

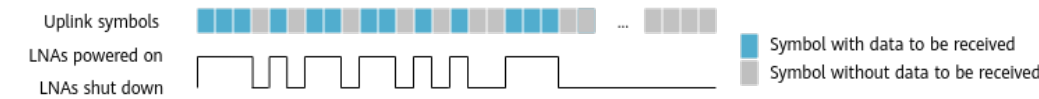
4.5 Uplink Symbol Power Saving (Low-Frequency TDD)

4.5.1 Principles

In a base station, low noise amplifiers (LNAs) cause static energy consumption even when there is no need to receive signals in the uplink. Uplink symbol power saving is introduced to reduce energy consumption while ensuring data reception integrity. [Figure 4-10](#) shows how it works.

- When no data needs to be received in an uplink symbol, the base station shuts down the LNAs in RF modules in real time to reduce energy consumption.
- When data needs to be received in an uplink symbol, the base station starts the LNAs in RF modules in real time to ensure data reception integrity.

Figure 4-10 Working principles for uplink symbol power saving



Uplink symbol power saving can be enabled by selecting the **UL_SYMBOL_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter.

[Table 4-10](#) describes the parameters in the **NRDUCellPowerSaving** MO that specify the effective period, load threshold, and other information under an energy saving policy of uplink symbol power saving.

Table 4-10 Parameters related to a policy of uplink symbol power saving

Parameter Name	Parameter ID	Description
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	This parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.

Parameter Name	Parameter ID	Description
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	This parameter specifies the energy saving type indicated by an energy saving policy index. Uplink symbol power saving requires that this parameter be set to UL_SYMBOL_SHUTDOWN .
Start Time	NRDUCellPowerSaving.StartTime	These parameters specify the effective period, evaluation period, and conditions for Entering the Uplink Symbol Power Saving State and Exiting the Uplink Symbol Power Saving State in a cell.
Stop Time	NRDUCellPowerSaving.StopTime	
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	
Uplink PRB Offset	NRDUCellPowerSaving.UlPrbOffset	
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	
User Number Offset	NRDUCellPowerSaving.UserNumOffset	

Whether uplink symbol power saving can take effect in multi-carrier scenarios depends on the setting of the **UL_SYM_SHUTDOWN_MULTI_CARR_SW** option of the **gNB PwrSavingAlgo.PwrSavingAlgoSwitch** parameter and the RF modules in use.

- When this option is selected and multi-band RF modules are used, uplink symbol power saving can take effect independently in NR cells in multi-carrier scenarios.
- In other cases, uplink symbol power saving can take effect in NR cells only in single-carrier scenarios, and **RRU.RS** must be set to **NO**.

When remote interference adaptive avoidance (controlled by the **RIM_ADAPT_AVOID_SW** option of **NRDUCellAlgoSwitch.RimAlgoSwitch**) or

enhanced remote interference adaptive avoidance (controlled by the **RIM_ADAPT_AVOID_ENH_SW** option of **NRDUCellAlgoSwitch.RimAlgoSwitch**) takes effect, the benefits of uplink symbol power saving decrease due to the reduced power saving duration. In this case, uplink symbol power saving optimization in remote interference avoidance scenarios can be enabled so that uplink symbol power saving can take effect in a higher proportion of cases, increasing energy saving gains. Uplink symbol power saving optimization in remote interference avoidance scenarios is controlled by the **RIM_UL_SYMBOL_SHUTDOWN_OPT_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter.

Entering the Uplink Symbol Power Saving State

Within the time period specified by **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** for uplink symbol power saving to take effect in a cell, the base station periodically checks whether the following conditions are met throughout a period specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter. If all the following conditions are met, the cell enters the uplink symbol power saving state:

- Uplink PRB usage of the cell $\leq$ **NRDUCellPowerSaving.UlPrbThld**
Uplink PRB usage of the cell = $\frac{\text{N.PRB.PUSCH.Used.Avg}}{\text{N.PRB.UL.PUSCH.Avail.Avg}} \times 100\%$
- Downlink PRB usage of the cell $\leq$ **NRDUCellPowerSaving.DlPrbThld**
Downlink PRB usage of the cell = $\frac{(\text{N.PRB.DL.DrbUsed.Avg} - \text{N.PRB.DL.ExtendUsed.Avg})}{\text{N.PRB.DL.Avail.Avg}} \times 100\%$
- Number of RRC_CONNECTED UEs in the cell (indicated by the **N.User.RRConn.Avg** counter) $\leq$ **NRDUCellPowerSaving.UserNumThld**

Prolonging the PRACH Period

For a cell in the uplink symbol power saving state, the PRACH period is automatically prolonged, so that more uplink slots have no PRACH configured in them and uplink symbol power saving takes effect in a higher proportion of cases.

Exiting the Uplink Symbol Power Saving State

After a cell enters the uplink symbol power saving state, the base station periodically checks whether the following conditions are met throughout a period specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter. If any of the conditions is met, the cell automatically exits this state.

- The current time is not within the effective period specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters for uplink symbol power saving.
- The number of RRC_CONNECTED UEs in the cell is greater than the sum of the **NRDUCellPowerSaving.UserNumThld** and **NRDUCellPowerSaving.UserNumOffset** parameter values. The **NRDUCellPowerSaving.UserNumOffset** parameter specifies the offset to the number of RRC_CONNECTED UEs and is used to prevent a cell from frequently entering and exiting the uplink symbol power saving state.

- The downlink PRB usage of the cell is greater than the sum of **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset**. The **NRDUCellPowerSaving.DlPrbOffset** parameter specifies the offset to the downlink PRB usage threshold and is used to prevent a cell from frequently entering and exiting the uplink symbol power saving state.
- The uplink PRB usage of the cell is greater than the sum of **NRDUCellPowerSaving.UlPrbThld** and **NRDUCellPowerSaving.UlPrbOffset**. The **NRDUCellPowerSaving.UlPrbOffset** parameter specifies the offset to the uplink PRB usage threshold and is used to prevent a cell from frequently entering and exiting the uplink symbol power saving state.

After the cell exits the uplink symbol power saving state, it cannot enter this state again within a period of 30 minutes, in which the cell is in the penalty state.

4.5.2 Network Analysis

4.5.2.1 Benefits

Uplink symbol power saving reduces energy consumption by shutting down LNAs in RF modules in real time when there is no need to receive signals in the uplink. The higher the network load, the smaller the energy saving gains.

When online fast Fourier transform (FFT) spectrum scanning is enabled, uplink symbol power saving does not shut down LNAs in RF modules. This reduces the energy saving gains.

When remote interference adaptive avoidance (controlled by the **RIM_ADAPT_AVOID_SW** option of **NRDUCellAlgoSwitch.RimAlgoSwitch**) or enhanced remote interference adaptive avoidance (controlled by the **RIM_ADAPT_AVOID_ENH_SW** option of **NRDUCellAlgoSwitch.RimAlgoSwitch**) takes effect, enabling uplink symbol power saving optimization in remote interference avoidance scenarios increases energy saving gains because uplink symbol power saving can take effect in a higher proportion of cases.

4.5.2.2 Impacts

The impacts between this function and other functions are described in [4.5.3.2.3 Function Impacts](#).

Uplink symbol power saving has the following impacts:

- The values of counters related to Received Total Wideband Power (RTWP) are small, such as **VS.MeanRTWP.Locell.SectorEqm.Ant0**, **VS.MeanRTWP.Locell.SectorEqm.Ant1**, and **VS.MeanRTWP.Locell.SectorEqm.Ant2**.
- The values of the following counters are invalid:
 - Counters that measure uplink interference, such as **N.UL.NI.Avg**, **N.UL.NI.Max**, **N.UL.NI.Avg.PRbX** ($X = 0$ to 272), and **N.UL.NI.Max.PRbX** ($X = 0$ to 272)
 - Counters that measure uplink received signal strength indicators (RSSIs), such as **N.UL.RSSI.Avg** and **N.UL.RSSI.Min**
 - Counters that measure the power in symbols, such as **N.UL.First.Symbol0.Pwr**, **N.UL.First.Symbol2.Pwr**,

**N.UL.First.Symbol4.Pwr, N.UL.Last.Symbol11.Pwr, and
N.UL.Last.Symbol13.Pwr**

- When 64T RF modules are used, the **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** decrease. In addition, the average rank for downlink scheduling, MCS index for downlink scheduling, IBLER, and CQI in a cell remain unchanged or slightly decrease, and the downlink PRB usage remains unchanged or slightly increases.
- When only a few UEs attempt to access the network or interference is severe, the contention-based random access success rate and non-contention-based random access success rate change.
- The uplink PRB usage slightly decreases.

4.5.3 Requirements

4.5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-081203	MetaAAU Deep Energy Saving	NR0S00MTDD00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PRACH configuration index	NRDUCellPrach.PrachConfigurationIndex	None	Uplink symbol power saving requires this parameter to be set to 65535 .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cross-link interference configuration	NRDUCellTimeIntf.CrossLinkIntfCfg	None	Uplink symbol power saving requires this parameter to be set to NOT_CONFIG or CROSS_LINK_SEQ_SEND .
Low-frequency TDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Uplink symbol power saving optimization in remote interference avoidance scenarios requires that the UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter be selected and the RIM_ADAPT_AVOID_SW or RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter be selected.
Low-frequency TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	
Low-frequency TDD	Enhanced remote interference adaptive avoidance	RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	

4.5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	Uplink symbol power saving is not supported in high-speed cells.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Uplink symbol power saving is not supported in Hyper Cell networking.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination /Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Uplink symbol power saving is not supported in Cell Combination or Joint Transmission Cell Combination networking.
Low-frequency TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	Uplink symbol power saving does not work with Fusion Cell.
Low-frequency TDD	Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_USLOT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgoSw parameter	<i>Positioning</i>	Uplink symbol power saving does not work with positioning-dedicated SRS transmission in uplink slots.
Low-frequency TDD	Extended Cell Range (TDD)	NRDUCell.CellRadius set to a value greater than 14500	<i>Extended Cell Range</i>	The Extended Cell Range function does not work with uplink symbol power saving.
Low-frequency TDD	RedCap identification strategy	NRDUCellRedCap.RedCapIdentifyStrategy	None	Uplink symbol power saving can be enabled only if this parameter is set to a value other than DEDICATED_RO .
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Uplink symbol power saving does not work with the low latency and high reliability function.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAIgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	Uplink symbol power saving does not work with distributed massive MIMO.
Low-frequency TDD	Light-load time sequence optimization	LIGHT_LOAD_TIME_SEQ_OPT_SW option of the NRDUCellAIgoSwitch.LowlatencySwitch parameter	None	Uplink symbol power saving does not work with light-load time sequence optimization.
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The period specified for timing carrier shutdown in a cell cannot overlap that specified for uplink symbol power saving in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.

4.5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTTI_CARR_SD_SW or INTER_GNB_MULTTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Uplink symbol power saving does not take effect when intelligent carrier shutdown is in effect.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCY_SW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Uplink symbol power saving does not take effect when RF module deep dormancy is in effect.
Low-frequency TDD	RF module super dormancy	RRU.DORMANCY_SW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Uplink symbol power saving does not take effect when RF module super dormancy is in effect.
Low-frequency TDD	LTE and NR intelligent carrier shutdown	NR: LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter LTE: CellAlgoExtSwitch.LnrSmartCarrierShutdownSw	<i>Multi-RAT Coordinated Energy Saving</i>	Uplink symbol power saving does not take effect when LTE and NR intelligent carrier shutdown is in effect.
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS_W (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	Uplink symbol power saving does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Clock out-of-synchronization detection	CLK_DETECT_SW option of the gNodeBParam.ClkOutofsyncDetSwitch parameter	<i>Synchronization</i>	When clock out-of-synchronization detection takes effect, uplink symbol power saving offers fewer energy saving gains.
Low-frequency TDD	PUSCH joint power control	PUSCH_JOINT_PC_SW option of the NRDUCellULPcConfig.ULPwrCtrlAlgoSwitch parameter	<i>Power Control</i>	PUSCH joint power control does not take effect when uplink symbol power saving is in effect.
Low-frequency TDD	Uplink-interference-based inter-frequency handover	UL_INTRF_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	Uplink-interference-based inter-frequency handover does not take effect when uplink symbol power saving is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUeIdentSw	None	Uplink symbol power saving fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
Low-frequency TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.AppIpv4 • gNBAsAppParam.AppIpv6	None	• Any option of the gNodeBParam.StUeIdentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.AppIpv4 parameter is set. • The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue: • Select the SPEED_TURBO_PWR_SAVING_SW option of

RAT	Function Name	Function Switch	Reference	Description
				the gNB PwrSavin gAlgo.PwrSavin gAlgoSwitch parameter. Uplink symbol power saving can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNB PwrSavin gAlgo.SpeedTurboPwrSavStartTime and gNB PwrSavin gAlgo.SpeedTurboPwrSavEndTime parameters, increasing the energy saving gains. • Select the STATIC_ST_PWR_SAVING_SW option of the gNB PwrSavin gAlgo.PwrSavin gAlgoSwitch parameter. Uplink symbol power saving can take effect for cells served by the base station in some of the preceding scenarios within the period

RAT	Function Name	Function Switch	Reference	Description
				specified by the gNB PwrSavin gAlgo.StaticST PwrSavStartTime and gNB PwrSavin gAlgo.StaticST PwrSavEndTime parameters, increasing the energy saving gains.
Low-frequency TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When remote interference adaptive avoidance is in effect, the benefits of uplink symbol power saving decrease due to the reduced power saving duration. In this scenario, the RIM_UL_SYMBOL_SHUTDOWN_OPTION_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter can be selected so that uplink symbol power saving can take effect in a higher proportion of cases, increasing energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced remote interference adaptive avoidance	RIM_ADAPT_AVO ID_ENH_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When enhanced remote interference adaptive avoidance is in effect, the benefits of uplink symbol power saving decrease due to the reduced power saving duration. In this scenario, the RIM_UL_SYMBOL_SHUTDOWN_OPTION_SW option of the NRDUCellPowerSavingAlgo.PowerSavingAlgoSwitch parameter can be selected so that uplink symbol power saving can take effect in a higher proportion of cases, increasing energy saving gains.
Low-frequency TDD	GP symbol shutdown in special slots	SS_GAP_SYMBOL_SHUTDN_SW option of the NRDUCellInterf.Trf.TimeIntrfAvoidSw parameter	<i>Interference Avoidance</i>	When both uplink symbol power saving and GP symbol shutdown in special slots are enabled, the value of VS.RADIOEQM.UL.SymbolShutdown.Dur increases due to GP symbol shutdown.

4.5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Only MetaAAUs support this function.

In addition, if the **UL_SYM_SHUTDOWN_MULTI_CARR_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected, multi-band MetaAAUs must be used if uplink symbol power saving needs to take effect independently in NR cells in multi-carrier scenarios. A multi-band RF module can work in multiple frequency bands. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.5.3.4 Networking

Only when the **UL_SYM_SHUTDOWN_MULTI_CARR_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected and multi-band RF modules are used, uplink symbol power saving can take effect independently in NR cells in multi-carrier scenarios. In other cases, uplink symbol power saving can take effect in NR cells only in single-carrier scenarios, and **RRU.RS** must be set to **NO**.

4.5.3.5 Cells

- The TX/RX mode of the cell must be 32T32R or higher.
- The **NRDUCell.FrequencyBand** parameter must be set to **N38, N39, N40, N41, N77, N78, or N79**.
- The **NRDUCell.SlotAssignment** parameter must be set to **4_1_DDDSU, 8_2_DDDDDDSUU, 8_2_DDDSUUDDDD, or 7_3_DDDSUDDSUU**.

4.5.3.6 Others

None

4.5.4 Operation and Maintenance

4.5.4.1 Data Configuration

4.5.4.1.1 Data Preparation

[Table 4-11](#) describes the parameters used for function activation.

Table 4-11 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellAlgoSwitch.NrDuCellId	Set this parameter based on the network plan.
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	Select the UL_SYMBOL_SHUTDOWN_SW option of this parameter when uplink symbol power saving is required.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: - For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. - For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.

Parameter Name	Parameter ID	Setting Notes
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan and the following requirements: - For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. - For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to UL_SYMBOL_SHUTDOWN when uplink symbol power saving is required.
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	You are advised to use the recommended value.
Uplink PRB Offset	NRDUCellPowerSaving.UlPrbOffset	You are advised to use the recommended value.
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	You are advised to use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	You are advised to use the recommended value.
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	You are advised to use the recommended value.
User Number Offset	NRDUCellPowerSaving.UserNumOffset	You are advised to use the recommended value.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	Retain the default value when uplink symbol power saving is required.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	You are advised to use the recommended value.
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	When multi-band RF modules are used, select the UL_SYM_SHUTDOWN_MULTI_CARR_SW option if uplink symbol power saving needs to take effect independently in NR cells in multi-carrier scenarios.
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	You are advised to select the RIM_UL_SYMBOL_SHUTDOWN_OPT_SW option if remote interference adaptive avoidance (controlled by the RIM_ADAPT_AVOID_SW option of NRDUCellAlgoSwitch.RimAlgoSwitch) or enhanced remote interference adaptive avoidance (controlled by the RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter) is enabled.

4.5.4.1.2 Using MML Commands

Before using MML commands, refer to [4.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on UL_SYMBOL_SHUTDOWN_SW  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=UL_SYMBOL_SHUTDOWN_SW-1;  
//Setting an energy saving policy of uplink symbol power saving  
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=00&00&00,  
StopTime=06&00&00, PowerSavingType=UL_SYMBOL_SHUTDOWN, EvalPeriodGranularity=MINUTE_LEVEL,  
ShutdownTrigJudgePeriod=5, UserNumThld=5, UserNumOffset=5, DlPrbThld=30, DlPrbOffset=10,  
UlPrbThld=30, UlPrbOffset=10;  
//(Optional) Turning on UL_SYM_SHUTDOWN_MULTI_CARR_SW  
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=UL_SYM_SHUTDOWN_MULTI_CARR_SW-1;  
//(Optional) Turning on RIM_UL_SYMBOL_SHUTDOWN_OPT_SW  
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,  
PowerSavingAlgoSwitch=RIM_UL_SYMBOL_SHUTDOWN_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//(Optional) Turning off UL_SYM_SHUTDOWN_MULTI_CARR_SW  
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=UL_SYM_SHUTDOWN_MULTI_CARR_SW-0;  
//(Optional) Turning off RIM_UL_SYMBOL_SHUTDOWN_OPT_SW  
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,  
PowerSavingAlgoSwitch=RIM_UL_SYMBOL_SHUTDOWN_OPT_SW-0;  
//Turning off UL_SYMBOL_SHUTDOWN_SW  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=UL_SYMBOL_SHUTDOWN_SW-0;  
//Removing the energy saving policy of uplink symbol power saving  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
```

4.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.5.4.2 Activation Verification

After uplink symbol power saving is enabled, use either of the following methods to check whether the function has taken effect:

- Method 1:
 - a. Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Switch** parameter is **UL Symbol Power Saving Switch:On** in the command output, uplink symbol power saving has been enabled.
 - b. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **Uplink Symbol Power Saving**.

 NOTE

If there are two or more records with **Power Saving Type** being **Uplink Symbol Power Saving**, uplink symbol power saving has been enabled in two or more time periods.

- c. Run the **DSP TIME** command. Check whether the system time is within an uplink symbol power saving period.
 - If it is, go to **d**.
 - If it is not, wait until the start time of uplink symbol power saving arrives.
 - d. Check whether the downlink PRB usage of the cell is less than or equal to the downlink PRB usage threshold for starting uplink symbol power saving (specified by the **NRDUCellPowerSaving.DlPrbThld** parameter), whether the uplink PRB usage of the cell is less than or equal to the uplink PRB usage threshold for starting this function (specified by the **NRDUCellPowerSaving.UlPrbThld** parameter), and whether the number of RRC_CONNECTED UEs in the cell (indicated by **N.User.RRCConn.Avg**) is less than or equal to the UE number threshold for starting this function (specified by the **NRDUCellPowerSaving.UserNumThld** parameter).
 - If they are, go to **e**.
 - If one or more are not, wait until the downlink PRB usage of the cell is less than or equal to the downlink PRB usage threshold for starting uplink symbol power saving, the uplink PRB usage of the cell is less than or equal to the uplink PRB usage threshold for starting this function, and the number of RRC_CONNECTED UEs in the cell is less than or equal to the UE number threshold for starting this function.
 - e. Wait for a period of time (specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter). Then, run the **DSP NRDUCELL** command. Verify that the value of the **Power Saving State** parameter is **UL Symbol Power Saving:Enabled**, which indicates that uplink symbol power saving has taken effect.
- Method 2:
Run the **DSP NRDUCELLPOWERSAVING** command to query the status of uplink symbol power saving.
 - If the value of **Power Saving Type** is **Uplink Symbol Power Saving** and the value of **Current State** is **Enabled**, uplink symbol power saving has taken effect.
 - If the value of **Power Saving Type** is **Uplink Symbol Power Saving** and the value of **Current State** is **Disabled**, check the **Current Disabled Cause** parameter value to determine the reason why uplink symbol power saving did not take effect. **Table 4-12** lists possible reasons.

Table 4-12 Possible reasons why uplink symbol power saving did not take effect

Reason	Description
Insufficient License	No license for this function is available.

Reason	Description
Switch Turned Off	The function switch is turned off.
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable for other reasons.
Triggering Time Not Reached	The current time is not within the range specified for uplink symbol power saving.
Impacted Feature Taken Effect	None
Downlink PRB Threshold Not Reached	None
Uplink PRB Threshold Not Reached	None
UE Number Threshold Not Reached	None
In Penalty State	A penalty is being imposed upon an exit from the uplink symbol power saving state.
RF Hardware Capability Not Supported	Other internal faults occur.
RF Channel Not Coordinated	• The channels of the RF module are faulty. • The antenna mode is incompatible.
RF Group Configuration Condition Not Met	Multiple carriers are set up on the RF module, and any of the following conditions is met: • The UL_SYM_SHUTDOWN_MULTI_CARR_SW option of the gNB PwrSavingAlgo.PwrSavingAlgoSwitch parameter is deselected. • The RF module is not a multi-band RF module.
Others	• Though it is currently within the range of time in which uplink symbol power saving can take effect, uplink symbol power saving is in its triggering evaluation period, which is specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter. • Internal software or other reasons cause the function to fail.

4.5.4.3 Network Monitoring

- Using counters
 - Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.
On the live network, the one-week base station energy consumption data obtained before and after uplink symbol power saving is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.
 - The duration of uplink symbol power saving can be observed using the counter **VS.RADIOEQM.UL.SymbolShutdown.Dur** (for an RF module) on the MAE-Access.
- Using monitoring items
When uplink symbol power saving is in effect:
 - On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management > NR > Cell Performance Monitoring > RSSI Statistic Monitoring**. Check that this monitoring item is present but has no reporting records.
 - On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management > NR > Cell Performance Monitoring > Interference Detect Monitoring**. Check that this monitoring item is present but has no reporting records.

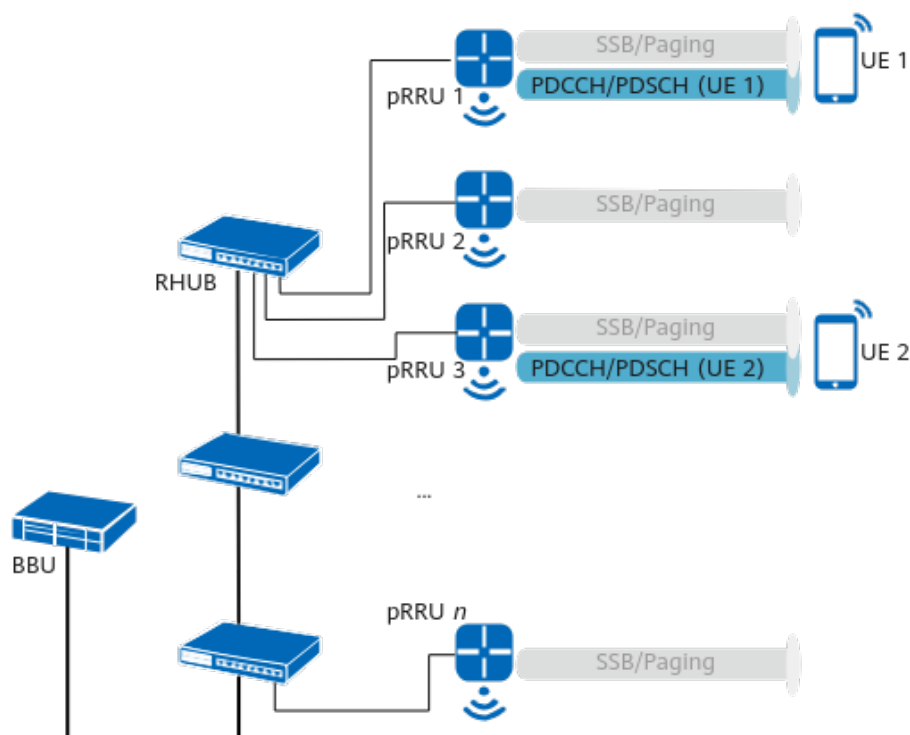
4.6 pRRU-Level Precise Symbol Power Saving (Low-Frequency TDD)

4.6.1 Principles

The LampSite solution adopts a three-level architecture composed of the BBU, RHUBs, and pRRUs. RF signals of multiple pRRUs are combined in an RHUB. All these pRRUs transmit downlink data to a UE even if the UE is covered by only one or two pRRUs in indoor coverage scenarios. Such transmission wastes energy. The function of pRRU-level precise symbol power saving is introduced to reduce energy consumption. As [Figure 4-11](#) illustrates, this function precisely controls the scope of pRRUs that transmit downlink data. Specifically:

- **If a pRRU does not serve any UEs, the pRRU does not send the PDSCH data. To do this, the base station shuts down the PA of this RF module in the PDSCH symbols to reduce energy consumption.**
- **If a pRRU is serving UEs, the pRRU sends UE-specific PDCCH and PDSCH data. To do this, the base station starts the PA of this RF module in real time to ensure data transmission integrity.**

Figure 4-11 Working principles for pRRU-level precise symbol power saving



pRRU-level precise symbol power saving can be enabled by selecting the **DL_PRECISE_SYMBOL_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter.

4.6.2 Network Analysis

4.6.2.1 Benefits

If a pRRU does not serve any UEs, the pRRU does not send the PDSCH data. This reduces energy consumption as the base station shuts down the PA of this RF module in the PDSCH symbols. The more concentrated the UEs, the greater the energy saving gains.

4.6.2.2 Impacts

The impacts between this function and other functions are described in [4.6.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.6.3 Requirements

4.6.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021203	gNodeB Power Saving	NR0S0GNBPS00	per Cell

NOTE

- For details about the consumption principles of feature license items in hyper cells or combined cells, see *Hyper Cell* or *Cell Combination*.
- The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	pRRU-level precise symbol power saving requires symbol power saving to be enabled.

4.6.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	VMIMO in DAS-based indoor coverage	NRDUCellAlgoSwitch.VmimoSwitch	<i>Indoor Coverage</i>	VMIMO in DAS-based indoor coverage does not work with pRRU-level precise symbol power saving.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAIgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	Distributed massive MIMO does not work with pRRU-level precise symbol power saving.
Low-frequency TDD	Distributed MIMO	D_MIMO_SERVICE_SWITCH option of the NRDUCellAIgoSwitch.DmMimoSwitch parameter	None	Distributed MIMO does not work with pRRU-level precise symbol power saving.
Low-frequency TDD	DIS-DAS TRP coordination	DIS_DAS_TRP_COORD_SW option of the NRDUCellMultiTrp.MultiTrpAlgoSwitch parameter	None	DIS-DAS TRP coordination does not work with pRRU-level precise symbol power saving.
Low-frequency TDD	Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to JOINT_MODE	<i>Cell Combination</i>	Joint Transmission Cell Combination does not work with pRRU-level precise symbol power saving.
Low-frequency TDD	Channel-based positioning for light pRRUs	LIGHT_PRRU_CHN_POS_SW option of the NRDUCellTrpAlgo.LcsPollingAlgoSw parameter	None	Channel-based positioning for light pRRUs does not work with pRRU-level precise symbol power saving.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Combining noise suppression	RF_COMBINING_NOISE_SUPPRESS_SW option of the NRDUCellULSch.RfCombSwitch parameter	None	pRRU-level precise symbol power saving does not work with combining noise suppression.
Low-frequency TDD	Rank selection based on the SRS power difference	SRS_DIFF_BASED_RANK_SELECTION_SW option of the NRDUCellULgoSwitch.RankSelectAlgorithmSw parameter	None	pRRU-level precise symbol power saving does not work with rank selection based on the SRS power difference.
Low-frequency TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	pRRU-level precise symbol power saving does not work with dual-BBU backup in dual-network redundancy networking.

4.6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellULgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	pRRU-level precise symbol power saving does not take effect when intelligent carrier shutdown is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	pRRU-level precise symbol power saving does not take effect when timing carrier shutdown is in effect.
Low-frequency TDD	pRRU deep dormancy in scheduled mode	RRU.DORMANCY_SW set to ON	<i>Multi-RAT Coordinated Energy Saving</i>	pRRU-level precise symbol power saving does not take effect when pRRU deep dormancy in scheduled mode is in effect.
Low-frequency TDD	LTE and NR intelligent carrier shutdown	NR: LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter LTE: CellAlgoExtSwitch.LnrSmartCarrierShutdownSw	<i>Multi-RAT Coordinated Energy Saving</i>	pRRU-level precise symbol power saving does not take effect when LTE and NR intelligent carrier shutdown is in effect.

4.6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

All baseband processing units except the UBBPfw support this function. In addition, one of the following configuration requirements must be met:

- The **NRDUCellTrp.BasebandEqmId** parameter is set to **255**.
- The **NRDUCellTrp.BasebandEqmId** parameter is set to a value other than **255**, and the baseband equipment works in NR-only mode.

RF Modules

All NR-capable pRRU565 xAB and pRRU575 xAB modules (x is a digit and A and B stand for different letters) support this function. The **RRU.RT** parameter must be set to **EPRRU**. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.6.3.4 Networking

- pRRU-level precise symbol power saving works only when the cells with this function enabled on the same CPRI chain share a baseband processing unit.
- There are no more than two values of the **NRDUCell.FrequencyBand** parameter for cells served by the same base station.
- pRRU-level precise symbol power saving does not work in an inter-board trunk chain topology. A trunk chain topology is indicated by the **RRUCHAIN.77** parameter setting of **TRUNK_CHAIN**.

4.6.3.5 Cells

- The **NRDUCellTrp.TxRxMode** parameter must be set to **4T4R** or **2T2R**.
- The **gNBFreqBandConfig.FrameOffset** parameter must be set to **4294967295**.
- The **NRDUCell.FrequencyBand** parameter cannot be set to **N257**, **N258**, **N260**, or **N261**.

4.6.3.6 Others

None

4.6.4 Operation and Maintenance

4.6.4.1 Data Configuration

4.6.4.1.1 Data Preparation

[Table 4-13](#) describes the parameters used for function activation.

Table 4-13 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	Select the DL_PRECISE_SYMBOL_SHUTDOWN_SW option of this parameter when pRRU-level precise symbol power saving is required.

4.6.4.1.2 Using MML Commands

Before using MML commands, refer to [4.6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the pRRU-level precise symbol power saving switch  
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=DL_PRECISE_SYMBOL_SHUTDOWN_SW-1;
```

Deactivation Command Examples

```
//Turning off the pRRU-level precise symbol power saving switch  
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=DL_PRECISE_SYMBOL_SHUTDOWN_SW-0;
```

4.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.6.4.2 Activation Verification

1. Run the **LST NRDUCCELLALGOSWITCH** command. If the value of the **Power Saving Extension Switch** parameter is **DL Precise Symbol Power Saving Switch:On** in the command output, pRRU-level precise symbol power saving has been enabled.
2. Run the **DSP NRDUCCELL** command to check the status of pRRU-level precise symbol power saving in the cell.

- If the value of the **Power Saving State** parameter is **DL Precise Symbol Power Saving:Enabled**, pRRU-level precise symbol power saving has taken effect in at least one TRP of the cell.
 - If the value of the **Power Saving State** parameter is **DL Precise Symbol Power Saving:Disabled**, pRRU-level precise symbol power saving has not taken effect in the cell. The possible reasons are as follows:
 - The RF module does not support pRRU-level precise symbol power saving. For details about the requirements of this function for RF modules, see [RF Modules](#).
 - The license item "gNodeB Power Saving" has not been acquired.
3. (Optional) For a hyper cell or combined cell, run the **DSP NRDUCELLTRP** command to query the status of pRRU-level precise symbol power saving of each TRP in the cell.
- If the value of the **Power Saving State** parameter is **DL Precise Symbol Power Saving:Enabled** for a TRP, pRRU-level precise symbol power saving has taken effect in this TRP.
 - If the value of the **Power Saving State** parameter is **DL Precise Symbol Power Saving:Disabled** for a TRP, pRRU-level precise symbol power saving has not taken effect in this TRP.

4.6.4.3 Network Monitoring

Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after pRRU-level precise symbol power saving is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases after this function is enabled.

The increase in the duration of symbol power saving brought by pRRU-level precise symbol power saving can be observed using the counter **VS.RADIOEQM.SymbolShutdown.Dur** (for an RF module) on the MAE-Access.

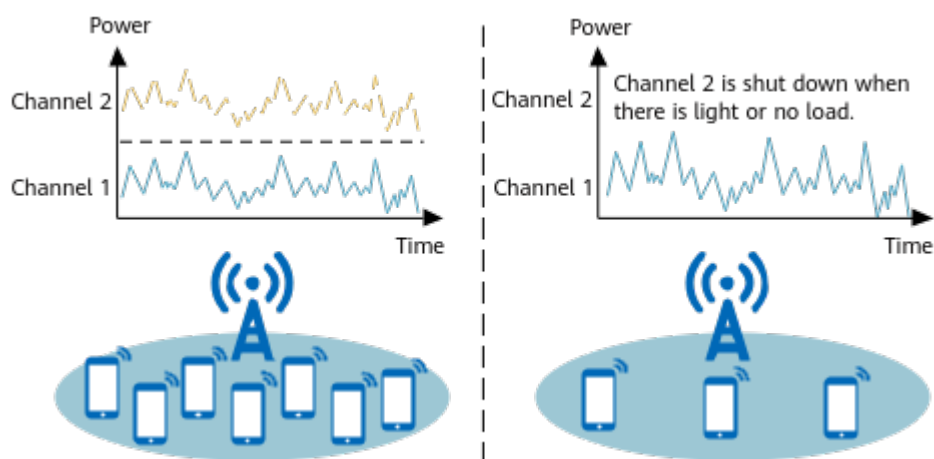
5 Device-Level Energy Saving (Space Domain)

5.1 RF Channel Intelligent Shutdown

5.1.1 Principles

RF channel intelligent shutdown enables a base station to automatically shut down some transmit channels of a cell when the network is under light or no load during preset periods. This way, this function helps save energy. [Figure 5-1](#) illustrates how this function works.

Figure 5-1 Working principles for RF channel intelligent shutdown



RF channel intelligent shutdown can be enabled by selecting the **RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter. [Table 5-1](#) describes the parameters in the **NRDUCellPowerSaving** MO that specify the effective period, load threshold, and other information under an energy saving policy of RF channel intelligent shutdown.

Table 5-1 Parameters related to a policy of RF channel intelligent shutdown

Parameter Name	Parameter ID	Description
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	This parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	This parameter specifies the energy saving type indicated by an energy saving policy index. Set this parameter to RF_SHUTDOWN to enable RF channel intelligent shutdown.
Start Time	NRDUCellPowerSaving.StartTime	These parameters specify the effective period, evaluation period granularity, evaluation periods, and conditions for Entering the RF Channel Intelligent Shutdown State and Exiting the RF Channel Intelligent Shutdown State in a cell.
Stop Time	NRDUCellPowerSaving.StopTime	
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	
Second-Level Exit Evaluation Period	NRDUCellPowerSaving.SecondLevelExitEvalPrd	
Second-Level Trigger Evaluation Period	NRDUCellPowerSaving.SecondLevelTrigEvalPrd	
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	

Parameter Name	Parameter ID	Description
User Number Offset	NRDUCellPowerSaving.UserNumOffset	
Power Saving Penalty Duration	NRDUCellPowerSaving.PwrSavingPenalty-Time	This parameter specifies the duration of power saving penalty on a cell upon Exiting the RF Channel Intelligent Shutdown State .
Coverage Compensation Mode	NRDUCellPowerSaving.CovCompensation-Mode	For NR low-frequency TDD: These parameters are used for transmit power compensation when RF channel intelligent shutdown takes effect. For details, see Transmit Power Compensation .
Power Compensation Offset	NRDUCellPowerSaving.PwrCompensationOffset	
Channel Power Algorithm Switch	BANDWIDTH_BASED_POWER_AGG_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter	For low-frequency NR TDD: RF channel intelligent shutdown takes effect only when this option is selected in certain scenarios. For details, see Transmit Power Compensation .
RF Channel Shutdown Mode	NRDUCellPowerSaving.RfChnShutdownMode	For low-frequency NR TDD: This parameter specifies the channel shutdown proportion when RF channel intelligent shutdown takes effect. For details, see Channel Shutdown Mode .
Channel Power Algorithm Switch	DL_CHN_PWR_CMP_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter	For low-frequency NR TDD: The controlled function increases the user-perceived rate when only one NR carrier is configured on a 32T32R or higher low-frequency RF module. For details, see Channel Shutdown Performance Improvement (Low-Frequency TDD) .

Entering the RF Channel Intelligent Shutdown State

Within the time period specified by **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** for RF channel intelligent shutdown to take effect for a cell, the base station periodically evaluates whether the cell enters the RF channel intelligent shutdown state. The granularity of the evaluation period is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.

- If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter.

- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** parameter.

For RF channel intelligent shutdown, the value of the **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** parameter must be greater than or equal to 3.

The cell enters the RF channel intelligent shutdown state if all of the following conditions are met:

- Downlink PRB usage:
For low-frequency TDD: The downlink PRB usage of the cell is less than or equal to the value of the **NRDUCellPowerSaving.DlPrbThld** parameter.
Downlink PRB usage of the cell = (Average number of PRBs used by PDSCH DRB services – Average number of extended PRBs used by the PDSCH)/
Average number of available downlink PRBs.
For high-frequency TDD: The sector-level downlink PRB usage is less than or equal to the value of the **NRDUCellPowerSaving.DlPrbThld** parameter.
Sector-level downlink PRB usage = Total number of PRBs used by PDSCH DRB services on all component carriers (CCs) in the sector/Number of available downlink PRBs in the sector.
- Number of RRC_CONNECTED UEs:
For low-frequency TDD: The number of RRC_CONNECTED UEs in the cell is less than or equal to the value of the **NRDUCellPowerSaving.UserNumThld** parameter. The number of RRC_CONNECTED UEs in a co-deployed SUL cell is the sum of **N.User.RRCCConn.Avg** and **N.User.SUL.UL.Avg**; and the number of RRC_CONNECTED UEs in other cells is indicated by **N.User.RRCCConn.Avg**.
For high-frequency TDD: The sector-level number of RRC_CONNECTED UEs is less than or equal to the value of the **NRDUCellPowerSaving.UserNumThld** parameter. This number is the sum of the **N.User.RRCCConn.Avg** counter values for all carriers in the sector.

Transmit Power Compensation

For a cell in the RF channel shutdown state, transmit power compensation is performed for the common channels of the cell to ensure the base station coverage and service provisioning.

For low-frequency TDD:

- For cells served by macro base stations, the transmit power compensation value for the common channels of a cell can be automatically determined or manually configured, depending on the setting of the **NRDUCellPowerSaving.CovCompensationMode** parameter:
 - If this parameter is set to **AUTO**, the transmit power compensation value for the common channels of the cell is automatically configured. In this scenario, for certain RF modules and certain power offset settings in NR TDD, RF channel intelligent shutdown takes effect only when bandwidth-based power aggregation (controlled by the **BANDWIDTH_BASED_POWER_AGG_SW** option of the **NRDUCellChnPwr.ChnPwrAlgoSwitch** parameter) is enabled. For specific RF modules and power offset settings, contact Huawei engineers.

- If this parameter is set to **MANUAL**, the transmit power compensation value for the common channels of the cell is manually configured using the **NRDUCellPowerSaving.PwrCompensationOffset** parameter.
- For cells served by LampSite base stations, the transmit power compensation value for the common channels of a cell can only be manually configured. That is, the **NRDUCellPowerSaving.CovCompensationMode** parameter can only be set to **MANUAL**. The transmit power compensation value for the common channels of a cell is manually configured by using the **NRDUCellPowerSaving.PwrCompensationOffset** parameter.

 NOTE

When the transmit power compensation value for the common channels of a cell is manually configured, it cannot exceed the RF module's maximum capabilities of aggregating the cell common channel transmit power. If it exceeds the RF module's maximum capabilities, the latter takes effect.

For high-frequency TDD: The transmit power compensation value for the common channels of a cell can only be automatically determined. That is, the **NRDUCellPowerSaving.CovCompensationMode** parameter can only be set to **AUTO**. After a cell enters the RF channel shutdown state, the following information is no longer transmitted on some CCs: SSB, remaining minimum system information (RMSI), paging messages, and tracking reference signal (TRS). Power aggregation is performed on the remaining CCs to ensure that the coverage of common channels remains unchanged after the RF channel shutdown.

Channel Shutdown Mode

For a cell served by a macro base station, the proportion of RF channels to be shut down can be configured using the **NRDUCellPowerSaving.RfShutdownNum** parameter. By default, half of the RF channels for the cell are shut down in the uplink or downlink.

- For high-frequency NR TDD: The base station supports downlink channel shutdown for a cell with the **NRDUCell.SlotAssignment** parameter set to **4_1_DDSU** only. In addition, only half of the downlink RF channels can be shut down.
- For low-frequency NR TDD: The **NRDUCellPowerSaving.RfChnShutdownMode** parameter specifies whether to shut down both uplink and downlink channels or shut down downlink channels.
 - If this parameter is set to **DL_UL_CHANNEL_SHUTDOWN**, both uplink and downlink channels are shut down. That is, half of the transmit channels and half of the receive channels can be shut down simultaneously. For RF modules that do not support simultaneous shutdown of uplink and downlink channels, only downlink channels are shut down.
 - If this parameter is set to **DL_CHANNEL_SHUTDOWN**, downlink channels are shut down.
- For NR SUL, the uplink channels can be shut down by setting the **NRDUCellPowerSaving.RfChnShutdownMode** parameter to **DL_UL_CHANNEL_SHUTDOWN** and selecting the **BI_DIR_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter.

A LampSite base station shuts down three out of four downlink channels of a 4T cell or one out of two downlink channels of a 2T cell by default.

Exiting the RF Channel Intelligent Shutdown State

After the cell enters the RF channel intelligent shutdown state, it periodically evaluates whether exit conditions are met. The granularity of the evaluation period is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.

- If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter.
- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelExitEvalPrd** parameter.

If any of the exit conditions is met, the cell automatically exits the RF channel intelligent shutdown state and will not enter the RF channel intelligent shutdown state again within the penalty time specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter, preventing frequent entry into and exit from RF channel intelligent shutdown. The exit conditions are as follows:

- The current time is not within the RF channel intelligent shutdown period specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
- Number of RRC_CONNECTED UEs:
For low-frequency TDD: Number of RRC_CONNECTED UEs in the cell > (**NRDUCellPowerSaving.UserNumThld** + **NRDUCellPowerSaving.UserNumOffset**)
For high-frequency NR TDD: Sector-level number of RRC_CONNECTED UEs > (**NRDUCellPowerSaving.UserNumThld** + **NRDUCellPowerSaving.UserNumOffset**)
The **NRDUCellPowerSaving.UserNumOffset** parameter specifies the offset to the number of RRC_CONNECTED UEs and is used to prevent a cell from frequently entering and exiting the RF channel intelligent shutdown state.
- Downlink PRB usage:
For low-frequency TDD: Downlink PRB usage of the cell > (**NRDUCellPowerSaving.DlPrbThld** + **NRDUCellPowerSaving.DlPrbOffset**)
For high-frequency NR TDD: Sector-level downlink PRB usage > (**NRDUCellPowerSaving.DlPrbThld** + **NRDUCellPowerSaving.DlPrbOffset**)
The **NRDUCellPowerSaving.DlPrbOffset** parameter specifies the offset to the downlink PRB usage threshold and is used to prevent a cell from frequently entering and exiting the RF channel intelligent shutdown state.
 - (TDD) A cell may fail to exit the RF channel intelligent shutdown state if the sum of the **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset** parameter values exceeds 35 in the scenarios listed in [Table 5-2](#).

Table 5-2 Beam coverage scenarios for some RF models in automatic coverage compensation mode

Model	TX/RX Mode	Supported Beam Coverage Scenario
AAU531x	32T32R	DEFAULT, SCENARIO_1, SCENARIO_6, SCENARIO_7, SCENARIO_8, SCENARIO_12, and SCENARIO_13
AAU561x	64T64R	DEFAULT, SCENARIO_1, SCENARIO_2, SCENARIO_3, SCENARIO_4, SCENARIO_6, SCENARIO_7, SCENARIO_8, SCENARIO_9, SCENARIO_12, and SCENARIO_13
5000 series RRUs	8T8R	DEFAULT, SCENARIO_3, and SCENARIO_4

- (TDD) A cell fails to exit the RF channel intelligent shutdown state if the sum of the **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset** parameter values exceeds 80 in the scenarios other than those listed in [Table 5-2](#).

Channel Shutdown Performance Improvement (Low-Frequency TDD)

When only one NR carrier is configured on a 32T32R or higher low-frequency NR TDD RF module, the downlink channel power compensation function can be used to increase the user-perceived rate that can be achieved after the cell enters the RF channel intelligent shutdown state. This function is controlled by the **DL_CHN_PWR_CMP_SW** option of the **NRDUCellChnPwr.ChnPwrAlgoSwitch** parameter. With this option selected, if RF channel intelligent shutdown is in effect and the maximum output power of the RF module is greater than the total channel power configured for the cell, the base station uses this power margin (the difference between the two) to supplement the transmit power of downlink channels, increasing the user-perceived rate.

Service-Type-based Dynamic Exit from RF Channel Intelligent Shutdown

For specific service types, cells can dynamically exit RF channel intelligent shutdown. This is known as service-type-based dynamic exit from RF channel intelligent shutdown. When there is any UE with a specific type of service in a cell,

the cell exits the RF channel intelligent shutdown state. This type of exit ensures the performance of UEs performing that type of service. The service types involved are specified by the **gNBQciBearer.ServiceType** parameter.

Service-type-based dynamic exit from RF channel intelligent shutdown takes effect when all of the following conditions are met:

- RF channel intelligent shutdown is enabled.
- The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.
- The **RF_SHUTDOWN_EXIT_SW** option of the **NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter is selected in the group identified by the **NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter that is set to the same value as the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter for a specific service type.

5.1.2 Network Analysis

5.1.2.1 Benefits

RF channel intelligent shutdown reduces energy consumption when the network is under light or no load. However, the energy saving gains are limited in the following scenarios:

- If an RF module is configured with multiple carriers, RF channel intelligent shutdown must be configured for all the carriers and the effective durations must overlap. In addition, the coverage compensation mode, channel shutdown mode, channel shutdown evaluation period granularity, and channel shutdown triggering and exit evaluation periods must be set consistently between these carriers. Otherwise:
 - For a macro base station, RF channel intelligent shutdown cannot take effect and therefore cannot yield energy saving gains.
 - For a LampSite base station, RF channel intelligent shutdown can still take effect but delivers reduced energy saving gains.
- Assume that RF channel intelligent shutdown has taken effect in single-carrier scenarios and another carrier needs to be added for capacity expansion. To enable this function to continue to work after capacity expansion, RF channel intelligent shutdown should be configured for the new carrier with the same effective duration, coverage compensation mode, channel shutdown mode, evaluation period granularity, entry evaluation period, and exit evaluation period as the existing carrier. Otherwise:
 - For a macro base station, RF channel intelligent shutdown no longer takes effect for the existing carrier, causing the energy saving gains to decrease, after capacity expansion.
 - For a LampSite base station, the energy saving gains decrease after capacity expansion, though RF channel intelligent shutdown can still take effect for the existing carrier.
- To prevent possible hardware faults caused by an excessively large temperature differential, this function may not take effect or may be

terminated when the temperature differential of RF modules is excessively large. Energy saving gains may decrease in this case.

- For low-frequency NR TDD: Downlink channel power compensation (controlled by the **DL_CHN_PWR_CMP_SW** option of the **NRDUCellChnPwr.ChnPwrAlgoSwitch** parameter) increases the user-perceived rate.
- If the value of the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter is greater than 0, energy saving gains decrease.
- Service-type-based dynamic exit from RF channel intelligent shutdown relieves the impact of RF channel intelligent shutdown on specific service types but reduces energy saving gains.

For high-frequency TDD: When both RF channel intelligent shutdown and downlink PDSCH coverage enhancement (controlled by the **PDSCH_COV_ENHANCE_SW** option of the **NRDUCellChnPwr.ChnPwrAlgoSwitch** parameter) are enabled and downlink PDSCH coverage enhancement takes effect, RF channel intelligent shutdown may be more likely to take effect, yielding more energy saving gains. For details about downlink PDSCH coverage enhancement, see *mmWave Super Coverage (High-Frequency TDD)*.

5.1.2.2 Impacts

The impacts between this function and other functions are described in [5.1.3.2.3 Function Impacts](#).

When RF channel intelligent shutdown starts or stops, the UE data rate may decrease and the service drop rate may increase for approximately 3 minutes.

When RF channel intelligent shutdown is in effect, downlink channel shutdown has the following impacts:

- The array and power gains decrease, exhibiting the impacts described in [Table 5-3](#).

Table 5-3 Impacts of array and power gain loss on the network

Scenario	Indicator	Description
Macro base station	Rank	For low-frequency TDD: • For 8T, 32T, and 64T RF modules, the average rank for downlink scheduling in individual cells remains almost unchanged or slightly decreases. • For 4T RF modules, the UE rank decreases to rank 2 or a lower one. • For 2T RF modules, the UE rank decreases to rank 1. For high-frequency TDD: No impact

Scenario	Indicator	Description
Macro base station	MCS	- If the average rank for downlink scheduling in a cell remains unchanged or slightly decreases, the MCS indexes for downlink scheduling may change due to factors such as the rank decrease rate, number of small-packet UEs, and UE location distribution. - In a cell serving a large number of small-packet UEs, the average MCS index of the cell remains almost unchanged because the MCS indexes for small-packet UEs are lowered before and after the feature takes effect. - In a cell serving a large number of UEs at the cell edge, the average MCS index of the cell decreases and the power loss has a greater impact on these UEs. - If the average rank for downlink scheduling in a cell decreases significantly and the SINR of UEs increases, the average MCS index of the cell increases.
Macro base station	IBLER	The IBLER changes with the MCS index.
Macro base station	CQI	The average CQI of a cell may decrease, depending on the UE distribution in the cell.
LampSite base station	Rank	The UE rank decreases to rank 1.
Macro or LampSite base station	Data rate and downlink PRB usage	The peak cell rate, peak UE rate, and user-perceived rate decrease; the scheduling delay increases. Given the same load, more downlink PRBs are required for transmitting the same amount of data, increasing the downlink PRB usage.
Macro or LampSite base station	Others	The mapping between the CSI-RS and the SSB or PRACH may change after shutdown, increasing the number of reconfigurations of the optimal beam for sending CSI-RSs.

- For low-frequency TDD: The processing delay increases, the scheduling capability deteriorates, the scheduling time is prolonged, and the uplink cell throughput and uplink UE throughput decrease by about 0% to 50%. In low-frequency TDD, in the scenarios listed in [Table 5-6](#), RF channel intelligent shutdown can take effect only if bandwidth-based power aggregation is enabled (by selecting the **BANDWIDTH_BASED_POWER_AGG_SW** option of the **NRDUCellChnPwr.ChnPwrAlgoSwitch** parameter). If it takes effect, the decreases in the cell throughput and UE throughput may increase.

For high-frequency TDD: The uplink MCS index, uplink cell data throughput, and uplink UE data throughput in the cell fluctuate.

For high-frequency TDD: In NSA networking, it is recommended that the **NR_FR2_MEAS_POLLING_FLAG** option of the **NrScgFreqConfig.AggregationAttribute** parameter be deselected when RF channel intelligent shutdown is enabled. If this option is selected in this case, the value of the **L.NsaDc.SgNB.Add.Att** counter will decrease.

Table 5-4 lists the amount of the decrease in the downlink cell and UE throughputs. The longer the distance between a UE and a base station, the larger the decrease in the downlink throughput. The larger the value of any parameter listed in **Table 5-5**, the larger the decrease in downlink throughput.

Table 5-4 Impacts on the downlink cell and UE throughputs in different scenarios

Scenario	Decrease in the Downlink Cell and UE Throughputs
Single base station networking except for scenarios listed in Table 5-2	0% to 70%
Continuous networking except for scenarios listed in Table 5-2	0% to 50%
Scenarios listed in Table 5-2 in single base station or continuous networking	60% to 70%

Table 5-5 Power offset parameters

Parameter Name	Parameter ID
Max SSB Power Offset	NRDUCellTrpBeam.MaxSsbPwrOffset
Maximum Common DCI Power Offset	NRDUCellChnPwr.MaxCommonDciPwrOffset
Max Paging Power Offset	NRDUCellChnPwr.PagingMaxPwrOffset
Max OSI Power Offset	NRDUCellChnPwr.OsiMaxPwrOffset
Max RMSI Power Offset	NRDUCellChnPwr.RmsiMaxPwrOffset
Max Msg2 Power Offset	NRDUCellChnPwr.Msg2MaxPwrOffset
Max Msg4 Power Offset	NRDUCellChnPwr.Msg4MaxPwrOffset

Table 5-6 RF channel intelligent shutdown scenario for certain low-frequency NR TDD AAUs/RRUs in automatic coverage compensation mode

Model	TX/RX Mode	Power Offset Value Range	Supported Beam Coverage Scenario
AAU531x, AAU561x, or AAU581x	64T64R or 32T32R	(0, 3]	DEFAULT, SCENARIO_1, SCENARIO_2, SCENARIO_4, SCENARIO_3, SCENARIO_6, SCENARIO_7, SCENARIO_8, SCENARIO_9, SCENARIO_12, SCENARIO_13, and CUSTOMIZED_BEAM_S CENARIO
AAU531x, AAU561x, or AAU581x	64T64R or 32T32R	(3, 6]	EXPAND_SCENARIO_1 and CUSTOMIZED_BEAM_S CENARIO
AAU533x, AAU563x, AAU5649, or AAU583x	64T64R or 32T32R	(0, 6]	DEFAULT, SCENARIO_1, SCENARIO_2, SCENARIO_3, SCENARIO_4, SCENARIO_6, SCENARIO_7, SCENARIO_8, SCENARIO_9, SCENARIO_12, SCENARIO_13, and CUSTOMIZED_BEAM_S CENARIO
AAU533x, AAU563x, AAU5649, or AAU583x	64T64R or 32T32R	(3, 9]	EXPAND_SCENARIO_1 and CUSTOMIZED_BEAM_S CENARIO
5000 series RRUs	8T8R	Any value	DEFAULT and SCENARIO_3
Note: The Power Offset Value Range column of this table refers to the value range of any parameter listed in Table 5-5 .			

- For low-frequency TDD: Further impacts may be imposed on the performance indicators such as the cell throughput and PRB usage when certain RF modules are used.

- For RF modules that support low-frequency NR TDD 2T2R/4T4R/8T8R, 256QAM does not work after channel shutdown. As a result, the cell throughput may decrease. The amount of the decrease may be greater in multi-carrier scenarios than in single-carrier scenarios.
- In a multi-carrier scenario where an NR TDD 8T8R RF module is used the amount of available PRB resources for a cell decreases when RF channel intelligent shutdown takes effect and the transmit power compensation value for the common channels of the cell is between the RF module's basic and maximum capabilities of aggregating the cell common channel transmit power. As a result, the PDCCH CCE usage, PRB usage, cell throughput, and UE throughput decrease.
- For a 2T or 4T RRU, as the RRU has a small number of channels, shutting down half of the channels decreases the rank used, causing a significant increase in the PRB usage and affecting the energy saving gains. To cope with this situation, it is recommended that:
For low-frequency NR TDD: The **NRDUCellPowerSaving.DlPrbThld** parameter be set to 2 and the **NRDUCellPowerSaving.DlPrbOffset** parameter be set to 3.
- There are impacts on KPIs related to handovers, access, and service drop rate.
For low-frequency TDD: The setting of the **NRDUCellPowerSaving.CovCompensationMode** parameter has the following impacts:
 - If this parameter is set to **AUTO**, the cell coverage may be affected, though power compensation is automatically performed for common channels to ensure that the coverage after RF channel shutdown is aligned as much as possible with the coverage before RF channel shutdown. If cell coverage alters, the number of handovers and the handover success rate may change, the access success rate may decrease, and the service drop rate may increase. In addition, the power compensation for common channels may increase the common-channel interference between base stations when the inter-base-station distance is small.
 - If this parameter is set to **MANUAL**, the coverage after RF channel shutdown may be still poorer than that before RF channel shutdown, even if the **NRDUCellPowerSaving.PwrCompensationOffset** parameter has been set to the maximum. In this case, the number of handovers and the handover success rate change, the access success rate decreases, and the service drop rate increases.
For high-frequency TDD: The service drop rate of UEs increases, and the number of handovers and the handover success rate fluctuate.
- Others
 - When the proportion of small packets is high:
 - The user-perceived rate may fluctuate.
 - The cell throughput may fluctuate when the total duration of data transmission becomes shorter due to insufficient CCE resources.
 - For low-frequency TDD: In medium- or heavy-load scenarios, the DRB usage may decrease. In addition, if MU-MIMO Basic Pairing takes effect, the number of paired layers may increase, the average uplink CCE usage

and average downlink CCE usage may increase, and the CCE aggregation level may decrease. In light-load scenarios, the CCE usage and the uplink and downlink DCI allocation success rates may decrease.

- For low-frequency TDD: When an RF module is configured with more than two NR carriers, the transmit power compensation for the common channels in a cell may not meet the coverage and service requirements of the base station. Therefore, downlink channel shutdown may not take effect, to prevent the coverage and services of the base station from being affected. This decreases the energy saving gains.

For low-frequency TDD: Uplink and downlink channel shutdown has the following impacts in addition to the preceding impacts:

- Half of receive antennas are shut down and the reception performance deteriorates. Therefore:
 - The uplink user-perceived rate decreases.
 - The uplink RB usage increases.
 - The uplink MCS index decreases.
 - The uplink interference measurement result decreases.
 - The uplink IBLER increases.
 - The PUCCH SINR and PUSCH SINR decrease.
 - The uplink HARQ retransmission rate increases.
- The uplink coverage is affected, which may have the following impacts:
 - The RRC connection setup success rate decreases.
 - The handover success rate decreases.
 - The number of RRC_CONNECTED UEs decreases.

For NR TDD: After downlink channel power compensation is enabled, the **Cell Downlink Average Throughput (DU)** changes and the energy saving gains decrease.

5.1.3 Requirements

5.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-031205	RF Channel Intelligent Shutdown	NR0S00RCIS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Service-type-based dynamic exit from RF channel intelligent shutdown takes effect only if RF channel intelligent shutdown has been enabled.

5.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL and NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>Hyper Cell</i>	- For a hyper cell served by a macro base station, RF channel intelligent shutdown can be enabled in high-speed scenarios when the HIGH_SPEED_CELL_MUTING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected. Otherwise, RF channel intelligent shutdown cannot be enabled. - For a hyper cell served by a macro base station, RF channel intelligent shutdown can be enabled in non-high-speed scenarios. After half of channels are shut down, the remaining TX/RX configuration must be compatible with the Hyper Cell feature.

RAT	Function Name	Function Switch	Reference	Description
				• For a hyper cell served by a LampSite base station, RF channel intelligent shutdown can be enabled. • Hyper Cell is incompatible with RF channel intelligent shutdown in high-frequency TDD.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	- NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE - NRDUCellMultiTrp.DataTransMode	Cell Combination	- When a combined cell served by a macro base station uses the basic or DPS transmission mode (with the NRDUCellMultiTrp.DataTransMode parameter set to BASIC_MODE or DPS_MODE), RF channel intelligent shutdown can be enabled in high-speed scenarios only if the HIGH_SPEED_CELL_MUTING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected. - When a combined cell served by a macro base station uses the basic or DPS transmission mode (with the NRDUCellMultiTrp.DataTransMode parameter set to

RAT	Function Name	Function Switch	Reference	Description
				BASIC_MODE or DPS_MODE), RF channel intelligent shutdown can be enabled in non-high-speed scenarios only if NRDUCellTrp.TxRxMode is set to 8T8R for all TRPs serving the cell. • When a combined cell served by a macro base station uses the joint transmission mode (with the NRDUCellMultiTrp.DataTransmissionMode parameter set to JOINT_MODE), RF channel intelligent shutdown can be enabled in high-speed scenarios only if the HIGH_SPEED_CELL_MUTING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
				- When a combined cell served by a macro base station uses the joint transmission mode (with the NRDUCellMultiTrp.DataTrans Mode parameter set to JOINT_MODE), RF channel intelligent shutdown can be enabled in non-high-speed scenarios. After half of channels are shut down, the remaining TX/RX configuration must be compatible with the Cell Combination feature. - For a combined cell served by a LampSite base station, RF channel intelligent shutdown can be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	RF channel intelligent shutdown is not supported in a distributed massive MIMO cell.
Low-frequency TDD High-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SWITCH option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	RF channel intelligent shutdown cannot take effect when the low latency and high reliability function is enabled.
Low-frequency TDD	VMIMO in DAS-based indoor coverage	NRDUCellAlgoSwitch.VmimoSwitch	<i>Indoor Coverage</i>	VMIMO in DAS-based indoor coverage is incompatible with RF channel intelligent shutdown.
Low-frequency TDD	Tunnel-based channel imbalance optimization	CSIRS_BEAM_POWER_BALANCE_SWITCH option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter	<i>Indoor Coverage</i>	RF channel intelligent shutdown cannot be enabled for a TDD macro cell with the NRDUCellTrp.TxRxMode parameter set to 2T2R or 4T4R and the CSIRS_BEAM_POWER_BALANCE_SWITCH option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TRP Antenna Type	NRDUCellTrpBeam.TrpAntType	None	When the NRDUCellTrpBeam.TrpAntType parameter is set to 8T_2222 , 8T_44 , or 8T_422 , RF channel intelligent shutdown cannot be enabled.
Low-frequency TDD	Intra-frequency assistant cell	NRDUCell.SupplementaryCellInd set to INTRA_FREQ_ASSISTANT_CELL	None	Intra-frequency assistant cells are incompatible with RF channel intelligent shutdown.
Low-frequency TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	RF channel intelligent shutdown is incompatible with the primary/backup function in intra-frequency dual-network redundancy networking.
Low-frequency TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	RF channel intelligent shutdown is incompatible with dual-BBU backup in dual-network redundancy networking.
Low-frequency TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	RF channel intelligent shutdown is incompatible with the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSB coverage pattern for emergency and normal services	NRDUCellTrpBeam.CoverageScenario	<i>Emergency Communications Solution (Low-Frequency TDD)</i>	If RF channel intelligent shutdown is enabled for a cell, the NRDUCellTrpBeam.CoverageScenario parameter cannot be set to SCENARIO_33 , SCENARIO_34 , or SCENARIO_35 for the cell.
Low-frequency TDD	Connected mode coverage pattern for emergency and normal services	NRDUCellTrpBeam.ConnModeCoverageScenarioExt	<i>Emergency Communications Solution (Low-Frequency TDD)</i>	If RF channel intelligent shutdown is enabled for a cell, the NRDUCellTrpBeam.ConnModeCoverageScenarioExt parameter must be set to DEFAULT for the cell.

5.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option (inter-base-station multi-carrier scenarios) of the NRDU CellAlg oSwitch.h.PowerSavingSwitc h parameter selected FAST_CARRIE R_SHU TDOW	<i>Energy Conservation and Emission Reduction</i>	When fast carrier shutdown takes effect: - If the local cell is a capacity-layer cell, RF channel intelligent shutdown cannot take effect in the scenarios listed in Table 5-6. - If the local cell is a basic-layer cell and RF channel intelligent shutdown takes effect in the scenarios listed in Table 5-6, the related capacity-layer cells may exit the fast carrier shutdown state.

RAT	Function Name	Function Switch	Reference	Description
		N_SW option of the NRDU CellAlg oSwitc h.<i>Power Saving ExtS witch</i> param eter selecte d • Basic- layer cells: FAST_CAR R_SHUT_S ERV_ASS UR_SW option of the NRDUCell AlgoSwit ch.<i>Power SavingExt Switch</i> parameter selected		
Low- frequency TDD High- frequency TDD	PDCCH symbol number adaptation	UE_PDCCH_S YM_NUM_A DAPT_SW option of the NRDUCellPd cch. <i>PdcchAl goExtSwitch</i> parameter	<i>Channel Management</i>	When RF channel intelligent shutdown is enabled, it is recommended that PDCCH symbol number adaptation be enabled to reduce the impact on the user-perceived rate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL 256QAM	NRDUCellAIgoSwitch.DL256QamSwitch	<i>Modulation Schemes</i>	TDD 2T2R/4T4R/8T8R RF modules do not support DL 256QAM after RF channel intelligent shutdown takes effect.
Low-frequency TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAIgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	RF channel intelligent shutdown does not take effect when remote interference adaptive avoidance is in effect.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAIgoSwitch.CoMpSwitch parameter	<i>CoMP</i>	If RF channel intelligent shutdown takes effect in a cell, intra-base-station joint transmission will become invalid for CoMP UEs that have been involved in intra-base-station joint transmission in the cell and will not take effect on UEs that are newly admitted to the cell. After the cell exits the RF channel intelligent shutdown state, intra-base-station joint transmission takes effect again.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Channel calibration	Macro base station: None LampSite base station: INTER_RRU_CHN_CALIB_SW option of the NRDUCellAIgoSwitch.ChnCalibSwitch parameter	None	If channel calibration starts before RF channel intelligent shutdown takes effect, RF channel intelligent shutdown cannot take effect. If channel calibration starts when RF channel intelligent shutdown is in effect, the RF channel intelligent shutdown status is not affected.
High-frequency TDD	Channel calibration	None	None	In 4T scenarios, channel calibration is not performed when RF channel intelligent shutdown is in effect.
Low-frequency TDD	Inter-cell cable sequence detection	AntPortOptDetPolicy parameter in the STR ANTENNAPORTOPTDET command set to INTER_CELL_DETECT	<i>MIMO (TDD)</i>	If RF channel intelligent shutdown is in effect during inter-cell cable sequence detection, the detection results will show that the detection fails in the corresponding cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	If the NRDUCellTrpBeam.CoverageScenario parameter is set to EXPAND_SCENARIO_1 for a cell enabled with Fusion Cell, SSBs cannot obtain power gains brought by Fusion Cell after RF channel intelligent shutdown takes effect.
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter, ENHANCED_NETWORK_SLICE_SW option of the NRCCellAlgoSwitch.NetworkSliceAlgoSwitch parameter, and BANDWIDTH_BASED_POWER_AGG_SW option of the NRDUCellChnPwr.ChnPowerAlgoSwitch parameter	<i>Network Slicing</i>	If NR DU cell resource management between network slices is in effect, RF channel intelligent shutdown cannot take effect in the scenarios listed in Table 5-6 .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	NRDUCellPdcch.OccupiedRbNum parameter and the PDCCH_RATEMATCH_SWITCH option of the NRDUCellPdcch.RateMatchingSwitch parameter	<i>Downlink Scheduling</i>	RF channel intelligent shutdown cannot take effect when rate-matching-pattern-configuration-based PDCCH rate matching is in effect.
Low-frequency TDD	TRS rate matching	TRS_RATEMATCH_SW option of the NRDUCellPdcch.RateMatchingSwitch parameter	<i>Downlink Scheduling</i>	When the TRS rate matching switch is off (that is, PDSCH data is transmitted in TRS symbols), TRS power aggregation cannot take effect in the scenarios listed in Table 5-6 .
Low-frequency TDD	RB allocation for downlink experience optimization	NRDUCellDLISchRes.DLExpOptRbIncreasePct	<i>Downlink Scheduling</i>	When both RB allocation for downlink experience optimization and RF channel intelligent shutdown take effect, an exit from the RF channel intelligent shutdown state may be performed earlier than expected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	RF channel intelligent shutdown does not take effect when RF module deep dormancy is in effect.
Low-frequency TDD High-frequency TDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	RF channel intelligent shutdown does not take effect when RF module super dormancy is in effect.
Low-frequency TDD	LTE and NR intelligent carrier shutdown	NR: LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter LTE: CellAlgoExtSwitch.LnrSmartCarrierShutdownSw	<i>Multi-RAT Coordinated Energy Saving</i>	If LTE and NR intelligent carrier shutdown is in effect, RF channel intelligent shutdown cannot take effect in the scenarios listed in Table 5-6 .
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over RF channel intelligent shutdown when both functions are enabled. RF channel intelligent shutdown does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	For low-frequency TDD: If timing carrier shutdown is in effect, RF channel intelligent shutdown cannot take effect in the scenarios listed in Table 5-6 . For high-frequency TDD: If timing carrier shutdown is in effect, RF channel intelligent shutdown does not take effect.
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect: - If the local cell is a capacity-layer cell, RF channel intelligent shutdown cannot take effect in the scenarios listed in Table 5-6. - If the local cell is a basic-layer cell and RF channel intelligent shutdown takes effect in the scenarios listed in Table 5-6, the related capacity-layer cells may exit the intelligent carrier shutdown state.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Intelligent carrier shutdown in non-precise mode	• INTRA_GNB_MULT I_CARR_S D_SW or INTER_GNB_MULT I_CARR_S D_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter selected • MULTI_CARR_PRECISE_SHD N_SW option of the NRDUCell AlgoSwitch.Power SavingExt Switch parameter deselected	<i>Energy Conservation and Emission Reduction</i>	RF channel intelligent shutdown does not take effect when intelligent carrier shutdown is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	If power saving scheduling is in effect and RF channel intelligent shutdown is enabled in the scenarios listed in Table 5-6 , power saving scheduling may stop. For 8T8R RF modules, when RF channel intelligent shutdown takes effect, beam-level adjustment of the RMSI broadcast interval in power saving scheduling does not take effect. Therefore, cell-level adjustment of the RMSI broadcast interval is recommended in such cases.
High-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both power saving scheduling and RF channel intelligent shutdown take effect, more symbols are required for transmitting a given amount of data. In this case, power saving scheduling delivers a lower symbol shutdown rate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	TTI-level channel shutdown does not take effect when RF channel intelligent shutdown is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low-power RF channel dynamic shutdown	LOW_POWER_RF_SHUT_DOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	• If low-power RF channel dynamic shutdown has been enabled, enabling RF channel intelligent shutdown brings low extra energy saving gains and causes performance deterioration. To ensure performance, it is recommended that RF channel intelligent shutdown be disabled. For details about performance deterioration caused by RF channel intelligent shutdown, see 5.1.2.2 Impacts. • If RF channel intelligent shutdown has been enabled, enabling low-power RF channel dynamic shutdown is allowed but extra energy saving gains are low with both functions enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Dynamic RF channel shutdown does not take effect when RF channel intelligent shutdown is in effect.
Low-frequency TDD	MetaAAU power reduction for cell-center UEs	NRDUCellChnPwr.ReductionValue not set to 0	<i>Energy Conservation and Emission Reduction</i>	When MetaAAU power reduction for cell-center UEs and RF channel intelligent shutdown take effect simultaneously, MetaAAU power reduction for cell-center UEs delivers fewer energy saving gains.
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	If RF channel intelligent shutdown is enabled in the scenarios listed in Table 5-6 , UEs switch from BWP2 to BWP1.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Virtual Grid-based Multi-Frequency Coordination	• RSRP prediction model – VG_MODEL_ALLOW_BUILD_FLAG option of the NRCellFreqRelationAggregationAttribute parameter – NR_IF_HO_MEAS_QTY_PRED_FUNCTION_SWITCH option of the NRCellSmartMultiCarrierAlgorithmSwitch parameter • Downlink intra-frequency spectral efficiency prediction model – DL_INTRAF_S	<i>Virtual Grid-based Multi-Frequency Coordination</i>	When RF channel intelligent shutdown and Virtual Grid-based Multi-Frequency Coordination are both enabled: • If RF channel intelligent shutdown is not in effect in a local cell and is in effect in its neighboring cells, building of virtual grid models in the local cell may be suspended, and existing virtual grid models of the local cell may go offline and become unavailable. To increase the proportion of online virtual grid models in this scenario, optimization of collaboration between virtual grid models and energy-saving-oriented channel shutdown functions can be enabled on the local and neighboring base stations by selecting the VIRTUAL_GRID_CH_SD_COOP

RAT	Function Name	Function Switch	Reference	Description
		E_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarrierModelBuildingSwitch parameter – DL_FREQUENCY_SPECTRUM_PRED_SWITCH option of the NRCellSmartMultiCarrierAlgorithmSwitch parameter Intra-frequency SRS beam prediction model INTRAF_SRS_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarrierModel		_OPT_SW option of the gNodeBAlgo.NrSmartMultiCarrierSw parameter. The optimization reduces the probability that building of virtual grid models in the local cell is suspended and the probability that existing virtual grid models of the local cell go offline and become unavailable. If RF channel intelligent shutdown is in effect in a local cell, building of virtual grid models in the local cell will be suspended and existing virtual grid models of the local cell will go offline and become unavailable.

RAT	Function Name	Function Switch	Reference	Description
		<i>Buildingswitch</i> parameter		
Low-frequency TDD	3D coverage pattern	NRDUCellTrpBeam.CoverageScenario set to CUSTOMIZED_BEAM_SCENARIO and NRDUCellTrpCustBeam.SsbBeamHorizontalWidth set to DEFAULT	<i>Beam Management</i>	In customized beam scenarios, if the horizontal beamwidth of an SSB beam is set to DEFAULT , the coverage cannot be completely aligned before and after RF channel shutdown takes effect. This may decrease the access success rate and handover success rate, and may increase the service drop rate. The coverage also changes the number of UEs in a cell. Specifically, larger coverage may increase the number of UEs and smaller coverage may result in the opposite effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAIgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	(TDD) The impacts are as follows: - If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is deselected for a cell, power sharing is not performed on the cell within the period specified for RF channel intelligent shutdown. - If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is selected for a cell, power sharing is not performed on the cell when the cell is in the energy saving state. When a cell enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter

RAT	Function Name	Function Switch	Reference	Description
				adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellIdIgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	Cells in the energy saving state do not participate in power adaptive scheduling. When a cell enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.
Low-frequency TDD	DL 1024QAM	DL_1024QAM_SW option of the NRDUCellIdIAmc.DlModulationSchemeSw parameter	<i>Modulation Schemes</i>	If RF channel intelligent shutdown takes effect, the proportion of occasions where DL 1024QAM takes effect decreases.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Full-buffer UE-specific CSI-RS for BM	USER_BM_C SIRS_SW option of the NRDUCellCsi rs.CsiSwitch parameter	<i>Channel Management</i>	If full-buffer UE-specific CSI-RS for BM has taken effect for UEs, RF channel intelligent shutdown cannot take effect. If RF channel intelligent shutdown has taken effect for UEs, full-buffer UE-specific CSI-RS for BM cannot take effect unless the UEs exit the RF channel intelligent shutdown state.
High-frequency TDD	UE-specific TRS resource management	USER_TRS_BEAM_SW option of the NRDUCellCsi rs.CsiSwitch parameter	<i>Channel Management</i>	If UE-specific TRS resource management has taken effect for UEs, RF channel intelligent shutdown cannot take effect. If RF channel intelligent shutdown has taken effect for UEs, UE-specific TRS resource management cannot take effect unless the UEs exit the RF channel intelligent shutdown state.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Intelligent beam selection	INTELLIGENT_BEAM_SELECTION_SW option of the NRDUCellBeamAlgo.BeamOptAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	Intelligent beam selection does not take effect when RF channel intelligent shutdown has shut down RF channels to have 2T left.
Low-frequency TDD	Multiple BWP1s for RedCap	NRDUCellRedCap.RedCapUeMaxAvailableBandwidth not set to 20M	<i>RedCap</i>	When RedCap Introduction is enabled and the NRDUCellRedCap.RedCapUeMaxAvailableBandwidth parameter is not set to 20M , bandwidth-based power aggregation (controlled by the BANDWIDTH_BASED_POWER_AGG_SW option of the NRDUCellChnPower.ChnPwrAlgoSwitch parameter) does not take effect though it is enabled, if the adjusted bandwidth does not cover all the BWP1s for RedCap. The purpose is to ensure that UEs on all the BWP1s for RedCap can perform services properly.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Multi-beam FDM	VOL_BASED_BEAM_MULTIPLEX_SW option of the NRDUCellAIgoSwitch.BeamMultiplexSwitch parameter	<i>MIMO (TDD)</i>	Multi-beam FDM does not take effect when RF channel intelligent shutdown has shut down RF channels to have 2T left.
High-frequency TDD	SRS-based weight scheme	NRDUCellPd sch.FixedWeightType set to SRS_WEIGHT .	None	The SRS-based weight scheme does not take effect when RF channel intelligent shutdown has shut down RF channels to have 2T left.
High-frequency TDD	Downlink beamforming weight filtering optimization	DL_BF_WEIGHT_FILTER_OPT_SW option of the NRDUCellAIgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>MIMO (TDD)</i>	Downlink beamforming weight filtering optimization does not take effect when RF channel intelligent shutdown has shut down RF channels to have 2T left.
High-frequency TDD	CSI-RS dynamic weight scheme	CSIRS_DYNAMIC_WEIGHT_SW option of the NRDUCellCsi rs.CsiSwitch parameter	None	The CSI-RS dynamic weight scheme does not take effect when RF channel intelligent shutdown has shut down RF channels to have 2T left.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Downlink single-UE robust mobility precoding	DL_SU_ROBUST_MOB_PRECODE_SW option of the NRDUCellAIgoswitch.SuMimoMultipleLayerSw parameter	None	Downlink single-UE robust mobility precoding does not take effect when RF channel intelligent shutdown has shut down RF channels to have 2T left.
Low-frequency TDD	Intelligent uplink multi-layer SINR prediction	UL_SU_SINR_INTEL_PREDICT_SW option of the NRDUCellAIAlgo.AiAmcAlgoSwitch parameter	<i>MIMO (TDD)</i>	Intelligent uplink multi-layer SINR prediction does not take effect when RF channel intelligent shutdown is in effect.
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCoIlabServ.MultipleCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When RF channel intelligent shutdown takes effect in a cell, Co-MM stops taking effect for UEs involved in Co-MM in the cell and will not take effect for UEs that are newly admitted to this cell. After the cell exits the RF channel intelligent shutdown state, Co-MM takes effect again.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR multi-band power adaptive scheduling	• LTE: LNR_MULTIBAND_PWR_ADPT_SCH_SW option of the CellDynamicPowerSharing.DynPowerSharingSwitch parameter • NR: LNR_MULTIBAND_PWR_ADPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	Cells that have entered the energy saving state do not participate in power adaptive scheduling. When a cell enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter adjustment. This causes changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	• LTE: LTE_NR_DYN_POWER_SHARING_SW option of the CellDynamicPowerSharing.DynPowerSharingSwitch parameter • NR: LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSshrCfg.DynPowerShrAlgoSwitch parameter is deselected for a cell, power sharing is not performed on the cell within the period specified for RF channel intelligent shutdown. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSshrCfg.DynPowerShrAlgoSwitch parameter is selected for a cell, power sharing is not performed on the cell when the cell is in the energy saving state. When a cell enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter adjustment. This may cause changes in the

RAT	Function Name	Function Switch	Reference	Description
				access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	L/NR Inter-band Deep Coordination	- LTE: LNR_INTE R_BAND_DEEP_COORD_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter - NR: LNR_INTE R_BAND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	- If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is deselected for a cell, power sharing is not performed on the cell within the period specified for RF channel intelligent shutdown. - If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is selected for a cell, power sharing is not performed on the cell when the cell is in the energy saving state. When a cell enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter adjustment. This may cause changes in the

RAT	Function Name	Function Switch	Reference	Description
				access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.
Low-frequency TDD	Paging Capacity Mode	NRDUCellPagingConfig.PagingCapacityMode	<i>5G Networking and Signaling</i>	After RF channel intelligent shutdown takes effect, beam weights may change, resulting in failures of UEs to receive paging messages.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Inter-FR CA	INTER_FR_F R1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When RF channel intelligent shutdown takes effect in high-frequency cells, some of the cells no longer send SSBs and may fail to be configured for inter-FR CA. To address this issue, full-frequency measurement (controlled by the FULL_FREQ_MEAS_SW option of the NRCCellAlgoSwitch.MeasPolicySwitch parameter) can be enabled for low-frequency cells. This function enables each low-frequency cell to deliver measurement configurations related to all high frequencies for addition of high-frequency cells, increasing the success rate of high-frequency cell configuration for inter-FR CA.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Smart PDCCH symbol allocation	PDCCH_SYM_SMART_ALL OC_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	When RF channel intelligent shutdown takes effect, enabling smart PDCCH symbol allocation may reduce the increase in the success rate of uplink CCE allocation to UEs and therefore may decrease the gains in the average uplink UE throughput.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUeIdentSw	None	RF channel intelligent shutdown fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
Low-frequency TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.AppIpv4 • gNBAsAppParam.AppIpv6	None	• Any option of the gNodeBParam.StUeIdentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.AppIpv4 parameter is set. • The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue: • Select the SPEED_TURBO_PWR_SAVING_SW option of

RAT	Function Name	Function Switch	Reference	Description
				the gNB Pwr Saving Algo. Pwr Saving Algo Switch parameter. RF channel intelligent shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNB Pwr Saving Algo. Speed Turbo Pwr Saving Start Time and gNB Pwr Saving Algo. Speed Turbo Pwr Saving End Time parameters, increasing the energy saving gains. • Select the STATIC_STATIC_PWR_SAVING_SWITCH option of the gNB Pwr Saving Algo. Pwr Saving Algo Switch parameter. RF channel intelligent shutdown can take effect for cells served by the base station in some of the preceding scenarios

RAT	Function Name	Function Switch	Reference	Description
				within the period specified by the gNB PwrSaving Algo.StaticSTPwrSavStartTime and gNB PwrSaving Algo.StaticSTPwrSavEndTime parameters, increasing the energy saving gains.
Low-frequency TDD	Multiple BWP1s for RedCap	NRDUCellRedCap.RedCapUeMaxAvailableBandwidth set to a value greater than 20M	<i>RedCap</i>	For a TDD cell, when RF channel intelligent shutdown is enabled and at least one RedCap BWP1 with the NCD-SSB is activated in the cell, power aggregation is required for each RedCap BWP1. In this case, RF channel intelligent shutdown does not take effect when SSB power aggregation is insufficient.

5.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

Low-frequency bands:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

High-frequency TDD bands:

5900 series base stations (macro base stations)

NOTE

LampSite base stations do not support service-type-based dynamic exit from RF channel intelligent shutdown.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Downlink channel shutdown has the following requirements for RF modules:

- Macro base station
For high-frequency TDD: Only HAAU5233w and HAAU5235w support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
For low-frequency TDD: All AAUs/RRUs that meet the requirements listed in [Table 5-7](#) support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
Downlink channel power compensation (controlled by the **DL_CHN_PWR_CMP_SW** option of the **NRDUCellChnPwr.ChnPwrAlgoSwitch** parameter) works only with 32T32R or higher low-frequency NR TDD RF modules that are each configured with only one NR carrier.

Table 5-7 Requirements of RF channel intelligent shutdown for AAUs/RRUs

RAT	Model	TX/RX Mode	Coverage Compensation Mode	Supported Beam Coverage Scenario	Description
Low-frequency TDD	All NR TDD-capable low-frequency RF modules with 32T32R or higher	64T64R or 32T32R	Automatic or manual	DEFAULT, SCENARIO_1, SCENARIO_2, SCENARIO_3, SCENARIO_4, SCENARIO_6, SCENARIO_7, SCENARIO_8, SCENARIO_9, SCENARIO_12, SCENARIO_13, SCENARIO_31, SCENARIO_36, EXPAND_SCENARIO_1, and CUSTOMIZED_BEAM_SCENARIO	RF channel intelligent shutdown is not supported by the AAU5613 with the value of Item in the command output of DSP ELABEL being 02312CHM .
Low-frequency TDD	5000 series RRUs and AAUs	8T8R	Automatic or manual	DEFAULT, SCENARIO_3, SCENARIO_4, and EXPAND_SCENARIO_1	Easy Macro and Book-type RF modules are included.
Low-frequency TDD	5000 series RRUs and AAUs	4T4R or 2T2R	Automatic or manual	DEFAULT and EXPAND_SCENARIO_1	Easy Macro and Book-type RF modules are included.

 NOTE

For low-frequency NR TDD: The supported beam coverage scenario is specified by the **NRDUCellTrpBeam.CoverageScenario** parameter.

For high-frequency NR TDD: The supported beam coverage scenario is specified by the **NRDUCellTrpMmwavBeam.CoverageScenario** parameter.

For the beam coverage scenarios supported by individual RF modules, see *Beam Management*.

For low-frequency NR TDD: When the transmit power compensation value for common channels in the cell is automatically determined by a macro base station (with the **NRDUCellPowerSaving.CovCompensationMode** parameter set to **AUTO**), certain RF modules are used, and power offset is configured, you must select the **BANDWIDTH_BASED_POWER_AGG_SW** option of the **NRDUCellChnPwr.ChnPwrAlgoSwitch** parameter for RF channel intelligent shutdown to take effect. For details, contact Huawei engineers.

- LampSite base station
 - For NR TDD:
 - In the LampSite Grid solution for indoor coverage, the corresponding pRRUs do not support RF channel intelligent shutdown.
 - In other scenarios, all NR TDD-capable 4T4R or 2T2R pRRUs support this function. For details, see the technical specifications of the related pRRUs in *3900 & 5900 Series Base Station Product Documentation*.

LampSite base stations do not allow uplink and downlink channels to be both shut down. Macro base stations allow uplink and downlink channels to be both shut down when RF modules meet the following requirements:

- For NR TDD: RF modules listed in [Table 5-8](#), which support simultaneous uplink and downlink channel shutdown, are used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Table 5-8 Requirements of simultaneous uplink and downlink channel shutdown for RF modules

RAT	Model	Supported Beam Coverage Scenario
Low-frequency TDD	3500 MHz AAU5636m	DEFAULT, SCENARIO_1, SCENARIO_2, SCENARIO_3, SCENARIO_4, SCENARIO_6, SCENARIO_7, SCENARIO_8, SCENARIO_9, SCENARIO_12, SCENARIO_13, and EXPAND_SCENARIO_1
Low-frequency TDD	2600 MHz AAU5636m	SCENARIO_6, SCENARIO_7, SCENARIO_8, SCENARIO_12, SCENARIO_13, and SCENARIO_31

5.1.3.4 Cells

- Duplex mode
This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**. This function can be configured but does not take effect in cells with the **NRDUCell.DuplexMode** parameter set to **CELL_SUL**.
- Cell bandwidth
This function has no requirements for the bandwidth of cells served by LampSite base stations. The cells served by macro base stations must meet the requirements described in [Table 5-9](#).

Table 5-9 Bandwidth requirements for cells served by macro base stations

Scenario	Cell Bandwidth Requirement
NR TDD	Both the downlink and uplink bandwidths must be 40 MHz or higher.

- TX/RX combination

Table 5-10 lists the TX/RX combinations required for Hyper Cell and Joint Transmission Cell Combination.

Table 5-10 TX/RX combinations compatible with RF channel intelligent shutdown

RAT	Feature or Function	TX/RX Combination Compatible with Channel Shutdown
Low-frequency TDD	Hyper Cell ^a	• 2T • 4T • 8T • 64T • 4T+8T hybrid configuration • 2T+8T hybrid configuration
TDD	Joint Transmission Cell Combination ^b	• 2T • 4T • 8T • 4T+8T hybrid configuration
a: The NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL. b: In Joint Transmission Cell Combination scenarios, all TRPs must have the same power. If the TRPs differ in their power compensation values, RF channel intelligent shutdown cannot take effect.		

- Others

This function requires that the number of carriers be not greater than two on each of the RF modules that support NR TDD 8T8R/4T4R/2T2R.

For high-frequency NR TDD: The **NRDUCell.SlotAssignment** parameter must be set to **4_1_DDDSU**.

For NR TDD: The **NRDUCell.DeployPosition** parameter cannot be set to **DEPLOY_IAB**.

5.1.3.5 Others

When the **SECTOREQM.ANTCFGMODE** parameter is set to **BEAM**, RF channel intelligent shutdown has the following requirements:

- If the **NRDUCell.DuplexMode** parameter is set to **CELL_TDD**, the **RRU.RS** parameter must be set to **NO**, **TN**, **LN**, **UN**, **ULN**, **GLN**, **GLFTDN**, or **LFTDN**.

- If the **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**, the **RRU.RS** parameter must be set to **NO**, **TN**, **LN**, **UN**, **ULN**, or **LFTDN**.

When the **SECTOREQM.ANTCFGMODE** parameter is set to **ANTENNAPORT**, it is recommended that the **RRU.RS** parameter be set to the same value as that when the **SECTOREQM.ANTCFGMODE** parameter is set to **BEAM**. If it is not set this way, RF channel intelligent shutdown may not take effect.

For low-frequency NR TDD: If the working modes configured using **RRU.RS** include both NR and LTE, and only NR cells are activated but LTE cells are not configured or activated, RF channel intelligent shutdown can take effect only when the **LNR_CARRIER_CHN_CO_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter and the **PSE_MULTI_MODE_RF_SHUTDOWN_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is selected.

For high-frequency TDD: The parameters related to RF channel intelligent shutdown must be set consistently for cells in the same sector. Otherwise, RF channel intelligent shutdown cannot take effect.

5.1.4 Operation and Maintenance

5.1.4.1 Data Configuration

5.1.4.1.1 Data Preparation

[Table 5-11](#) and [Table 5-12](#) describe the parameters used for function activation and optimization, respectively. This section does not describe parameters related to cell establishment.

Table 5-11 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellAlgoSwitch.NrDuCellId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	• Select the RF_SHUTDOWN_SW option of this parameter when RF channel intelligent shutdown is required. • RF channel intelligent shutdown is not recommended in distributed antenna coverage scenarios, as enabling this function may affect the coverage on some floors. • For NR TDD, if the working modes configured using RRU.RS include both NR and LTE, and only NR cells are activated but LTE cells are not configured or activated, it is good practice to select the LNR_CARRIER_CHN_CO_SHUTDOWN_SW option of this parameter and the PSE_MULTI_MODE_RF_SHUTDOWN_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter.
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	For NR TDD, if the working modes configured using RRU.RS include both NR and LTE, and only NR cells are activated but LTE cells are not configured or activated, it is good practice to select the PSE_MULTI_MODE_RF_SHUTDOWN_SW option of this parameter. If this option is selected, the LNR_CARRIER_CHN_CO_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter must be selected.

Parameter Name	Parameter ID	Setting Notes
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: • For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. • For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.

Parameter Name	Parameter ID	Setting Notes
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan and the following requirements: • For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. • For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to RF_SHUTDOWN when RF channel intelligent shutdown is required.

Parameter Name	Parameter ID	Setting Notes
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	For low-frequency TDD: - Set this parameter to the recommended value if energy saving is preferred and set this parameter to 10 to ensure both energy saving and performance. - For a 2T or 4T RRU, as the RRU has a small number of channels, shutting down half of the channels decreases the rank used, causing a significant increase in the PRB usage and affecting the energy saving gains. For NR TDD, it is recommended that the downlink PRB usage threshold be set to 2 and the downlink PRB usage offset be set to 3. For high-frequency TDD: Set this parameter to its recommended value.
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	For low-frequency TDD: - Set this parameter to the recommended value if energy saving is preferred and set this parameter to 10 to ensure both energy saving and performance. - For a 2T or 4T RRU, as the RRU has a small number of channels, shutting down half of the channels decreases the rank used, causing a significant increase in the PRB usage and affecting the energy saving gains. For NR TDD, it is recommended that the downlink PRB usage threshold be set to 2 and the downlink PRB usage offset be set to 3. For high-frequency TDD: Set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	For low-frequency TDD: Set this parameter to the recommended value if energy saving is preferred and set this parameter to 5 to ensure both energy saving and performance. For high-frequency TDD: Set this parameter to its recommended value.
User Number Offset	NRDUCellPowerSaving.UserNumOffset	For low-frequency TDD: Set this parameter to the recommended value if energy saving is preferred and set this parameter to 5 to ensure both energy saving and performance. For high-frequency TDD: Set this parameter to its recommended value.
Coverage Compensation Mode	NRDUCellPowerSaving.CovCompensationMode	For low-frequency NR TDD: The value AUTO is recommended. For high-frequency NR TDD: This parameter must be set to AUTO .
Coverage Compensation Power Offset	NRDUCellPowerSaving.PwrCompensationOffset	Set this parameter based on the network plan and the RF module's maximum capabilities of aggregating the cell common channel transmit power. This parameter applies only to low-frequency NR TDD.
RF Shutdown Number	NRDUCellPowerSaving.RfShutdownNum	The default value is recommended.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	You are advised to use the recommended value.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	You are advised to use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Second-Level Trigger Evaluation Period	NRDUCellPowerSaving.SecondLevelTrigEvalPrd	You are advised to use the recommended value. For RF channel intelligent shutdown, the value of the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter must be greater than or equal to 3 .
Second-Level Exit Evaluation Period	NRDUCellPowerSaving.SecondLevelExitEvalPrd	You are advised to use the recommended value.
Power Saving Penalty Duration	NRDUCellPowerSaving.PwrSavingPenaltyTime	You are advised to use the recommended value.
Channel Power Algorithm Switch	NRDUCellChnPwr.ChnPwrAlgoSwitch	For low-frequency TDD: When the transmit power compensation value for common channels in the cell is automatically determined by a macro base station (with the NRDUCellPowerSaving.CovCompensationMode parameter set to AUTO), certain RF modules are used, and power offset is configured, you must select the BANDWIDTH_BASED_POWER_AGG_SW option of this parameter for RF channel intelligent shutdown to take effect. For details, contact Huawei engineers. If this option is selected, the power spectral density increases, but the number of available downlink RBs may decrease. As a result, the downlink user-perceived rate may decrease.

Parameter Name	Parameter ID	Setting Notes
Channel Power Algorithm Switch	NRDUCellChnPwr.ChnPwrAlgoSwitch	For low-frequency TDD: For RF modules that support both FDD and TDD (including AAU5243 and AAU5942), select the RF_POWER_OPT_SW option of this parameter.
RF Channel Shutdown Mode	NRDUCellPowerSaving.RfChnShutdownMode	For TDD: - Set this parameter to DL_UL_CHANNEL_SHUTDOWN if both uplink and downlink channels need to be shut down for supported RF modules. - Set this parameter to DL_CHANNEL_SHUTDOWN if only downlink channels need to be shut down.
Channel Power Algorithm Switch	NRDUCellChnPwr.ChnPwrAlgoSwitch	For low-frequency TDD: To enable PDSCH and PDCCH power compensation, select the DL_CHN_PWR_CMP_SW option of this parameter.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	It is recommended that the SERV_TYPE_DYN_EXIT_PWR_SAVE_SW option be selected when there are specific types of services and user experience of these services needs to be ensured.
Power Saving Policy Group Index	gNBQciBearer.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Power Saving Policy Group Index	NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan. The value of this parameter must be the same as that of the gNBQciBearer.PowerSavingPolicyGrpIndex parameter for services to be guaranteed. Before setting this parameter, ensure that the QCI for which the gNBQciBearer.PowerSavingPolicyGrpIndex parameter is set has been configured.
User Identification Method	NRDUCellPsPolicyGrp.UserIdentMethod	Set this parameter to SERVICE_TYPE .
Service Type Dynamic Exit Power Saving Switch	NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw	To enable service-type-based dynamic exit from RF channel intelligent shutdown, select the RF_SHUTDOWN_EXIT_SW option.

Table 5-12 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
PDCCH Algorithm Extension Switch	NRDUCellPdcch.PdcchAlgoExtSwitch	Select the UE_PDCCH_SYM_NUM_ADAPT_SW option of this parameter to enable PDCCH symbol number adaptation.

 NOTE

This function requires that the RF module associated with the NR cell work in NR mode. If the RF module works in another mode, this function is configurable but does not take effect.

5.1.4.1.2 Using MML Commands

Before using MML commands, refer to [5.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Examples for low-frequency NR TDD cells

Scenario 1: RF channel intelligent shutdown with minute-level evaluation periods

```
//Enabling RF channel intelligent shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=RF_SHUTDOWN_SW-1;
//Turning on LNR_CARRIER_CHN_CO_SHUTDOWN_SW and PSE_MULTI_MODE_RF_SHUTDOWN_SW if the
working modes configured using RRU.RS include both NR and LTE, and only NR cells are activated but LTE
cells are not configured or activated
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingSwitch=LNR_CARRIER_CHN_CO_SHUTDOWN_SW-1;
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=PSE_MULTI_MODE_RF_SHUTDOWN_SW-1;
//Setting an energy saving policy of RF channel intelligent shutdown for an NR TDD cell
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=01&00&00,
StopTime=03&00&00, PowerSavingType=RF_SHUTDOWN, DlPrbThld=10, DlPrbOffset=10, UserNumThld=5,
UserNumOffset=5, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5,
CovCompensationMode=AUTO, PwrSavingPenaltyTime=5;
//((Optional) Enabling bandwidth-based power aggregation when the transmit power compensation value
for common channels in the cell is automatically determined by a macro base station
MOD NRDUCELLCHNPWR: NrDuCellId=0, ChnPwrAlgoSwitch=BANDWIDTH_BASED_POWER_AGG_SW-1;
//((Optional) Turning on the RF power optimization switch when RF modules (such as AAU5243 and
AAU5942) support both FDD and TDD
MOD NRDUCELLCHNPWR: NrDuCellId=0, ChnPwrAlgoSwitch=RF_POWER_OPT_SW-1;
//((Optional) Setting the energy saving policy of uplink and downlink RF channel intelligent shutdown for
the NR TDD cell
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=01&00&00,
StopTime=03&00&00, PowerSavingType=RF_SHUTDOWN, DlPrbThld=10, DlPrbOffset=10, UserNumThld=5,
UserNumOffset=5, EvalPeriodGranularity=MINUTE_LEVEL,
ShutdownTrigJudgePeriod=5, CovCompensationMode=AUTO, PwrSavingPenaltyTime=5,
RfChnShutdownMode=DL_UL_CHANNEL_SHUTDOWN;
//((Optional) Setting another energy saving policy of RF channel intelligent shutdown for the NR TDD cell
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1, StartTime=04&00&00,
StopTime=05&00&00, PowerSavingType=RF_SHUTDOWN, DlPrbThld=10, DlPrbOffset=10, UserNumThld=5,
UserNumOffset=5, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5,
CovCompensationMode=MANUAL, PwrCompensationOffset=60, PwrSavingPenaltyTime=5;
//((Optional) Enabling downlink channel power compensation for the NR TDD cell
MOD NRDUCELLCHNPWR: NrDuCellId=0, ChnPwrAlgoSwitch=DL_CHN_PWR_CMP_SW-1;
//((Optional) Enabling service-type-based dynamic exit from RF channel intelligent shutdown
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;
ADD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=RF_SHUTDOWN_EXIT_SW-1;
```

Scenario 2: (optional) RF channel intelligent shutdown with second-level evaluation periods

```
//Enabling RF channel intelligent shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=RF_SHUTDOWN_SW-1;
//Turning on LNR_CARRIER_CHN_CO_SHUTDOWN_SW and PSE_MULTI_MODE_RF_SHUTDOWN_SW if the
working modes configured using RRU.RS include both NR and LTE, and only NR cells are activated but LTE
cells are not configured or activated
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingSwitch=LNR_CARRIER_CHN_CO_SHUTDOWN_SW-1;
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=PSE_MULTI_MODE_RF_SHUTDOWN_SW-1;
```

```
//Setting an energy saving policy of RF channel intelligent shutdown for an NR TDD cell
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=01&00&00,
StopTime=03&00&00, PowerSavingType=RF_SHUTDOWN, DlPrbThld=10, DlPrbOffset=10, UserNumThld=5,
UserNumOffset=5, EvalPeriodGranularity=SECOND_LEVEL, SecondLevelTrigEvalPrd=6,
SecondLevelExitEvalPrd=12, CovCompensationMode=AUTO, PwrSavingPenaltyTime=5;
//((Optional) Enabling bandwidth-based power aggregation when the transmit power compensation value
for common channels in the cell is automatically determined by a macro base station
MOD NRDUCELLCHNPWR: NrDuCellId=0, ChnPwrAlgoSwitch=BANDWIDTH_BASED_POWER_AGG_SW-1;
//((Optional) Turning on the RF power optimization switch when RF modules (such as AAU5243 and
AAU5942) support both FDD and TDD
MOD NRDUCELLCHNPWR: NrDuCellId=0, ChnPwrAlgoSwitch=RF_POWER_OPT_SW-1;
//((Optional) Setting the energy saving policy of uplink and downlink RF channel intelligent shutdown for
the NR TDD cell
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=01&00&00,
StopTime=03&00&00, PowerSavingType=RF_SHUTDOWN, DlPrbThld=10, DlPrbOffset=10, UserNumThld=5,
UserNumOffset=5, EvalPeriodGranularity=SECOND_LEVEL, SecondLevelTrigEvalPrd=6,
SecondLevelExitEvalPrd=12, CovCompensationMode=AUTO, PwrSavingPenaltyTime=5,
RfChnShutdownMode=DL_UL_CHANNEL_SHUTDOWN;
//((Optional) Setting another energy saving policy of RF channel intelligent shutdown for the NR TDD cell
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1, StartTime=04&00&00,
StopTime=05&00&00, PowerSavingType=RF_SHUTDOWN, DlPrbThld=10, DlPrbOffset=10, UserNumThld=5,
UserNumOffset=5, EvalPeriodGranularity=SECOND_LEVEL, SecondLevelTrigEvalPrd=6,
SecondLevelExitEvalPrd=12, CovCompensationMode=MANUAL, PwrCompensationOffset=60,
PwrSavingPenaltyTime=5;
//((Optional) Enabling downlink channel power compensation for the NR TDD cell
MOD NRDUCELLCHNPWR: NrDuCellId=0, ChnPwrAlgoSwitch=DL_CHN_PWR_CMP_SW-1;
//((Optional) Enabling service-type-based dynamic exit from RF channel intelligent shutdown
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;
ADD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=RF_SHUTDOWN_EXIT_SW-1;
```

Examples for high-frequency NR TDD cells

Scenario 1: RF channel intelligent shutdown with minute-level evaluation periods

```
//Enabling RF channel intelligent shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=RF_SHUTDOWN_SW-1;
//Setting an energy saving policy of RF channel intelligent shutdown for an NR TDD cell
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=01&00&00,
StopTime=03&00&00, PowerSavingType=RF_SHUTDOWN, DlPrbThld=10, DlPrbOffset=15, UserNumThld=20,
UserNumOffset=20, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5,
CovCompensationMode=AUTO, PwrSavingPenaltyTime=5;
//((Optional) Setting another energy saving policy of RF channel intelligent shutdown for the NR TDD cell
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1, StartTime=04&00&00,
StopTime=05&00&00, PowerSavingType=RF_SHUTDOWN, DlPrbThld=10, DlPrbOffset=15, UserNumThld=20,
UserNumOffset=20, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5,
CovCompensationMode=AUTO, PwrSavingPenaltyTime=5;
```

Scenario 2: (optional) RF channel intelligent shutdown with second-level evaluation periods

```
//Enabling RF channel intelligent shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=RF_SHUTDOWN_SW-1;
//Setting an energy saving policy of RF channel intelligent shutdown for an NR TDD cell
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=01&00&00,
StopTime=03&00&00, PowerSavingType=RF_SHUTDOWN, DlPrbThld=10, DlPrbOffset=15, UserNumThld=20,
UserNumOffset=20, EvalPeriodGranularity=SECOND_LEVEL, SecondLevelTrigEvalPrd=6,
SecondLevelExitEvalPrd=12, CovCompensationMode=AUTO, PwrSavingPenaltyTime=5;
//((Optional) Setting another energy saving policy of RF channel intelligent shutdown for the NR TDD cell
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1, StartTime=04&00&00,
StopTime=05&00&00, PowerSavingType=RF_SHUTDOWN, DlPrbThld=10, DlPrbOffset=15, UserNumThld=20,
UserNumOffset=20, EvalPeriodGranularity=SECOND_LEVEL, SecondLevelTrigEvalPrd=6,
SecondLevelExitEvalPrd=12, CovCompensationMode=AUTO, PwrSavingPenaltyTime=5;
```

Optimization Command Examples

```
//Enabling PDCCH symbol number adaptation  
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=UE_PDCCH_SYM_NUM_ADAPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Examples for low-frequency NR TDD cells

```
//(Optional) Disabling service-type-based dynamic exit from RF channel intelligent shutdown  
MOD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,  
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=RF_SHUTDOWN_EXIT_SW-0;  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;  
//(Optional) Disabling downlink channel power compensation for an NR TDD cell  
MOD NRDUCELLCHNPWR: NrDuCellId=0, ChnPwrAlgoSwitch=DL_CHN_PWR_CMP_SW-0;  
//Disabling RF channel intelligent shutdown  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=RF_SHUTDOWN_SW-0;  
//Disabling pseudo-multimode RF channel shutdown  
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,  
PowerSavingAlgoSwitch=PSE_MULTI_MODE_RF_SHUTDOWN_SW-0;  
//Disabling LNR coordinated carrier-channel shutdown  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingSwitch=LNR_CARRIER_CHN_CO_SHUTDOWN_SW-0;  
//Removing an energy saving policy of RF channel intelligent shutdown  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;  
//(Optional) Removing the other energy saving policy of RF channel intelligent shutdown  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1;  
//(Optional) Disabling bandwidth-based power aggregation when the transmit power compensation value  
for common channels in the cell is automatically determined by a macro base station  
MOD NRDUCELLCHNPWR: NrDuCellId=0, ChnPwrAlgoSwitch=BANDWIDTH_BASED_POWER_AGG_SW-0;
```

Examples for high-frequency NR TDD cells

```
//Disabling RF channel intelligent shutdown  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=RF_SHUTDOWN_SW-0;  
//Removing an energy saving policy of RF channel intelligent shutdown  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;  
//(Optional) Removing the other energy saving policy of RF channel intelligent shutdown  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1;
```

5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.2 Activation Verification

After activating RF channel intelligent shutdown, use either of the following methods to check whether the function has taken effect:

- Method 1:
 - a. Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Switch** parameter is **RF Channel Power Saving Switch:On** in the command output, RF channel intelligent shutdown has been enabled.

- b. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **RF Channel Power Saving**.

 NOTE

If there are two or more records with **Power Saving Type** being **RF Channel Power Saving**, RF channel intelligent shutdown has been enabled in two or more time periods.

- c. Run the **DSP TIME** command. Check whether the system time is within a period for RF channel intelligent shutdown.
- If it is, go to **d**.
 - If it is not, wait until the start time of RF channel intelligent shutdown arrives.
- d. Check whether the downlink PRB usage of the cell is less than or equal to the downlink PRB usage threshold for starting RF channel intelligent shutdown (specified by the **NRDUCellPowerSaving.DlPrbThld** parameter). In addition, check whether the number of RRC_CONNECTED UEs in the cell (indicated by **N.User.RRCConn.Avg**) is less than or equal to the UE number threshold for starting this function (specified by the **NRDUCellPowerSaving.UserNumThld** parameter).
- If they are, go to **e**.
 - If one or both are not, wait until the downlink PRB usage of the cell becomes less than or equal to the downlink PRB usage threshold for starting RF channel intelligent shutdown and the number of RRC_CONNECTED UEs in the cell becomes less than or equal to the UE number threshold for starting this function.
- e. Wait for a certain period of time. Then, run the **DSP NRDUCELL** command. If the **Power Saving State** parameter indicates **DL Channel Shutdown:Enabled**, downlink RF channel intelligent shutdown has taken effect. For low-frequency TDD, if the value of the **Power Saving State** parameter is **DL and UL Channel Shutdown:Enabled**, uplink channel shutdown has taken effect.
- In single-carrier scenarios, the granularity of the wait time is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.
 - If this parameter is set to **MINUTE_LEVEL**, the wait time is equal to $\max(\text{NRDUCellPowerSaving.ShutdownTrigJudgePeriod}, \text{NRDUCellPowerSaving.PwrSavingPenaltyTime})$.
 - If this parameter is set to **SECOND_LEVEL**, the wait time is equal to $\max(\text{NRDUCellPowerSaving.SecondLevelTrigEvalPrd}, \text{NRDUCellPowerSaving.PwrSavingPenaltyTime})$.
 - In multi-carrier or split scenarios, the length of the wait time varies depending on whether the conditions for starting RF channel intelligent shutdown are met for the first time after this function is enabled.
 - If they are, wait for 5 to 20 minutes.

- If they are not, the granularity of the wait time is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter. If this parameter is set to **MINUTE_LEVEL**, the wait time is equal to $\max(\text{NRDUCellPowerSaving.ShutdownTrigJudgePeriod}, \text{NRDUCellPowerSaving.PwrSavingPenaltyTime})$. If this parameter is set to **SECOND_LEVEL**, the wait time is equal to $\max(\text{NRDUCellPowerSaving.SecondLevelTrigEvalPrd}, \text{NRDUCellPowerSaving.PwrSavingPenaltyTime})$.
- Method 2:
Wait for a certain period of time. Then, run the **DSP NRDUCELLPOWERSAVING** command to query the status of RF channel intelligent shutdown.
 - In single-carrier scenarios, the granularity of the wait time is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.
 - If this parameter is set to **MINUTE_LEVEL**, the wait time is equal to $\max(\text{NRDUCellPowerSaving.ShutdownTrigJudgePeriod}, \text{NRDUCellPowerSaving.PwrSavingPenaltyTime})$.
 - If this parameter is set to **SECOND_LEVEL**, the wait time is equal to $\max(\text{NRDUCellPowerSaving.SecondLevelTrigEvalPrd}, \text{NRDUCellPowerSaving.PwrSavingPenaltyTime})$.
 - In multi-carrier or split scenarios, the length of the wait time varies depending on whether the conditions for starting RF channel intelligent shutdown are met for the first time after this function is enabled.
 - If they are, wait for 5 to 20 minutes.
 - If they are not, the granularity of the wait time is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter. If this parameter is set to **MINUTE_LEVEL**, the wait time is equal to $\max(\text{NRDUCellPowerSaving.ShutdownTrigJudgePeriod}, \text{NRDUCellPowerSaving.PwrSavingPenaltyTime})$. If this parameter is set to **SECOND_LEVEL**, the wait time is equal to $\max(\text{NRDUCellPowerSaving.SecondLevelTrigEvalPrd}, \text{NRDUCellPowerSaving.PwrSavingPenaltyTime})$.

The following describes how to determine the status of RF channel intelligent shutdown from the MML command output:

- If the value of **Power Saving Type** is **RF Channel Power Saving** and the value of **Current State** is **Enabled**, RF channel intelligent shutdown has taken effect.
- If the value of **Power Saving Type** is **RF Channel Power Saving** and the value of **Current State** is **Disabled**, check the **Current Disabled Cause**, **Latest Exit Cause**, and **Latest Disabled Cause** parameter values to find out the possible reason why RF channel intelligent shutdown did not take effect. [Table 5-13](#) lists possible reasons. You can also check **Latest Exit Time** or **Latest Disabled Time** in the command output for the time the latest exit or the latest application failure occurred, respectively.
Latest Exit Time, **Latest Exit Cause**, **Latest Disabled Time**, and **Latest Disabled Cause** are used to query the time and reasons for the latest exit and the latest application failure within the specified period. The query may be performed outside the specified period. For example, if the

specified period is from 00:00:00 to 06:00:00, **Latest Exit Cause** or **Latest Disabled Cause** indicates the reason for the latest exit or the latest application failure that occurred within the period from 00:00:00 to 06:00:00. This reason, together with the time the latest exit or application failure occurred, can be queried even by an operation performed within the time from 06:00:00 to 23:59:59.

Table 5-13 Possible reasons why RF channel intelligent shutdown did not take effect

Reason	Description
Insufficient License	No license for this function is available.
Switch Turned Off	None
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable for other reasons.
Triggering Time Not Reached	The current time is not within the effective period of RF channel intelligent shutdown.
Impacted Feature Taken Effect	- Service-type-based dynamic exit from RF channel intelligent shutdown has taken effect. - Any other impacted feature has taken effect.
Downlink PRB Threshold Not Reached	None
UE Number Threshold Not Reached	None
In Penalty State	A penalty is being imposed upon an exit from the RF channel intelligent shutdown state.
RF Hardware Capability Not Supported	- The temperature differential of the RF module is too large. - Other internal faults occur.

Reason	Description
Power Aggregation Configuration Not Met	The settings of the parameters related to transmit power compensation for common channels do not meet the requirements for RF channel intelligent shutdown to take effect. - If the parameters listed in Table 5-5 are set to larger values than expected, the power aggregation capability may be insufficient. For details, contact Huawei engineers. - When the NRDUCellPowerSaving.CovCompensationMode parameter is set to AUTO, RF channel intelligent shutdown cannot take effect for certain RF modules or under certain power offset configurations. For specific RF modules and power offset configurations, contact Huawei engineers. With this parameter setting in NR TDD, RF channel intelligent shutdown can take effect for certain RF modules or under certain power offset configurations only when bandwidth-based power aggregation (controlled by the BANDWIDTH_BASED_POWER_AGG_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter) is enabled. In this case, you can also change the NRDUCellPowerSaving.CovCompensationMode parameter value to MANUAL so that RF channel intelligent shutdown can take effect. However, the common channel coverage performance cannot be guaranteed after this function takes effect.

Reason	Description
Shared Channel Not Coordinated	In multi-carrier co-PA scenarios, not all carriers meet the conditions for entering RF channel intelligent shutdown.
Multi-Carrier Not Coordinated	- In multi-carrier co-RF module scenarios: - The effective durations of RF channel shutdown under energy saving policies for different carriers do not overlap. - The RF channel shutdown modes of different carriers are inconsistent. - The effective durations for different carriers overlap, but the configurations of coverage compensation mode are inconsistent between these carriers. - The effective durations for different carriers overlap, but the settings of the triggering evaluation period (specified by the NRDUCellPowerSaving.ShutdownTrigJudge-Period parameter) for channel shutdown are inconsistent between these carriers. - The negotiation on the power compensation values required for different carriers fails. - An activated carrier is added in the scenarios listed in Table 5-6.
RF Channel Not Coordinated	- The channels of the RF module are faulty. - The antenna mode is incompatible.
Channel Calibration in Progress	None
Channel Calibration Info Being Queried	None

Reason	Description
Detecting for Port Mapping Optimization	None
Channel Calibration Resource Request Failed	None
Parameter Inconsistency Between Cells	For high-frequency TDD: The parameters in the RF channel intelligent shutdown policies for different cells are inconsistent.
RF Unit Working Standard Not Met	The working modes configured for the RF module include both NR and LTE. - For low-frequency NR TDD, only NR cells are activated, LTE cells are not configured or activated, and both the following options are deselected: LNR_CARRIER_CHN_CO_SHUT DOWN_SW of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter and PSE_MULTI_MODE_RF_SHUT DOWN_SW of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter. - Single-RAT channel shutdown does not take effect when LNR coordinated channel shutdown is in effect.
Others	- Though it is currently within the range of time in which channel shutdown can take effect, RF channel intelligent shutdown is in its triggering evaluation period, which is specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter. - RRC connection reconfiguration fails for a batch of UEs. - Internal software or other reasons cause the function to fail.

 NOTE

For high-frequency TDD: In a multi-CC sector, where a sector serves multiple cells:

- If the parameter setting policies for RF channel intelligent shutdown are changed from inconsistent to consistent between the cells in the same sector, the reason why this function does not take effect within the first evaluation period may be inaccurate.
- If the activation status of a cell in the sector changes, the reason why this function does not take effect within the first evaluation period may be inconsistent between the cells in the same sector.

5.1.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.
On the live network, the one-week base station energy consumption data obtained before and after RF channel intelligent shutdown is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.
- The duration of RF channel shutdown caused by the activation of RF channel intelligent shutdown can be obtained by observing the **N.PowerSaving.RFShutdown.Dur** counter on the MAE-Access. The policy of counter measurement upon carrier shutdown is determined by the function of energy saving counter optimization upon carrier shutdown, which is controlled by the **COUNTER_OPT_IN_CARROFF_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter. If this option is deselected and a carrier has been shut down, the **N.PowerSaving.RFShutdown.Dur** counter counts in the measured RF channel shutdown duration of the carrier. If this option is selected and a carrier has been shut down, the **N.PowerSaving.RFShutdown.Dur** counter does not count in the RF channel shutdown duration of the carrier.
- When channel shutdown performance improvement (low-frequency TDD) takes effect, the RRU/RFU output power monitoring result may indicate that the single-channel power exceeds the maximum single-channel transmit power, which is specified by the **NRDUCellTrp.MaxTransmitPower** parameter. RRU/RFU output power monitoring is performed on the LMT. For details, see *3900 & 5900 Series Base Station LMT User Guide* in *3900 & 5900 Series Base Station Product Documentation*.

5.2 MetaAAU Deep Energy Saving (Low-Frequency TDD)

5.2.1 Principles

MetaAAU deep energy saving **enables a base station to take energy saving measures for the digital processors corresponding to the MetaAAU channels that have been shut down through RF channel intelligent shutdown** (detailed

in [5.1 RF Channel Intelligent Shutdown](#)), when the network is under light or no load within specified periods. This function helps produce more energy saving gains.

MetaAAU deep energy saving can be enabled by selecting the **DEEP_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter. In addition, the **NRDUCellPowerSaving.RfChnShutdownMode** parameter must be set to **DL_UL_CHANNEL_CHIP_PWR_SAVING**.

5.2.2 Network Analysis

5.2.2.1 Benefits

When the network is under light or no load, MetaAAU deep energy saving further reduces energy consumption on the basis of RF channel intelligent shutdown.

5.2.2.2 Impacts

The impacts between this function and other functions are described in [5.2.3.2.3 Function Impacts](#).

MetaAAU deep energy saving has the same impacts on network performance as RF channel intelligent shutdown. For details, see [5.1.2.2 Impacts](#).

Offline fast Fourier transform (FFT) spectrum scanning triggers an exit from MetaAAU deep energy saving. You are advised to perform the scanning manually after MetaAAU deep energy saving stops taking effect.

When an RF module is configured to work in LNR mode:

- If no LTE TDD cell has been set up, this function can take effect.
- If LTE TDD cells have been set up, this function cannot take effect on the NR side.
- If no LTE TDD cell has been set up and this function is in effect on the NR side, setting up an LTE TDD cell will cause this function to stop taking effect.

In the dual-star topology, if the eCPRI link for NR is disconnected after MetaAAU deep energy saving has taken effect, setting up an LTE TDD cell on the other eCPRI link will fail until sometime (at most 10 minutes) later.

5.2.3 Requirements

5.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-081203	MetaAAU Deep Energy Saving	NR0S00MTDD00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	MetaAAU deep energy saving requires RF channel intelligent shutdown to be enabled and the NRDUCellPowerSaving.RfChnShutdownMode parameter to be set to DL_UL_CHANNEL_CHIP_PWR_SAVING .

5.2.3.2.2 Mutually Exclusive Functions

None

5.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SLEEP	<i>Multi-RAT Coordinated Energy Saving</i>	When MetaAAU deep energy saving and RF module deep dormancy are both enabled, RF module deep dormancy preferentially takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	MetaAAU deep energy saving does not work in high-speed cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUelidentSw	None	MetaAAU deep energy saving fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
Low-frequency TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.AppIpv4 • gNBAsAppParam.AppIpv6	None	• Any option of the gNodeBParam.StUelidentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.AppIpv4 parameter is set. • The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue: • Select the SPEED_TURBO_PWR_SAVING_SW option of

RAT	Function Name	Function Switch	Reference	Description
				the gNB PwrSavin gAlgo.PwrSavin gAlgoSwitch parameter. MetaAAU deep energy saving can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNB PwrSavin gAlgo.SpeedTurboPwrSavStartTime and gNB PwrSavin gAlgo.SpeedTurboPwrSavEndTime parameters, increasing the energy saving gains. • Select the STATIC_STATIC_PWR_SAVING_SWITCH option of the gNB PwrSavin gAlgo.PwrSavin gAlgoSwitch parameter. MetaAAU deep energy saving can take effect for cells served by the base station in some of the preceding scenarios within the period

RAT	Function Name	Function Switch	Reference	Description
				specified by the gNBPwrSavin gAlgo.StaticST PwrSavStartTime and gNBPwrSavin gAlgo.StaticST PwrSavEndTime parameters, increasing the energy saving gains.

5.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Only 64T64R MetaAAUs support this function.

5.2.3.4 Cells

This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**.

The **NRDUCell.FrequencyBand** parameter must be set to **N38**, **N40**, **N41**, **N77**, **N78**, or **N79**.

When an RF module is configured to work in LNR mode:

- If no LTE TDD cell has been set up, this function can take effect.
- If LTE TDD cells have been set up, this function cannot take effect on the NR side.

- If no LTE TDD cell has been set up and this function is in effect on the NR side, setting up an LTE TDD cell will cause this function to stop taking effect.

5.2.3.5 Others

None

5.2.4 Operation and Maintenance

5.2.4.1 Data Configuration

5.2.4.1.1 Data Preparation

Table 5-14 describes the parameters used for function activation.

Table 5-14 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	Select the DEEP_RF_SHUTDOWN_SW option of this parameter when MetaAAU deep energy saving is required.
RF Channel Shutdown Mode	NRDUCellPowerSaving.RfChnShutdownMode	Set this parameter to DL_UL_CHANNEL_CHIP_PWR_SAVING when MetaAAU deep energy saving is required.

5.2.4.1.2 Using MML Commands

Before using MML commands, refer to [5.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring a policy of MetaAAU deep energy saving with the RF channel shutdown mode set to DL_UL_CHANNEL_CHIP_PWR_SAVING
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=RF_SHUTDOWN, StartTime=00&00&00, StopTime=06&00&00,
RfChnShutdownMode=DL_UL_CHANNEL_CHIP_PWR_SAVING;
//Turning on the MetaAAU deep energy saving switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=DEEP_RF_SHUTDOWN_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the MetaAAU deep energy saving switch  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=DEEP_RF_SHUTDOWN_SW-0;  
//Removing the energy saving policy of MetaAAU deep energy saving  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
```

5.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.4.2 Activation Verification

To check whether MetaAAU deep energy saving has taken effect, perform the following steps:

1. Run the **DSP NRDUCELL** command to query the status of MetaAAU deep energy saving.
 - If the value of the **Power Saving State** parameter is **DL and UL Channel and Chip Power Saving:Enabled**, MetaAAU deep energy saving has taken effect in at least one TRP of the cell.
 - If the value of the **Power Saving State** parameter is **DL and UL Channel and Chip Power Saving:Disabled**, run the **DSP NRDUCELLPOWERSAVING** command and check the **Current Disabled Cause** parameter value to determine the reason why MetaAAU deep energy saving did not take effect. The possible reasons are the same as those listed in [Table 5-13](#) for RF channel intelligent shutdown.
2. (Optional) For a hyper cell or combined cell, run the **DSP NRDUCELLTRP** command to query the status of MetaAAU deep energy saving in each TRP of the cell.
 - If the value of the **Power Saving State** parameter is **DL and UL Channel and Chip Power Saving:Enabled** for a TRP, MetaAAU deep energy saving has taken effect in this TRP.
 - If the value of the **Power Saving State** parameter is **DL and UL Channel and Chip Power Saving:Disabled** for a TRP, MetaAAU deep energy saving has not taken effect in this TRP.

5.2.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after MetaAAU deep energy saving is enabled can be

compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.

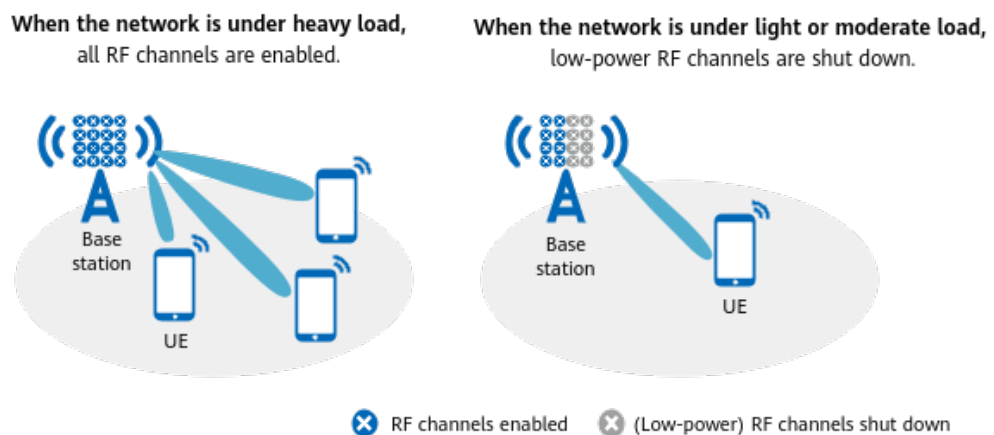
- The RF channel shutdown duration caused by the activation of MetaAAU deep energy saving can be obtained by observing the **N.PowerSaving.RFShutdown.Dur** counter on the MAE-Access. The policy of counter measurement upon carrier shutdown is determined by the function of energy saving counter optimization upon carrier shutdown, which is controlled by the **COUNTER_OPT_IN_CARROFF_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter. If this option is deselected and a carrier has been shut down, the **N.PowerSaving.RFShutdown.Dur** counter counts in the measured RF channel shutdown duration of the carrier. If this option is selected and a carrier has been shut down, the **N.PowerSaving.RFShutdown.Dur** counter does not count in the RF channel shutdown duration of the carrier.

5.3 Low-Power RF Channel Dynamic Shutdown (Low-Frequency TDD)

5.3.1 Principles

Low-power RF channel dynamic shutdown enables a base station to shut down the low-power transmit channels (if any) where only data services of SU-MIMO UEs are present, when the network is under light or moderate load during preset period. As such, this function produces energy saving gains. [Figure 5-2](#) illustrates how this function works.

Figure 5-2 Working principles for low-power RF channel dynamic shutdown



Low-power RF channel dynamic shutdown can be enabled by selecting the **LOW_POWER_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter.

[Table 5-15](#) describes the parameters in the **NRDUCellPowerSaving** MO that specify the time period, load threshold, and other information under an energy saving policy of low-power RF channel dynamic shutdown.

Table 5-15 Parameters related to a policy of low-power RF channel dynamic shutdown

Parameter Name	Parameter ID	Description
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	This parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	This parameter specifies the energy saving type for an energy saving policy index. Low-power RF channel dynamic shutdown requires that this parameter be set to LOW_POWER_RF_SHUTDOWN .

Parameter Name	Parameter ID	Description
Start Time	NRDUCellPowerSaving.StartTime	These parameters specify the time period in which low-power RF channel dynamic shutdown takes effect, as well as the evaluation period and conditions for a cell to enter and exit the low-power RF channel dynamic shutdown state. Within the time period specified by NRDUCellPowerSaving.StartTime and NRDUCellPowerSaving.StopTime for low-power RF channel dynamic shutdown to take effect, the base station periodically determines whether the cell enters the state of low-power RF channel dynamic shutdown at an interval specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter. • If the downlink PRB usage of a cell or transmission reception point (TRP)^a is not greater than the value of NRDUCellPowerSaving.DlPrbThld after low-power RF channel dynamic shutdown is enabled, the cell or TRP automatically enters the state of low-power RF channel dynamic shutdown^b. • If the downlink PRB usage of a cell or TRP^a is greater than the sum of the values of NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset or the current time is not within the time period for this function to take effect, the cell or TRP automatically exits the state of low-power RF channel dynamic shutdown^c.
Stop Time	NRDUCellPowerSaving.StopTime	
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	

Parameter Name	Parameter ID	Description
Low Power Threshold	NRDUCellPowerSaving.LowPowerThld	After a cell enters the low-power RF channel dynamic shutdown state, the base station determines the channels to be shut down based on this threshold. • If the normalized power of a channel is less than or equal to this threshold, low-power RF channel dynamic shutdown takes effect for this channel. • Otherwise, low-power RF channel dynamic shutdown does not take effect for this channel.

Parameter Name	Parameter ID	Description
		a. The downlink PRB usage of a cell or TRP is calculated in the following ways. - Downlink PRB usage of a common cell = (Average number of PRBs used by PDSCH DRB services – Average number of extended PRBs used by the PDSCH)/Average number of available downlink PRBs - Downlink PRB usage of a TRP of a hyper cell or combined cell in joint transmission mode = (Average number of downlink PRBs used by the TRP – Average number of downlink extended PRBs used by the TRP)/Average number of available downlink PRBs for the TRP b: In a common cell, hyper cell, or combined cell in joint transmission mode scenario, the cell or a TRP enters the low-power RF channel dynamic shutdown state. - For a common cell, if the downlink PRB usage of the cell is less than or equal to the downlink PRB threshold for starting low-power RF channel dynamic shutdown, the common cell automatically enters the channel shutdown state. - For a hyper cell, if the downlink PRB usage of a TRP is less than or equal to the downlink PRB threshold for starting low-power RF channel dynamic shutdown, the TRP automatically enters the channel shutdown state. - For a combined cell in joint transmission mode, if the downlink PRB usage of all TRPs is less than or equal to the downlink PRB threshold for starting low-power RF channel dynamic shutdown, the cell automatically enters the channel shutdown state. c: The cell or TRP automatically exits the state of low-power RF channel dynamic shutdown. - When the downlink PRB usage of a common cell is greater than the sum of the values of NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset, the cell exits the state of low-power RF channel dynamic shutdown. - When the downlink PRB usage of a TRP of a hyper cell is greater than the sum of the values of NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset, the TRP automatically exits the state of low-power RF channel dynamic shutdown. The hyper cell exits the state of low-power RF channel dynamic shutdown only when all its TRPs exit the state. - When the downlink PRB usage of each TRP of a combined cell in joint transmission mode is greater than the sum of the values of NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset, the cell automatically exits the state of low-power RF channel dynamic shutdown.

5.3.2 Network Analysis

5.3.2.1 Benefits

When the network is under light or moderate load, low-power RF channel dynamic shutdown can shut down some channels with low transmit power to reduce the energy consumption of the entire base station. If only a few low-power channels can be shut down, energy saving gains decrease.

If there are UEs involved in the following services on the network, low-power RF channel dynamic shutdown cannot take effect and therefore energy saving gains may decrease: CoMP, CS, CBF, MU-MIMO, FWA services with a guaranteed rate configured but not actually in effect, VoNR, Non-Slot, or high-reliability services.

5.3.2.2 Impacts

The impacts between this function and other functions are described in [5.3.3.2.3 Function Impacts](#).

After low-power RF channel dynamic shutdown takes effect:

- The IBLER increases.
- The user-perceived rate decreases. The decrease varies depending on the low power threshold. The higher the low power threshold, the more the user-perceived rate decreases.
- The average MCS index for downlink scheduling decreases.
- The values of the following counters related to downlink PRB usage increase in common cells:
 - **N.PR.B.DL.DrbUsed.Avg**
 - **N.PR.B.DL.Used.Avg**
 - **N.PR.B.DL.ExtendUsed.Avg**
- The values of the following counters related to the downlink PRB usage of TRPs increase in a Hyper Cell and Joint Transmission Cell Combination scenario:
 - **N.HyperCell.TRP.PR.B.DL.Used.Avg**
 - **N.HyperCell.TRP.PR.B.DL.ExtendUsed.Avg**

5.3.3 Requirements

5.3.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-061203	Intelligent Channel Dynamic Shutdown	NR0S00DRCS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	Low-power RF channel dynamic shutdown depends on symbol power saving.

5.3.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDualCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransmissionMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>	If a combined cell served by a macro base station uses the basic or DPS transmission mode, low-power RF channel dynamic shutdown cannot be enabled.
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	When distributed massive MIMO is enabled, low-power RF channel dynamic shutdown cannot be enabled.
Low-frequency TDD	Intra-frequency assistant cell	NRDUCell.SupplementaryCellInd set to INTRA_FREQ_ASSISTANT_CELL	None	Intra-frequency assistant cells are incompatible with low-power RF channel dynamic shutdown.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel power adaptation	RF_CHN_PWR_ADAPT_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	RF channel power adaptation does not work with low-power RF channel dynamic shutdown.
Low-frequency TDD	Dynamic MIMO layer number decrease	DYNAMIC_MIMO_LAYER_DECREASE_SW option of the gNodeBAlgo.DedicatedNetworkFuncSw parameter	None	In 8T8R scenarios where the NRDUCell.NrDuCellNetworkingMode parameter is set to NORMAL_CELL , dynamic MIMO layer number decrease does not work with low-power RF channel dynamic shutdown.

5.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Low-power RF channel dynamic shutdown does not take effect when timing carrier shutdown is in effect.
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTICARRIER_SHUTDOWN_SW or INTER_GNB_MULTICARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Low-power RF channel dynamic shutdown does not take effect when intelligent carrier shutdown is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option (inter-base-station multi-carrier scenarios) of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected - Basic-layer cells: FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected	<i>Energy Conservation and Emission Reduction</i>	Low-power RF channel dynamic shutdown does not take effect when fast carrier shutdown is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	- If low-power RF channel dynamic shutdown has been enabled, enabling RF channel intelligent shutdown brings low extra energy saving gains and causes performance deterioration. To ensure performance, it is recommended that RF channel intelligent shutdown be disabled. For details about performance deterioration caused by RF channel intelligent shutdown, see 5.1.2.2 Impacts. - If RF channel intelligent shutdown has been enabled, enabling low-power RF channel dynamic shutdown is allowed but extra energy saving gains are low with both functions enabled.
Low-frequency TDD	MetaAAU power reduction for cell-center UEs	NRDUCellChnPwr.PwrReductionValue not set to 0	<i>Energy Conservation and Emission Reduction</i>	When MetaAAU power reduction for cell-center UEs and low-power RF channel dynamic shutdown take effect simultaneously, MetaAAU power reduction for cell-center UEs delivers fewer energy saving gains.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCY_W (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Low-power RF channel dynamic shutdown does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	RF module super dormancy takes precedence over low-power RF channel dynamic shutdown when both functions are enabled. Low-power RF channel dynamic shutdown does not take effect when RF module super dormancy is in effect.
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRU DOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over low-power RF channel dynamic shutdown when both functions are enabled. Low-power RF channel dynamic shutdown does not take effect when RF module fast deep dormancy is in effect.
Low-frequency TDD	PDSC H MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Low-power RF channel dynamic shutdown is not performed in slots with MU-MIMO pairing.
Low-frequency TDD	Paging Capacity Mode	NRDUCellPagingConfig.PagingCapacityMode	<i>5G Networking and Signaling</i>	After low-power RF channel dynamic shutdown takes effect, beam weights may change, resulting in failures of UEs to receive paging messages.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUelidentSw	None	Low-power RF channel dynamic shutdown fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
Low-frequency TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.AppIpv4 • gNBAsAppParam.AppIpv6	None	• Any option of the gNodeBParam.StUelidentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.AppIpv4 parameter is set. • The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue: • Select the SPEED_TURBO_PWR_SAVING_SW option of the gNBPwrSavingAlgo.PwrSavingAlgoSwitch parameter. Low-power RF channel dynamic shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBPwrSavingAlgo.SpeedTurboPwrSavStartTime and gNBPwrSavingAlgo.SpeedTurboPwrSavEndTime parameters, increasing the energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
				Select the STATIC_ST_PWR_SAVING_SW option of the gNBWprSavingAlgo.PwrSavingAlgoSwitch parameter. Low-power RF channel dynamic shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBWprSavingAlgo.StaticSTPwrSavStartTime and gNBWprSavingAlgo.StaticSTPwrSavEndTime parameters, increasing the energy saving gains.

5.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All 32T32R and higher RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.3.4 Cells

This function requires that the duplex mode of cells be NR TDD, **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**.

This function requires that the **NRDUCellTrp. *TxRxMode*** parameter be set to **32T32R**, **64T64R**, or **128T128R**.

5.3.3.5 Others

To ensure energy saving gains, it is recommended that this function be enabled together with TTI-level channel shutdown. For details, see [5.4 TTI-Level Channel Shutdown](#).

5.3.4 Operation and Maintenance

5.3.4.1 Data Configuration

5.3.4.1.1 Data Preparation

[Table 5-16](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-16 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellAlgoSwitch. <i>NrDuCellId</i>	Set this parameter based on the network plan.
Power Saving Switch	NRDUCellAlgoSwitch. <i>PowerSavingSwitch</i>	Select the LOW_POWER_RF_SHUTDOWN_SW option of this parameter when low-power RF channel dynamic shutdown is required.
Power Saving Policy Index	NRDUCellPowerSaving. <i>PowerSavingPolicyIndex</i>	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: - For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. - For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.

Parameter Name	Parameter ID	Setting Notes
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan and the following requirements: - For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. - For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to LOW_POWER_RF_SHUTDOWN when low-power RF channel dynamic shutdown is required.
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	You are advised to use the recommended value.
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	You are advised to use the recommended value.
Low Power Threshold	NRDUCellPowerSaving.LowPowerThld	You are advised to use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	For low-power RF channel dynamic shutdown, this parameter can only be set to its default value.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	You are advised to use the recommended value.
Downlink PDSCH Algorithm Switch	NRDUCellPdsch.DlPdschAlgorithmSwitch	When low-power RF channel dynamic shutdown is enabled, select the RB_ESTIMATE_OPT_SW option of this parameter to enable downlink RB estimation optimization. For details about the impact of downlink RB estimation optimization on network performance, see <i>Downlink Scheduling</i> .

5.3.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling low-power RF channel dynamic shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=LOW_POWER_RF_SHUTDOWN_SW-1;
//Enabling downlink RB estimation optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPdschAlgoSwitch=RB_ESTIMATE_OPT_SW-1;
//Setting an energy saving policy of low-power RF channel dynamic shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=01&00&00,
StopTime=03&00&00, PowerSavingType=LOW_POWER_RF_SHUTDOWN, DlPrbThId=50, DlPrbOffset=15,
LowPowerThId=10, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling low-power RF channel dynamic shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=LOW_POWER_RF_SHUTDOWN_SW-0;
```

```
//Removing an energy saving policy of low-power RF channel dynamic shutdown  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
```

5.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.3.4.2 Activation Verification

After activating low-power RF channel dynamic shutdown, use either of the following methods to check whether the function has taken effect:

- Method 1:
 - a. Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Switch** parameter is **Low Power Channel Shutdown Sw:On** in the command output, low-power RF channel dynamic shutdown has been enabled.
 - b. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **Low Power Channel Shutdown**.

NOTE

- If there are two or more records with **Power Saving Type** being **Low Power Channel Shutdown**, low-power RF channel dynamic shutdown has been enabled in two or more time periods.
 - c. Run the **DSP TIME** command. Check whether the system time is within a period for low-power RF channel dynamic shutdown to take effect.
 - If it is, go to **d**.
 - If it is not, wait until the start time of low-power RF channel dynamic shutdown arrives.
 - d. Check whether the downlink PRB usage of the cell is less than or equal to the downlink PRB usage threshold for starting low-power RF channel dynamic shutdown (specified by the **NRDUCellPowerSaving.DlPrbThld** parameter).
 - If it is, go to **e**.
 - If it is not, wait until the downlink PRB usage of the cell becomes less than or equal to the downlink PRB usage threshold for starting low-power RF channel dynamic shutdown.
 - e. Wait about 10 minutes and then run the **DSP NRDUCELL** command. Verify that the value of the **Power Saving State** parameter is **Low Power Channel Shutdown:Enabled**, which indicates that low-power RF channel dynamic shutdown has taken effect.
- Method 2:

Run the **DSP NRDUCELLPOWERSAVING** command to query the status of low-power RF channel dynamic shutdown.

- If the value of the **Power Saving Type** parameter is **Low Power Channel Shutdown** and the value of the **Current State** parameter is **Enabled**, low-power RF channel dynamic shutdown has taken effect.
- If the value of the **Power Saving Type** parameter is **Low Power Channel Shutdown** and the value of the **Current State** parameter is **Disabled**, you can check the **Current Disabled Cause** parameter value to determine the reason why low-power RF channel dynamic shutdown does not take effect. [Table 5-17](#) provides possible reasons.

Table 5-17 Possible reasons why low-power RF channel dynamic shutdown does not take effect

Reason	Description
Insufficient License	No license for this function is available.
Switch Turned Off	None
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable for other reasons.
Triggering Time Not Reached	The current time is not within the time period specified for low-power RF channel dynamic shutdown to take effect.
Impacted Feature Taken Effect	None
Prerequisite Feature Not Taken Effect	Symbol power saving has not taken effect.
Downlink PRB Threshold Not Reached	None
In Penalty State	A penalty is being imposed upon an exit from the state of low-power RF channel dynamic shutdown.
Ineffective for All TRPs	Low-power RF channel dynamic shutdown does not take effect in any TRP.
TRP Unavailable	A TRP of a hyper cell is blocked or is deactivated, or is unavailable due to other reasons.

Reason	Description
Others	- The current time is within the time period specified for low-power RF channel dynamic shutdown to take effect, but this function is in its evaluation period, which is specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter. - Internal software or other reasons cause the function to fail.

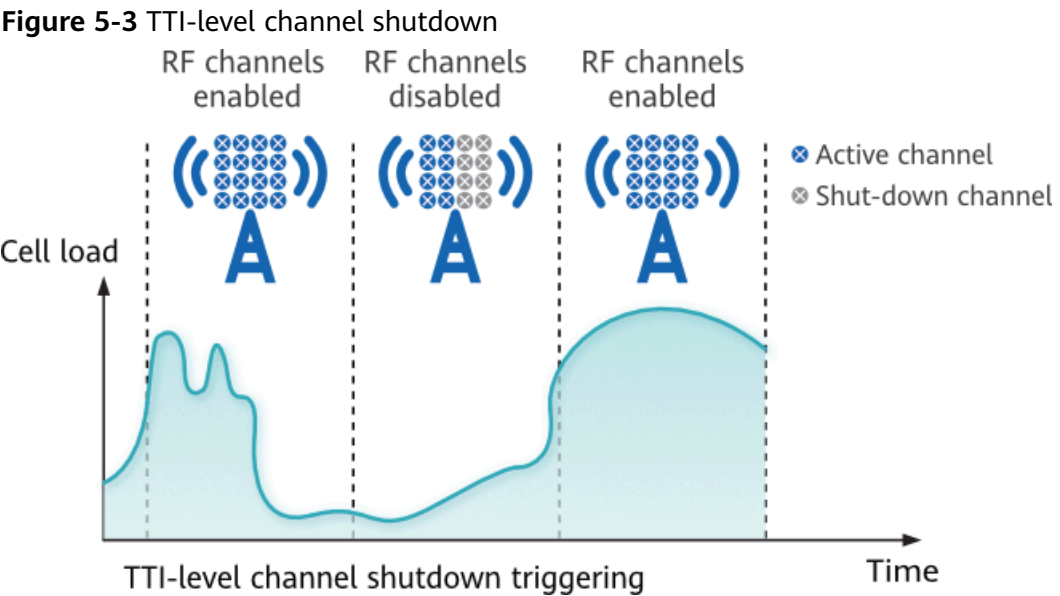
5.3.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.
- On the live network, the one-week base station energy consumption data obtained before and after low-power RF channel dynamic shutdown is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.
- The duration of dynamic RF channel shutdown caused by the activation of low-power RF channel dynamic shutdown can be obtained by observing the **N.PowerSaving.DynamicRFShutdown.Dur** counter on the MAE-Access.

5.4 TTI-Level Channel Shutdown

5.4.1 Principles

TTI-level channel shutdown determines whether to shut down RF channels based on the TTI-level load to obtain energy saving gains, as shown in [Figure 5-3](#).



In a specified period, when the network is under light or moderate load and the TTI-level cell load is low, the base station shuts down the corresponding transmit channels for energy saving if the data- or common-channel condition described in Table 5-18 is met. In the case of common-channel dynamic shutdown, the base station automatically performs power compensation on the remaining transmit channels to ensure coverage.

The base station quickly restores all transmit channels to ensure user experience when the TTI-level cell load is high.

Table 5-18 Conditions for TTI-level channel shutdown

Shutdown Type	Shutdown Condition
Data-channel dynamic shutdown	The transmit channels only transmit data services of SU-MIMO UEs.
Common-channel dynamic shutdown	(TDD) The transmit channels only transmit SSBs, RMSI, OSI, paging messages, and SU-MIMO UEs' data services.

TTI-level channel shutdown can be enabled by selecting the **TTI_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter.

Table 5-19 describes the parameters used to specify the time period, load threshold, and other information under an energy saving policy of TTI-level channel shutdown.

Table 5-19 Parameters for the TTI-level channel shutdown policy

Parameter Name	Parameter ID	Description
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	This parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	This parameter specifies the energy saving type indicated by an energy saving policy index. Set this parameter to TTI_RF_SHUTDOWN to enable TTI-level channel shutdown.
Start Time	NRDUCellPowerSaving.StartTime	These parameters specify the time period in which TTI-level channel shutdown takes effect, as well as the evaluation period and conditions for a cell to enter and exit the TTI-level channel shutdown state. Within the time period specified by NRDUCellPowerSaving.StartTime and NRDUCellPowerSaving.StopTime for TTI-level channel shutdown to take effect for a cell, the base station periodically determines whether the cell enters the TTI-level channel shutdown state. The period is specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter. - If the downlink PRB usage of a cell or TRP^b is not greater than the value of NRDUCellPowerSaving.DlPrbThld, the cell or TRP automatically enters the TTI-level channel shutdown state^a. - If the downlink PRB usage of a cell or TRP^b is greater than the sum of the values of NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset or the current time is not within the time period for this function to take effect, the cell or TRP automatically exits the TTI-level channel shutdown state.
Stop Time	NRDUCellPowerSaving.StopTime	
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	

Parameter Name	Parameter ID	Description
TTI RF Channel Shutdown Mode	NRDUCellPowerSaving.TtiRfShutdownMode	After a cell or TRP enters the TTI-level channel shutdown state: • If the NRDUCellPowerSaving.TtiRfShutdownMode parameter is set to BASIC, fixed mode of TTI-level channel shutdown is used. – In TDD, when the slot-level PRB usage in a TTI is less than or equal to the value of NRDUCellPowerSaving.SlotPrbThld, channel shutdown is performed in this TTI. • If the NRDUCellPowerSaving.TtiRfShutdownMode parameter is set to ADAPTIVE, adaptive mode of TTI-level channel shutdown is used. The cell determines whether to enable channel shutdown and the number of channels to be shut down based on the remaining available frequency-domain resources in the current slot. For TDD, when the NRDUCellPowerSaving.ComChnDynShutdownSwitch parameter is set to ON, channel shutdown falls back to fixed mode (the same as that with NRDUCellPowerSaving.TtiRfShutdownMode set to BASIC) in the slots where RMSI, OSI, paging messages, or SSBs reside. (For paging messages, this requires the PAGING_DYN_SHUTDOWN_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter be selected; for SSBs, this requires that NRDUCellPowerSaving.SsbDynShutdownSwitch be set to ON.)
Slot PRB Threshold	NRDUCellPowerSaving.SlotPrbThld	This parameter applies only to TDD. If the NRDUCellPowerSaving.TtiRfShutdownMode parameter is set to BASIC, fixed mode of TTI-level channel shutdown is used. When the slot-level PRB usage in a TTI is less than or equal to the value of NRDUCellPowerSaving.SlotPrbThld, channel shutdown is performed in this TTI.

Parameter Name	Parameter ID	Description
PMI Weight UE Dynamic Shutdown Switch	NRDUCellPowerSaving.PmiWtUeDynShutdownSwitch	This parameter specifies whether to enable TTI-level channel shutdown for UEs using PMI-based weights. In TDD, TTI-level channel shutdown takes effect by default for UEs using SRS-based weights. For TDD: • If this parameter is set to ON, TTI-level channel shutdown takes effect for UEs using PMI-based weights. • If this parameter is set to OFF, TTI-level channel shutdown does not take effect for UEs using PMI-based weights.

Parameter Name	Parameter ID	Description
Common Channel Dynamic Shutdown Switch	NRDUCellPowerSaving.ComChnDynShutdownSwitch	The NRDUCellPowerSaving.ComChnDynShutdownSwitch parameter specifies whether to enable common-channel dynamic shutdown. - If this parameter is set to ON, common-channel dynamic shutdown is enabled and takes effect for RMSI and OSI. In this case, the following subfunctions can be configured to enable common-channel dynamic shutdown for SSBs and paging messages in addition to RMSI and OSI: - SSB common-channel dynamic shutdown: This function allows common-channel dynamic shutdown to take effect for SSBs. It is controlled by NRDUCellPowerSaving.SsbDynShutdownSwitch, which is configurable only when NRDUCellPowerSaving.ComChnDynShutdownSwitch is set to ON. - Dynamic paging shutdown: This function allows common-channel dynamic shutdown to take effect for paging messages. This function is controlled by the PAGING_DYN_SHUTDOWN_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter and can take effect only when NRDUCellPowerSaving.ComChnDynShutdownSwitch is set to ON. - If this parameter is set to OFF, common-channel dynamic shutdown is disabled. In the case of coexistence with other RATs, common-channel dynamic shutdown takes effect only in certain beam coverage scenarios of TRPs, for example, EXPAND_SCENARIO_1 for SSB beams.
SSB Dynamic Shutdown Switch	NRDUCellPowerSaving.SsbDynShutdownSwitch	
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	

Parameter Name	Parameter ID	Description
NR Multi-Carrier Protection Switch	NRDUCellPowerSaving.NrMultiCarrProtSwitch	This parameter applies only to TDD. It specifies whether to enable NR multi-carrier protection for common-channel dynamic shutdown. It can be configured only when the NRDUCellPowerSaving.ComChnDynShutdownSwitch parameter is set to ON in NR multi-carrier scenarios and applies only to low-frequency TDD. • If this parameter is set to ON, NR multi-carrier protection takes effect. Common-channel dynamic shutdown takes effect only in certain beam coverage scenarios of TRPs^c, for example, EXPAND_SCENARIO_1. The energy saving gains may decrease and the impact on network KPIs also decreases. • If this parameter is set to OFF, NR multi-carrier protection is disabled. In this case, no restriction is imposed on beam coverage scenarios of TRPs^c for common-channel dynamic shutdown. The energy saving gains may increase, the coverage slightly deteriorates, the service drop rate slightly increases, and the access success rate and handover success rate slightly decrease.

Parameter Name	Parameter ID	Description
		a. A cell or TRP enters the TTI-level channel shutdown state. - A common cell automatically enters the TTI-level channel shutdown state if its downlink PRB usage is less than or equal to the value of NRDUCellPowerSaving.DlPrbThld. - A TRP in a hyper cell automatically enters the TTI-level channel shutdown state if its downlink PRB usage is less than or equal to the value of NRDUCellPowerSaving.DlPrbThld. - A combined cell automatically enters the TTI-level channel shutdown state if the downlink PRB usage of each TRP is less than or equal to the value of NRDUCellPowerSaving.DlPrbThld. b. The following formulas are used to calculate the downlink PRB usage of a cell or TRP. - Downlink PRB usage of a common cell = (Average number of PRBs used by PDSCH DRB services – Average number of extended PRBs used by the PDSCH)/Average number of available downlink PRBs - Downlink PRB usage of a TRP of a hyper cell or combined cell = (Average number of downlink PRBs used by the TRP – Average number of downlink extended PRBs used by the TRP)/Average number of available downlink PRBs for the TRP c. In TDD, the beam coverage scenario of a TRP is specified by the NRDUCellTrpBeam.CoverageScenario parameter.

After TTI-level channel shutdown (controlled by the **TTI_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter) is enabled, data-channel dynamic shutdown and common-channel dynamic shutdown in different configuration combinations take effect as described in [Table 5-20](#) and [Table 5-21](#).

The recommended enabling policies are as follows:

- If energy saving is preferred, you are advised to turn on all of **NRDUCellPowerSaving.PmiWtUeDynShutdownSwitch**, **NRDUCellPowerSaving.ComChnDynShutdownSwitch**, **NRDUCellPowerSaving.SsbDynShutdownSwitch**, and **PAGING_DYN_SHUTDOWN_SW** option of **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch**.
- If coverage and network performance are preferred, you are advised to turn on only **NRDUCellPowerSaving.PmiWtUeDynShutdownSwitch**.

Table 5-20 Data-channel dynamic shutdown in different configuration combinations

NRDUCellPowerSaving.PmiWtUeDynShutdownSwitch	UEs for Which Data-Channel Dynamic Shutdown Takes Effect (TDD)
On	UEs using SRS-based weights and UEs using PMI-based weights

NRDUCellPowerSav- ing.PmiWtUeDynShutdownSwitch	UEs for Which Data-Channel Dynamic Shutdown Takes Effect (TDD)
Off	UEs using SRS-based weights

Table 5-21 Common-channel dynamic shutdown in different configuration combinations

NRDUCellPowerSav- ing.ComChnDynShut downSwitch	NRDUCellPowerSav- ing.SsbDynShutdown Switch	PAGING_DYN_SHUTD OWN_SW Option of NRDUCellPwrSaveAl- go.PowerSavingAlgoS witch	Signaling for Which Common-Channel Dynamic Shutdown Takes Effect
On	On	On	SSB, RMSI, OSI, and paging
On	On	Off	SSB, RMSI, and OSI
On	Off	On	RMSI, OSI, and paging
On	Off	Off	RMSI and OSI
Off	N/A	N/A	None

For specific service types, cells can dynamically exit TTI-level channel shutdown. This is known as service-type-based dynamic exit from TTI-level channel shutdown. When there is any UE with a specific type of service in a cell, the cell exits the TTI-level channel shutdown state. This type of exit ensures the performance of UEs performing that type of service. The service types involved are specified by the **gNBQciBearer.ServiceType** parameter.

Service-type-based dynamic exit from TTI-level channel shutdown takes effect when all of the following conditions are met:

- TTI-level channel shutdown is enabled.
- The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.
- The **TTI_RF_SHUTDOWN_EXIT_SW** option of the **NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter is selected in the group identified by the **NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter that is set to the same value as the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter for a specific service type.

5.4.2 Network Analysis

5.4.2.1 Benefits

When the network is under light or moderate load and the instantaneous cell load is light, some RF channels can be shut down by TTI-level channel shutdown. This reduces the overall energy consumption of the base station.

Energy saving gains may decrease in the following scenarios:

- If there are UEs involved in the following services on the network, TTI-level channel shutdown cannot take effect and therefore energy saving gains may decrease: CoMP, CS, CBF, MU-MIMO, FWA services with a guaranteed rate configured but not actually in effect, VoNR, Non-Slot, or high-reliability services.
- In NR multi-carrier scenarios, the energy saving gains of TTI-level channel shutdown decrease. It is recommended that the same energy saving policy be configured for TTI-level channel shutdown between multiple carriers to obtain more energy saving gains.
- When there are no real UEs in a cell with false alarms, the SSB channel cannot be shut down. Therefore, the energy saving gains decrease.
- Service-type-based dynamic exit from TTI-level channel shutdown relieves the impact of TTI-level channel shutdown on specific service types but reduces energy saving gains.

5.4.2.2 Impacts

The impacts between this function and other functions are described in [5.4.3.2.3 Function Impacts](#).

- After data-channel dynamic shutdown in TTI-level channel shutdown takes effect:
 - The IBLER may increase.
 - The average CCE usage of the cell and the CCE aggregation level may increase.
 - The average MCS index for downlink scheduling decreases.
 - The downlink cell throughput may slightly decrease.
 - The values of the following counters related to downlink PRB usage increase in common cells:
 - **N.PRB.DL.DrbUsed.Avg**
 - **N.PRB.DL.Used.Avg**
 - **N.PRB.DL.ExtendUsed.Avg**
 - The values of the following counters related to the downlink PRB usage of TRPs increase in hyper cells and combined cells:
 - **N.HyperCell.TRP.PRB.DL.Used.Avg**
 - **N.HyperCell.TRP.PRB.DL.ExtendUsed.Avg**
 - In NR TDD, when the **NRDUCellPowerSaving.TtiRfShutdownMode** parameter is set to **BASIC**, the higher the slot-level PRB usage threshold, the larger the decrease in the user-perceived rate.

- When the **NRDUCellPowerSaving.TtiRfShutdownMode** parameter is set to **ADAPTIVE**, the user-perceived rate slightly decreases.
- The values of counters related to downlink user-plane delay may increase.
- After common-channel dynamic shutdown in TTI-level channel shutdown takes effect, the coverage slightly deteriorates, the service drop rate slightly increases, and the access success rate and handover success rate slightly decrease. In addition, when dynamic paging shutdown takes effect, the paging success rate on the core network slightly decreases. In TDD, when common-channel dynamic shutdown takes effect, data-channel dynamic shutdown also takes effect. Therefore, impacts are generated by both functions.

5.4.3 Requirements

5.4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-061203	Intelligent Channel Dynamic Shutdown	NR0S00DRCS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	TTI-level channel shutdown depends on symbol power saving.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	FR1 Max Cell CSI-RS Port Num	NRDUCellCsirs.FR1MaxCellCsirsPortNum	<i>Channel Management</i>	When the NRDUCellPowerSaving.PmiWtUeDynShutdownSwitch parameter is set to ON and the NRDUCellTrp.TxRxMode parameter is set to 64T64R or 32T32R , the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter must be set to 8PORT .
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Service-type-based dynamic exit from TTI-level channel shutdown takes effect only if TTI-level channel shutdown has been enabled.

5.4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransmissionMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>	In TDD, TTI-level channel shutdown (including data-channel dynamic shutdown and common-channel dynamic shutdown) is not supported in combined cells in basic or DPS transmission mode.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Common-channel dynamic shutdown cannot be enabled for hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	VMIMO in DAS-based indoor coverage	NRDUCellAlgoSwitch.VmimoSwitch	<i>Indoor Coverage</i>	Common-channel dynamic shutdown cannot be enabled if VMIMO in DAS-based indoor coverage has been enabled.
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Common-channel dynamic shutdown cannot be enabled if the function of low latency and high reliability has been enabled.
Low-frequency TDD	Distributed MIMO	D_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	None	Common-channel dynamic shutdown cannot be enabled if Distributed MIMO has been enabled.
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	TDD: <i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	TTI-level channel shutdown (including data-channel dynamic shutdown and common-channel dynamic shutdown) cannot be enabled if distributed massive MIMO has been enabled.
Low-frequency TDD	Dynamic SSB	DYNAMIC_SSB_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	None	Common-channel dynamic shutdown cannot be enabled if dynamic SSB has been enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BAC KUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	TTI-level channel shutdown (including data-channel dynamic shutdown and common-channel dynamic shutdown) cannot be enabled if the primary/backup function in intra-frequency dual-network redundancy networking has been enabled.
Low-frequency TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BA CKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	TTI-level channel shutdown (including data-channel dynamic shutdown and common-channel dynamic shutdown) cannot be enabled if dual-BBU backup in dual-network redundancy networking has been enabled.
Low-frequency TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	TTI-level channel shutdown is incompatible with the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios.
Low-frequency TDD	TRP Antenna Type	NRDUCellTrpBeam.TrpAntType	None	TTI-level channel shutdown cannot be enabled if the NRDUCellTrpBeam.TrpAntType parameter is set to 8T_2222 , 8T_44 , or 8T_422 .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Weight optimization in scenarios of aggregation into 2T	AGG_2T_WEIGHT_OPT_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter	None	TTI-level channel shutdown does not work with weight optimization in scenarios of aggregation into 2T.
Low-frequency TDD	Intra-frequency assistant cell	NRDUCell.SupplementaryCellInd set to INTRA_FREQ_ASSISTANT_CELL	None	Intra-frequency assistant cells are incompatible with TTI-level channel shutdown.
Low-frequency TDD	Dynamic MIMO layer number decrease	DYNAMIC_MIMO_LAYER_DECREASE_SW option of the gNodeBAlgo.DedicatedNetworkFuncSw parameter	None	In 8T8R scenarios where the NRDUCell.NrDuCellNetworkingMode parameter is set to NORMAL_CELL , dynamic MIMO layer number decrease does not work with TTI-level channel shutdown.

5.4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When both remote interference adaptive avoidance and TTI-level channel shutdown take effect, the energy saving gains of TTI-level channel shutdown decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Target IBLER adaptation based on interference from neighboring cells	DL_INTRF_BASED_IBLER_ADAPT_SW option of the NRDUCellDlAmc.DlMcsSelAlgoSw parameter	<i>Downlink Scheduling</i>	When both TTI-level channel shutdown and target IBLER adaptation based on interference from neighboring cells take effect, the latter function may not deliver any gains.
Low-frequency TDD	Downlink RB estimation optimization	RB_ESTIMATE_OPT_SW option of the NRDUCellPdsch.DlPdschAlgoSwitch parameter	<i>Downlink Scheduling</i>	If TTI-level channel shutdown takes effect but downlink RB estimation optimization does not take effect, the user-perceived rate may decrease. It is recommended that downlink RB estimation optimization be enabled when TTI-level channel shutdown is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink tail packet scheduling optimization	TAIL_PKT_SCH_OPT_SW option of the NRDUCellPdsch.DLPdschAlgoSwitch parameter	<i>Downlink Scheduling</i>	If data-channel dynamic shutdown in TTI-level channel shutdown is in effect but downlink tail packet scheduling optimization does not take effect, the user-perceived rate may decrease. It is recommended that downlink tail packet scheduling optimization be enabled when data-channel dynamic shutdown in TTI-level channel shutdown is in effect.
Low-frequency TDD	Two-level BWP	None	None	When both two-level BWP and TTI-level channel shutdown take effect, the user-perceived rate decrease caused by TTI-level channel shutdown rises.
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When both NR inter-carrier dynamic power sharing and TTI-level channel shutdown take effect, the energy saving gains of TTI-level channel shutdown decrease and the downlink UE throughput may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SC H_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When NR multi-band power adaptive scheduling and TTI-level channel shutdown both take effect, the energy saving gains of TTI-level channel shutdown decrease and the downlink UE throughput may decrease.
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	• LTE: LTE_NR_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynmicPowerSharingSwitch parameter • NR: LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When both LTE TDD and NR Flash Dynamic Power Sharing and TTI-level channel shutdown take effect, the energy saving gains of TTI-level channel shutdown decrease and the downlink UE throughput may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR multi-band power adaptive scheduling	• LTE: LNR_MULTI_BAND_PWR_ADPT_SCH_SW option of the CellDynPowerSharing.DynMicPowerSharingSwitch parameter • NR: LNR_MULTI_BAND_PWR_ADPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When LTE and NR multi-band power adaptive scheduling takes effect together with TTI-level channel shutdown, the energy saving benefits of TTI-level channel shutdown decrease and downlink UE throughput may decrease.
Low-frequency TDD	L/NR Inter-band Deep Coordination	• LTE: LNR_INTER_BAND_DEEP_COORD_SW option of the CellDynPowerSharing.DynMicPowerSharingSwitch parameter • NR: LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	When both L/NR Inter-band Deep Coordination and TTI-level channel shutdown take effect, the energy saving gains of TTI-level channel shutdown decrease and the downlink UE throughput may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to JOINT_MODE	<i>Cell Combination</i>	When both NRDUCellPowerSaving.PmiWtUeDynShutdownSwitch and NRDUCellPowerSaving.ComChnDynShutdownSwitch are turned off, TTI-level channel shutdown has no energy saving gains in Joint Transmission Cell Combination scenarios.
Low-frequency TDD	Intelligent scheduling for power saving	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both power saving scheduling of intelligent scheduling for power saving and TTI-level channel shutdown take effect, the energy saving gains of TTI-level channel shutdown decrease.
Low-frequency TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both TTI-level channel shutdown and energy saving based on flexible frequency-domain scheduling take effect, the gains of energy saving based on flexible frequency-domain scheduling decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	TTI-level channel shutdown does not take effect when RF channel intelligent shutdown is in effect.
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	TTI-level channel shutdown does not take effect when timing carrier shutdown is in effect.
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTTI_CARR_SD_SW or INTER_GNB_MULTTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	TTI-level channel shutdown does not take effect when intelligent carrier shutdown is in effect.
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When TTI-level channel shutdown and dynamic RF channel shutdown are both enabled, the latter function takes effect preferentially.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option (inter-base-station multi-carrier scenarios) of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected - Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected	<i>Energy Conservation and Emission Reduction</i>	TTI-level channel shutdown does not take effect when fast carrier shutdown is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MetaAAU power reduction for cell-center UEs	NRDUCellChnPwr.PwrReductionValue not set to 0	<i>Energy Conservation and Emission Reduction</i>	When MetaAAU power reduction for cell-center UEs and TTI-level channel shutdown take effect simultaneously, MetaAAU power reduction for cell-center UEs delivers fewer energy saving gains.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCY SW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	TTI-level channel shutdown does not take effect when RF module deep dormancy is in effect.
Low-frequency TDD	RF module super dormancy	RRU.DORMANCY SW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	TTI-level channel shutdown does not take effect when RF module super dormancy is in effect.
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS W (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over TTI-level channel shutdown when both functions are enabled. TTI-level channel shutdown does not take effect when RF module fast deep dormancy is in effect.
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	TTI-level channel shutdown does not take effect in slots with MU-MIMO pairing.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL Blanking	NRDUCELL.PeerCellSlotAssignment set to 4_1_DDDSU or 8_2_DDDDDDSUU	<i>Standards Compliance</i>	With the DL Blanking function enabled, the maximum number of SSBs decreases, and common-channel dynamic shutdown (a sub-function of TTI-level channel shutdown) may take effect in more scenarios.
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDLRank.DlRankSelAlgoSw parameter	<i>FWA</i>	If downlink 8-layer transmission for a single FWA user has taken effect, the energy saving gains of TTI-level channel shutdown decrease.
Low-frequency TDD	Cell load simulation	STRNRDUCELLSIMULATELOAD	None	When cell load simulation and TTI-level channel shutdown take effect simultaneously, TTI-level channel shutdown delivers fewer energy saving gains.
Low-frequency TDD	Paging Capacity Mode	NRDUCellPagingConfig.PagingCapacityMode	<i>5G Networking and Signaling</i>	After TTI-level channel shutdown takes effect, beam coverage areas alter. As a result, UEs may fail to receive paging messages.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUeIdentSw	None	TTI-level channel shutdown fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
Low-frequency TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.AppIpv4 • gNBAsAppParam.AppIpv6	None	• Any option of the gNodeBParam.StUeIdentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.AppIpv4 parameter is set. • The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue: • Select the SPEED_TURBO_PWR_SAVING_SW option of the

RAT	Function Name	Function Switch	Reference	Description
				gNBPwrSavingAlgo.PwrSavin gAlgoSwitch parameter. TTI-level channel shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBPwrSavingAlgo.SpeedTurboPwrSavStartTime and gNBPwrSavingAlgo.SpeedTurboPwrSavEndTime parameters, increasing the energy saving gains. • Select the STATIC_ST_PWR_SAVING_SW option of the gNBPwrSavingAlgo.PwrSavin gAlgoSwitch parameter. TTI-level channel shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBPwrSaving

RAT	Function Name	Function Switch	Reference	Description
				Algo.StaticSTPwrSavStartTime and gNBPwrSavingAlgo.StaticSTPwrSavEndTime parameters, increasing the energy saving gains.
Low-frequency TDD	Multiple BWP1s for RedCap	NRDUCellRedCap.RedCapUeMaxAvailableBandwidth set to a value greater than 20M	<i>RedCap</i>	For a TDD cell, when TTI-level channel shutdown is enabled and at least one RedCap BWP1 with the NCD-SSB is activated in the cell, power aggregation is required for each RedCap BWP1. In this case, TTI-level channel shutdown does not take effect when SSB power aggregation is insufficient.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Virtual Grid-based Multi-Frequency Coordination	• RSRP prediction model – VG_MODEL_ALLOW_BUILD_FLAG option of the NRCellFreqRelation.AggregationAttribute parameter – NR_IF_HO_MEAS_QTY_PRED_FUNCTION option of the NRCellSmartMultiCarrier.NrMultiCarrierAlgoSwitch parameter • Downlink intra-frequency spectral efficiency prediction model – DL_INTRA_SE_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarrier.ModelBuildingSwitch parameter – DL_INTRA_FREQ_SPCT EFF_PRED_SW option of the NRCellSmartMultiCarrier.	<i>Virtual Grid-based Multi-Frequency Coordination</i>	After TTI-level channel shutdown takes effect, the building of virtual grid models will be suspended and existing virtual grid models will be brought offline and cannot be used. If virtual grid models need to be used while TTI-level channel shutdown is in effect, the VIRTUAL_GRID_C H_SD_COEXIST_SW option of the gNB PwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected for both the local and neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
		NrMultiCarrierAlgoSwitch parameter • Intra-frequency SRS beam prediction model INTRAF_SRS_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarr.ModelBuildingSwitch parameter • Downlink-frequency spectral efficiency model DL_FREQ_SE_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarr.ModelBuildingSwitch parameter		

5.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

(TDD) The following low-frequency AAUs/RRUs support TTI-level channel shutdown:

- Data-channel dynamic shutdown in TTI-level channel shutdown: 64T64R, 32T32R, or 8T8R RF modules
- Common-channel dynamic shutdown in TTI-level channel shutdown: 32T32R or higher RF modules complying with the eCPRI protocol and 8T8R 5000 series RRUs

Whether the common-channel dynamic shutdown function can take effect is subject to the power aggregation capability of RF modules, beam coverage scenario, and settings of parameters related to power offset. For example, this function can take effect for AAU531x, AAU561x, and 5000 series RRUs only when the beam coverage scenario is **EXPAND_SCENARIO_1** and the power-offset parameters are all set to 0. For details about these parameters, see [Table 5-5](#).

5.4.3.4 Cells

This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**.

(TDD) The cell TX/RX mode (specified by the **NRDUCellTrp.TxRxMode** parameter) must be 64T64R, 32T32R, or 8T8R.

(TDD) Common-channel dynamic shutdown in TTI-level channel shutdown has the following additional requirements:

- The total number of NR carriers configured on an RF module must be less than or equal to 2.
- The **NRDUCell.FrequencyBand** parameter must be set to **N38**, **N40**, **N41**, **N77**, **N78**, or **N79**.
- The radius of cells configured on an 8T8R RF module must be less than or equal to 14500 m.
- The downlink bandwidth must be 100 MHz, 90 MHz, 80 MHz, 70 MHz, 60 MHz, 50 MHz, or 40 MHz.
- The **NRDUCellTrpBeam.CoverageScenario** parameter must be set to **DEFAULT**, **SCENARIO_1**, **SCENARIO_2**, **SCENARIO_3**, **SCENARIO_4**, **SCENARIO_6**, **SCENARIO_7**, **SCENARIO_8**, **SCENARIO_9**, **SCENARIO_12**, **SCENARIO_13**, **EXPAND_SCENARIO_1**, **EXPAND_SCENARIO_5**, **EXPAND_SCENARIO_7**, **EXPAND_SCENARIO_8**, **SCENARIO_27**, **SCENARIO_28**, **SCENARIO_31**, **SCENARIO_36** or **CUSTOMIZED_BEAM_SCENARIO**.
- When the **NRDUCellTrpBeam.CoverageScenario** parameter is set to **CUSTOMIZED_BEAM_SCENARIO** for a TRP, the **NRDUCellTrpCustBeam.SsbBeamHorizWidth** parameter of each customized scenario-based beam specified by the **NRDUCellTrpCustBeam.BeamId** parameter must be set to **WIDTH_30** for the TRP.
- When the **SECTOREQM.ANTCFGMODE** parameter is set to **BEAM**, the **RRU.RS** parameter must be set to **NO**, **TN**, **LN**, **UN**, **ULN**, **GLN**, **GLFTDN**, or **LFTDN** for all sector equipment serving each cell. The cabinet number, subrack number, and slot number of each RF module in the sector equipment identified by the **NRDUCellCoverage.SectorEqmId** parameter are specified by

the **SECTOREQM.RRUCN**, **SECTOREQM.RRUSRN**, and **SECTOREQM.RRUSN** parameters, respectively.

- When the **SECTOREQM.ANTCFGMODE** parameter is set to **ANTENNAPORT**, it is recommended that the **RRU.RS** parameter be set to the same value as that when the **SECTOREQM.ANTCFGMODE** parameter is set to **BEAM**. If it is not set this way, common-channel dynamic shutdown in TTI-level channel shutdown may not take effect.
- The **NRDUCell.DeployPosition** parameter cannot be set to **DEPLOY_IAB**.

5.4.3.5 Others

None

5.4.4 Operation and Maintenance

5.4.4.1 Data Configuration

5.4.4.1.1 Data Preparation

[Table 5-22](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-22 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellAlgoSwitch.NrDuCellId	Set this parameter based on the network plan.
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	Select the TTI_RF_SHUTDOWN_SW option of this parameter when TTI-level channel shutdown is required.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: • For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example, if the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. • For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.

Parameter Name	Parameter ID	Setting Notes
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan and the following requirements: • For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example, if the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. • For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to TTI_RF_SHUTDOWN when TTI-level channel shutdown is required.
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	You are advised to use the recommended value.
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	You are advised to use the recommended value.
Slot PRB Threshold	NRDUCellPowerSaving.SlotPrbThld	You are advised to use the recommended value.

Parameter Name	Parameter ID	Setting Notes
TTI RF Channel Shutdown Mode	NRDUCellPowerSaving.TtiRfShutdownMode	You are advised to use the recommended value.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	For TTI-level channel shutdown, this parameter can only be set to its default value.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	You are advised to use the recommended value.
PMI Weight UE Dynamic Shutdown Switch	NRDUCellPowerSaving.PmiWtUeDynShutdownSwitch	You are advised to use the recommended value.
Common Channel Dynamic Shutdown Switch	NRDUCellPowerSaving.ComChnDynShutdownSwitch	If coverage and network performance are preferred, the value OFF is recommended. If energy saving is preferred, the value ON is recommended.
SSB Dynamic Shutdown Switch	NRDUCellPowerSaving.SsbDynShutdownSwitch	This parameter can be set only when the NRDUCellPowerSaving.ComChnDynShutdownSwitch parameter is set to ON . Its default value is recommended.
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	Set the PAGING_DYN_SHUTDOWN_SW option of this parameter to its recommended value if NRDUCellPowerSaving.ComChnDynShutdownSwitch is set to ON .
NR Multi-Carrier Protection Switch	NRDUCellPowerSaving.NrMultiCarrProtSwitch	In TDD, this parameter can be set only when the NRDUCellPowerSaving.ComChnDynShutdownSwitch parameter is set to ON . Its default value is recommended. This parameter takes effect in NR multi-carrier scenarios and applies only to low-frequency TDD.

Parameter Name	Parameter ID	Setting Notes
Downlink PDSCH Algorithm Switch	NRDUCellPdsch.DLPdschAlgoSwitch	When TTI-level channel shutdown is enabled, select the RB_ESTIMATE_OPT_SW option of this parameter to enable downlink RB estimation optimization. For details about the impact of downlink RB estimation optimization on network performance, see <i>Downlink Scheduling</i> .
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	It is recommended that the SERV_TYPE_DYN_EXIT_PWR_SAVE_SW option be selected when there are specific types of services and user experience of these services needs to be ensured.
Power Saving Policy Group Index	gNBQciBearer.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan.
Power Saving Policy Group Index	NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan. The value of this parameter must be the same as that of the gNBQciBearer.PowerSavingPolicyGrpIndex parameter for services to be guaranteed. Before setting this parameter, ensure that the QCI for which the gNBQciBearer.PowerSavingPolicyGrpIndex parameter is set has been configured.
User Identification Method	NRDUCellPsPolicyGrp.UserIdentMethod	Set this parameter to SERVICE_TYPE .
Service Type Dynamic Exit Power Saving Switch	NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw	To enable service-type-based dynamic exit from TTI-level channel shutdown, select the TTI_RF_SHUTDOWN_EXIT_SW option.

5.4.4.1.2 Using MML Commands

Before using MML commands, refer to [5.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- The following are command examples for function activation in TDD.


```
//Turning on the energy saving switch for TTI-level channel shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=TTI_RF_SHUTDOWN_SW-1;
//Enabling downlink RB estimation optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DiPdschAlgoSwitch=RB_ESTIMATE_OPT_SW-1;
```

 - Scenario 1:


```
//Setting an energy saving policy of common-channel dynamic shutdown and data-channel
dynamic shutdown in TTI-level channel shutdown (Take adaptive mode in TDD as an example.)
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
StartTime=04&00&00, StopTime=05&00&00, PowerSavingType=TTI_RF_SHUTDOWN,
DiPrbThld=30, DiPrbOffset=15, TtiRfShutdownMode=ADAPTIVE, SlotPrbThld=50,
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5,
ComChnDynShutdownSwitch=ON, PmiWtUeDynShutdownSwitch=ON,
SsbDynShutdownSwitch=OFF, NrMultiCarrProtSwitch=ON;
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=PAGING_DYN_SHUTDOWN_SW-0;
```
 - Scenario 2:


```
///(Optional) Setting an energy saving policy of only data-channel dynamic shutdown in TTI-
level channel shutdown (Take adaptive mode in TDD as an example.)
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
StartTime=04&00&00, StopTime=05&00&00, PowerSavingType=TTI_RF_SHUTDOWN,
DiPrbThld=30, DiPrbOffset=15, TtiRfShutdownMode=ADAPTIVE,
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5,
PmiWtUeDynShutdownSwitch=ON;
```
 - Scenario 3:


```
///(Optional) Setting an energy saving policy of only common-channel dynamic shutdown in TTI-
level channel shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
StartTime=04&00&00, StopTime=05&00&00, PowerSavingType=TTI_RF_SHUTDOWN,
DiPrbThld=30, DiPrbOffset=15, TtiRfShutdownMode=ADAPTIVE, SlotPrbThld=50,
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5,
ComChnDynShutdownSwitch=ON, SsbDynShutdownSwitch=OFF, NrMultiCarrProtSwitch=ON;
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=PAGING_DYN_SHUTDOWN_SW-0;

///(Optional) Enabling service-type-based dynamic exit from TTI-level channel shutdown
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;
ADD NRDUCELLPSPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=TTI_RF_SHUTDOWN_EXIT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
///(Optional) Disabling service-type-based dynamic exit from TTI-level channel shutdown  
MOD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,  
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=TTI_RF_SHUTDOWN_EXIT_SW-0;  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;  
//Turning off the energy saving switch for TTI-level channel shutdown  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=TTI_RF_SHUTDOWN_SW-0;  
//Removing the energy saving policy of TTI-level channel shutdown  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
```

5.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.4.2 Activation Verification

After activating TTI-level channel shutdown, use either of the following methods to check whether the function has taken effect:

- Method 1:
 - a. Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Switch** parameter is **TTI Channel Shutdown Sw:On** in the command output, TTI-level channel shutdown has been enabled.
 - b. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **TTI Channel Shutdown**.

NOTE

If there are two or more records with **Power Saving Type** being **TTI Channel Shutdown**, TTI-level channel shutdown has been enabled in two or more time periods.

In NR multi-carrier scenarios, the energy saving gains of TTI-level channel shutdown decrease. In NR multi-carrier scenarios, it is recommended that the same energy saving policy of TTI-level channel shutdown be configured for multiple carriers to obtain more energy saving gains.

- c. Run the **DSP TIME** command to check whether the system time is within a period for TTI-level channel shutdown to take effect.
 - If it is, go to step **d**.
 - If it is not, wait until the start time of TTI-level channel shutdown arrives.
- d. Check whether the downlink PRB usage of the cell is less than or equal to the downlink PRB usage threshold for starting TTI-level channel shutdown (specified by the **NRDUCellPowerSaving.DlPrbThld** parameter).

- If it is, go to step e.
- If it is not, wait until the downlink PRB usage of the cell is less than or equal to the downlink PRB usage threshold for starting TTI-level channel shutdown.
- e. Wait up to 10 minutes and run the **DSP NRDUCELL** command. Verify that the value of the **Power Saving State** parameter is **TTI Channel Shutdown:Enabled**, which indicates that TTI-level channel shutdown has taken effect.
- Method 2:
Run the **DSP NRDUCELLPOWERSAVING** command to query the status of TTI-level channel shutdown. The **DSP NRDUCELLPOWERSAVING** command output is displayed in two rows, indicating whether data-channel dynamic shutdown and common-channel dynamic shutdown in TTI-level channel shutdown have taken effect respectively.
 - Verify whether data-channel dynamic shutdown has taken effect:
 - If the value of **Power Saving Type** is **TTI Channel Shutdown** and the value of **Current State** is **PDSCH Dynamic Shutdown Enabled**, data-channel dynamic shutdown in TTI-level channel shutdown has taken effect.
 - If the value of **Power Saving Type** is **TTI Channel Shutdown** and the value of **Current State** is **PDSCH Dynamic Shutdown Disabled**, check the value of **Current Disabled Cause** to determine the reason why data-channel dynamic shutdown in TTI-level channel shutdown did not take effect. [Table 5-23](#) lists possible reasons.
 - Verify whether common-channel dynamic shutdown has taken effect:
 - If the value of **Power Saving Type** is **TTI Channel Shutdown** and the value of **Current State** is **Common-Channel Dynamic Shutdown Enabled**, common-channel dynamic shutdown in TTI-level channel shutdown has taken effect.
 - If the value of **Power Saving Type** is **TTI Channel Shutdown** and the value of **Current State** is **Common-Channel Dynamic Shutdown Disabled**, check the value of **Current Disabled Cause** to determine the reason why common-channel dynamic shutdown in TTI-level channel shutdown did not take effect. [Table 5-23](#) lists possible reasons.

Table 5-23 Possible reasons why TTI-level channel shutdown does not take effect

Reason	Description
Insufficient License	No license for this function is available.
Switch Turned Off	None
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable for other reasons.

Reason	Description
Triggering Time Not Reached	The current time is not within the effective period of TTI-level channel shutdown.
Impacted Feature Taken Effect	- Service-type-based dynamic exit from TTI-level channel shutdown has taken effect. - Any other impacted feature has taken effect.
Prerequisite Feature Not Taken Effect	None
Downlink PRB Threshold Not Reached	None
In Penalty State	A penalty is being imposed upon an exit from the TTI-level channel shutdown state.
RF Hardware Capability Not Supported	None
Power Aggregation Configuration Not Met	Whether the common-channel dynamic shutdown function can take effect is subject to the power aggregation capability of RF modules, beam coverage scenario, and settings of parameters related to power offset. (The power-offset parameters are described in Table 5-5 , and setting them to overly large values may cause this function to fail). You can contact Huawei engineers for more information about scenarios where this function cannot take effect.
RF Channel Not Coordinated	- The channels of the RF module are faulty. - The antenna mode is incompatible.
Ineffective for All TRPs	TTI-level channel shutdown does not take effect for all TRPs.
TRP Unavailable	A TRP of a hyper cell or combined cell is blocked or is deactivated, or is unavailable due to other reasons.

Reason	Description
Multi-Carrier Not Coordinated	When multiple NR carriers share the same RF module: • For TDD: – The effective durations of common-channel dynamic shutdown policies for different carriers do not overlap. – The value of the NRDUCellPowerSav-ing.ComChnDynShutdownSwitch parameter varies for different carriers. – The NRDUCellPowerSav-ing.ShutdownTrigJudgePeriod parameter is set to different values for different carriers. – (For common-channel dynamic shutdown only) NRDUCellPowerSav-ing.NrMultiCarrProtSwitch is set to ON. – (For common-channel dynamic shutdown only) In multimode scenarios, non-NR carriers are activated. When multiple LTE and NR carriers share the same RF module: TTI-level channel shutdown does not take effect in certain beam coverage scenarios. This is because RF channels cannot be shut down when some LTE carriers are not shut down and the power compensation in configured beam coverage scenarios is large. Consequently, energy-saving efforts might not yield benefits.
Shared Channel Not Coordinated	When multiple NR carriers share the same RF module, inter-carrier negotiation of shared channels fails.

Reason	Description
Others	- The current time is within the time period specified for TTI-level channel shutdown, but this function is in its evaluation period, which is specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter. - In a combined cell, if any TRP does not meet the conditions for this function to take effect, the entire cell exits the TTI-level channel shutdown state. For TRPs that do not meet the conditions, the specific cause is displayed. For TRPs that meet the conditions, OTHERS is displayed. - Internal software or other reasons cause the function to fail.

NOTE

The preceding values are causes displayed while the function is in effect stably.

5.4.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after TTI-level channel shutdown is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.

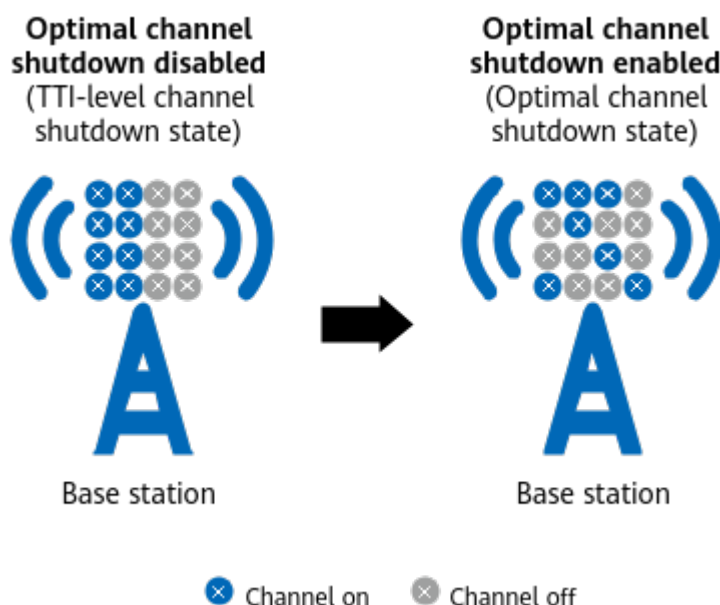
- The duration of RF channel dynamic shutdown caused by the activation of data-channel dynamic shutdown in TTI-level channel shutdown can be obtained by using the **N.PowerSaving.DynamicRFShutdown.Dur** counter on the MAE-Access.
- The duration of RF channel dynamic shutdown caused by the activation of common-channel dynamic shutdown in TTI-level channel shutdown can be obtained by using the **N.PowerSaving.ComChnDynamicRFShutdown.Dur** counter on the MAE-Access.

5.5 Optimal Channel Shutdown (Low-Frequency TDD)

5.5.1 Principles

On the basis of TTI-level channel shutdown (described in [5.4 TTI-Level Channel Shutdown](#)), optimal channel shutdown adjusts the positions of channels to shut down in order to increase spectral efficiency. This way, additional RF channels can be shut down, while ensuring service experience of UEs. In other words, this function selects the optimal number of channels to shut down based on spectral efficiency. [Figure 5-4](#) illustrates how this function works. Optimal channel shutdown can yield additional energy saving gains on the basis of TTI-level channel shutdown.

Figure 5-4 Optimal channel shutdown



Optimal channel shutdown can take effect when all the following conditions are met:

- A 32T32R or higher RF module is used.
- There is only one carrier.
- TTI-level channel shutdown takes effect in the current TTI as follows: The shutdown mode is adaptive, as indicated by the value **ADAPTIVE** of the **NRDUCellPowerSaving.TtiRfShutdownMode** parameter. Data-channel dynamic shutdown takes effect for UEs using SRS-based weights.

Optimal channel shutdown can be enabled by selecting the **TTI_OPTIMAL_RF_CHN_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter.

5.5.2 Network Analysis

5.5.2.1 Benefits

When the network is under light or moderate load, optimal channel shutdown enables additional RF channels to be shut down on the basis of TTI-level channel shutdown. This reduces the overall energy consumption of the base station.

5.5.2.2 Impacts

The impacts between this function and other functions are described in [5.5.3.2.3 Function Impacts](#).

The network impacts of optimal channel shutdown are the same as those of TTI-level channel shutdown. For details, see [5.4.2.2 Impacts](#).

5.5.3 Requirements

5.5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-061203	Intelligent Channel Dynamic Shutdown	NR0S00DRCS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Optimal channel shutdown requires TTI-level channel shutdown to be enabled and the NRDUCellPowerSaving.TtiRfShutdownMode parameter to be set to ADAPTIVE .

5.5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	Optimal channel shutdown does not work with Fusion Cell.
Low-frequency TDD	Dynamic MIMO layer number decrease	DYNAMIC_MIMO_LAYER_DECREASE_SW option of the gNodeBAlgo.DedicatedNetworkFuncSw parameter	None	In 8T8R scenarios where the NRDUCell.NrDuCellNetworkingMode parameter is set to NORMAL_CELL , dynamic MIMO layer number decrease does not work with optimal channel shutdown.

5.5.3.2.3 Function Impacts

None

5.5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All 32T32R and higher RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.5.3.4 Cells

This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**.

5.5.3.5 Others

None

5.5.4 Operation and Maintenance

5.5.4.1 Data Configuration

5.5.4.1.1 Data Preparation

[Table 5-24](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-24 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	To enable optimal channel shutdown, select the TTI_OPTIMAL_RF_CHN_SHUTDOWN_SW option of this parameter.

5.5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling optimal channel shutdown  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=TTI_OPTIMAL_RF_CHN_SHUTDOWN_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling optimal channel shutdown  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=TTI_OPTIMAL_RF_CHN_SHUTDOWN_SW-0;
```

5.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.5.4.2 Activation Verification

After optimal channel shutdown is activated, use the following method to check whether the function has been enabled:

Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Extension Switch** parameter is **TTI Optimal RF Channel Shutdown Switch:On** in the command output, optimal channel shutdown has been enabled.

5.5.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after optimal channel shutdown is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.

- The duration of dynamic RF channel shutdown caused by the activation of optimal channel shutdown can be obtained by observing the **N.PowerSaving.DynamicRFShutdown.Dur** counter on the MAE-Access.

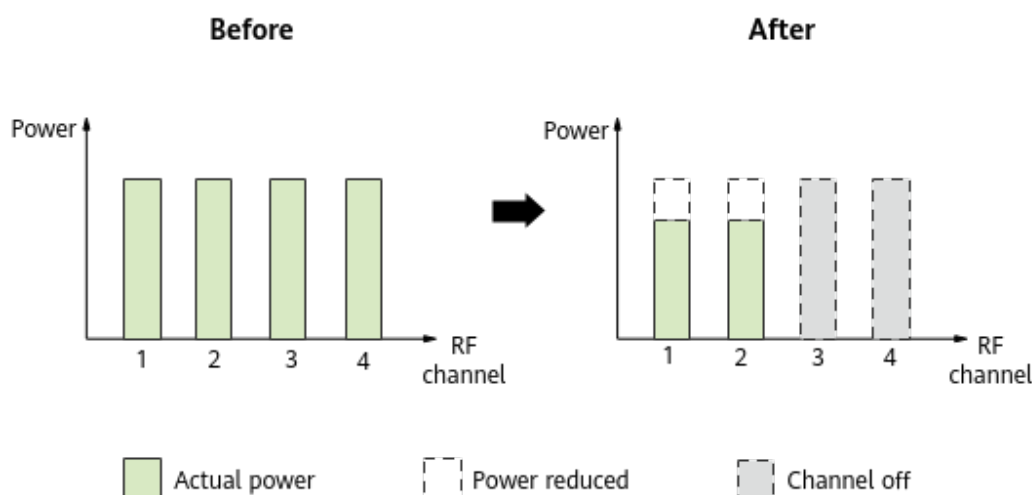
5.6 RF Channel Power Adaptation (Low-Frequency TDD)

5.6.1 Principles

RF channel power adaptation works on the basis of TTI-level channel shutdown (see [5.4 TTI-Level Channel Shutdown](#)) and energy saving based on flexible frequency-domain scheduling (see [7.1 Energy Saving Based on Flexible Frequency-Domain Scheduling](#)). Considering the traffic demand of UEs, the base station calculates the number of downlink transmit channels to be shut down and the PDSCH transmit power. The result of such adaptive calculation minimizes the radio power consumption given the same data for transmission. This function is controlled by the **RF_CHN_PWR_ADAPT_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter.

[Figure 5-5](#) shows an example of RF channel power adaptation by presenting a change in the number of downlink transmit channels shut down and the PDSCH transmit power.

Figure 5-5 RF channel power adaptation



5.6.2 Network Analysis

5.6.2.1 Benefits

When the network load is light or medium, RF channel power adaptation optimizes the number of RF channels shut down and reduces PDSCH power on the basis of TTI-level channel shutdown and energy saving based on flexible frequency-domain scheduling. This function reduces the overall power consumption of the base station.

However, RF channel power adaptation does not take effect in TTIs with any of the following types of UEs and therefore the energy saving gains may decrease: Coordinated multipoint transmission/reception (CoMP), coordinated scheduling (CS), coordinated beamforming (CBF), multi-user MIMO (MU-MIMO), fixed wireless access (FWA) with a guaranteed rate configured but not actually in effect, Voice over NR (VoNR), Non-Slot, or high-reliability services are running; non-SRS-based weights or ranks higher than 4 are used; more than one carrier is scheduled for a single UE; the UE is served by a network slice.

5.6.2.2 Impacts

The impacts between this function and other functions are described in [5.6.3.2.3 Function Impacts](#).

RF channel power adaptation alters the average downlink UE throughput and IBLER, reduces the downlink MCS index, and increases the downlink PRB usage and the value of the **VS.RADIOEQM.SymbolShutdown.Dur** counter.

5.6.3 Requirements

5.6.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-061203	Intelligent Channel Dynamic Shutdown	NR0S00DRCS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.6.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
TTI-level channel shutdown	TTI_RF_SHUTDO WN_SW option of the NRDUCellAlgoSwitch.PowerSaving Switch parameter	<i>Energy Conservation and Emission Reduction</i>	RF channel power adaptation requires TTI-level channel shutdown to be enabled and the NRDUCellPowerSaving.TtiRfShutdownMode parameter to be set to ADAPTIVE .

Function Name	Function Switch	Reference	Description
Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	RF channel power adaptation requires energy saving based on flexible frequency-domain scheduling.
Optimal channel shutdown	TTI_OPTIMAL_RF_CHN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	RF channel power adaptation requires optimal channel shutdown.

5.6.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Intra-TTI time-frequency-power collaboration	INTRA_TTI_TFP_COLLAB_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	None	RF channel power adaptation does not work with intra-TTI time-frequency-power collaboration.
Low-power RF channel dynamic shutdown	LOW_POWER_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	RF channel power adaptation does not work with low-power RF channel dynamic shutdown.

5.6.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Downlink anti-interference scheduling	DL_ANTI_INTRF_SCH_SW option of the NRDUCELLPDSCH.DI PdschAlgoSwitch parameter	<i>MIMO (TDD)</i>	Only downlink anti-interference scheduling takes effect when it is enabled together with RF channel power adaptation.

5.6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules with at least 32T32R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.6.3.4 Networking

This function is supported only in NR single-carrier scenarios.

5.6.3.5 Cells

This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**.

5.6.3.6 Others

None

5.6.4 Operation and Maintenance

5.6.4.1 Data Configuration

5.6.4.1.1 Data Preparation

Table 5-25 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-25 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	To enable RF channel power adaptation, select the RF_CHN_PWR_ADAPT_SW option.

5.6.4.1.2 Using MML Commands

Before using MML commands, refer to **5.6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling RF channel power adaptation  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=RF_CHN_PWR_ADAPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling RF channel power adaptation  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=RF_CHN_PWR_ADAPT_SW-0;
```

5.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.6.4.2 Activation Verification

After activating RF channel power adaptation, perform the following operation to check whether it has taken effect:

Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Extension Switch** parameter is **RF Channel Power Adaptation Switch:On** in the command output, RF channel power adaptation has been enabled.

5.6.4.3 Network Monitoring

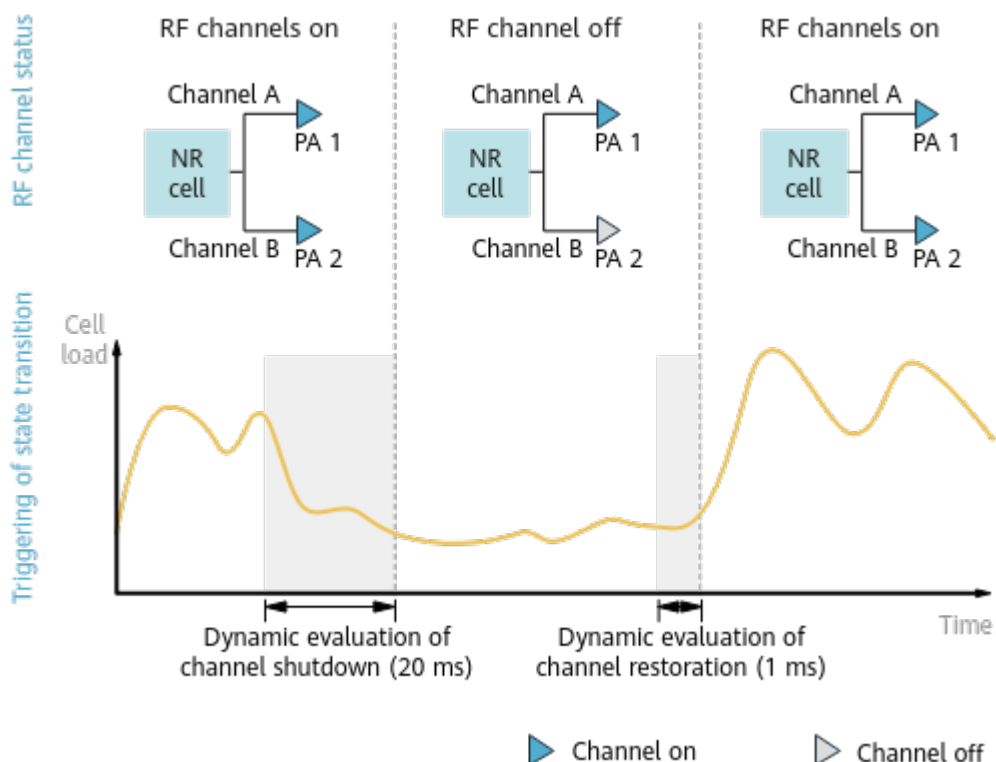
- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.
On the live network, the one-week base station energy consumption data obtained before and after RF channel power adaptation is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.
- The duration of RF channel shutdown caused by the activation of RF channel power adaptation can be obtained by observing the **N.PowerSaving.DynamicRFShutdown.Dur** counter on the MAE-Access.
- The PRB increase caused by RF channel power adaptation can be obtained by observing the **N.PRBDL.ExtendUsed.Avg** and **N.HyperCell.TRP.PRBDL.ExtendUsed.Avg** counters.

5.7 Dynamic RF Channel Shutdown

5.7.1 Principles

Within a specified period, dynamic RF channel shutdown enables a base station to shut down some transmit channels of a cell in order to reduce the power consumption of RF modules when the cell load is light, and to quickly restore all transmit channels in order to ensure user experience when the cell load increases. [Figure 5-6](#) illustrates how this function works.

Figure 5-6 Dynamic RF channel shutdown



Dynamic RF channel shutdown can be enabled by selecting the **RF_CHN_DYN_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter.

Dynamic RF channel shutdown includes data-channel dynamic shutdown and common-channel dynamic shutdown.

- If the **RF_CHN_DYN_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected and the **NRDUCellPowerSaving.ComChnDynShutdownSwitch** parameter is set to **OFF**, only data-channel dynamic shutdown is enabled.
- If the **RF_CHN_DYN_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected and the **NRDUCellPowerSaving.ComChnDynShutdownSwitch** parameter is set to **ON**, both data- and common-channel dynamic shutdown are enabled.

In addition, SSB common-channel dynamic shutdown (controlled by **NRDUCellPowerSaving.SsbDynShutdownSwitch**) can be performed in SSB slots. For TDD, if the **RF_CHN_DYN_SHUTDOWN_SSB_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is deselected, the shutdown is supported in SSB slots without UE scheduling only; if this option is selected, the shutdown is supported in SSB slots, either with or without UE scheduling.

For TDD, dynamic RF channel shutdown is supported in multi-carrier scenarios, but the support in single-carrier scenarios depends on the setting of the **RF_CHN_DYN_SHDN_SGL_CARR_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter.

- If this option is deselected, dynamic RF channel shutdown does not take effect in single-carrier scenarios.
- If this option is selected, dynamic RF channel shutdown can take effect in single-carrier scenarios.

Table 5-26 describes the parameters used to specify the effective period, load threshold, and other information under an energy saving policy of dynamic RF channel shutdown.

Table 5-26 Parameters related to a policy of dynamic RF channel shutdown

Parameter Name	Parameter ID	Description
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	This parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	This parameter specifies the energy saving type indicated by an energy saving policy index. Set this parameter to RF_CHN_DYN_SHUTDOWN for dynamic RF channel shutdown.
Start Time	NRDUCellPowerSaving.StartTime	These parameters specify the effective period, evaluation period, and conditions for Entering the Dynamic RF Channel Shutdown State and Exiting the Dynamic RF Channel Shutdown State in a cell.
Stop Time	NRDUCellPowerSaving.StopTime	
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	
User Number Offset	NRDUCellPowerSaving.UserNumOffset	
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTriggerPeriod	

Parameter Name	Parameter ID	Description
RF Restore DL PRB Usage Thld	NRDUCellPowerSaving.RfRestoreDLPrbUsageThld	
RF Shutdown DL PRB Usage Thld	NRDUCellPowerSaving.RfShutdownDLPrbUsageThld	
RF Restore DL High Load TTI Num Thld	NRDUCellPowerSaving.RfRstrDLHighLoadTTI NumThld	
RF Shutdown DL High Load TTI Num Thld	NRDUCellPowerSaving.RfShdnDLHighLoadTTI NumThld	

Parameter Name	Parameter ID	Description
Common Channel Dynamic Shutdown Switch	NRDUCellPowerSaving.ComChnDynShutdownSwitch	The NRDUCellPowerSaving.ComChnDynShutdownSwitch parameter specifies whether to enable common-channel dynamic shutdown. - If this parameter is set to ON, common-channel dynamic shutdown is enabled and takes effect for RMSI and OSI. In this case, the following subfunctions can be configured to enable common-channel dynamic shutdown for SSBs and paging messages in addition to RMSI and OSI: - SSB common-channel dynamic shutdown: This function allows common-channel dynamic shutdown to take effect for SSBs. It is controlled by NRDUCellPowerSaving.SsbDynShutdownSwitch, which is configurable only when NRDUCellPowerSaving.ComChnDynShutdownSwitch is set to ON. - Dynamic paging shutdown: This function allows common-channel dynamic shutdown to take effect for paging messages. This function is controlled by the PAGING_DYN_SHUTDOWN_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter and can take effect only when NRDUCellPowerSaving.ComChnDynShutdownSwitch is set to ON. - If this parameter is set to OFF, common-channel dynamic shutdown is disabled. In the case of coexistence with other RATs, common-channel dynamic shutdown takes effect only in certain beam coverage scenarios of TRPs, for example, EXPAND_SCENARIO_1 for SSB beams. The beam coverage scenario of a TRP is specified by the NRDUCellTrpBeam.CoverageScenario parameter.
SSB Dynamic Shutdown Switch	NRDUCellPowerSaving.SsbDynShutdownSwitch	
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	

Parameter Name	Parameter ID	Description
NR Multi-Carrier Protection Switch	NRDUCellPowerSaving.NrMultiCarrProtSwitch	This parameter applies only to TDD. It specifies whether to enable NR multi-carrier protection for common-channel dynamic shutdown. It can be configured only when the NRDUCellPowerSaving.ComChnDynShutdownSwitch parameter is set to ON in NR multi-carrier scenarios and applies only to low-frequency TDD. - If this parameter is set to ON, NR multi-carrier protection is enabled. In this case, common-channel dynamic shutdown takes effect only in certain beam coverage scenarios of TRPs, for example, EXPAND_SCENARIO_1 for SSB beams. The energy saving gains may decrease and the impact on network KPIs also decreases. - If this parameter is set to OFF, NR multi-carrier protection is disabled. In this case, no restriction is imposed on beam coverage scenarios of TRPs for common-channel dynamic shutdown. The energy saving gains may increase, the coverage slightly deteriorates, the service drop rate slightly increases, and the access success rate and handover success rate slightly decrease.

In NR multi-carrier scenarios, dynamic RF channel shutdown delivers fewer-than-expected energy saving gains. To increase energy saving gains in these scenarios, it is recommended that an identical energy saving policy of dynamic RF channel shutdown be configured for these carriers.

Entering the Dynamic RF Channel Shutdown State

Within the time period specified by **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** for dynamic RF channel shutdown to take effect for a cell:

- The cell periodically evaluates whether it meets the following conditions throughout a period specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter and, if it does, triggers the dynamic RF channel shutdown function and goes to 2:
 - Downlink PRB usage of the cell < **NRDUCellPowerSaving.DlPrbThld**
 - Number of UEs in the cell < **NRDUCellPowerSaving.UserNumThld**. The number of UEs in a co-deployed SUL cell is the sum of

- N.User.RRCCConn.Avg** and **N.User.SUL.UL.Avg**; and the number of UEs in other cells is indicated by **N.User.RRCCConn.Avg**.
2. The cell periodically (at an interval of 20 ms) evaluates whether it meets the following conditions and, if it does, enters the dynamic RF channel shutdown state:
- Number of downlink TTIs used in the cell < **NRDUCellPowerSaving.RfShdnDLHighLoadTtiNumThld**
 - Downlink PRB usage of the cell < **NRDUCellPowerSaving.RfShutdownDLPrbUsageThld**

Exiting the Dynamic RF Channel Shutdown State

A cell in the dynamic RF channel shutdown state periodically evaluates whether it meets any of the following conditions throughout a period specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter and, if it does, triggers an exit from the dynamic RF channel shutdown function:

- The current time is not within the dynamic RF channel shutdown period specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
- Downlink PRB usage of the cell > **NRDUCellPowerSaving.DLPrbThld** + **NRDUCellPowerSaving.DLPrbOffset**
- Number of UEs in the cell > **NRDUCellPowerSaving.UserNumThld** + **NRDUCellPowerSaving.UserNumOffset**

The cell then evaluates whether it meets any of the following conditions and, if it does, exits the RF channel shutdown state in the next TTI (if none of the conditions is met, RF channels remain in the shutdown state):

- Either of the following conditions is met, as detected in periodic evaluation at an interval of 20 ms:
 - Number of downlink TTIs used in the cell > **NRDUCellPowerSaving.RfRstrDLHighLoadTtiNumThld**
 - Downlink PRB usage of the cell > **NRDUCellPowerSaving.RfRestoreDLPrbUsageThld**
- A high-traffic data transmission request occurs in the cell, as detected in periodic evaluation at an interval of 1 ms.

Service-Type-based Dynamic Exit from Dynamic RF Channel Shutdown

For specific service types, cells can dynamically exit dynamic RF channel shutdown. This is known as service-type-based dynamic exit from dynamic RF channel shutdown. When there is any UE with a specific type of service in a cell, the cell exits the dynamic RF channel shutdown state. This type of exit ensures the performance of UEs performing that type of service. The service types involved are specified by the **gNBQciBearer.ServiceType** parameter.

Service-type-based dynamic exit from dynamic RF channel shutdown takes effect when all of the following conditions are met:

- Dynamic RF channel shutdown is enabled.
- The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.

- The **RF_CHN_DYN_SHUTDOWN_EXIT_SW** option of the **NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter is selected in the group identified by the **NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter that is set to the same value as the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter for a specific service type.

Parameter Optimization for Dynamic RF Channel Shutdown

This function uses reinforcement learning to adaptively adjust the threshold parameter settings for dynamic RF channel shutdown, to adapt to traffic loads. This maximizes the proportion of TTIs with dynamic RF channel shutdown in effect while meeting the downlink user-perceived rate and first packet delay requirements.

Parameter optimization for dynamic RF channel shutdown is controlled by the **RF_CHN_DYN_SHDN_PARAM_OPT_SW** option of the **NRDUCellReinfLrnAlgo.ReinfLrnAlgoSwitch** parameter.

- If this option is selected, parameter optimization for dynamic RF channel shutdown is performed for NR cells that share a PA. Note that this option must be selected for all the NR cells that share the PA.
- If this option is deselected, this function does not take effect.

When parameter optimization for dynamic RF channel shutdown is enabled, the NR cells sharing a PA start a reinforcement learning procedure for a maximum of 48 hours, during which the reinforcement learning mechanism is adopted to find the threshold parameter settings that maximize the proportion of TTIs with dynamic RF channel shutdown in effect while meeting the downlink user-perceived rate and first packet delay requirements under various traffic loads. When the procedure is complete, these NR cells build reinforcement learning models and use them to apply these threshold parameter settings based on loads.

The downlink user-perceived rate and first packet delay requirements are determined by the **NRDUCellReinfLrnAlgo.DlPerceivedRateThld** and **NRDUCellReinfLrnAlgo.DlPdcPFirstPktLatThld** parameters, respectively. The following thresholds are involved:

- **NRDUCellPowerSaving.RfRestoreDlPrbUsageThld**
- **NRDUCellPowerSaving.RfShutdownDlPrbUsageThld**
- **NRDUCellPowerSaving.RfRstrDlHighLoadTtiNumThld**
- **NRDUCellPowerSaving.RfShdnDlHighLoadTtiNumThld**

A reinforcement learning model may stay in one of the following states, which can be queried by running the **DSP NRDUCELLREINFLRNMDL** command with **ReinfLrnCase** set to **RF_CHN_DYN_SHDN_PARAM_OPT**.

Table 5-27 Reinforcement learning model states

Model State	Description
Initializing	The model is working properly and waiting for parameter optimization.

Model State	Description
Learning	The model is performing parameter optimization.
In Use	The model is using the optimized parameter settings.
Suspended	The model cannot be used for learning or parameter application.

5.7.2 Network Analysis

5.7.2.1 Benefits

When the network is under light or moderate load and the instantaneous cell load is light, some RF channels can be shut down by dynamic RF channel shutdown. This reduces the overall energy consumption of the base station.

The energy saving gains delivered by dynamic RF channel shutdown may decrease in the following scenarios:

- Dynamic RF channel shutdown cannot take effect and therefore energy saving gains may decrease, if there are UEs involved in any of the following services on the network: CoMP, coordinated scheduling, coordinated beamforming, MU-MIMO, FWA with a guaranteed rate configured but not actually in effect, VoNR, Non-Slot, high-reliability services, and services with specific 5QIs or QCI (whose **gNB5qiConfig.PwrSavingProtectMode** is set to **USER_LEVEL** or **gNBQciBearer.PwrSavingProtectMode** is set to **USER_LEVEL**).
- In NR multi-carrier scenarios, the energy saving gains decrease. To increase energy saving gains in these scenarios, it is recommended that an identical energy saving policy be configured for these carriers.
- When there are no real UEs in a cell with false alarms, the SSB channel cannot be shut down. Therefore, the energy saving gains decrease.
- Service-type-based dynamic exit from dynamic RF channel shutdown relieves the impact of dynamic RF channel shutdown on specific service types but reduces energy saving gains.

Larger values of the following parameters result in higher energy saving gains but also have a greater impact on the network after the energy saving function takes effect; smaller values of these parameters produce the opposite effects:

NRDUCellPowerSaving.RfRestoreDLPrbUsageThld,
NRDUCellPowerSaving.RfShutdownDLPrbUsageThld,
NRDUCellPowerSaving.RfRstrDLHighLoadTtiNumThld, and
NRDUCellPowerSaving.RfShdnDLHighLoadTtiNumThld. For details about the impact of energy saving on the network, see [5.7.2.2 Impacts](#).

For TDD, when the **RF_CHN_DYN_SHUTDOWN_SSB_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is selected, dynamic RF channel shutdown can be performed in SSB slots with UE scheduling to deliver energy saving gains in this scenario.

For TDD, when the **RF_CHN_DYN_SHDN_SGL_CARR_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is selected, dynamic

RF channel shutdown can be performed in single-carrier scenarios to deliver energy saving gains in this scenario.

If the proportion of TTIs with dynamic RF channel shutdown in effect is low due to inappropriate threshold settings, parameter optimization for dynamic RF channel shutdown is recommended to increase energy saving gains.

5.7.2.2 Impacts

The impacts between this function and other functions are described in [5.7.3.2.3 Function Impacts](#).

After data-channel dynamic shutdown in dynamic RF channel shutdown takes effect:

- The IBLER may increase.
- The average CCE usage of the cell and the CCE aggregation level may increase.
- The average MCS index for downlink scheduling decreases.
- The proportion of high-rank UEs decreases.
- The downlink cell throughput may slightly decrease.
- The values of the following counters related to downlink PRB usage increase in common cells:
 - **N.PRB.DL.DrbUsed.Avg**
 - **N.PRB.DL.Used.Avg**
 - **N.PRB.DL.ExtendUsed.Avg**
- The values of counters related to downlink user-plane delay may increase.
- The user-perceived rate may change as follows:
 - If scheduling occurs in more TTIs after channel shutdown, the user-perceived rate decreases.
 - If the percentage of tail packets (the percentage of last TTIs before the downlink buffer is empty) decreases after channel shutdown, the user-perceived rate increases.

After common-channel dynamic shutdown in dynamic RF channel shutdown takes effect, the coverage slightly deteriorates, the service drop rate slightly increases, and the access success rate and handover success rate slightly change. For TDD, when the **RF_CHN_DYN_SHUTDOWN_SSB_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is selected, shutdown can be performed in SSB slots with UE scheduling. This intensifies the impacts of common-channel dynamic shutdown, decreasing the downlink user-perceived rate and average downlink cell throughput. In addition, when dynamic paging shutdown takes effect, the paging success rate on the core network slightly decreases.

When common-channel dynamic shutdown and data-channel dynamic shutdown are both in effect, the impacts of both of them are present.

Larger values of the following parameters result in higher energy saving gains but also have a greater impact on the network after the energy saving function takes effect; smaller values of these parameters produce the opposite effects:

NRDUCellPowerSaving.RfRestoreDlPrbUsageThld,

NRDUCellPowerSaving.RfShutdownDLPrbUsageThld,
NRDUCellPowerSaving.RfRstrDLHighLoadTtiNumThld, and
NRDUCellPowerSaving.RfShdnDLHighLoadTtiNumThld.

For TDD, when the **RF_CHN_DYN_SHDN_SGL_CARR_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is selected, dynamic RF channel shutdown can take effect in single-carrier scenarios and the preceding impacts apply.

Parameter optimization for dynamic RF channel shutdown may further worsen the preceding performance indicators, as this function adjusts thresholds to maximize the proportion of TTIs in which dynamic RF channel shutdown takes effect. The values of the following performance indicators will fluctuate during model building and launch, but they meet the constraints defined by the **NRDUCellReinfLrnAlgo.DlPerceivedRateThld** and **NRDUCellReinfLrnAlgo.DlPdcPFirstPktLatThld** parameters during model application:

- Downlink user-perceived rate = $(N.ThpVol.DL - N.ThpVol.DL.LastSlot) / N.ThpTime.DL.RmvLastSlot$
- First packet delay = $N.Traffic.DL.RlcFirstPktDelay.Time / N.Qos.DL.RlcFirstPktDelay.Num$

After parameter optimization for dynamic RF channel shutdown is enabled, its energy saving gains may decrease if strict requirements are imposed on the downlink user-perceived rate and first packet delay. In some cases, the proportion of TTIs with dynamic RF channel shutdown in effect could even be lower than that when this function is disabled.

5.7.3 Requirements

5.7.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-061203	Intelligent Channel Dynamic Shutdown	NR0S00DRCS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	Dynamic RF channel shutdown requires symbol power saving.
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Service-type-based dynamic exit from dynamic RF channel shutdown and parameter optimization for dynamic RF channel shutdown take effect only if dynamic RF channel shutdown has been enabled.

5.7.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Dynamic RF channel shutdown does not work with Cell Combination.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Dynamic RF channel shutdown cannot be enabled for hyper cells.
Low-frequency TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	Dynamic RF channel shutdown does not work with Fusion Cell.
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>TDD: Distributed Antenna Solutions (Low-Frequency TDD)</i>	Dynamic RF channel shutdown cannot be enabled if distributed massive MIMO has been enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BAC KUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Dynamic RF channel shutdown cannot be enabled if the primary/backup function in intra-frequency dual-network redundancy networking has been enabled.
Low-frequency TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BAC KUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Dynamic RF channel shutdown cannot be enabled if dual-BBU backup in dual-network redundancy networking has been enabled.
Low-frequency TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	Dynamic RF channel shutdown is incompatible with the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios.
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	High-speed Railway Superior Experience does not work with dynamic RF channel shutdown.
Low-frequency TDD	SSB beam enhancement	SSB_BEAM_ENH_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter	None	Common-channel dynamic shutdown cannot be enabled if SSB beam enhancement has been enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Common-channel dynamic shutdown cannot be enabled if the function of low latency and high reliability has been enabled.
Low-frequency TDD	VMIMO in DAS-based indoor coverage	NRDUCellAlgoSwitch.VmimoSwitch	<i>Indoor Coverage</i>	Common-channel dynamic shutdown cannot be enabled if VMIMO in DAS-based indoor coverage has been enabled.
Low-frequency TDD	Distributed MIMO	D_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	None	Common-channel dynamic shutdown cannot be enabled if Distributed MIMO has been enabled.
Low-frequency TDD	LTE TDD and NR dynamic spectrum sharing	LTE_NR_TDD_SPECT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	None	Common-channel dynamic shutdown cannot be enabled if LTE TDD and NR dynamic spectrum sharing has been enabled.
Low-frequency TDD	Dynamic SSB	DYNAMIC_SSB_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	None	Common-channel dynamic shutdown cannot be enabled if dynamic SSB has been enabled.
Low-frequency TDD	Weight optimization in scenarios of aggregation into 2T	AGG_2T_WEIGHT_OPT_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter	None	Weight optimization in scenarios of aggregation into 2T does not work with dynamic RF channel shutdown.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TRP Antenna Type	NRDUCellTrpBeam.TrpAntType	None	Dynamic RF channel shutdown cannot be enabled if the NRDUCellTrpBeam.TrpAntType parameter is set to 8T_2222 , 8T_44 , or 8T_422 .
Low-frequency TDD	Intra-frequency assistant cell	NRDUCell.SupplementaryCellInd set to INTRA_FREQ_ASSISTANT_CELL	None	Intra-frequency assistant cells are incompatible with dynamic RF channel shutdown.
Low-frequency TDD	Dynamic MIMO layer number decrease	DYNAMIC_MIMO_LAYER_DECREASE_SW option of the gNodeBAlgo.DedicatedNetworkFuncSw parameter	None	In 8T8R scenarios where the NRDUCell.NrDuCellNetworkingMode parameter is set to NORMAL_CELL , dynamic MIMO layer number decrease does not work with dynamic RF channel shutdown.

5.7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	Inter-carrier dynamic power sharing is not performed during RF channel shutdown.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCHEME_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	NR multi-band power adaptive scheduling is not performed when dynamic RF channel shutdown is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent scheduling for power saving	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When power saving scheduling (a sub-function of intelligent scheduling for power saving) and dynamic RF channel shutdown are both enabled, the latter function takes effect preferentially.
Low-frequency TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both energy saving based on flexible frequency-domain scheduling and dynamic RF channel shutdown are enabled, energy saving based on flexible frequency-domain scheduling will deliver fewer energy saving gains.
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Dynamic RF channel shutdown does not take effect when RF channel intelligent shutdown is in effect.
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When timing carrier shutdown takes effect, dynamic RF channel shutdown will deliver fewer energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect, dynamic RF channel shutdown will deliver fewer energy saving gains.
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When TTI-level channel shutdown and dynamic RF channel shutdown are both enabled, the latter function takes effect preferentially.
Low-frequency TDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When LTE and NR intelligent carrier shutdown takes effect, dynamic RF channel shutdown will deliver fewer energy saving gains.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCY_SW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Dynamic RF channel shutdown does not take effect when RF module deep dormancy is in effect.
Low-frequency TDD	RF module super dormancy	RRU.DORMANCY_SW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Dynamic RF channel shutdown does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS W (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over dynamic RF channel shutdown when both functions are enabled. Dynamic RF channel shutdown does not take effect when RF module fast deep dormancy is in effect.
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Dynamic RF channel shutdown is not performed in slots with MU-MIMO pairing.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	Dynamic RF channel shutdown does not take effect in a cell serving DL CoMP UEs.
Low-frequency TDD	Basic VoNR functions	VONR_SW option of the NRCCellAlgoSwitch.VonrSwitch parameter	<i>VoNR</i>	Dynamic RF channel shutdown does not take effect for VoNR UEs.
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Dynamic RF channel shutdown does not take effect for UEs running low latency and high reliability services.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell load simulation	STR NRDUCELLSIMUL ATELOAD	None	When load simulation and dynamic RF channel shutdown take effect simultaneously, dynamic RF channel shutdown delivers fewer energy saving gains.
Low-frequency TDD	Inter-carrier frequency selective scheduling	CROSS_CARRIER_FREQ_SEL_SCH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	When inter-carrier frequency selective scheduling and dynamic RF channel shutdown are both enabled, inter-carrier frequency selective scheduling gains and energy saving gains both decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUeIdentSw	None	Dynamic RF channel shutdown fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
Low-frequency TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.AppIpv4 • gNBAsAppParam.AppIpv6	None	• Any option of the gNodeBParam.StUeIdentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.AppIpv4 parameter is set. • The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue: • Select the SPEED_TURBO_PWR_SAVING_SW option of

RAT	Function Name	Function Switch	Reference	Description
				the gNBPwrSaving Algo.PwrSavin gAlgoSwitch parameter. Dynamic RF channel shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBPwrSaving Algo.SpeedTurboPwrSavStartTime and gNBPwrSaving Algo.SpeedTurboPwrSavEndTime parameters, increasing the energy saving gains. • Select the STATIC_ST_PWR_SAVING_SW option of the gNBPwrSaving Algo.PwrSavin gAlgoSwitch parameter. Dynamic RF channel shutdown can take effect for cells served by the base station in some of the preceding scenarios within the

RAT	Function Name	Function Switch	Reference	Description
				period specified by the gNBPwrSavingAlgo.StaticSTPwrSavStartTime and gNBPwrSavingAlgo.StaticSTPwrSavEndTime parameters, increasing the energy saving gains.
Low-frequency TDD	Multiple BWP1s for RedCap	NRDUCellRedCap.RedCapUeMaxAvailableBandwidth set to a value greater than 20M	<i>RedCap</i>	For a TDD cell, when dynamic RF channel shutdown is enabled and at least one RedCap BWP1 with the NCD-SSB is activated in the cell, power aggregation is required for each RedCap BWP1. In this case, dynamic RF channel shutdown does not take effect when SSB power aggregation is insufficient.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Virtual Grid-based Multi-Frequency Coordination	• RSRP prediction model – VG_MODEL_ALLOW_BUILD_FLAG option of the NRCellFreqRelation.AggregationAttribute parameter – NR_IF_HO_MEAS_QTY_PRED_FUNCTION option of the NRCellSmartMultiCarrierAlgoSwitch parameter • Downlink intra-frequency spectral efficiency prediction model – DL_INTRA_SE_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarrierModelBuildingSwitch parameter – DL_INTRA_FREQ_SPCT EFF_PRED_SW option of the NRCellSmartMultiCarrier	<i>Virtual Grid-based Multi-Frequency Coordination</i>	After dynamic RF channel shutdown takes effect, the building of virtual grid models will be suspended and existing virtual grid models will be brought offline and cannot be used. If virtual grid models need to be used while dynamic RF channel shutdown is in effect, the VIRTUAL_GRID_C H_SD_COEXIST_SW option of the gNB PwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected for both the local and neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
		<i>NrMultiCarrierAlgoSwitch</i> parameter • Intra-frequency SRS beam prediction model INTRA_F_SRS_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarr.ModelBuildingSwitch parameter • Downlink-frequency spectral efficiency model DL_FREQ_SE_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarr.ModelBuildingSwitch parameter		

5.7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

BBUs

Book BBUs do not support parameter optimization for dynamic RF channel shutdown.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

Parameter optimization for dynamic RF channel shutdown requires that the main control board configuration should not be either of the following:

- The **MPT.BRDSPEC** parameter is set to **UMPTa**, **UMPTb**, or **UMPTa/b**.
- The **MPT.BRDSPEC** parameter is set to **UMPTe** when the **NODE.WM** parameter is set to **CONCURRENT**.

RF Modules

(TDD) Only 8T8R, 32T32R, and 64T64R low-frequency TDD RF modules support this function.

Whether the common-channel dynamic shutdown function can take effect is subject to the power aggregation capability of RF modules, beam coverage scenario, and the settings of parameters related to power offset. For example, this function can take effect for AAU531x, AAU561x, and 5000 series RRUs only when the beam coverage scenario is **EXPAND_SCENARIO_1** and the power-offset parameters are all set to **0**. For details about these parameters, see [Table 5-5](#).

5.7.3.4 Cells

This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**.

(TDD) The following requirements must also be met:

- The cell TX/RX mode (specified by the **NRDUCellTrp.TxRxMode** parameter) must be 8T8R, 32T32R, or 64T64R.
- The total number of LTE and NR carriers within a transmit channel must not exceed four. If the **RF_CHN_DYN_SHDN_SGL_CARR_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is deselected, the number of NR carriers must be greater than or equal to 2.
- The **NRDUCell.FrequencyBand** parameter must be set to **N38**, **N40**, **N41**, **N77**, **N78**, or **N79**.
- The radius of cells configured on an 8T8R RF module must be less than or equal to 14500 m.
- The downlink bandwidth must be 100 MHz, 90 MHz, 80 MHz, 70 MHz, 60 MHz, 50 MHz, or 40 MHz.
- The **NRDUCellTrpBeam.CoverageScenario** parameter must be set to **DEFAULT**, **SCENARIO_1**, **SCENARIO_2**, **SCENARIO_3**, **SCENARIO_4**, **SCENARIO_6**, **SCENARIO_7**, **SCENARIO_8**, **SCENARIO_9**, **SCENARIO_12**, **SCENARIO_13**, **EXPAND_SCENARIO_1**, **EXPAND_SCENARIO_5**, **EXPAND_SCENARIO_7**, **EXPAND_SCENARIO_8**, **SCENARIO_27**, **SCENARIO_28**, **SCENARIO_31**, **SCENARIO_36**, or **CUSTOMIZED_BEAM_SCENARIO**.
- When the **NRDUCellTrpBeam.CoverageScenario** parameter is set to **CUSTOMIZED_BEAM_SCENARIO** for a TRP, the

NRDUCellTrpCustBeam.SsbBeamHorizWidth parameter of each customized scenario-based beam specified by the **NRDUCellTrpCustBeam.BeamId** parameter must be set to **WIDTH_30** for the TRP.

- When the **SECTOREQM.ANTCFGMODE** parameter is set to **BEAM** and the **NRDUCellPowerSaving.ComChnDynShutdownSwitch** parameter is set to **ON**, the **RRU.RS** parameter must be set to **NO**, **TN**, **LN**, **UN**, **ULN**, **GLN**, **GLFTDN**, or **LFTDN** for all sector equipment serving each cell. The cabinet number, subrack number, and slot number of each RF module in the sector equipment identified by the **NRDUCellCoverage.SectorEqmId** parameter are specified by the **SECTOREQM.RRUCN**, **SECTOREQM.RRUSRN**, and **SECTOREQM.RRUSN** parameters, respectively.
- When the **SECTOREQM.ANTCFGMODE** parameter is set to **ANTENNAPORT**, it is recommended that the **RRU.RS** parameter be set to the same value as that when the **SECTOREQM.ANTCFGMODE** parameter is set to **BEAM**. If it is not set this way, common-channel dynamic shutdown in dynamic RF channel shutdown may not take effect.
- The **NRDUCell.DeployPosition** parameter cannot be set to **DEPLOY_IAB**.

5.7.3.5 Others

None

5.7.4 Operation and Maintenance

5.7.4.1 Data Configuration

5.7.4.1.1 Data Preparation

Table 5-28 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-28 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellAlgoSwitch.NrDuCellId	Set this parameter based on the network plan.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	To enable dynamic RF channel shutdown, select the RF_CHN_DYN_SHUTDOWN_SW option of this parameter.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: • For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example, if the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. • For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.

Parameter Name	Parameter ID	Setting Notes
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan and the following requirements: • For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example, if the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. • For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to RF_CHN_DYN_SHUTDOWN when dynamic RF channel shutdown is required.
Common Channel Dynamic Shutdown Switch	NRDUCellPowerSaving.ComChnDynShutdownSwitch	If coverage and network performance are preferred, the value OFF is recommended. If energy saving is preferred, the value ON is recommended.

Parameter Name	Parameter ID	Setting Notes
SSB Dynamic Shutdown Switch	NRDUCellPowerSaving.SsbDynShutdownSwitch	This parameter can be set only when the NRDUCellPowerSaving.ComChnDynShutdownSwitch parameter is set to ON . Its default value is recommended.
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	Set the PAGING_DYN_SHUTDOWN_SW option of this parameter to its recommended value if NRDUCellPowerSaving.ComChnDynShutdownSwitch is set to ON .
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	For dynamic RF channel shutdown, this parameter can only be set to its default value.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	You are advised to use the recommended value.
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	You are advised to use the recommended value. When the NRDUCellPowerSaving.PowerSavingType parameter is set to RF_CHN_DYN_SHUTDOWN , the following condition must be met: NRDUCellPowerSaving.DlPrbThld + NRDUCellPowerSaving.DlPrbOffset ≤ 100.
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	You are advised to use the recommended value. When the NRDUCellPowerSaving.PowerSavingType parameter is set to RF_CHN_DYN_SHUTDOWN , the following condition must be met: NRDUCellPowerSaving.DlPrbThld + NRDUCellPowerSaving.DlPrbOffset ≤ 100.

Parameter Name	Parameter ID	Setting Notes
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	You are advised to use the recommended value. When the NRDUCellPowerSaving.PowerSavingType parameter is set to RF_CHN_DYN_SHUTDOWN , the following condition must be met: NRDUCellPowerSaving.UserNumThld + NRDUCellPowerSaving.UserNumOffset ≤ 100.
User Number Offset	NRDUCellPowerSaving.UserNumOffset	You are advised to use the recommended value. When the NRDUCellPowerSaving.PowerSavingType parameter is set to RF_CHN_DYN_SHUTDOWN , the following condition must be met: NRDUCellPowerSaving.UserNumThld + NRDUCellPowerSaving.UserNumOffset ≤ 100.
RF Restore DL PRB Usage Thld	NRDUCellPowerSaving.RfRestoreDLPrbUsageThld	You are advised to use the recommended value.
RF Shutdown DL PRB Usage Thld	NRDUCellPowerSaving.RfShutdownDLPrbUsageThld	You are advised to use the recommended value.
RF Restore DL High Load TTI Num Thld	NRDUCellPowerSaving.RfRstrDLHighLoadTTI-NumThld	You are advised to use the recommended value.
RF Shutdown DL High Load TTI Num Thld	NRDUCellPowerSaving.RfShdnDLHighLoadTTI-NumThld	You are advised to use the recommended value.

Parameter Name	Parameter ID	Setting Notes
NR Multi-Carrier Protection Switch	NRDUCellPowerSaving.NrMultiCarrProtSwitch	In TDD, this parameter can be set only when the NRDUCellPowerSaving.ComChnDynShutdownSwitch parameter is set to ON . Its default value is recommended. This parameter takes effect in NR multi-carrier scenarios and applies only to low-frequency TDD.
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	For TDD, select the RF_CHN_DYN_SHUTDOWN_SSB_SW option if shutdown needs to be performed in SSB slots with UE scheduling.
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	For TDD, select the RF_CHN_DYN_SHDN_SGL_CARRIER_SW option if dynamic RF channel shutdown needs to take effect in single-carrier scenarios.
Downlink PDSCH Algorithm Switch	NRDUCellPdsch.DlPdschAlgoSwitch	When dynamic RF channel shutdown is enabled, select the RB_ESTIMATE_OPT_SW option of this parameter to enable downlink RB estimation optimization. For details about the impact of downlink RB estimation optimization on network performance, see <i>Downlink Scheduling</i> .
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	It is recommended that the SERV_TYPE_DYN_EXIT_PWR_SAVE_SW option be selected when there are specific types of services and user experience of these services needs to be ensured.
Power Saving Policy Group Index	gNBQciBearer.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Power Saving Policy Group Index	NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan. The value of this parameter must be the same as that of the gNBQciBearer.PowerSavingPolicyGrpIndex parameter for services to be guaranteed. Before setting this parameter, ensure that the QCI for which the gNBQciBearer.PowerSavingPolicyGrpIndex parameter is set has been configured.
User Identification Method	NRDUCellPsPolicyGrp.UserIdentMethod	Set this parameter to SERVICE_TYPE .
Service Type Dynamic Exit Power Saving Switch	NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw	To enable service-type-based dynamic exit from dynamic RF channel shutdown, select the RF_CHN_DYN_SHUTDOWN_EXIT_SW option.
Reinforcement Learning Algorithm Switch	NRDUCellReinLrnAlgo.ReinLrnAlgoSwitch	To enable parameter optimization for dynamic RF channel shutdown, select the RF_CHN_DYN_SHDN_PARAM_OPT_SW option of this parameter.
DL Perceived Rate Threshold	NRDUCellReinLrnAlgo.DLPerceivedRateThld	Use the default value.
DL PDCP First Packet Latency Thld	NRDUCellReinLrnAlgo.DLPdcpFirstPktLatThld	Use the default value.

5.7.4.1.2 Using MML Commands

Before using MML commands, refer to [5.7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

The following is an example of function activation in TDD, in two-carrier scenarios. The MML-based configurations for the two carriers are the same.

```
//Enabling dynamic RF channel shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=RF_CHN_DYN_SHUTDOWN_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, PowerSavingExtSwitch=RF_CHN_DYN_SHUTDOWN_SW-1;
//Setting an energy saving policy of common-channel dynamic shutdown and data-channel dynamic
shutdown in dynamic RF channel shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=RF_CHN_DYN_SHUTDOWN, StartTime=00&00&00, StopTime=23&59&59, DlPrbThld=80,
DlPrbOffset=20, UserNumThld=80, UserNumOffset=20, EvalPeriodGranularity=MINUTE_LEVEL,
ShutdownTrigJudgePeriod=5, ComChnDynShutdownSwitch=ON, SsbDynShutdownSwitch=OFF,
RfShutdownDlPrbUsageThld=10, RfRestoreDlPrbUsageThld=15, RfShdnDlHighLoadTtiNumThld=4,
RfRstrDlHighLoadTtiNumThld=6, NrMultiCarrProtSwitch=ON;
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0, PowerSavingAlgoSwitch=PAGING_DYN_SHUTDOWN_SW-0;
ADD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0,
PowerSavingType=RF_CHN_DYN_SHUTDOWN, StartTime=00&00&00, StopTime=23&59&59, DlPrbThld=80,
DlPrbOffset=20, UserNumThld=80, UserNumOffset=20, EvalPeriodGranularity=MINUTE_LEVEL,
ShutdownTrigJudgePeriod=5, ComChnDynShutdownSwitch=ON, SsbDynShutdownSwitch=OFF,
RfShutdownDlPrbUsageThld=10, RfRestoreDlPrbUsageThld=15, RfShdnDlHighLoadTtiNumThld=4,
RfRstrDlHighLoadTtiNumThld=6, NrMultiCarrProtSwitch=ON;
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=1, PowerSavingAlgoSwitch=PAGING_DYN_SHUTDOWN_SW-0;
//Enabling downlink RB estimation optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPdschAlgoSwitch=RB_ESTIMATE_OPT_SW-1;
MOD NRDUCELLPDSCH: NrDuCellId=1, DlPdschAlgoSwitch=RB_ESTIMATE_OPT_SW-1;
//(Optional) Turning on RF_CHN_DYN_SHUTDOWN_SSB_SW if shutdown needs to be performed in SSB
slots with UE scheduling
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=RF_CHN_DYN_SHUTDOWN_SSB_SW-1;
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=1,
PowerSavingAlgoSwitch=RF_CHN_DYN_SHUTDOWN_SSB_SW-1;
//(Optional) Turning on RF_CHN_DYN_SHDN_SGL_CARR_SW if dynamic RF channel shutdown needs to
take effect in single-carrier scenarios
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=RF_CHN_DYN_SHDN_SGL_CARR_SW-1;
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=1,
PowerSavingAlgoSwitch=RF_CHN_DYN_SHDN_SGL_CARR_SW-1;
//(Optional) Enabling service-type-based dynamic exit from dynamic RF channel shutdown (using cell 0 as
an example)
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;
ADD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=RF_CHN_DYN_SHUTDOWN_EXIT_SW-1;
//(Optional) Enabling parameter optimization for dynamic RF channel shutdown (using cell 0 as an
example)
MOD NRDUCELLREINFLRNALGO: NrDuCellId=0,
ReinfLrnAlgoSwitch=RF_CHN_DYN_SHDN_PARAM_OPT_SW-1, DlPerceivedRateThld=10,
DlPdcPFirstPktLatThld=10;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

The following is an example of function deactivation in TDD, in two-carrier scenarios. The MML-based configurations for the two carriers are the same.

```
/(Optional) Disabling parameter optimization for dynamic RF channel shutdown
MOD NRDUCELLREINFLRNALGO: NrDuCellId=0,
ReinfLrnAlgoSwitch=RF_CHN_DYN_SHDN_PARAM_OPT_SW-0;
/(Optional) Disabling service-type-based dynamic exit from dynamic RF channel shutdown
MOD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=RF_CHN_DYN_SHUTDOWN_EXIT_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;
/(Optional) Turning off RF_CHN_DYN_SHUTDOWN_SSB_SW
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=RF_CHN_DYN_SHUTDOWN_SSB_SW-0;
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=1,
PowerSavingAlgoSwitch=RF_CHN_DYN_SHUTDOWN_SSB_SW-0;
/(Optional) Turning off RF_CHN_DYN_SHDN_SGL_CARR_SW
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=RF_CHN_DYN_SHDN_SGL_CARR_SW-0;
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=1,
PowerSavingAlgoSwitch=RF_CHN_DYN_SHDN_SGL_CARR_SW-0;
//Disabling dynamic RF channel shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=RF_CHN_DYN_SHUTDOWN_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, PowerSavingExtSwitch=RF_CHN_DYN_SHUTDOWN_SW-0;
//Removing the energy saving policy of dynamic RF channel shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
RMV NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0;
```

5.7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.7.4.2 Activation Verification

Dynamic RF Channel Shutdown

After activating dynamic RF channel shutdown, use either of the following methods to check whether the function has taken effect:

- Method 1:
 - a. Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Extension Switch** parameter is **Dynamic RF Channel Shutdown Switch:On** in the command output, dynamic RF channel shutdown has been enabled.
 - b. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **Dynamic RF Channel Shutdown**.

NOTE

If there are two or more records with **Power Saving Type** being **Dynamic RF Channel Shutdown**, dynamic RF channel shutdown has been enabled in two or more time periods.

In NR multi-carrier scenarios, dynamic RF channel shutdown delivers fewer-than-expected energy saving gains. To increase energy saving gains in these scenarios, it is recommended that an identical energy saving policy of dynamic RF channel shutdown be configured for these carriers.

- c. Run the **DSP TIME** command. Check whether the system time is within a period for dynamic RF channel shutdown to take effect.
 - If it is, go to **d**.
 - If it is not, wait until the start time of dynamic RF channel shutdown arrives.
- d. Check whether the downlink PRB usage of the cell is less than or equal to the downlink PRB usage threshold for starting dynamic RF channel shutdown (specified by the **NRDUCellPowerSaving.DlPrbThld** parameter).
 - If it is, go to **e**.
 - If it is not, wait until the downlink PRB usage of the cell becomes less than or equal to the downlink PRB usage threshold for starting dynamic RF channel shutdown.
- e. Wait up to 10 minutes. Then, run the **DSP NRDUCELL** command. If the value of **Power Saving State** is **Dynamic RF Channel Shutdown:Enabled** and:
 - If the value of the **N.PowerSaving.DynamicRFShutdown.Dur** counter on the MAE-Access is not 0, then data-channel dynamic shutdown in dynamic RF channel shutdown has taken effect.
 - If the value of the **N.PowerSaving.ComChnDynamicRFShutdown.Dur** counter on the MAE-Access is not 0, then common-channel dynamic shutdown in dynamic RF channel shutdown has taken effect.
- Method 2:

Run the **DSP NRDUCELLPOWERSAVING** command to query the status of dynamic RF channel shutdown.

 - Verify whether data-channel dynamic shutdown has taken effect:
 - If the value of **Power Saving Type** is **Dynamic RF Channel Shutdown**, the value of **Current State** is **PDSCH Dynamic Shutdown Enabled**, and the value of the **N.PowerSaving.DynamicRFShutdown.Dur** counter on the MAE-Access is not 0, data-channel dynamic shutdown in dynamic RF channel shutdown has taken effect.
 - If the value of **Power Saving Type** is **Dynamic RF Channel Shutdown** and the value of **Current State** is **PDSCH Dynamic Shutdown Disabled**, check the value of **Current Disabled Cause** to determine the reason why data-channel dynamic shutdown in dynamic RF channel shutdown did not take effect. [Table 5-29](#) lists possible reasons.
 - Verify whether common-channel dynamic shutdown has taken effect:
 - If the value of **Power Saving Type** is **Dynamic RF Channel Shutdown**, the value of **Current State** is **Common-Channel Dynamic Shutdown Enabled**, and the value of the **N.PowerSaving.ComChnDynamicRFShutdown.Dur** counter on the

MAE-Access is not 0, then common-channel dynamic shutdown in dynamic RF channel shutdown has taken effect.

- If the value of **Power Saving Type** is **Dynamic RF Channel Shutdown** and the value of **Current State** is **Common-Channel Dynamic Shutdown Disabled**, check the value of **Current Disabled Cause** to determine the reason why common-channel dynamic shutdown in dynamic RF channel shutdown did not take effect. [Table 5-29](#) lists possible reasons.

Table 5-29 Possible reasons why dynamic RF channel shutdown did not take effect

Reason	Description
Insufficient License	No license for this function is available.
Switch Turned Off	None
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable for other reasons.
Triggering Time Not Reached	The current time is not within the effective period of dynamic RF channel shutdown.
Impacted Feature Taken Effect	• Service-type-based dynamic exit from dynamic RF channel shutdown has taken effect. • Any other impacted feature has taken effect.
Prerequisite Feature Not Taken Effect	None
Downlink PRB Threshold Not Reached	None
UE Number Threshold Not Reached	None
RF Hardware Capability Not Supported	None
Power Aggregation Configuration Not Met	The settings of the parameters related to transmit power compensation for common channels do not meet the requirements for dynamic RF channel shutdown to take effect. If the parameters listed in Table 5-5 are set to larger values than expected, the power aggregation capability may be insufficient. For details, contact Huawei engineers.

Reason	Description
Multi-Carrier Not Met	When the RF_CHN_DYN_SHDN_SGL_CARR_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter is deselected, dynamic RF channel shutdown does not take effect in NR TDD single-carrier scenarios.
In Penalty State	A penalty is being imposed upon an exit from the dynamic RF channel shutdown state.
RF Channel Not Coordinated	- The channels of the RF module are faulty. - The antenna mode is incompatible.
Multi-Carrier Not Coordinated	When multiple NR carriers share the same RF module: - The effective periods of dynamic RF channel shutdown policies for different carriers do not overlap. - The switch status of common-channel dynamic shutdown under dynamic RF channel shutdown differs between the carriers. - The negotiation on the power compensation values required for different carriers fails. When multiple LTE and NR carriers share the same RF module: Dynamic RF channel shutdown does not take effect in certain beam coverage scenarios. This is because RF channels cannot be shut down when some LTE carriers are not shut down and the power compensation in configured beam coverage scenarios is large. Consequently, energy-saving efforts might not yield benefits.
RF Group Configuration Condition Not Met	When multiple NR carriers share the same RF module: - The license is insufficient for another carrier. - The switch status of dynamic RF channel shutdown differs between the carriers.
Shared Channel Not Coordinated	When multiple NR carriers share the same RF module, inter-carrier negotiation of shared channels fails.

Reason	Description
Others	- The current time is within the time period specified for dynamic RF channel shutdown, but this function is in its evaluation period, which is specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter. - Internal software or other reasons cause the function to fail.

NOTE

The preceding values are causes displayed while the function is in effect stably.

For TDD, observe the values of

N.PowerSaving.ComChnDynamicRFShutdown.Dur (measured during the same time period) before and after the **RF_CHN_DYN_SHUTDOWN_SSB_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** is selected. If the **N.PowerSaving.ComChnDynamicRFShutdown.Dur** value increases after the option is selected, shutdown is performed in SSB slots with UE scheduling.

Parameter Optimization for Dynamic RF Channel Shutdown

After activating parameter optimization for dynamic RF channel shutdown, you can run the **DSP NRDUCELLREINFLRNMDL** command with **ReinflrnCase** set to **RF_CHN_DYN_SHDN_PARAM_OPT** to observe the status of the reinforcement learning model. If **Reinforcement Learning Model Status** is displayed, the reinforcement learning model has taken effect. If its value is **In Use**, the model has finished learning and been put into use.

5.7.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after dynamic RF channel shutdown is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.

- The duration of dynamic RF channel shutdown caused by data-channel dynamic shutdown can be obtained by observing the **N.PowerSaving.DynamicRFShutdown.Dur** counter on the MAE-Access.
- The duration of dynamic RF channel shutdown caused by common-channel dynamic shutdown can be obtained by observing the

N.PowerSaving.ComChnDynamicRFShutdown.Dur counter on the MAE-Access.

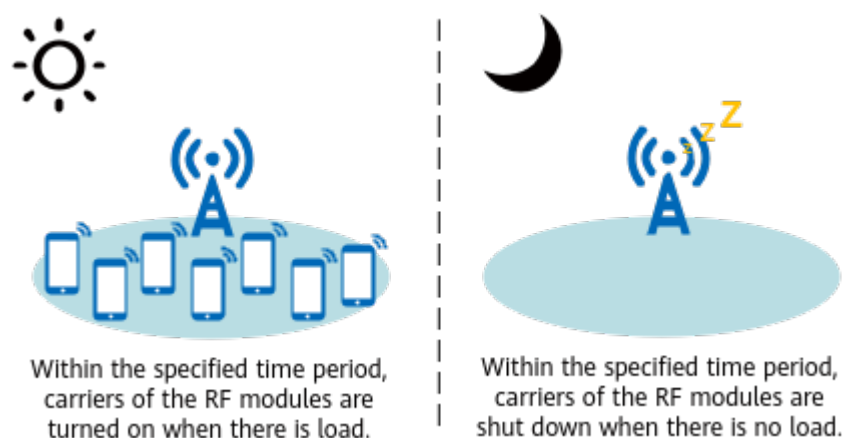
6 Device-Level Energy Saving (Frequency Domain)

6.1 Timing Carrier Shutdown

6.1.1 Principles

Maintaining all PAs of RF modules in the operational state when the network is under no load wastes base station energy. **Timing carrier shutdown enables carriers to be shut down within specified periods. If the PA (of an RF module) serving a shut-down carrier is not serving any other operating carrier, the PA will also be shut down to save energy.** Figure 6-1 illustrates how this function works.

Figure 6-1 Working principles for timing carrier shutdown



Timing carrier shutdown can be enabled by selecting the **TIMING_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter. The information under an energy saving policy of timing carrier shutdown is specified by the parameters in the **NRDUCellPowerSaving MO**.

Parameter Name	Parameter ID	Description
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	This parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	This parameter specifies the energy saving type indicated by an energy saving policy index. Set this parameter to TIMING_CARRIER_SHUTDOWN to enable timing carrier shutdown.
Start Time	NRDUCellPowerSaving.StartTime	These parameters specify the effective period for timing carrier shutdown. For details, see 6.1.2 Energy Saving Procedure .
Stop Time	NRDUCellPowerSaving.StopTime	
Carrier Shutdown Judge Mode	NRDUCellPowerSaving.CarrierShutdownJudgeMode	This parameter specifies the evaluation mode for a cell to enter the timing carrier shutdown state. • If this parameter is set to CARR_SHDN_BY_TIME, time-based evaluation applies. The cell enters the carrier shutdown state within the configured power saving period • If this parameter is set to CARR_SHDN_BY_LOAD, load-based evaluation applies. The cell enters the carrier shutdown state when the load conditions are met within the configured power saving period. For high-frequency TDD, this parameter cannot be set to CARR_SHDN_BY_LOAD.

Parameter Name	Parameter ID	Description
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	These parameters specify the load conditions and evaluation period for a cell to enter the carrier shutdown state when NRDUCellPowerSaving.CarrierShutdownJudgeMode is set to CARR_SHDN_BY_LOAD. For timing carrier shutdown, the NRDUCellPowerSaving.EvalPeriodGranularity parameter can only be set to MINUTE_LEVEL.
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	
SUL User Number Threshold	NRDUCellPowerSaving.SulUserNumThld	
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	
Rru Chip Power Saving	NRDUCellPowerSaving.RruChipPowerSaving	For a low-frequency TDD cell served by a LampSite base station, if this parameter is set to ENHANCE_MODE, intermediate frequency (IF) digital processor shutdown for RF modules is enabled. This function shuts down IF digital processors in RF modules on condition that timing carrier shutdown has taken effect. If this parameter is set to NORMAL_MODE, only timing carrier shutdown takes effect.

For high-frequency TDD: The parameters related to the timing carrier shutdown policy must be set consistently between the cells in the same sector. Otherwise, the sector cannot enter the carrier shutdown state.

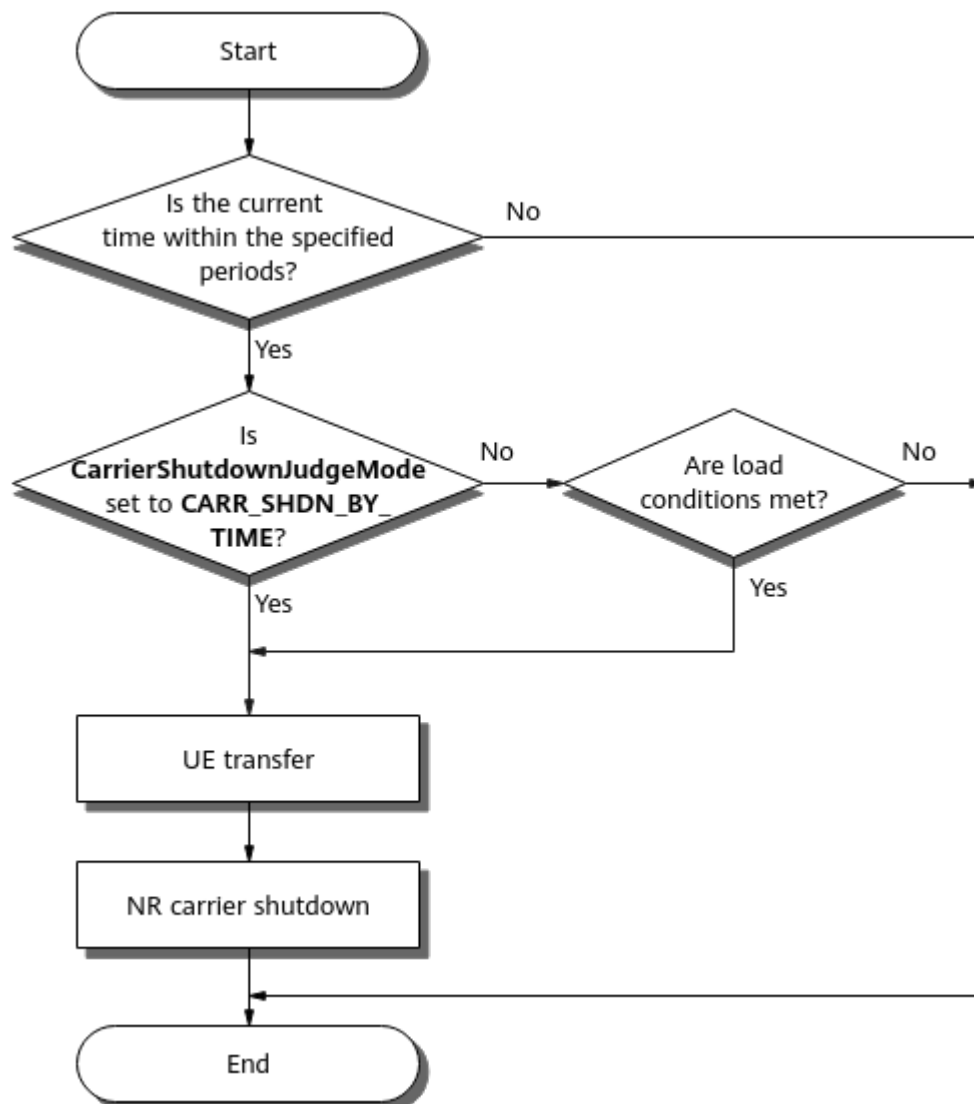
NOTE

- If timing carrier shutdown takes effect for a hyper cell or combined cell, carriers of all TRPs are shut down.
- A cell in the carrier shutdown state exits this state if it is deactivated and then reactivated or the base station software version is upgraded. Once either action is complete, the cell re-enters the carrier shutdown state if the current time is still within the period during which timing carrier shutdown takes effect.

6.1.2 Energy Saving Procedure

Figure 6-2 shows the procedure for timing carrier shutdown.

Figure 6-2 Timing carrier shutdown procedure



1. Determines whether the current time is within the periods specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters. If yes, the procedure goes to 2. If not, shutdown is not performed.
2. Checks the value of the **NRDUCellPowerSaving.CarrierShutdownJudgeMode** parameter.
 - If the value is **CARR_SHDN_BY_TIME**, the procedure goes to 3.
 - If the value is **CARR_SHDN_BY_LOAD**, the cell periodically evaluates whether the load conditions are met. The evaluation period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter.
 - If the following load conditions are all met, the procedure goes to 3:
 - Downlink PRB usage of the cell $\leq$ **NRDUCellPowerSaving.DlPrbThld** Downlink PRB usage of the

$$\text{cell} = (\text{N.PRB.DL.DrbUsed.Avg} - \text{N.PRB.DL.ExtendUsed.Avg}) / \text{N.PRB.DL.Actual.Avail.Avg} \times 100\%$$

- Uplink PRB usage of the cell $\leq$ **NRDUCellPowerSaving.UlPrbThld** Uplink PRB usage of the cell $= \text{N.PRB.PUSCH.Used.Avg} / \text{N.PRB.UL.PUSCH.Actual.Avail.Avg} \times 100\%$
 - Number of RRC_CONNECTED UEs in the cell $\leq$ **NRDUCellPowerSaving.UserNumThld**. The number of RRC_CONNECTED UEs in a co-deployed SUL cell is the sum of **N.User.RRCConn.Avg** and **N.User.SUL.UL.Avg**; and the number of RRC_CONNECTED UEs in other cells is indicated by **N.User.RRCConn.Avg**.
 - In SUL co-deployment scenarios: Number of SUL UEs in the cell $\leq$ **NRDUCellPowerSaving.SulUserNumThld**
 - Otherwise, shutdown is not performed.
3. Performs UE transfer.
- In SA networking, UEs are transferred from the local cell to neighboring cells as follows:
 - Non-CA UEs
 - UEs without voice services are transferred to inter-frequency or inter-RAT neighboring cells through blind redirections.
 - Voice service UEs are transferred to inter-frequency or inter-RAT neighboring cells first. If the handover fails, an inter-frequency or inter-RAT redirection is performed.
 - CA UEs
 - For CA UEs that treat the capacity-layer cell as their secondary cell (SCell), the SCell is removed.
 - For CA UEs that treat the capacity-layer cell as their primary cell (PCell), they are transferred to neighboring cells in the same way as non-CA UEs.
 - In NSA networking, UE connections on the NR side are released as follows:
 - If timing carrier shutdown takes effect in the primary secondary cell (PSCell) of a UE, all connections of the UE on the NR side are released.
 - If timing carrier shutdown takes effect in an SCell of a UE, its connection to this SCell on the NR side is released but its other connections on the NR side are retained.

Proceeds to 4 ten seconds after the UE transfer procedure starts.

4. Shuts down the NR carrier.

If the current time is not within the time period specified by **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime**, the timing carrier shutdown function is terminated.

6.1.3 Network Analysis

6.1.3.1 Benefits

Timing carrier shutdown reduces energy consumption when the network is under no load. However, the energy saving gains are limited in the following scenarios:

- When multiple carriers share the same PA of an RF module, the PA cannot be shut down if an NR carrier is shut down while the other carriers are still working. Energy saving gains decrease in this case.
- To prevent possible hardware faults caused by an excessively large temperature differential, this function may not take effect or may be terminated when the temperature differential of RF modules is excessively large. Energy saving gains may decrease in this case.

When the **NRDUCellPowerSaving.CarrierShutdownJudgeMode** parameter is set to **CARR_SHDN_BY_LOAD**, timing carrier shutdown takes effect only when the specified time and cell load requirements are both met. This reduces the probability of triggering carrier shutdown and may deliver fewer energy saving gains compared to using **CARR_SHDN_BY_TIME**.

6.1.3.2 Impacts

The impacts between this function and other functions are described in [6.1.4.2.3 Function Impacts](#).

- After timing carrier shutdown takes effect in a cell:
 - UEs that fail to be transferred from the cell will not be able to perform services. The service drop rate slightly increases.
 - New UEs in the cell coverage cannot access the network.
 - During NR carrier shutdown, new UEs are prohibited from accessing an NR cell. If UEs attempt to be handed over to the NR cell, the values of the counters related to handover preparation failures (such as **N.HO.IntraFreq.FailOut.PrepareFailure**) slightly increase. Such counters are not used for calculating Huawei-defined KPIs on outgoing intra- and inter-frequency handover success rates, but may be used for calculating user-defined KPIs. When carrier shutdown is set to take effect on the hour, no UEs can be handed over to the carrier in the shutdown state. Therefore, even slight increases in the counter values may significantly affect the user-defined outgoing handover success rates. If carrier shutdown is scheduled to take effect on the hour, you are advised to set the start time to 5 minutes earlier and the lengths of penalty timers applied to handover failures to 30s (using the **MOD GNBMOBILITYCOMMPARAM** command).
 - During NR carrier shutdown, if UEs fail to be handed over to target cells, the values of counters related to the number of intra-NR incoming handover failures (including **N.HO.FailIn.HOCancel**, **N.HO.FailIn.AdmitFail**, **N.HO.FailIn.AdmitFail.TNL**, **N.HO.FailIn.AdmitFail.UserLic**, and **N.HO.FailIn.AdmitFail.Load**), or the values of the LTE counters related to the number of inter-RAT incoming handover failures for different causes (including **L.IRATHO.N2E.PrepareInFail**, **L.IRATHO.N2E.PrepareInFail.AdmitFail**,

L.IRATHO.N2E.PreplnFail.TgtNotAllow, and **L.IRATHO.N2E.PreplnFail.HOCancel**) slightly increase.

- This function saves energy by shutting down carriers, but the cells remain activated. As a result, when this function takes effect, the performance counters and KPIs listed in [Table 6-1](#) are still in measurement, which has no impact on the energy saving effect. The exact time when carrier shutdown actually starts taking effect may not be synchronized with that when related performance counters are reported. Consequently, if carrier shutdown starts taking effect in a measurement period, there will be a measurement error of less than 5 minutes between the reported results and the actual results for the shut-down cell in the measurement period.

Table 6-1 Related performance counters and KPIs

Function Subset	Performance Counter and KPI	Description
• PRB Measurement • Hyper Cell TRP PRB Measurement • DU Cell Operator PRB Measurement	• Counters related to the number of available PRBs • Counters related to the number of available PDCCH CCEs • PRB usage	None
DU Cell Algorithm Measurement	• Counters related to the total number of physical uplink shared channel (PUSCH) transmission time intervals (TTIs) • Counters related to the total number of PDSCH TTIs	None
DU Cell SI Message Measurement	Number of SI transmissions	None

Function Subset	Performance Counter and KPI	Description
- Channel Quality Measurement - DU Cell TRP Channel Quality Measurement - Uplink Interference Measurement on PRB 0 to PRB 99 - Uplink Interference Measurement on PRB 100 to PRB 199 - Uplink Interference Measurement on PRB 200 to PRB 272	- Counters related to interference and noise received in the uplink - Counters related to uplink received signal strength indicator (RSSI) - Counters related to uplink power - Counters related to the average interference and noise detected on a single PRB - Counters related to the maximum interference and noise detected on a single PRB	The PWR_SAVING_COUNTER_CORR_SW option of the gNB0amParam.StatisticsStrategySwitch parameter specifies whether to report invalid uplink interference counter values during carrier shutdown. - If this option is selected, invalid uplink interference counter values are reported for carriers that have been shut down. - If this option is deselected, measured uplink interference counter values are reported for such carriers.

- In the case of false physical random access channel (PRACH) detection where the base station mistakenly regards interference or noise as the random access preambles of UEs, measurement for the following counters and KPIs will continue:
 - Counters related to the number of UEs scheduled in the uplink and downlink in a cell
 - Counters related to uplink data transmission duration
 - Counters related to the number of transport blocks (TBs) used for uplink data transmission
 - Counters related to MCS index distribution for scheduling on the PUSCH
 - Counters related to the number of RBs used for scheduling on the PUSCH
 - PUSCH RB usage
- As the cell in the carrier shutdown state does not send downlink paging messages, the counter listed in [Table 6-2](#) is no longer measured.

Table 6-2 Counter related to paging

Function Subset	Counter
DU Cell Paging Measurement	Number of paged UEs over the Uu interface in a cell

- Assuming that the neighboring base stations are in the carrier shutdown state, cell-edge UEs in weak coverage areas may access the cells served by the local base station. This affects KPIs related to the aspects such as access and service drops of the local base station. For example, the value of the **N.NsaDc.DRB.AbnormRel** counter may increase.

6.1.4 Requirements

6.1.4.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021203	gNodeB Power Saving	NR0S0GNBPS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.1.4.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.1.4.2.1 Prerequisite Functions

None

6.1.4.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The period specified for timing carrier shutdown in a cell cannot overlap that specified for intelligent carrier shutdown in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.
High-frequency TDD	Intelligent carrier shutdown in non-precise mode	• INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter selected • MULTI_CARR_PRECISE_SHDN_SW option of the NRDUCellAIgoSwitch.PowerSavingExtSwitch parameter deselected	<i>Energy Conservation and Emission Reduction</i>	The period specified for timing carrier shutdown in a cell cannot overlap that specified for intelligent carrier shutdown in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option (inter-base-station multi-carrier scenarios) of the NRDU CellAlg oSwitch.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SW option	<i>Energy Conservation and Emission Reduction</i>	The period specified for fast carrier shutdown in a cell cannot overlap that specified for timing carrier shutdown in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.

RAT	Function Name	Function Switch	Reference	Description
		of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected Basic-layer cells: FAST_CARRIER_SHUT_DOWN_ASSUR_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected		
Low-frequency TDD	Intelligent scheduling for power saving	POWER_SAVING_SCHEDULE_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The period specified for timing carrier shutdown cannot overlap that specified for power saving scheduling in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The period specified for power saving scheduling in a cell cannot overlap that specified for timing carrier shutdown in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.
Low-frequency TDD	Intelligent power control energy saving	SMART_POWER_CONTROL_ENERGY_SAVE_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The period specified for timing carrier shutdown in a cell cannot overlap that specified for intelligent power control energy saving in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.
Low-frequency TDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	The period specified for timing carrier shutdown in a cell cannot overlap that specified for LTE and NR intelligent carrier shutdown in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR bundled scheduling	LNR_BUNDLED_SCHEDULING_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	The period specified for timing carrier shutdown in a cell cannot overlap that specified for LTE and NR bundled scheduling in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.
Low-frequency TDD	LNR coordinated channel shutdown	NR: MULTI_RAT_RF_SHUTDOWN_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter LTE: CellRfShutdown.MultiRATJointChnShutdownSw	<i>Multi-RAT Coordinated Energy Saving</i>	The period specified for timing carrier shutdown in a cell cannot overlap that specified for LNR coordinated channel shutdown in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	The period specified for timing carrier shutdown cannot overlap that specified for the following: NR or LNR load-based RF module deep dormancy, a sub-function of RF module deep dormancy of the local cell; optimized evaluation of NR RF module deep dormancy, a sub-function in NR local-cell-load-based RF module deep dormancy. You are advised to preferentially plan the periods during which timing carrier shutdown takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	The period specified for timing carrier shutdown cannot overlap that specified for the following: NR or LNR load-based RF module super dormancy, a sub-function of RF module super dormancy of the local cell; optimized evaluation of NR RF module super dormancy, a sub-function in NR local-cell-load-based RF module super dormancy. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	The time period specified for timing carrier shutdown cannot overlap with that specified for NR or LNR load-based RF module fast deep dormancy, a sub-function of RF module fast deep dormancy of the local cell. You are advised to preferentially plan the periods during which timing carrier shutdown takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellIgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	The low latency and high reliability function does not work with timing carrier shutdown.
Low-frequency TDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The period specified for timing carrier shutdown in a cell cannot overlap that specified for uplink symbol power saving in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.
Low-frequency TDD	Intra-frequency assistant cell	NRDUCell.SupplementaryCellInd set to INTRA_FREQ_ASSISTANT_CELL	None	Intra-frequency assistant cells are incompatible with timing carrier shutdown.
Low-frequency TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Timing carrier shutdown is incompatible with the primary/backup function in intra-frequency dual-network redundancy networking.
Low-frequency TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Timing carrier shutdown is incompatible with dual-BBU backup in dual-network redundancy networking.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	Timing carrier shutdown is incompatible with the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios.
Low-frequency TDD	Dynamic MIMO layer number decrease	DYNAMIC_MIMO_LAYER_DEC_SW option of the gNodeBAlgo.DedicatedNetworkFuncSw parameter	None	In 8T8R scenarios where the NRDUCell.NrDualCellNetworkingMode parameter is set to NORMAL_CELL , dynamic MIMO layer number decrease does not work with timing carrier shutdown.

6.1.4.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Symbol power saving	NRDUCellAIgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	When timing carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of symbol power saving is not affected.
Low-frequency TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When timing carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, deep symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of deep symbol power saving is not affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	pRRU-level precise symbol power saving	DL_PRECISE_SYMBOL_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	pRRU-level precise symbol power saving does not take effect when timing carrier shutdown is in effect.
Low-frequency TDD High-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	For low-frequency TDD: If timing carrier shutdown is in effect, RF channel intelligent shutdown cannot take effect in the scenarios listed in Table 5-6 . For high-frequency TDD: If timing carrier shutdown is in effect, RF channel intelligent shutdown does not take effect.
Low-frequency TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both timing carrier shutdown and energy saving based on flexible frequency-domain scheduling take effect, the latter function does not deliver energy saving gains.
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	TTI-level channel shutdown does not take effect when timing carrier shutdown is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low-power RF channel dynamic shutdown	LOW_POWER_RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Low-power RF channel dynamic shutdown does not take effect when timing carrier shutdown is in effect.
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHANNEL_DYNAMIC_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When timing carrier shutdown takes effect, dynamic RF channel shutdown will deliver fewer energy saving gains.
Low-frequency TDD	Initial transmission power reduction	INIT_TRANSMISSION_POWER_REDUCE_SW option of the NRDUCellPowerSavingAlgorithmPowerSavingAlgorithmSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both initial transmission power reduction and timing carrier shutdown take effect, the former function does not deliver any gains.
Low-frequency TDD	Retransmission power reduction	RETRANSMISSION_POWER_REDUCE_SW option of the NRDUCellPowerSavingAlgorithmPowerSavingAlgorithmSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both retransmission power reduction and timing carrier shutdown take effect, the former function does not deliver any gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR coordinated symbol power saving	MULTI_RAT_SYMBOL_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When timing carrier shutdown takes effect: - If the RF module is not serving any other NR carrier, LTE and NR coordinated symbol power saving does not take effect. - If the RF module is serving other NR carriers, the status of LTE and NR coordinated symbol power saving is not affected.
Low-frequency TDD High-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Timing carrier shutdown does not take effect when scheduled RF module deep dormancy, which is a sub-function of RF module deep dormancy, is in effect.
Low-frequency TDD High-frequency TDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Timing carrier shutdown does not take effect when scheduled RF module super dormancy, which is a sub-function of RF module super dormancy, is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CMAS	gNBsibConfig.SibType set to SIB_TYPE_8	<i>CMAS</i>	When an NR carrier is shut down, the corresponding NR cell is blocked and commercial mobile alert system (CMAS) warning notifications cannot be properly sent in this cell. However, UEs can still receive CMAS warning notifications from other carriers.
Low-frequency TDD	Super uplink with SUL and NR FDD co-deployment	FLEX_SUPER_UPLINK_SWITCH option of the NRDUCellAIgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	After an NR FDD carrier is shut down, SUL scheduling cannot be performed for super uplink UEs.
Low-frequency TDD	UL and DL Decoupling	NRDUCellAIgoSwitch.UlDlDecouplingSwitch	<i>UL and DL Decoupling</i>	After an NR FDD carrier is shut down, SUL scheduling cannot be performed for SUL UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Air-interface-based network-wide synchronization deviation detection	GAP_BASED_SYNC_DIFF_DET_SW option of the gNodeBParam.SyncDiffDetectSwitch parameter	<i>Synchronization</i>	If sequence detection is performed on all cells of a to-be-detected base station during the period when timing carrier shutdown takes effect, the sequence detection fails. In this case, the possible clock fault of the base station may fail to be detected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Ultra-long holdover for sites with abnormal clock sources	CLK_OTASY NC_PROC_SW option of the TASM.MSCLKDIAGPROC ESSSW parameter	<i>Synchronization</i>	Assume that the Ultra-Long Hold Switch is turned on but the Ultra-Long Hold Enhancement Switch is turned off. If sequence detection is performed on a to-be-detected base station during the period when timing carrier shutdown takes effect for all cells of the base station, the sequence detection fails. In this case, the possible clock fault may cause the base station to be unable to enter the ultra-long holdover state, or to exit the ultra-long holdover state if sequence detection cannot be performed for a long time due to energy conservation. If the Ultra-Long Hold Enhancement Switch is turned on, timing carrier shutdown does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell load simulation	STR NRDUCELLSIMULATELOAD	None	Cell load simulation cannot be started when timing carrier shutdown is enabled. If cell load simulation has started, it will stop when timing carrier shutdown is enabled. Therefore, enabling both functions is not recommended.
Low-frequency TDD	Inter-cell cable sequence detection	AntPortOptDetPolicy parameter in the STR ANTENNAPORTOPTDETECT command set to INTER_CELL_DETECT	<i>MIMO (TDD)</i>	If timing carrier shutdown takes effect during inter-cell cable sequence detection, the detection results will show that the detection fails in the corresponding cells.
Low-frequency TDD	Channel calibration	Macro base station: None LampSite base station: INTER_RRU_CHN_CALIB_SW option of the NRDUCellAIgoSwitch.ChnCalibSwitch parameter	None	If channel calibration is being performed when timing carrier shutdown takes effect, channel calibration fails and will not be started again. You can run the DSP NRDUCELLCHNCALIB command to check whether channel calibration is successful.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell-level MRO	MRO_SCENARIO_IDENTIFY_SWITCH option of the gNBMinMroScenarioIdentifySwitch parameter and the INTRA_FREQ_MRO_SWITCH option of the NRCellMinMroMroSwitch parameter	<i>MRO</i>	When both timing carrier shutdown and intra-frequency MRO of cell-level MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, intra-frequency MRO may take effect later than expected.
Low-frequency TDD	UE-level MRO	UE_MRO_SWITCH option of the NRCellMinMroMroSwitch parameter	<i>MRO</i>	When both timing carrier shutdown and UE-level MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO may be affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAIgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	(TDD) The impacts are as follows: • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResS hrCfg.DynPwr ShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of timing carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResS hrCfg.DynPwr ShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control

RAT	Function Name	Function Switch	Reference	Description
				information changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SWITCH option of the NRDUCellInfoSwitch.DynamicPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	A carrier exits a power adaptive scheduling group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-RAT redirection back to the HPLMN based on roaming UE identification	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	In timing carrier shutdown, when the gNodeB receives the measurement report of the home 4G network, only redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 4G network.
Low-frequency TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	CA Performance Boost Based on Multi-dimensional Optimization does not take effect for cells in the shutdown state.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR multi-band power adaptive scheduling	• LTE: LNR_MULTIBAND_PWR_ADPT_SCH_SW option of the CellDynamicPowerSharing.<i>DynPowerSharingSwitch</i> parameter • NR: LNR_MULTIBAND_PWR_ADPT_SCH_SW option of the NRDUCellAlgoSwitch.<i>DynPowerSharingSwitch</i> parameter	<i>LTE and NR Power Allocation</i>	A carrier exits a power adaptive scheduling group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter adjustment. This causes changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	• LTE: LTE_NR_DYN_POWER_SHARING_SW option of the CellDynamicPowerSharing.DynPowerSharingSwitch parameter • NR: LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSbrCfg.DynPwrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of timing carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSbrCfg.DynPwrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter

RAT	Function Name	Function Switch	Reference	Description
				adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	L/NR Inter-band Deep Coordination	• LTE: LNR_INTE R_BAND_DEEP_COORD_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter • NR: LNR_INTE R_BAND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPowerShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPowerShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of timing carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPowerShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter

RAT	Function Name	Function Switch	Reference	Description
				adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUelidentSw	None	Timing carrier shutdown fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect: - Any option of the gNodeBParam.StUelidentSw parameter is selected. - The gNBAsAppParam.AppDomainName parameter is set. - The gNBAsAppParam.AppIpv4 parameter is set. - The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue: - Select the SPEED_TURBO_PWR_SAVING_SW option of the
Low-frequency TDD	gNodeB acceleration service application	gNBAsAppParam.AppDomainName gNBAsAppParam.AppIpv4 gNBAsAppParam.AppIpv6	None	

RAT	Function Name	Function Switch	Reference	Description
				gNB PwrSavin gAlgo.PwrSavin gAlgoSwitch parameter. Timing carrier shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNB PwrSavin gAlgo.SpeedTurboPwrSavStartTime and gNB PwrSavin gAlgo.SpeedTurboPwrSavEndTime parameters, increasing the energy saving gains. • Select the STATIC_STATIC_PWR_SAVING_SWITCH option of the gNB PwrSavin gAlgo.PwrSavin gAlgoSwitch parameter. Timing carrier shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by

RAT	Function Name	Function Switch	Reference	Description
				the gNB PwrSavin gAlgo.StaticST PwrSavStartTime and gNB PwrSavin gAlgo.StaticST PwrSavEndTime parameters, increasing the energy saving gains.

6.1.4.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

Low-frequency bands:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

High-frequency TDD bands:

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

For low-frequency TDD: All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

IF digital processor shutdown for RF modules is compatible only with pRRU5733GR and pRRU5633GR.

For high-frequency TDD: This function is compatible only with HAAU5233w and HAAU5235w.

6.1.4.4 Cells

The **NRDUCell.DuplexMode** parameter cannot be set to **CELL_SUL**.
For TDD: The **NRDUCell.DeployPosition** parameter cannot be set to **DEPLOY_IAB**.

6.1.4.5 Others

None

6.1.5 Operation and Maintenance

6.1.5.1 Data Configuration

6.1.5.1.1 Data Preparation

Table 6-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-3 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellAlgoSwitch.NrDuCellId	Set this parameter based on the network plan.
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	Select the TIMING_CARRIER_SHUTDOWN_SW option of this parameter when timing carrier shutdown is required.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: • For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. • For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. If timing carrier shutdown is scheduled to take effect on the hour, you are advised to set the start time to 5 minutes earlier to prevent the values of the counters related to handover preparation failures (such as N.HO.IntraFreq.FailOut.PreparationFailure) from increasing. For example, if timing carrier shutdown is scheduled to take effect at 00:00:00, you are advised to set the NRDUCellPowerSaving.StartTime to 23:55:00.

Parameter Name	Parameter ID	Setting Notes
		ing. <i>StartTime</i> parameter to 23:55:00.

Parameter Name	Parameter ID	Setting Notes
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan and the following requirements: • For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. • For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. • The period specified for timing carrier shutdown in a cell cannot overlap that specified for any of the functions with the NRDUCellPowerSaving.PowerSavingType parameter set to INTRA_GNB_MULTI_CARR_SD, INTER_GNB_MULTI_CARR_SD, POWER_SAVING_SCHEDULE, SMART_SSB_BEAM, DL_POWER_OPTIMIZE,

Parameter Name	Parameter ID	Setting Notes
		LNR_SMART_CARRIER_SHUTDOWN , LNR_BUNDLED_SCHEDULING , MULTI_RAT_RF_SHUTDOWN , and INTRA_NR_DORMANCY_BY_LOAD in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to TIMING_CARRIER_SHUTDOWN when timing carrier shutdown is required.
Carrier Shutdown Judge Mode	NRDUCellPowerSaving.CarrierShutdownJudgeMode	Set this parameter based on the network plan. - Set this parameter to CARR_SHDN_BY_TIME when time-based carrier shutdown is required. - Set this parameter to CARR_SHDN_BY_LOAD when load-based carrier shutdown is required. For high-frequency TDD, this parameter cannot be set to CARR_SHDN_BY_LOAD.
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	Use the recommended value. This parameter is mandatory only when NRDUCellPowerSaving.CarrierShutdownJudgeMode is set to CARR_SHDN_BY_LOAD .
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	Use the recommended value. This parameter is mandatory only when NRDUCellPowerSaving.CarrierShutdownJudgeMode is set to CARR_SHDN_BY_LOAD .

Parameter Name	Parameter ID	Setting Notes
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	Use the recommended value. This parameter is mandatory only when NRDUCellPowerSaving.CarrierShutdownJudgeMode is set to CARR_SHDN_BY_LOAD . The value of this parameter must be less than or equal to 150 .
SUL User Number Threshold	NRDUCellPowerSaving.SulUserNumThld	Use the recommended value. This parameter is mandatory only when NRDUCellPowerSaving.CarrierShutdownJudgeMode is set to CARR_SHDN_BY_LOAD . - This parameter is invalid when it is set to 65535. - When the FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is deselected and there is a record with NRDUCellPowerSaving.PowerSavingType set to TIMING_CARRIER_SHUTDOWN and NRDUCellPowerSaving.CarrierShutdownJudgeMode set to CARR_SHDN_BY_LOAD for a cell, the NRDUCellPowerSaving.SulUserNumThld parameter must be set to 65535 for the record.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	Only the default value can be used when timing carrier shutdown is required. This parameter is mandatory only when NRDUCellPowerSaving.CarrierShutdownJudgeMode is set to CARR_SHDN_BY_LOAD .

Parameter Name	Parameter ID	Setting Notes
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	Use the recommended value. This parameter is mandatory only when NRDUCellPowerSaving.CarrierShutdownJudgeMode is set to CARR_SHDN_BY_LOAD .
Rru Chip Power Saving	NRDUCellPowerSaving.RruChipPowerSaving	For a low-frequency TDD cell served by a LampSite base station, set this parameter to ENHANCE_MODE if IF digital processor shutdown for RF modules is required.
Necessary HO Prepare Fail Punish Timer	gNBMobilityCommParam.NecHoPrepFailPunishTimer	The value 30 is recommended when timing carrier shutdown is enabled.
Optimized HO Prepare Fail Punish Timer	gNBMobilityCommParam.OptHoPrepFailPunishTimer	The value 30 is recommended when timing carrier shutdown is enabled.

6.1.5.1.2 Using MML Commands

Before using MML commands, refer to [6.1.4 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Scenario 1: time-based carrier shutdown

```
//Setting an energy saving policy of timing carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=01&00&00,
StopTime=03&00&00, PowerSavingType=TIMING_CARRIER_SHUTDOWN,
CarrierShutdownJudgeMode=CARR_SHDN_BY_TIME;
//(Optional) Setting another energy saving policy of timing carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1, StartTime=04&00&00,
StopTime=05&00&00, PowerSavingType=TIMING_CARRIER_SHUTDOWN,
CarrierShutdownJudgeMode=CARR_SHDN_BY_TIME;
//Turning on the timing carrier shutdown switch
```

```
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=TIMING_CARRIER_SHUTDOWN_SW-1;
//Enabling IF digital processor shutdown for RF modules for a low-frequency TDD cell served by a LampSite
base station
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=2,
PowerSavingType=TIMING_CARRIER_SHUTDOWN, StartTime=01&00&00, StopTime=03&00&00,
RruChipPowerSaving=ENHANCE_MODE;
```

Scenario 2: load-based carrier shutdown (optional)

```
//Setting an energy saving policy of timing carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=01&00&00,
StopTime=05&00&00, PowerSavingType=TIMING_CARRIER_SHUTDOWN,
CarrierShutdownJudgeMode=CARR_SHDN_BY_LOAD, DIPrbThld=5, UIPrbThld=5, UserNumThld=2,
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5;
//(Optional) Setting the SUL UE number threshold in SUL co-deployment scenarios
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=TIMING_CARRIER_SHUTDOWN, SulUserNumThld=2;
//Turning on the timing carrier shutdown switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=TIMING_CARRIER_SHUTDOWN_SW-1;
//Enabling IF digital processor shutdown for RF modules for a low-frequency TDD cell served by a LampSite
base station
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=2,
PowerSavingType=TIMING_CARRIER_SHUTDOWN, StartTime=01&00&00, StopTime=03&00&00,
RruChipPowerSaving=ENHANCE_MODE;
```

Optimization Command Examples

```
//Changing the length of the penalty timers applied to resource-related handovers to 30s when the
handover success rate decreases due to timing carrier shutdown
MOD GNBMOBILITYCOMMPARAM: NecHoPrepFailPunishTimer=30, OptHoPrepFailPunishTimer=30;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Scenario 1: time-based carrier shutdown

```
//Disabling timing carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=TIMING_CARRIER_SHUTDOWN_SW-0;
//Disabling IF digital processor shutdown for RF modules for a low-frequency TDD cell served by a
LampSite base station
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=2,
PowerSavingType=TIMING_CARRIER_SHUTDOWN, StartTime=01&00&00, StopTime=03&00&00,
RruChipPowerSaving=NORMAL_MODE;
//Removing an energy saving policy of timing carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
//(Optional) Removing the other energy saving policy of timing carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1;
```

Scenario 2: load-based carrier shutdown (optional)

```
//Disabling timing carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=TIMING_CARRIER_SHUTDOWN_SW-0;
//Disabling IF digital processor shutdown for RF modules for a low-frequency TDD cell served by a
LampSite base station
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=2,
PowerSavingType=TIMING_CARRIER_SHUTDOWN, StartTime=01&00&00, StopTime=03&00&00,
RruChipPowerSaving=NORMAL_MODE;
//Removing an energy saving policy of timing carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
```

6.1.5.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.1.5.2 Activation Verification

After the timing carrier shutdown function is activated, use either of the following methods to check whether the function has taken effect:

- Method 1:
 - a. Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Switch** parameter is **Timing Carrier Shutdown Switch:On** in the command output, timing carrier shutdown has been enabled.
 - b. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **Scheduled Carrier Shutdown**.

 NOTE

If there are two or more records with **Power Saving Type** being **Scheduled Carrier Shutdown**, timing carrier shutdown has been enabled in two or more time periods.

- c. Run the **DSP TIME** command. Check whether the system time is within a timing carrier shutdown period.
 - If it is, go to **d**.
 - If it is not, wait until the start time of timing carrier shutdown arrives.
 - d. Wait 3 minutes and then run the **DSP NRDUCELL** command. Verify that the value of the **Power Saving State** parameter is **Scheduled Carrier Shutdown:Enabled**, which indicates that timing carrier shutdown has taken effect.
 - e. If the cell is a low-frequency TDD cell served by a LampSite base station, run the **DSP NRDUCELL** command. If the **Extension Power Saving Status** parameter indicates **RF Module Chip Power Saving:Enabled**, IF digital processor shutdown for RF modules has taken effect.
- Method 2:

Run the **DSP NRDUCELLPOWERSAVING** command to query the status of timing carrier shutdown.

 - If the value of **Power Saving Type** is **Scheduled Carrier Shutdown** and the value of **Current State** is **Enabled**, timing carrier shutdown has taken effect.
 - If the value of **Power Saving Type** is **Scheduled Carrier Shutdown** and the value of **Current State** is **Disabled**, check the **Current Disabled Cause**, **Latest Exit Cause**, and **Latest Disabled Cause** parameter values to find out the possible reason why timing carrier shutdown did not take effect. **Table 6-4** lists possible reasons. You can also check **Latest Exit Time** or **Latest Disabled Time** in the command output for the time the latest exit or the latest application failure occurred, respectively.

Latest Exit Time, **Latest Exit Cause**, **Latest Disabled Time**, and **Latest Disabled Cause** are used to query the time and reasons for the latest exit

and the latest application failure within the specified period. The query may be performed outside the specified period. For example, if the specified period is from 00:00:00 to 06:00:00, **Latest Exit Cause** or **Latest Disabled Cause** indicates the reason for the latest exit or the latest application failure that occurred within the period from 00:00:00 to 06:00:00. This reason, together with the time the latest exit or application failure occurred, can be queried even by an operation performed within the time from 06:00:00 to 23:59:59.

Table 6-4 Possible reasons why timing carrier shutdown did not take effect

Reason	Description
Insufficient License	No license for this function is available.
Switch Turned Off	None
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable for other reasons.
Triggering Time Not Reached	The current time is not within the range specified for timing carrier shutdown.
Impacted Feature Taken Effect	None
In Penalty State	A penalty is being imposed upon an exit from the timing carrier shutdown state.
RF Hardware Capability Not Supported	None
Parameter Inconsistency Between Cells	For high-frequency TDD: The parameters in the timing carrier shutdown policies for different cells are inconsistent.
Multi-Carrier Not Coordinated	For high-frequency TDD: Some carriers do not meet the conditions for timing carrier shutdown to take effect, in which case timing carrier shutdown cannot take effect for other carriers that meet the conditions.
Ultra-Long Holdover in Progress	None

Reason	Description
Downlink PRB Threshold Not Reached	When NRDUCellPowerSaving.CarrierShutdownJudgeMode is set to CARR_SHDN_BY_LOAD , the downlink PRB usage of the cell does not meet the threshold requirements.
Uplink PRB Threshold Not Reached	When NRDUCellPowerSaving.CarrierShutdownJudgeMode is set to CARR_SHDN_BY_LOAD , the uplink PRB usage of the cell does not meet the threshold requirements.
UE Number Threshold Not Reached	When NRDUCellPowerSaving.CarrierShutdownJudgeMode is set to CARR_SHDN_BY_LOAD , the number of RRC_CONNECTED UEs or SUL UEs in the cell does not meet the threshold requirements.
Others	• The current time is within the range specified for timing carrier shutdown, but this function is in its detection period, which lasts for 3 minutes. • When NRDUCellPowerSaving.CarrierShutdownJudgeMode is set to CARR_SHDN_BY_LOAD, the current time is within the effective period specified for timing carrier shutdown, and this function is not in its detection period. However, the current time is within the evaluation period for triggering timing carrier shutdown (specified by NRDUCellPowerSaving.ShutdownTrigJudgePeriod). • Internal software or other reasons cause the function to fail.

6.1.5.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after timing carrier shutdown is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.

- The duration of cell unavailability due to timing carrier shutdown can be obtained using the **N.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access.

There is delay in making decisions on whether to enter or exit the timing carrier shutdown state. Therefore, the duration measured by the **N.Cell.Unavail.Dur.EnergySaving** counter has an approximately 1 minute deviation from the timing carrier shutdown period set by the operator.

- The duration in which the cell stays in the carrier shutdown state due to timing carrier shutdown can be obtained using the **N.PowerSaving.CarrierShutdown.Dur** counter on the MAE-Access. There is delay in making decisions on whether to enter or exit the timing carrier shutdown state. Therefore, the duration measured by the **N.PowerSaving.CarrierShutdown.Dur** counter has an approximately one minute deviation from the timing carrier shutdown period set by the operator.
- Counters related to handovers from NG-RAN to E-UTRAN triggered by timing carrier shutdown are listed in [Table 6-5](#). In multi-operator sharing scenarios, operator-specific counters listed in [Table 6-6](#) are measured.

Table 6-5 Counters related to power-saving-based NR-to-EUTRAN handovers

Counter ID	Counter Name
1911840487	N.HO.InterRAT.N2E.PowerSaving.PrepAttOut
1911840488	N.HO.InterRAT.N2E.PowerSaving.ExecAttOut
1911840489	N.HO.InterRAT.N2E.PowerSaving.ExecSuccOut
1911840490	N.HO.InterRAT.N2E.PowerSaving.PrepAttOut.Voice
1911840491	N.HO.InterRAT.N2E.PowerSaving.ExecAttOut.Voice
1911840492	N.HO.InterRAT.N2E.PowerSaving.ExecSuccOut.Voice

Table 6-6 Counters related to power-saving-based NR-to-EUTRAN handovers in multi-operator sharing scenarios

Counter ID	Counter Name
1911840493	N.HO.InterRAT.N2E.PowerSaving.PrepAttOut.PLMN
1911840494	N.HO.InterRAT.N2E.PowerSaving.ExecAttOut.PLMN
1911840495	N.HO.InterRAT.N2E.PowerSaving.ExecSuccOut.PLMN
1911840496	N.HO.InterRAT.N2E.PowerSaving.PrepAttOut.Voice.PLMN
1911840497	N.HO.InterRAT.N2E.PowerSaving.ExecAttOut.Voice.PLMN
1911840498	N.HO.InterRAT.N2E.PowerSaving.ExecSuccOut.Voice.PLMN

6.2 Intelligent Carrier Shutdown

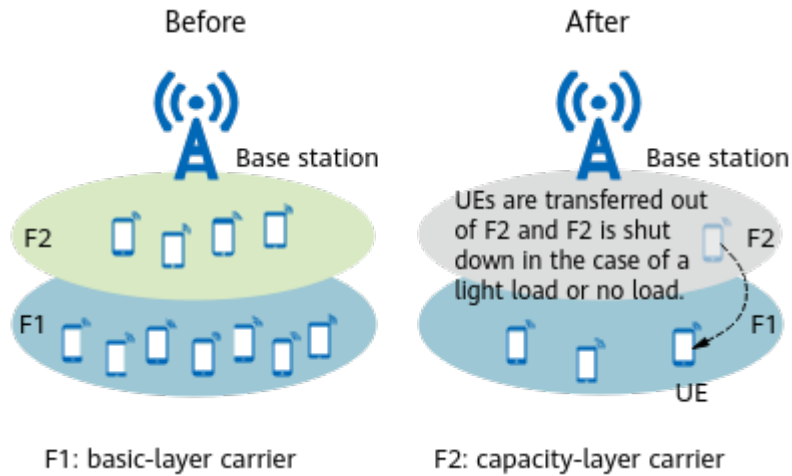
6.2.1 Principles

6.2.1.1 Overview

Multi-carrier co-coverage networking provides a larger network capacity, as well as a higher data throughput on a heavily-loaded network, than single-carrier networking. However, it wastes base station energy when the network is under light or no load.

Against this backdrop, intelligent carrier shutdown has been introduced. **This function enables the gNodeB to transfer UEs from capacity-layer cells to basic-layer cells and to shut down the capacity-layer carriers when the network is under light or no load within specified periods. In addition, if the PA (of an RF module) serving the shut-down capacity-layer carriers is not serving any other operating carrier, the PA will also be shut down.** [Figure 6-3](#) illustrates how this function works.

Figure 6-3 Working principles for intelligent carrier shutdown



Intelligent carrier shutdown works in two modes: precise and non-precise. In high-frequency TDD, this function works only in non-precise shutdown mode.

Precise shutdown mode maximizes the probability of capacity-layer cells entering the carrier shutdown state and minimizes the probability of these cells exiting this state, prolonging the carrier shutdown duration. [Figure 6-4](#) shows how it works. In precise shutdown mode, the base station takes the UE distribution in capacity- and basic-layer cells into consideration and precisely identifies capacity-layer cells that can enter or exit carrier shutdown, when all the following conditions are met:

- A basic-layer cell covers a large area, whereas capacity-layer cells cover small areas.
- One basic-layer cell corresponds to multiple capacity-layer cells.
- UEs are unevenly distributed.

Figure 6-4 Precise shutdown mode

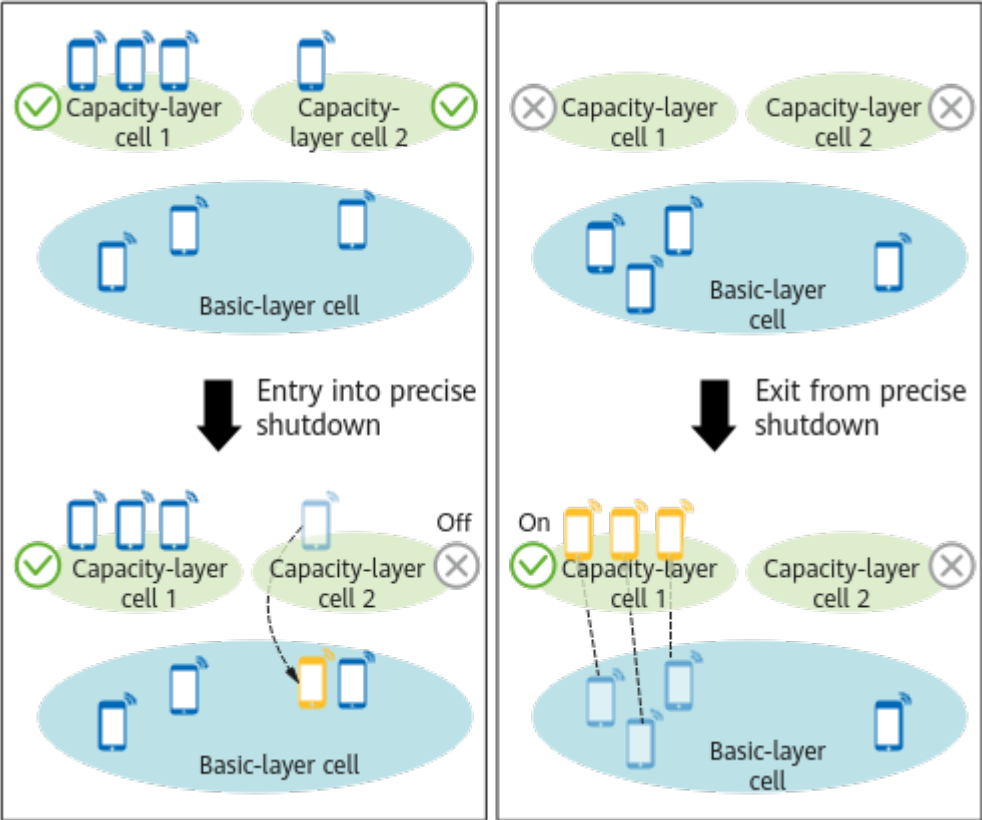


Table 6-7 describes how to set precise and non-precise shutdown.

Table 6-7 Modes of intelligent carrier shutdown

Mode	Parameter Settings for Capacity-Layer Cells	Description
Non-precise	- The INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected. - The MULTI_CARR_PRECISE_SHDN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter is deselected.	- The INTRA_GNB_MULTI_CARR_SD_SW option specifies whether to enable shutdown in intra-base-station multi-carrier scenarios. - The INTER_GNB_MULTI_CARR_SD_SW option specifies whether to enable shutdown in inter-base-station multi-carrier scenarios.

Mode	Parameter Settings for Capacity-Layer Cells	Description
Precise	- The INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected. - The MULTI_CARR_PRECISE_SHDN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter is selected.	

Related Concepts

- Basic-layer carriers (usually low-frequency carriers) provide basic coverage for the network and will not be shut down by the intelligent carrier shutdown function. For example, **F1** in **Figure 6-3** is a basic-layer carrier. The cells served by basic-layer carriers are referred to as basic-layer cells.
- Capacity-layer carriers (usually high-frequency carriers) support large traffic volumes during peak hours and can be shut down by the intelligent carrier shutdown function. For example, **F2** in **Figure 6-3** is a capacity-layer carrier. The cells served by capacity-layer carriers are referred to as capacity-layer cells.

NOTE

Intelligent carrier shutdown is enabled in capacity-layer cells. One capacity-layer cell can correspond to one or more basic-layer cells. Basic-layer cells and capacity-layer cells must be inter-frequency neighboring cells of each other.

Basic-layer and capacity-layer cells are a pair of relative concepts. They are defined by the **NRCellRelation.PowerSavingCellFlag** parameter during neighbor relationship configuration.

Table 6-8 describes the meanings of different values.

Table 6-8 Meanings of **NRCellRelation.PowerSavingCellFlag** values

Value	Meaning
COVERAGE	The neighboring cell is a basic-layer cell, and the serving cell is a capacity-layer cell.
CAPACITY	The neighboring cell is a capacity-layer cell, and the serving cell is a basic-layer cell.

Value	Meaning
NOT_CONFIG	The neighboring cell is not considered in intelligent carrier shutdown.

The following table provides an example of configuring **Power Saving Cell Flag** for cells in [Figure 6-3](#).

Table 6-9 Configuring **Power Saving Cell Flag**
(NRCellRelation.PowerSavingCellFlag)

Value of NRCellRelation.PowerSavingCellFlag	NRCellRelation.NrCellId	NRCellRelation.CellId
COVERAGE	ID of the cell operating on F2	ID of the cell operating on F1
CAPACITY	ID of the cell operating on F1	ID of the cell operating on F2

In NR multi-carrier scenarios, it is recommended that capacity- and basic-layer cells be configured according to [Table 6-10](#).

Table 6-10 Suggestions on designation of capacity- and basic-layer cells

Networking	Capacity-Layer Cell	Basic-Layer Cell
NR TDD (low-frequency) sub-6 GHz and NR TDD (low-frequency) sub-3 GHz coordinated networking	NR TDD (low-frequency) sub-6 GHz cells	NR TDD (low-frequency) sub-3 GHz cells
Intra-band multi-carrier networking	Cells with large frequency numbers	Cells with small frequency numbers
NR TDD (high-frequency) and NR TDD (low-frequency) coordinated networking	NR TDD (high-frequency) cells	NR TDD (low-frequency) cells

NOTE

On the MAE, functions are available for automatic identification of co-coverage neighbor relationships (with basic-layer cells) and configuration of the power saving cell flag. With these functions, the neighbor relationships between capacity-layer cells and basic-layer cells can be automatically and accurately identified and configured. For details, contact Huawei engineers.

6.2.1.2 Energy Saving Procedure

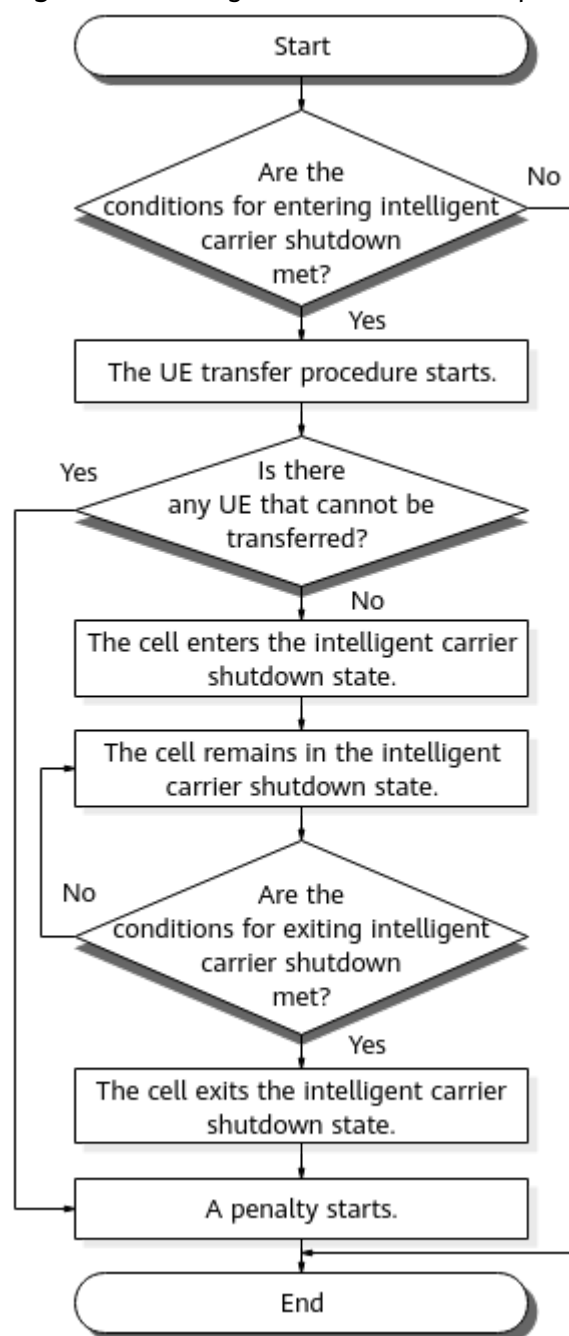
The effective periods, load threshold, and other information under an energy saving policy of intelligent carrier shutdown are specified by the parameters in the **NRDUCellPowerSaving** MO.

- The **NRDUCellPowerSaving.PowerSavingPolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.
- The **NRDUCellPowerSaving.PowerSavingType** parameter specifies the energy saving type indicated by an energy saving policy index. Intelligent carrier shutdown requires that this parameter be set to **INTRA_GNB_MULTI_CARR_SD** (in intra-base-station multi-carrier scenarios) or **INTER_GNB_MULTI_CARR_SD** (in inter-base-station multi-carrier scenarios).
- The **NRDUCellPowerSaving.CarrierShutdownMode** parameter specifies the carrier shutdown mode indicated by an energy saving policy index. For intelligent carrier shutdown, set this parameter to **LOAD_BASED_CARR_SHUTDOWN**.

For high-frequency TDD: The parameters related to the intelligent carrier shutdown policy must be set consistently between the cells in the same sector. Otherwise, the sector cannot enter the intelligent carrier shutdown state.

Figure 6-5 shows the procedure for intelligent carrier shutdown.

Figure 6-5 Intelligent carrier shutdown procedure



Conditions for Entering Intelligent Carrier Shutdown

The base station periodically evaluates whether a cell meets the conditions for entering the intelligent carrier shutdown state listed below. The granularity of the evaluation period is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.

- If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter. In the case of high-frequency TDD, the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter must be set to **MINUTE_LEVEL**.

- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** parameter.

If all of the conditions are met, a UE transfer procedure starts in a capacity-layer cell. If any of the conditions is not met, the base station waits for the next carrier shutdown evaluation period.

- The capacity-layer cell must meet all of the following conditions:
 - The current time is within the intelligent carrier shutdown period specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
 - The capacity-layer cell is not in the state of penalty on intelligent carrier shutdown. For details about the penalty state, see [UE Transfer](#) and [Conditions for Exiting Intelligent Carrier Shutdown](#).
- The basic-layer cells corresponding to the capacity-layer cell must meet one of the following conditions:
 - All basic-layer cells are available.
 - Some basic-layer cells are available, and all the unavailable cells are in the energy saving state. An example of this case is that an unavailable cell acts as a capacity-layer cell for another basic-layer cell in an energy saving function, such as NR intelligent carrier shutdown, LTE and NR intelligent carrier shutdown, NR load-based RF module deep dormancy, or LNR load-based RF module dormancy.
- The capacity-layer cell and the corresponding basic-layer cells must meet all the following conditions:
 - The total downlink PRB usage of the capacity-layer cell and basic-layer cells is less than or equal to the value of the **NRDUCellPowerSaving.DlPrbThld** parameter. For the formula used to calculate the total downlink PRB usage of the capacity-layer cell and basic-layer cells, see [Table 6-11](#).
 - The total uplink PRB usage of the capacity-layer cell and basic-layer cells is less than or equal to the value of the **NRDUCellPowerSaving.UlPrbThld** parameter. For the formula used to calculate the total uplink PRB usage of the capacity-layer cell and basic-layer cells, see [Table 6-11](#).
 - In precise shutdown mode, all the following conditions must be met:
 - The uplink PRB usage of each basic-layer cell corresponding to the capacity-layer cell is less than the result of $(\text{NRDUCellPowerSaving.ProtectUlPrbThld} - \text{NRDUCellPowerSaving.ProtectUlPrbOffset})$.
 - The downlink PRB usage of each basic-layer cell corresponding to the capacity-layer cell is less than the result of $(\text{NRDUCellPowerSaving.ProtectDlPrbThld} - \text{NRDUCellPowerSaving.ProtectDlPrbOffset})$.
For the formulas used to calculate the uplink and downlink PRB usage of each basic-layer cell corresponding to the capacity-layer cell, see [Table 6-11](#).

- The total number of RRC_CONNECTED UEs on each pair of the capacity-layer cell and a basic-layer cell is less than or equal to the value of the **NRDUCellPowerSaving.UserNumThld** parameter.

If multiple basic-layer cells correspond to one capacity-layer cell, each pair of the basic- and capacity-layer cells must meet this condition.

In the case of high-frequency TDD, the term "capacity-layer cell" refers to the combination of all cells in a high-frequency sector.

- For each operator, the operator-specific total number of RRC_CONNECTED UEs on each pair of the capacity-layer cell and a basic-layer cell is less than or equal to the value of the **NRDUCellPowerSaving.OpUserNumThld** parameter for the relevant operator.

If multiple basic-layer cells correspond to one capacity-layer cell, each pair of the basic- and capacity-layer cells must meet this condition.

- In multi-operator sharing scenarios:
 - The non-reserved operators of the capacity-layer cell are a subset of the non-reserved operators of any basic-layer cell.
 - The reserved operators of the capacity-layer cell are a subset of the reserved and non-reserved operators of any basic-layer cell.

For each of the capacity-layer and basic-layer cells, operators are classified into non-reserved operators and reserved operators. A non-reserved operator is an operator with the **NRExternalNCellPlmn.PlmnReservedFlag** parameter set to **FALSE**, and a reserved operator is an operator with the **NRExternalNCellPlmn.PlmnReservedFlag** parameter set to **TRUE**.

- In SUL co-deployment scenarios, the total number of SUL UEs in the capacity-layer and basic-layer cells is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter.

If multiple basic-layer cells correspond to one capacity-layer cell, each pair of the basic- and capacity-layer cells must meet this condition.

- When all of the following conditions are met, the number of UEs in each energy saving policy group is checked against the following rule: Total number of UEs in the energy saving policy groups of capacity-layer and basic-layer cells $\leq$ Threshold specified by **NRCellPsPolicyGrp.GrpUserNumThld**.
 - The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.
 - For the capacity-layer cell, the **NRCellPsPolicyGrp** MO is configured, and the **USER_NUM_CSD_EXIT_SW** option of the **NRCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter corresponding to the **NRCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter is selected. The **NRCellPsPolicyGrp.UserIdentMethod** parameter is set to **SERVICE_TYPE**.

UEs in an energy saving policy group refer to those who have established QCI-specific bearers bound to this **NRCellPsPolicyGrp.PowerSavingPolicyGrpIndex**. QCIs are bound to

energy saving policy groups through the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter. The following rules must also be followed for checking the number of UEs in an energy saving policy group:

- If multiple basic-layer cells correspond to one capacity-layer cell, the total number of UEs in energy saving groups of each pair of the basic- and capacity-layer cells must meet the threshold requirements.
 - If the capacity-layer cell is configured with multiple energy saving policy groups with the **USER_NUM_CSD_EXIT_SW** option of the **NRCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter selected, the number of UEs in each energy saving policy group must meet the requirements.
 - If the capacity-layer cell is configured with **NRCellPsPolicyGrp.PowerSavingPolicyGrpIndex** for which the **USER_NUM_CSD_EXIT_SW** option of the **NRCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter is selected, all basic-layer cells corresponding to the capacity-layer cell must be configured. Otherwise, the number of UEs in a power saving policy group does not meet the requirements.
- In inter-base-station multi-carrier scenarios, the Xn interface is in the normal state.

 NOTE

If the capacity-layer cell is a hyper cell or combined cell:

- The total downlink PRB usage and total uplink PRB usage of the capacity-layer cell and basic-layer cells are calculated based on the maximum number of used PRBs among all TRPs. When intelligent carrier shutdown takes effect, cells of all TRPs are shut down.
- It is recommended that basic-layer cells also be hyper cells or combined cells. Common cells are not recommended as basic-layer cells in this case.

The cell bandwidth, slot configuration, and subcarrier spacing are converted for evaluation of the total uplink or downlink PRB usage of the capacity- and basic-layer cells against the configured uplink or downlink PRB threshold, respectively.

Table 6-11 Formulas for calculating various PRB usage statistics

PRB Usage	Mode	Formula	Description
Total downlink PRB usage of the capacity-layer cell and basic-layer cells	Non-precise	For low-frequency TDD: (Number of PRBs used by PDSCH DRBs of the capacity-layer cell + Total number of used downlink PRBs of the basic-layer cells)/Total number of available downlink PRBs of the basic-layer cells High-frequency TDD: (Number of PRBs used by PDSCH DRBs in the high-frequency sector x (Subcarrier spacing used in the high-frequency sector/ Subcarrier spacing used in the basic-layer cells) + Total number of used downlink PRBs of the basic-layer cells)/Total number of available downlink PRBs of the basic-layer cells	The base station automatically obtains the following factors: number of PRBs used by PDSCH DRBs in the high-frequency sector, subcarrier spacing used in the high-frequency sector, subcarrier spacing used in the basic-layer cells, number of PRBs used by PDSCH DRBs of the capacity-layer cell, number of used downlink PRBs of each basic-layer cell, and number of available downlink PRBs of each basic-layer cell. It then calculates the total downlink PRB usage of the capacity-layer cell and basic-layer cells.
	Precise	(Number of PRBs used by PDSCH DRBs of the capacity-layer cell + Overlap rate of each neighboring cell x Number of used downlink PRBs of each basic-layer cell)/ (Number of available downlink PRBs of the capacity-layer cell + Overlap rate of each neighboring cell x Number of available downlink PRBs of each respective basic-layer cell)	The overlap rate of each neighboring cell is specified by the NRCellRelation.OverlapRate parameter at the capacity-layer cell side.

PRB Usage	Mode	Formula	Description
Total uplink PRB usage of the capacity-layer cell and basic-layer cells	Non-precise	For low-frequency TDD: (Number of used uplink PRBs of the capacity-layer cell + Total number of used uplink PRBs of the basic-layer cells)/ Total number of available uplink PRBs of the basic-layer cells High-frequency TDD: (Number of used uplink PRBs of the high-frequency sector x (Subcarrier spacing used in the high-frequency sector/ Subcarrier spacing used in the basic-layer cells) + Total number of used uplink PRBs of the basic-layer cells)/ Total number of available uplink PRBs of the basic-layer cells	None
	Precise	(Number of used uplink PRBs of the capacity-layer cell + Overlap rate of each neighboring cell x Number of used uplink PRBs of each basic-layer cell)/ (Number of available uplink PRBs of the capacity-layer cell + Overlap rate of each neighboring cell x Number of available uplink PRBs of each respective basic-layer cell)	The overlap rate of each neighboring cell is specified by the NRCellRelation.OverlapRate parameter at the capacity-layer cell side.

PRB Usage	Mode	Formula	Description
Uplink PRB usage of each basic-layer cell corresponding to the capacity-layer cell	Precise	$\frac{(\text{Number of used uplink PRBs of a basic-layer cell} + \text{Overlap rate} \times \text{Number of used uplink PRBs of the capacity-layer cell})}{\text{Number of available uplink PRBs of the basic-layer cell}}$	The overlap rate is specified by the NRCellRelation.OverlapRate parameter at the basic-layer cell side.
Downlink PRB usage of each basic-layer cell corresponding to the capacity-layer cell	Precise	$\frac{(\text{Number of used downlink PRBs of a basic-layer cell} + \text{Overlap rate} \times \text{Number of PRBs used by PDSCH DRBs of the capacity-layer cell})}{\text{Number of available downlink PRBs of the basic-layer cell}}$	The overlap rate is specified by the NRCellRelation.OverlapRate parameter at the basic-layer cell side.

UE Transfer

UE transfer from the capacity-layer cell varies as follows:

- In SA networking, UEs are transferred from the capacity-layer cell to basic-layer cells as follows:
 - Non-CA UEs
 - UEs not performing voice services are transferred to basic-layer cells through inter-frequency handovers. If the handovers fail, the UEs are transferred to inter-frequency basic-layer cells through blind redirections.
 - UEs performing voice services are transferred to basic-layer cells through inter-frequency handovers.
 - CA UEs
 - For CA UEs that treat the capacity-layer cell as their secondary cell (SCell), the SCell is removed.
 - For CA UEs that treat the capacity-layer cell as their primary cell (PCell), they are transferred to the basic-layer cells in the same way as non-CA UEs.

If UEs cannot be transferred out of the capacity-layer cell 30s after the UE transfer procedure starts (for example, they do not support the operating band of the target basic-layer cell or the handover fails), the intelligent carrier shutdown procedure stops and cannot be triggered for the capacity-layer cell again within a period specified by the

NRDUCellPowerSaving.PwrSavingPenaltyTime parameter, in which the capacity-layer cell is in the penalty state.

- In NSA networking, UE connections on the NR side are released as follows:
 - If intelligent carrier shutdown takes effect in the primary secondary cell (PSCell) of a UE, all connections of the UE on the NR side are released.
 - If intelligent carrier shutdown takes effect in an SCell of a UE, its connection to this SCell on the NR side is released but its other connections on the NR side are retained.

NOTE

If an RFSP-specific handover policy conflicts with a handover policy for intelligent carrier shutdown, the former takes precedence. (RFSP is short for RAT/Frequency Selection Priority.)

Entering Intelligent Carrier Shutdown

Once all UEs are transferred out of the capacity-layer cell, no RRC_CONNECTED UEs remain in the cell, and the cell enters the intelligent carrier shutdown state. If the PA (of an RF module) serving the shut-down carrier is not serving any other operating carrier, the PA will also be shut down.

Conditions for Exiting Intelligent Carrier Shutdown

After the capacity-layer cell enters the intelligent carrier shutdown state, the base station periodically evaluates the conditions for exiting this state. The granularity of the evaluation period is specified by the

NRDUCellPowerSaving.EvalPeriodGranularity parameter.

- If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ExitEvaluationPeriod** parameter.
- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelExitEvalPrd** parameter.

The capacity-layer cell automatically exits the intelligent carrier shutdown state and cannot enter this state again within a period specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter if any of the conditions listed below is met. (The cell is in the penalty state during this period.) If none of these conditions is met, the cell remains in the intelligent carrier shutdown state.

- The current time is not within the intelligent carrier shutdown period specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
- The total downlink PRB usage of the basic-layer cells is greater than the sum of **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset**. In the case of high-frequency TDD, an additional condition is that the number of high-frequency-capable UEs in any of the basic-layer cells is greater than the value of the **NRDUCellPowerSaving.MmwaveUserNumThld** parameter.

If the sum of **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset** is greater than 100, the cell cannot exit the intelligent carrier shutdown state.

- The total uplink PRB usage of the basic-layer cells is greater than the sum of **NRDUCellPowerSaving.UlPrbThld** and **NRDUCellPowerSaving.NCellUlPrbOffset**. In the case of high-frequency TDD, an additional condition is that the number of high-frequency-capable UEs in any of the basic-layer cells is greater than the value of the **NRDUCellPowerSaving.MmwaveUserNumThld** parameter.

If the sum of **NRDUCellPowerSaving.UlPrbThld** and **NRDUCellPowerSaving.NCellUlPrbOffset** is greater than 100, the cell cannot exit the intelligent carrier shutdown state.

- In precise shutdown mode, the uplink PRB usage of any basic-layer cell corresponding to the capacity-layer cell is greater than or equal to the value of the **NRDUCellPowerSaving.ProtectUlPrbThld** parameter.

Uplink PRB usage of a basic-layer cell corresponding to the capacity-layer cell = Number of used uplink PRBs of the basic-layer cell/Total number of available uplink PRBs of the basic-layer cell

- In precise shutdown mode, the downlink PRB usage of any basic-layer cell corresponding to the capacity-layer cell is greater than or equal to the value of the **NRDUCellPowerSaving.ProtectDlPrbThld** parameter.

Downlink PRB usage of a basic-layer cell corresponding to the capacity-layer cell = Number of used downlink PRBs of the basic-layer cell/Total number of available downlink PRBs of the basic-layer cell

- The number of RRC_CONNECTED UEs in a basic-layer cell is greater than the sum of **NRDUCellPowerSaving.UserNumThld** and **NRDUCellPowerSaving.UserNumOffset**. In the case of high-frequency TDD, an additional condition is that the number of high-frequency-capable UEs in a basic-layer cell is greater than the value of the **NRDUCellPowerSaving.MmwaveUserNumThld** parameter.

If multiple basic-layer cells correspond to one capacity-layer cell, only one basic-layer cell needs to meet the present condition on the number of RRC_CONNECTED UEs. In the case of high-frequency TDD, the requirement on the number of high-frequency-capable UEs is fulfilled as long as the number in any of these basic-layer cells meets the threshold condition.

- The operator-specific number of RRC_CONNECTED UEs in a basic-layer cell is greater than the sum of **NRDUCellPowerSaving.OpUserNumThld** and **NRDUCellPowerSaving.OpUserNumOffset**. In the case of high-frequency TDD, an additional condition is that the number of high-frequency-capable UEs in a basic-layer cell is greater than the value of the **NRDUCellPowerSaving.MmwaveUserNumThld** parameter.

If multiple basic-layer cells correspond to one capacity-layer cell, only one basic-layer cell needs to meet the present condition on the operator-specific number of RRC_CONNECTED UEs. In the case of high-frequency TDD, the requirement on the number of high-frequency-capable UEs is fulfilled as long as the number in any of these basic-layer cells meets the threshold condition.

- In multi-operator sharing scenarios:
 - The non-reserved operators of the capacity-layer cell are not a subset of the non-reserved operators of any basic-layer cell.

- The reserved operators of the capacity-layer cell are not a subset of the reserved and non-reserved operators of any basic-layer cell.
- In SUL co-deployment scenarios, the number of SUL UEs in a basic-layer cell is greater than the sum of the **NRDUCellPowerSaving.SulUserNumThld** and **NRDUCellPowerSaving.SulUserNumOffset** parameter values.
If multiple basic-layer cells correspond to one capacity-layer cell, only one basic-layer cell needs to meet this condition.
- When all of the following conditions are met, the number of UEs in each energy saving policy group is checked against the following rule: Number of UEs in a power saving policy group of a basic-layer cell > Threshold specified by **NRCellPsPolicyGrp.GrpUserNumThld** + Offset specified by **NRCellPsPolicyGrp.GrpUserNumOffset**.
 - The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.
 - For the capacity-layer cell, the **NRCellPsPolicyGrp** MO is configured, and the **USER_NUM_CSD_EXIT_SW** option of the **NRCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter corresponding to the **NRCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter is selected. The **NRCellPsPolicyGrp.UserIdentMethod** parameter is set to **SERVICE_TYPE**.

UEs in an energy saving policy group refer to those who have established QCI-specific bearers bound to this **NRCellPsPolicyGrp.PowerSavingPolicyGrpIndex**. QCIs are bound to energy saving policy groups through the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter. The following rules must also be followed for checking the number of UEs in an energy saving policy group:

 - If multiple basic-layer cells correspond to one capacity-layer cell, the number of UEs in the energy saving policy group of only one basic-layer cell needs to meet the requirements.
 - If the capacity-layer cell is configured with multiple energy saving policy groups with the **USER_NUM_CSD_EXIT_SW** option of the **NRCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter selected, the number of UEs in one of the energy saving policy groups needs to meet the requirements.
- In inter-base-station multi-carrier scenarios, the Xn interface is in the abnormal state.
- Any of the basic-layer cells becomes unavailable for a reason other than intelligent carrier shutdown, LTE and NR intelligent carrier shutdown, and NR load-based RF module deep dormancy.

 NOTE

- When a basic-layer cell becomes unavailable in inter-base-station scenarios, there may be a minute-level delay for the wake-up of the capacity-layer cell. This is because the messages need to be frequently transmitted over the Xn interface to indicate the energy saving status and capacity-layer cell wake-up notifications, leading to possible flow control.
- If a basic-layer cell is a hyper cell or combined cell and some of its TRP carriers become unavailable, the capacity-layer cell also automatically exits the intelligent carrier shutdown state.
- The penalty is not imposed if the capacity-layer cell exits the intelligent carrier shutdown state for a reason that the intelligent carrier shutdown switch is turned off, the cell is deactivated and then reactivated, or the base station software version is upgraded. After the capacity-layer cell is deactivated and then reactivated or the base station software version upgrade is complete, the base station evaluates whether the cell enters the intelligent carrier shutdown state again based on related entering conditions.
- The capacity-layer cell may exit the intelligent carrier shutdown state because the exit conditions are met during evaluation of basic-layer neighboring cell addition for the capacity-layer cell.
- A low-frequency basic-layer cell served by a base station with a software version no later than V100R020C00 does not report the number of high-frequency-capable UEs.
- If all the low-frequency basic-layer cells are served by base stations with software versions no later than V100R020C00, the high-frequency capacity-layer cell does not need to check the number of high-frequency-capable UEs. In this situation, either one of the PRB usage and UE number conditions of the basic-layer cells can trigger an exit from intelligent carrier shutdown.
- If some of the low-frequency basic-layer cells are served by base stations with software versions no later than V100R020C00, the high-frequency capacity-layer cell needs to check the number of high-frequency-capable UEs.

6.2.2 Network Analysis

6.2.2.1 Benefits

In multi-frequency multi-carrier co-coverage scenarios, enabling intelligent carrier shutdown reduces energy consumption when the network is under light or no load. However, the energy saving gains are limited in the following scenarios:

- When multiple carriers share the same PA of an RF module, the PA cannot be shut down if an NR carrier is shut down while the other carriers are still working. Energy saving gains decrease in this case.
- To prevent possible hardware faults caused by an excessively large temperature differential, this function may not take effect or may be terminated when the temperature differential of RF modules is excessively large. Energy saving gains may decrease in this case.

6.2.2.2 Impacts

The impacts between this function and other functions are described in [6.2.3.2.3 Function Impacts](#).

- When intelligent carrier shutdown takes effect, the values of counters related to inter-frequency handovers and redirections change.

If a UE attempts to be handed over to a capacity-layer cell whose carrier is to be shut down when a UE transfer process (which lasts 30s at most) is ongoing for this cell, the source cell does not initiate the handover. In this case, the values of counters related to outgoing and incoming handover attempts increase in the source and capacity-layer cell, respectively. However, the values of counters related to handover executions and successful handovers do not increase. As Huawei-defined KPIs **Inter-Frequency Handover Out Success Rate (CU)** and **Intra-RAT Handover In Success Rate (CU)** are calculated using the number of outgoing/incoming handover executions as the denominator, these KPIs are not affected in this scenario. However, if the user-defined KPIs are calculated using the number of outgoing/incoming handover attempts as the denominator, these KPIs may be slightly affected in this scenario.

- During NR carrier shutdown, new UEs are prohibited from accessing an NR cell. If UEs attempt to be handed over to the NR cell, the values of the counters related to handover preparation failures (such as **N.HO.IntraFreq.FailOut.PrepFailure**) slightly increase.
- During NR carrier shutdown, if UEs fail to be handed over to target cells, the values of counters related to the number of intra-NR incoming handover failures (including **N.HO.FailIn.HOCancel**, **N.HO.FailIn.AdmitFail**, **N.HO.FailIn.AdmitFail.TNL**, **N.HO.FailIn.AdmitFail.UserLic**, and **N.HO.FailIn.AdmitFail.Load**) slightly increase.
- This function saves energy by shutting down carriers, but the cells remain activated. As a result, when this function takes effect, the performance counters and KPIs listed in [Table 6-12](#) are still in measurement, which has no impact on the energy saving effect. The exact time when carrier shutdown actually starts taking effect may not be synchronized with that when related performance counters are reported. Consequently, if carrier shutdown starts taking effect in a measurement period, there will be a measurement error of less than 5 minutes between the reported results and the actual results for the shut-down cell in the measurement period.

Table 6-12 Related performance counters and KPIs

Function Subset	Performance Counter and KPI	Description
• PRB Measurement • Hyper Cell TRP PRB Measurement • DU Cell Operator PRB Measurement	• Counters related to the number of available PRBs • Counters related to the number of available PDCCH CCEs • PRB usage	None

Function Subset	Performance Counter and KPI	Description
DU Cell Algorithm Measurement	- Counters related to the total number of physical uplink shared channel (PUSCH) transmission time intervals (TTIs) - Counters related to the total number of PDSCH TTIs	None
DU Cell SI Message Measurement	Number of SI transmissions	None
- Channel Quality Measurement - DU Cell TRP Channel Quality Measurement - Uplink Interference Measurement on PRB 0 to PRB 99 - Uplink Interference Measurement on PRB 100 to PRB 199 - Uplink Interference Measurement on PRB 200 to PRB 272	- Counters related to interference and noise received in the uplink - Counters related to uplink received signal strength indicator (RSSI) - Counters related to uplink power - Counters related to average interference and noise detected on a single PRB - Counters related to the maximum interference and noise detected on a single PRB	The PWR_SAVING_COUNTER_CORR_SW option of the gNBOamParam.StatisticsStrategySwitch parameter specifies whether to report invalid uplink interference counter values during carrier shutdown. - If this option is selected, invalid uplink interference counter values are reported for carriers that have been shut down. - If this option is deselected, measured uplink interference counter values are reported for such carriers.

- In the case of false physical random access channel (PRACH) detection where the base station mistakenly regards interference or noise as the random access preambles of UEs, measurement for the following counters and KPIs will continue:
 - Counters related to the number of UEs scheduled in the uplink and downlink in a cell
 - Counters related to uplink data transmission duration

- Counters related to the number of TBs used for uplink data transmission
- Counters related to modulation and coding scheme (MCS) index distribution for scheduling on the PUSCH
- Counters related to the number of resource blocks (RBs) used for scheduling on the PUSCH
- PUSCH RB usage
- As the cell in the carrier shutdown state does not send downlink paging messages, the counter listed in [Table 6-13](#) is no longer measured.

Table 6-13 Counter related to paging

Function Subset	Counter
DU Cell Paging Measurement	Number of paged UEs over the Uu interface in a cell

- When the throughput fluctuates greatly, intelligent carrier shutdown may repeatedly start to take effect and stop in an NR cell if the **NRDUCellPowerSaving.CellAvgDlThptThld** and **NRDUCellPowerSaving.UeAvgDlThptThld** parameters are set to values other than **65535** for the cell. To mitigate this issue, the value **65535** is recommended for these parameters.
- Enabling precise shutdown mode increases the carrier shutdown duration and therefore reduces the capacity of the base station. If precise shutdown mode is enabled and overlap rates are calculated based on grids, the CPU usage slightly increases.
- Assuming that the neighboring base stations are in the carrier shutdown state, cell-edge UEs in weak coverage areas may access the cells served by the local base station. This affects KPIs related to the aspects such as access and service drops of the local base station. For example, the value of the **N.NsaDc.DRB.AbnormRel** counter may increase.

6.2.3 Requirements

6.2.3.1 Licenses

For 3900 and 5900 series base stations, the license requirements are as follows.

Feature ID	Feature Name	Model	Sales Unit
FOFD-050203	Intelligent Carrier Shutdown	NR0S00ICSC00	per Cell

For DBS3900 and DBS5900 LampSite base stations, the license requirements are as follows.

Feature ID	Feature Name	Model	Sales Unit
FOFD-050203	Intelligent Carrier Shutdown	NR0SSHUTDW00	per Carrier per pRRU

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Virtual grid model building and maintenance	VG_MODEL_ALL OW_BUILD_FLAG option of the NRCellFreqRelation.AggregationAttribute parameter and NR_IF_HO_MEAS_QTY_PRED_FUNCTION_SW option of the NRCellSmartMultiCarr.NrMultiCarrierAlgoSwitch parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	When precise shutdown is enabled and overlap rates need to be calculated based on grids, virtual grid model building and maintenance must be enabled. To further improve the accuracy of grid models, perform data configuration according to <i>Virtual Grid-based Multi-Frequency Coordination</i> .

6.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch . <i>PowerSavingSwitch</i> parameter	<i>Energy Conservation and Emission Reduction</i>	The period specified for intelligent carrier shutdown in a cell cannot overlap that specified for timing carrier shutdown in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch . <i>HighReliabilitySwitch</i> parameter	<i>URLLC</i>	The low latency and high reliability function does not work with intelligent carrier shutdown.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option (inter-base-station multi-carrier scenarios) of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected - Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected	<i>Energy Conservation and Emission Reduction</i>	- The period specified for fast carrier shutdown in a cell cannot overlap that specified for non-precise intelligent carrier shutdown in the same cell. - Fast carrier shutdown and precise intelligent carrier shutdown cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-carrier joint carrier shutdown	MULTI_CARR_JOINT_CARR_SHDN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i> <i>Multi-RAT Coordinated Energy Saving</i>	If the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL and the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter is set to a value less than 6 for intelligent carrier shutdown, multi-carrier joint carrier shutdown must be disabled.
Low-frequency TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Intelligent carrier shutdown is incompatible with the primary/backup function in intra-frequency dual-network redundancy networking.
Low-frequency TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Intelligent carrier shutdown is incompatible with dual-BBU backup in dual-network redundancy networking.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TO_B_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	Intelligent carrier shutdown is incompatible with the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios.
Low-frequency TDD	Dynamic MIMO layer number decrease	DYNAMIC_MIMO_LAYER_DEC_SW option of the gNodeBAlgo.DedicatedNetworkFunctionSw parameter	None	In 8T8R scenarios where the NRDUCell.NrDUCellNetworkingMode parameter is set to NORMAL_CELL , dynamic MIMO layer number decrease does not work with intelligent carrier shutdown in precise mode.

6.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Uplink symbol power saving does not take effect when intelligent carrier shutdown is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	After intelligent carrier shutdown takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	After intelligent carrier shutdown takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.
Low-frequency TDD	Inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	After intelligent carrier shutdown takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.
Low-frequency TDD High-frequency TDD	Symbol power saving	NRDUCellAIgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of symbol power saving is not affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, deep symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of deep symbol power saving is not affected.
Low-frequency TDD	pRRU-level precise symbol power saving	DL_PRECISE_SYMBOL_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	pRRU-level precise symbol power saving does not take effect when intelligent carrier shutdown is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, power saving scheduling does not take effect. • If the RF module is serving other NR carriers, the status of power saving scheduling is not affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	For low-frequency TDD: When intelligent carrier shutdown is in effect: • If the local cell is a capacity-layer cell, RF channel intelligent shutdown cannot take effect in the scenarios listed in Table 5-6. • If the local cell is a basic-layer cell and RF channel intelligent shutdown takes effect in the scenarios listed in Table 5-6, the related capacity-layer cells may exit the intelligent carrier shutdown state. For high-frequency TDD: When intelligent carrier shutdown is in effect, RF channel intelligent shutdown does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both intelligent carrier shutdown and energy saving based on flexible frequency-domain scheduling take effect, the latter function does not deliver energy saving gains.
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	TTI-level channel shutdown does not take effect when intelligent carrier shutdown is in effect.
Low-frequency TDD	Low-power RF channel dynamic shutdown	LOW_POWER_RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Low-power RF channel dynamic shutdown does not take effect when intelligent carrier shutdown is in effect.
Low-frequency TDD	Intelligent power control energy saving	SMART_POWER_CTRL_ENERGY_SAVE_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both intelligent carrier shutdown and intelligent power control energy saving take effect, the latter function does not deliver energy saving gains.
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHANNEL_DYNAMIC_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect, dynamic RF channel shutdown will deliver fewer energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Initial transmission power reduction	INIT_TRANS_POWER_REDUCE_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both initial transmission power reduction and intelligent carrier shutdown take effect, the former function does not deliver any gains.
Low-frequency TDD	Retransmission power reduction	RETRANS_POWER_REDUCE_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both retransmission power reduction and intelligent carrier shutdown take effect, the former function does not deliver any gains.
Low-frequency TDD	LTE and NR coordinated symbol power saving	MULTI_RAT_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When intelligent carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, LTE and NR coordinated symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of LTE and NR coordinated symbol power saving is not affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR bundled scheduling	LNR_BUNDLED_SCHEDULING_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When intelligent carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, LTE and NR bundled scheduling does not take effect. • If the RF module is serving other NR carriers, the status of LTE and NR bundled scheduling is not affected.
Low-frequency TDD High-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Intelligent carrier shutdown does not take effect when RF module deep dormancy is in effect.
Low-frequency TDD High-frequency TDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Intelligent carrier shutdown does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When both intelligent carrier shutdown and LTE and NR intelligent carrier shutdown are enabled for a cell, this cell enters the carrier shutdown state only if the entering conditions for both functions are met.
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over intelligent carrier shutdown when both functions are enabled. Intelligent carrier shutdown does not take effect when RF module fast deep dormancy is in effect.
Low-frequency TDD	Cell load simulation	STRNRDUCELLSIMULATELOAD	None	Cell load simulation cannot be started when intelligent carrier shutdown is enabled. If cell load simulation has started, it will stop when intelligent carrier shutdown is enabled. Therefore, enabling both functions is not recommended.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Automatic removal of neighboring cells	NR_NR_ANR_DEL_SW option of the NRCellAlgoSwitch.AnrSwitch parameter	<i>ANR</i>	When automatic removal of neighboring cells in ANR takes effect, intelligent carrier shutdown does not take effect if the neighbor relationships with the basic-layer cells are automatically removed for capacity-layer cells.
Low-frequency TDD	CMAS	gNBSibConfig.SibType set to SIB_TYPE_8	<i>CMAS</i>	When an NR carrier is shut down, the corresponding NR cell is blocked and CMAS warning notifications cannot be properly sent in this cell. However, UEs can still receive CMAS warning notifications from other carriers.
Low-frequency TDD	Super uplink with SUL and NR FDD co-deployment	FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	After an NR FDD carrier is shut down, SUL scheduling cannot be performed for super uplink UEs.
Low-frequency TDD	UL and DL Decoupling	NRDUCellAlgoSwitch.UlDlDecouplingSwitch	<i>UL and DL Decoupling</i>	After an NR FDD carrier is shut down, SUL scheduling cannot be performed for SUL UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-cell cable sequence detection	AntPortOptDetPolicy parameter in the STR ANTENNAP ORTOPTDET command set to INTER_CELL_DETECT	TDD: <i>MIMO (TDD)</i>	If intelligent carrier shutdown takes effect during inter-cell cable sequence detection, the detection results will show that the detection fails in the corresponding cells.
Low-frequency TDD	Basic VoNR functions	VONR_SW option of the NRCellAlgoSwitch . <i>VonrSwitch</i> parameter	<i>VoNR</i>	Intelligent carrier shutdown does not take effect when there are VoNR UEs that cannot be handed over from the capacity-layer cell during a UE transfer procedure.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Virtual Grid-based Multi-Frequency Coordination	• RSRP prediction model – VG_MODEL_ALLOW_BUILD_FLAG option of the NRCellFreqRelation. <i>Aggregation Attribute</i> parameter – NR_IF_HO_MEAS_QTY_PRED_FUNCTION_SWITCH option of the NRCellSmartMultiCarrierAlgorithmSwitch parameter • Downlink intra-frequency spectral efficiency prediction model – DL_INTERF_S	<i>Virtual Grid-based Multi-Frequency Coordination</i>	After intelligent carrier shutdown takes effect, the building of virtual grid models will be suspended and existing virtual grid models will be brought offline and cannot be used.

RAT	Function Name	Function Switch	Reference	Description
		E_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarrier.ModelBuildingSwitch parameter - DL_INTRAFREQ_EFFECT_PRED_SW option of the NRCellSmartMultiCarrier.NrMultiCarrierAlgorithmSwitch parameter • Intra-frequency SRS beam prediction model INTRAFREQSRS_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarrier.Model		

RAT	Function Name	Function Switch	Reference	Description
		<i>Buildingswitch</i> parameter		
Low-frequency TDD	Channel calibration	Macro base station: None LampSite base station: INTER_RRU_CHN_CALIB_SW option of the NRDUCellAIgoSwitch.ChnCalibSwitch parameter	None	If channel calibration is being performed when intelligent carrier shutdown takes effect, channel calibration fails and will not be started again. You can run the DSP NRDUCELLCHNCALIB command to check whether channel calibration is successful.
Low-frequency TDD	RB allocation for downlink experience optimization	NRDUCellDLSchRes.DLExpOptRbIncreasePct	<i>Downlink Scheduling</i>	When both RB allocation for downlink experience optimization and intelligent carrier shutdown take effect, an exit from intelligent carrier shutdown may be performed earlier than expected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell-level MRO	MRO_SCENARIO_IDENTIFY_SWITCH option of the gNB-MRO-ScenarioIdentifySwitch parameter and the INTRA_FREQ_MRO_SWITCH option of the NRCell-MRO-MroSwitch parameter	<i>MRO</i>	When both intelligent carrier shutdown and intra-frequency MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.
Low-frequency TDD	UE-level MRO	UE_MRO_SWITCH option of the NRCell-MRO-MroSwitch parameter	<i>MRO</i>	When both intelligent carrier shutdown and UE-level MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO may be affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAIgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	(TDD) The impacts are as follows: • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of intelligent carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH,

RAT	Function Name	Function Switch	Reference	Description
				and control information changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SWITCH option of the NRDUCellAIgoSwitch.DynamicPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	A carrier exits a power adaptive scheduling group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-RAT redirection back to the HPLMN based on roaming UE identification	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgorithmSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	When intelligent carrier shutdown is enabled, only blind redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 5G network.
Low-frequency TDD	Clock out-of-synchronization detection	CLK_DETECT_SW option of the gNodeBParam.ClkOutofSyncDetectionSwitch parameter	<i>Synchronization</i>	When intelligent carrier shutdown is in effect, any instance of clock out-of-synchronization may fail to be detected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Ultra-long holdover for sites with abnormal clock sources	CLK_OTASY NC_PROC_SW option of the TASM.MSCLKDIAGPROC ESSSW parameter	<i>Synchronization</i>	Assume that the Ultra-Long Hold Switch is turned on but the Ultra-Long Hold Enhancement Switch is turned off. If sequence detection is performed on a to-be-detected base station during the period when intelligent carrier shutdown takes effect for all cells of the base station, the sequence detection fails. In this case, a possible clock fault may cause the base station to be unable to enter the ultra-long holdover state, or to exit the ultra-long holdover state if sequence detection cannot be performed for a long time due to energy conservation. If the Ultra-Long Hold Enhancement Switch is turned on, intelligent carrier shutdown does not take effect for the base station in the ultra-long holdover state or its reference

RAT	Function Name	Function Switch	Reference	Description
				neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	In the handover decision-making phase for intelligent carrier shutdown where the specific timer has not expired, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterFHoFlag parameter is set to TRUE for the target cell's frequency in the frequency list group bound to the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports

RAT	Function Name	Function Switch	Reference	Description
				handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, intelligent carrier shutdown-based inter-frequency handover does not take effect for the roaming UE. • When the gNBFreqPriorityGroup.RoamingUeHplmnInterFHoFlag parameter is set to FALSE for the target cell's frequency in the frequency group that is associated with the roaming UE, intelligent carrier shutdown-based inter-frequency handover does not take effect for the roaming UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrierMgmt.CAEnhancedAlgorithmExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	CA Performance Boost Based on Multi-dimensional Optimization does not take effect for cells in the shutdown state.
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SWITCH option of the NRDUCellCoLabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When intelligent carrier shutdown takes effect, the probability that Co-MM takes effect decreases.
Low-frequency TDD	LTE and NR multi-band power adaptive scheduling	- LTE: LNR_MULTIBAND_PWR_ADPT_SCH_SWITCH option of the CellDynaPowerSharing.DynaPowerSharingSwitch parameter - NR: LNR_MULTIBAND_PWR_ADPT_SCH_SWITCH option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	A carrier exits a power adaptive scheduling group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter adjustment. This causes changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	• LTE: LTE_NR_DYN_POWER_SHARING_SW option of the CellDynamicPowerSharing.DynPowerSharingSwitch parameter • NR: LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSrrCfg.DynPwrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of intelligent carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSrrCfg.DynPwrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to

RAT	Function Name	Function Switch	Reference	Description
				parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	L/NR Inter-band Deep Coordination	• LTE: LNR_INTE R_BAND_DEEP_COORD_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter • NR: LNR_INTE R_BAND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPowerShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPowerShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of intelligent carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPowerShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to

RAT	Function Name	Function Switch	Reference	Description
				parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUelidentSw	None	Intelligent carrier shutdown fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect: - Any option of the gNodeBParam.StUelidentSw parameter is selected. - The gNBAsAppParam.AppDomainName parameter is set. - The gNBAsAppParam.AppIpv4 parameter is set. - The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue: - Select the SPEED_TURBO_PWR_SAVING_SW option of the
Low-frequency TDD	gNodeB acceleration service application	gNBAsAppParam.AppDomainName gNBAsAppParam.AppIpv4 gNBAsAppParam.AppIpv6	None	

RAT	Function Name	Function Switch	Reference	Description
				gNB PwrSavin gAlgo.PwrSavin gAlgoSwitch parameter. Intelligent carrier shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNB PwrSavin gAlgo.SpeedTurboPwrSavStartTime and gNB PwrSavin gAlgo.SpeedTurboPwrSavEndTime parameters, increasing the energy saving gains. • Select the STATIC_STATIC_PWR_SAVING_SWITCH option of the gNB PwrSavin gAlgo.PwrSavin gAlgoSwitch parameter. Intelligent carrier shutdown can take effect for cells served by the base station in some of the preceding scenarios within the

RAT	Function Name	Function Switch	Reference	Description
				period specified by the gNB PwrSavin gAlgo.StaticST PwrSavStartTime and gNB PwrSavin gAlgo.StaticST PwrSavEndTime parameters, increasing the energy saving gains.

6.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

Low-frequency bands:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

High-frequency TDD bands:

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

For low-frequency TDD: All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

For high-frequency TDD: This function is compatible only with HAAU5233w and HAAU5235w.

6.2.3.4 Cells

This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**. This function can be configured but does not take effect in cells with the **NRDUCell.DuplexMode** parameter set to **CELL_SUL**.

In intra-base-station multi-carrier scenarios (where the **INTRA_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter is selected), if the networking option is NSA for capacity-layer cells, it must be either NSA or SA_NSA for basic-layer cells; if the networking option is SA or SA_NSA for capacity-layer cells, it must be SA or SA_NSA for basic-layer cells. In multi-operator sharing scenarios, the networking option must be SA or SA_NSA for capacity-layer cells.

In inter-base-station multi-carrier scenarios (where the **INTER_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter is selected), the networking option must be SA or SA_NSA for both capacity-layer cells and basic-layer cells.

6.2.3.5 Others

For high-frequency TDD: It is recommended that the base stations serving low-frequency basic-layer cells run the software version of V100R020C10 or later. If the software version is V100R020C00 or earlier, certain conditions for exiting this function will become invalid.

For low frequency bands: None

6.2.4 Operation and Maintenance

6.2.4.1 Data Configuration

6.2.4.1.1 Data Preparation

Table 6-14 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-14 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellAlgoSwitch.NrDuCellId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	When intelligent carrier shutdown is required: • In intra-base-station multi-carrier scenarios, select the INTRA_GNB_MULTI_CARR_S D_SW option of this parameter for capacity-layer cells. • In inter-base-station multi-carrier scenarios, select the INTER_GNB_MULTI_CARR_S D_SW option of this parameter for capacity-layer cells.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	• To enable precise shutdown, select the MULTI_CARR_PRECISE_SHD N_SW option of this parameter. • To use grid models to calculate the overlap rates for precise shutdown, select the GRID_MODE_SW option of this parameter. – If this option is selected, virtual grid model building and maintenance are enabled, and grid models are successfully built, then the overlap rates calculated using these models are used. – If this option is selected but virtual grid model building and maintenance are disabled or grid models fail to be built, then the overlap rates specified by the NRCellRelation.OverlapRate parameter are used. • If the GRID_MODE_SW option is deselected, the overlap rates are determined by the NRCellRelation.OverlapRate parameter.

Parameter Name	Parameter ID	Setting Notes
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: • For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. • For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. • The period specified for intelligent carrier shutdown in a cell cannot overlap that specified for the function with the NRDUCellPowerSaving.PowerSavingType parameter set to TIMING_CARRIER_SHUTDOWN for the same cell.

Parameter Name	Parameter ID	Setting Notes
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan and the following requirements: - For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. - For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. - The period specified for intelligent carrier shutdown in a cell cannot overlap that specified for the function with the NRDUCellPowerSaving.PowerSavingType parameter set to TIMING_CARRIER_SHUTDOWN for the same cell.

Parameter Name	Parameter ID	Setting Notes
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	When intelligent carrier shutdown is required: • In intra-base-station multi-carrier scenarios, set this parameter to INTRA_GNB_MULTI_CARR_SD for capacity-layer cells. • In inter-base-station multi-carrier scenarios, set this parameter to INTER_GNB_MULTI_CARR_SD for capacity-layer cells.
Carrier Shutdown Mode	NRDUCellPowerSaving.CarrierShutdownMode	When intelligent carrier shutdown is required, set this parameter to LOAD_BASED_CARR_SHUTDOWN for capacity-layer cells.
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	You are advised to use the recommended value. It is recommended that the sum of NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset be less than or equal to 100. If the sum exceeds 100, the cell cannot exit the intelligent carrier shutdown state.
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	You are advised to use the recommended value. It is recommended that the sum of NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset be less than or equal to 100. If the sum exceeds 100, the cell cannot exit the intelligent carrier shutdown state.

Parameter Name	Parameter ID	Setting Notes
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	You are advised to use the recommended value. It is recommended that the sum of NRDUCellPowerSaving.UlPrbThld and NRDUCellPowerSaving.NCellUlPrbOffset be less than or equal to 100. If the sum exceeds 100, the cell cannot exit the intelligent carrier shutdown state.
Neighboring Cell Uplink PRB Offset	NRDUCellPowerSaving.NCellUlPrbOffset	You are advised to use the recommended value. It is recommended that the sum of NRDUCellPowerSaving.UlPrbThld and NRDUCellPowerSaving.NCellUlPrbOffset be less than or equal to 100. If the sum exceeds 100, the cell cannot exit the intelligent carrier shutdown state.
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	You are advised to use the recommended value. Intelligent carrier shutdown requires that the NRDUCellPowerSaving.UserNumThld parameter be set to a value less than or equal to 150 .
User Number Offset	NRDUCellPowerSaving.UserNumOffset	You are advised to use the recommended value.
Operator-specific User Number Threshold	NRDUCellPowerSaving.OpUserNumThld	You are advised to use the recommended value. If this parameter is set to 65535 , this parameter and the NRDUCellPowerSaving.OpUserNumOffset parameter are invalid.
Operator-specific User Number Offset	NRDUCellPowerSaving.OpUserNumOffset	You are advised to use the recommended value.

Parameter Name	Parameter ID	Setting Notes
SUL User Number Threshold	NRDUCellPowerSaving.SulUserNumThld	You are advised to use the recommended value. • This parameter is invalid when it is set to 65535. • If the value of NRDUCellPowerSaving.SulUserNumThld is not 65535, the sum of the NRDUCellPowerSaving.SulUserNumThld and NRDUCellPowerSaving.SulUserNumOffset parameter values must be less than 65535, and the value of NRDUCellPowerSaving.SulUserNumThld must be less than that of NRDUCellPowerSaving.UserNumThld. • When the FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is deselected and there is a record with NRDUCellPowerSaving.PowerSavingType set to INTRA_GNB_MULTI_CARR_S D or INTER_GNB_MULTI_CARR_S D_SW for the cell, the NRDUCellPowerSaving.SulUserNumThld parameter must be set to 65535 for this record.

Parameter Name	Parameter ID	Setting Notes
SUL User Number Offset	NRDUCellPowerSaving.SulUserNumOffset	You are advised to use the recommended value. - This parameter is invalid when it is set to 65535. - When the value of NRDUCellPowerSaving.SulUserNumThld is not 65535, the sum of the NRDUCellPowerSaving.SulUserNumThld and NRDUCellPowerSaving.SulUserNumOffset parameter values must be less than 65535.
Power Saving Penalty Duration	NRDUCellPowerSaving.PwrSavingPenaltyTime	You are advised to use the recommended value. If the setting of this parameter is too small and some UEs cannot be transferred, other UEs may be frequently transferred. In this case, the carrier frequently attempts to enter the carrier shutdown state, and therefore UEs cannot access it for a long period of time.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	You are advised to use the recommended value. For high-frequency TDD: This parameter must be set to MINUTE_LEVEL.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	You are advised to use the recommended value.
Second-Level Trigger Evaluation Period	NRDUCellPowerSaving.SecondLevelTrigEvalPrd	You are advised to use the recommended value.
Exit Evaluation Period	NRDUCellPowerSaving.ExitEvaluationPeriod	You are advised to use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Second-Level Exit Evaluation Period	NRDUCellPowerSaving.SecondLevelExitEvalPrd	You are advised to use the recommended value.
UL PRB Protect Thld for Power Saving	NRDUCellPowerSaving.ProtectUlPrbThld	You are advised to use the recommended value.
DL PRB Protect Thld for Power Saving	NRDUCellPowerSaving.ProtectDlPrbThld	You are advised to use the recommended value.
Uplink PRB Protection Offset	NRDUCellPowerSaving.ProtectUlPrbOffset	You are advised to use the recommended value.
Downlink PRB Protection Offset	NRDUCellPowerSaving.ProtectDlPrbOffset	You are advised to use the recommended value.
Power Saving Cell Flag	NRCellRelation.PowerSavingCellFlag	This parameter indicates the flag of intelligent carrier shutdown for a neighboring cell. Set this parameter based on the network plan. Neither the number of neighboring cells for which this parameter is set to COVERAGE nor that of neighboring cells for which this parameter is set to CAPACITY can be greater than 16 when the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter is set to SECOND_LEVEL.

Parameter Name	Parameter ID	Setting Notes
No Removal Flag	NRCellRelation.NoRmvFlag	When automatic removal of neighboring cells in ANR is enabled (with the NR_NR_ANR_DEL_SW option of the NRCellAlgoSwitch.AnrSwitch parameter selected), it is recommended that the NRCellRelation.NoRmvFlag parameter be set to FORBID_ANR_RMV for the bidirectional neighbor relationships between capacity-layer and basic-layer cells.
Overlap Rate	NRCellRelation.OverlapRate	This parameter specifies the degree of the overlap between the specified neighboring cell and the local cell. When precise shutdown is enabled, set this parameter based on the co-coverage degree of respective neighboring cells.
PLMN Reserved Flag	NRExternalNCellPlmn.PlmnReservedFlag	In multi-operator sharing scenarios, set this parameter for capacity-layer cells based on the network plan.
mmWave User Number Threshold	NRDUCellPowerSaving.MmwaveUserNumThld	Use the recommended value.
Necessary HO Prepare Fail Punish Timer	gNBMobilityCommParam.NecHoPrepFailPunishTimer	You are advised to set this parameter to 30 when the handover success rate decreases due to intelligent carrier shutdown.
Optimized HO Prepare Fail Punish Timer	gNBMobilityCommParam.OptHoPrepFailPunishTimer	You are advised to set this parameter to 30 when the handover success rate decreases due to intelligent carrier shutdown.

Parameter Name	Parameter ID	Setting Notes
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	Select the SERV_TYPE_DYN_EXIT_PWR_SAVE_SW option if the number of UEs in an energy saving policy group needs to be checked for entering or exiting intelligent carrier shutdown.
Power Saving Policy Group Index	gNBQciBearer.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan.
Power Saving Policy Group Index	NRCeIPsPolicyGrp.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan. The value of this parameter must be the same as that of the gNBQciBearer.PowerSavingPolicyGrpIndex parameter for services to be guaranteed. Before setting this parameter, ensure that the QCI for which the gNBQciBearer.PowerSavingPolicyGrpIndex parameter is set has been configured.
User Identification Method	NRCeIPsPolicyGrp.UserIdentificationMethod	Set this parameter to SERVICE_TYPE .
Service Type Dynamic Exit Power Saving Switch	NRCeIPsPolicyGrp.ServTypeDynamicExitPwrSavingSw	Select the USER_NUM_CSD_EXIT_SW option if the number of UEs in an energy saving policy group needs to be checked for entering or exiting intelligent carrier shutdown.
PS Policy Group User Number Threshold	NRCeIPsPolicyGrp.GrpUserNumThld	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
PS Policy Group User Number Offset	NRCeLLPsPolicyGrp.GrpUserNumOffset	Use the recommended value.

6.2.4.1.2 Using MML Commands

Before using MML commands, refer to [6.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- **Intra-base-station multi-carrier scenario**

Assume that a cell with **NrCellId** being **0** is a capacity-layer cell and another cell with **NrCellId** being **10** is a basic-layer cell in an intra-base-station multi-carrier scenario. The **PowerSavingCellFlag** parameter in the **NRCeLLRelation** MO must be set in both capacity- and basic-layer cells for intelligent carrier shutdown. For details about how to configure intra-base-station neighboring cells, see *SA Mobility Management in Connected Mode*.

Scenario 1: intelligent carrier shutdown with minute-level evaluation periods

```
//Enabling intelligent carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingSwitch=INTRA_GNB_MULTI_CARR_SD_SW-1;
//Setting an energy saving policy of intelligent carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, StartTime=00&00&00, StopTime=06&00&00,
DlPrbThld=25, DlPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
OpUserNumOffset=65535, PwrSavingPenaltyTime=30, UlPrbThld=25, NCellUlPrbOffset=25,
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1, ExitEvaluationPeriod=1,
CarrierShutdownMode=LOAD_BASED_CARR_SHUTDOWN;
//(Optional) Setting another energy saving policy of intelligent carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, StartTime=12&00&00, StopTime=14&00&00,
DlPrbThld=25, DlPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
OpUserNumOffset=65535, PwrSavingPenaltyTime=30, UlPrbThld=25, NCellUlPrbOffset=25,
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1, ExitEvaluationPeriod=1,
CarrierShutdownMode=LOAD_BASED_CARR_SHUTDOWN;
//For high-frequency TDD: Setting an energy saving policy of intelligent carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=2,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, StartTime=00&00&00, StopTime=06&00&00,
DlPrbThld=25, DlPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
OpUserNumOffset=65535, PwrSavingPenaltyTime=30, UlPrbThld=25, NCellUlPrbOffset=25,
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1, ExitEvaluationPeriod=1,
CarrierShutdownMode=LOAD_BASED_CARR_SHUTDOWN, MmwaveUserNumThld=20;
```

```
//Setting PowerSavingCellFlag on the local base station, and (recommended) setting NoRmvFlag to
FORBID_ANR_RMV for the bidirectional neighbor relationships between capacity-layer and basic-layer
cells when automatic removal of neighboring cells in ANR takes effect
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=10,
PowerSavingCellFlag=COVERAGE, NoRmvFlag=FORBID_ANR_RMV;
MOD NRCELLRELATION: NrCellId=10, Mcc="302", Mnc="220", gNBId=0, CellId=0,
PowerSavingCellFlag=CAPACITY, NoRmvFlag=FORBID_ANR_RMV;
//((Optional) Setting the precise shutdown switch and an energy saving policy
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=MULTI_CARR_PRECISE_SHDN_SW-1;
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, ProtectDlPrbThld=50, ProtectUlPrbThld=50,
ProtectUlPrbOffset=20, ProtectDlPrbOffset=20;
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=10, OverlapRate=60;
//((Optional) Enabling grid-based precise carrier shutdown, with which overlap rates are calculated
based on grids
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=GRID_MODE_SW-1;
```

Scenario 2: (optional) intelligent carrier shutdown with second-level evaluation periods

```
//Enabling intelligent carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingSwitch=INTRA_GNB_MULTI_CARR_SD_SW-1;
//Setting an energy saving policy of intelligent carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, StartTime=00&00&00, StopTime=06&00&00,
DlPrbThld=25, DlPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
OpUserNumOffset=65535, SulUserNumThld=65535, SulUserNumOffset=65535,
PwrSavingPenaltyTime=3, UlPrbThld=25, NCellUlPrbOffset=25,
EvalPeriodGranularity=SECOND_LEVEL, SecondLevelTrigEvalPrd=6, SecondLevelExitEvalPrd=12,
CarrierShutdownMode=LOAD_BASED_CARR_SHUTDOWN;
//((Optional) Setting another energy saving policy of intelligent carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, StartTime=12&00&00, StopTime=14&00&00,
DlPrbThld=25, DlPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
OpUserNumOffset=65535, SulUserNumThld=65535, SulUserNumOffset=65535,
PwrSavingPenaltyTime=30, UlPrbThld=25, NCellUlPrbOffset=25,
EvalPeriodGranularity=SECOND_LEVEL, SecondLevelTrigEvalPrd=6, SecondLevelExitEvalPrd=12,
CarrierShutdownMode=LOAD_BASED_CARR_SHUTDOWN;
//Setting PowerSavingCellFlag on the local base station, and (recommended) setting NoRmvFlag to
FORBID_ANR_RMV for the bidirectional neighbor relationships between capacity-layer and basic-layer
cells when automatic removal of neighboring cells in ANR takes effect
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=10,
PowerSavingCellFlag=COVERAGE, NoRmvFlag=FORBID_ANR_RMV;
MOD NRCELLRELATION: NrCellId=10, Mcc="302", Mnc="220", gNBId=0, CellId=0,
PowerSavingCellFlag=CAPACITY, NoRmvFlag=FORBID_ANR_RMV;
//((Optional) Setting the precise shutdown switch and an energy saving policy
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=MULTI_CARR_PRECISE_SHDN_SW-1;
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, ProtectDlPrbThld=50, ProtectUlPrbThld=50,
ProtectUlPrbOffset=20, ProtectDlPrbOffset=20;
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=10, OverlapRate=60;
//((Optional) Enabling grid-based precise carrier shutdown, with which overlap rates are calculated
based on grids
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=GRID_MODE_SW-1;
```

(Optional) Multi-operator sharing scenario

```
//((Optional) Modifying an energy saving policy of intelligent carrier shutdown
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, OpUserNumThld=12, OpUserNumOffset=12;
//((Optional) Modifying the other energy saving policy of intelligent carrier shutdown
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, OpUserNumThld=12, OpUserNumOffset=12;
```

(Optional) SUL co-deployment scenario

```
//((Optional) Modifying an energy saving policy of intelligent carrier shutdown
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, SulUserNumThld=4, SulUserNumOffset=4;
//((Optional) Modifying the other energy saving policy of intelligent carrier shutdown
```

```
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, SulUserNumThld=4, SulUserNumOffset=4;
```

(Optional) Scenario where the number of UEs in an energy saving policy group needs to be checked for entering or exiting intelligent carrier shutdown

//(Optional) Performing the following configurations if the number of UEs in an energy saving policy group needs to be checked for entering or exiting intelligent carrier shutdown

//For a capacity-layer cell

```
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
```

```
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
```

```
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;
```

```
ADD NRCELLPOLICYGRP: NrCellId=0, PowerSavingPolicyGrpIndex=0,
```

```
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=USER_NUM_CSD_EXIT_SW-1,
```

```
GrpUserNumThld=7, GrpUserNumOffset=4;
```

//For a basic-layer cell

```
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
```

```
ADD NRCELLPOLICYGRP: NrCellId=10, PowerSavingPolicyGrpIndex=0,
```

```
UserIdentMethod=SERVICE_TYPE;
```

• Inter-base-station multi-carrier scenario

Assume that, in an inter-base-station multi-carrier scenario, a cell with **NrCellId** being **1** is a capacity-layer cell served by a base station with **gNBId** being **0** and another cell with **NrCellId** being **11** is a basic-layer cell served by a base station with **gNBId** being **1**. The **PowerSavingCellFlag** parameter in the **NRCellRelation** MO must be set in both capacity- and basic-layer cells for intelligent carrier shutdown. For details about how to configure inter-base-station neighboring cells, see *SA Mobility Management in Connected Mode*.

Scenario 1: intelligent carrier shutdown with minute-level evaluation periods

//Enabling intelligent carrier shutdown

```
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
```

```
PowerSavingSwitch=INTER_GNB_MULTI_CARR_SD_SW-1;
```

//Setting an energy saving policy of intelligent carrier shutdown

```
ADD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0,
```

```
PowerSavingType=INTER_GNB_MULTI_CARR_SD, StartTime=12&00&00, StopTime=14&00&00,
```

```
DIPrbThld=25, DIPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
```

```
OpUserNumOffset=65535, SulUserNumThld=65535, SulUserNumOffset=65535,
```

```
PwrSavingPenaltyTime=30, UIPrbThld=25, NCellUIPrbOffset=25,
```

```
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1, ExitEvaluationPeriod=1,
```

```
CarrierShutdownMode=LOAD_BASED_CARR_SHUTDOWN;
```

//(Optional) Setting another energy saving policy of intelligent carrier shutdown

```
ADD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=1,
```

```
PowerSavingType=INTER_GNB_MULTI_CARR_SD, StartTime=12&00&00, StopTime=14&00&00,
```

```
DIPrbThld=25, DIPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
```

```
OpUserNumOffset=65535, SulUserNumThld=65535, SulUserNumOffset=65535,
```

```
PwrSavingPenaltyTime=30, UIPrbThld=25, NCellUIPrbOffset=25,
```

```
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1, ExitEvaluationPeriod=1,
```

```
CarrierShutdownMode=LOAD_BASED_CARR_SHUTDOWN;
```

//For high-frequency TDD: Setting an energy saving policy of intelligent carrier shutdown

```
ADD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=2,
```

```
PowerSavingType=INTER_GNB_MULTI_CARR_SD, StartTime=12&00&00, StopTime=14&00&00,
```

```
DIPrbThld=25, DIPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
```

```
OpUserNumOffset=65535, PwrSavingPenaltyTime=30, UIPrbThld=25, NCellUIPrbOffset=25,
```

```
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1, ExitEvaluationPeriod=1,
```

```
CarrierShutdownMode=LOAD_BASED_CARR_SHUTDOWN, MmwaveUserNumThld=20;
```

//Setting PowerSavingCellFlag on the local base station, and (recommended) setting NoRmvFlag to FORBID_ANR_RMV for the neighbor relationships with the basic-layer cell when automatic removal of neighboring cells in ANR takes effect

```
MOD NRCELLRELATION: NrCellId=1, Mcc="302", Mnc="220", gNBId=1, CellId=11,
```

```
PowerSavingCellFlag=COVERAGE, NoRmvFlag=FORBID_ANR_RMV;
```

//Setting PowerSavingCellFlag on the peer base station, and (recommended) setting NoRmvFlag to FORBID_ANR_RMV for the neighbor relationships with the capacity-layer cell when automatic removal of neighboring cells in ANR takes effect

```
MOD NRCELLRELATION: NrCellId=11, Mcc="302", Mnc="220", gNBId=0, CellId=1,
```

```
PowerSavingCellFlag=CAPACITY, NoRmvFlag=FORBID_ANR_RMV;
```

//(Optional) Setting the precise shutdown switch and an energy saving policy

```
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
```

```
PowerSavingExtSwitch=MULTI_CARR_PRECISE_SHDN_SW-1;
MOD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0,
PowerSavingType=INTER_GNB_MULTI_CARR_SD, ProtectDlPrbThld=50, ProtectUlPrbThld=50,
ProtectUlPrbOffset=20, ProtectDlPrbOffset=20;
MOD NRCELLRELATION: NrCellId=1, Mcc="302", Mnc="220", gNBId=1, CellId=11, OverlapRate=60;
//(Optional) Disabling grid-based precise carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, PowerSavingExtSwitch=GRID_MODE_SW-1;
```

Scenario 2: (optional) intelligent carrier shutdown with second-level evaluation periods

```
//Enabling intelligent carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
PowerSavingSwitch=INTER_GNB_MULTI_CARR_SD_SW-1;
//Setting an energy saving policy of intelligent carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0,
PowerSavingType=INTER_GNB_MULTI_CARR_SD, StartTime=00&00&00, StopTime=06&00&00,
DlPrbThld=25, DlPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
OpUserNumOffset=65535, PwrSavingPenaltyTime=30, UlPrbThld=25, NCellUlPrbOffset=25,
EvalPeriodGranularity=SECOND_LEVEL, SecondLevelTrigEvalPrd=6, SecondLevelExitEvalPrd=12,
CarrierShutdownMode=LOAD_BASED_CARR_SHUTDOWN;
//(Optional) Setting another energy saving policy of intelligent carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=1,
PowerSavingType=INTER_GNB_MULTI_CARR_SD, StartTime=12&00&00, StopTime=14&00&00,
DlPrbThld=25, DlPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
OpUserNumOffset=65535, PwrSavingPenaltyTime=30, UlPrbThld=25, NCellUlPrbOffset=25,
EvalPeriodGranularity=SECOND_LEVEL, SecondLevelTrigEvalPrd=6, SecondLevelExitEvalPrd=12,
CarrierShutdownMode=LOAD_BASED_CARR_SHUTDOWN;
//Setting PowerSavingCellFlag on the local base station, and (recommended) setting NoRmvFlag to
FORBID_ANR_RMV for the neighbor relationships with the basic-layer cell when automatic removal of
neighboring cells in ANR takes effect
MOD NRCELLRELATION: NrCellId=1, Mcc="302", Mnc="220", gNBId=1, CellId=11,
PowerSavingCellFlag=COVERAGE, NoRmvFlag=FORBID_ANR_RMV;
//Setting PowerSavingCellFlag on the peer base station, and (recommended) setting NoRmvFlag to
FORBID_ANR_RMV for the neighbor relationships with the capacity-layer cell when automatic
removal of neighboring cells in ANR takes effect
MOD NRCELLRELATION: NrCellId=11, Mcc="302", Mnc="220", gNBId=0, CellId=1,
PowerSavingCellFlag=CAPACITY, NoRmvFlag=FORBID_ANR_RMV;
//(Optional) Setting the precise shutdown switch and an energy saving policy
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
PowerSavingExtSwitch=MULTI_CARR_PRECISE_SHDN_SW-1;
MOD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0,
PowerSavingType=INTER_GNB_MULTI_CARR_SD, ProtectDlPrbThld=50, ProtectUlPrbThld=50,
ProtectUlPrbOffset=20, ProtectDlPrbOffset=20;
MOD NRCELLRELATION: NrCellId=1, Mcc="302", Mnc="220", gNBId=1, CellId=11, OverlapRate=60;
//(Optional) Disabling grid-based precise carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, PowerSavingExtSwitch=GRID_MODE_SW-1;
```

(Optional) Multi-operator sharing scenario

```
//Setting PlmnReservedFlag of the basic-layer cell based on the network plan at the base station
serving the capacity-layer cell
MOD NREXTERNALNCELLPLMN: Mcc="302", Mnc="220", gNBId=1, CellId=11, SharedMcc="302",
SharedMnc="610", PlmnReservedFlag=FALSE;
//(Optional) Modifying an energy saving policy of intelligent carrier shutdown
MOD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0,
PowerSavingType=INTER_GNB_MULTI_CARR_SD, OpUserNumThld=12, OpUserNumOffset=12;
//(Optional) Modifying the other energy saving policy of intelligent carrier shutdown
MOD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=1,
PowerSavingType=INTER_GNB_MULTI_CARR_SD, OpUserNumThld=12, OpUserNumOffset=12;
```

(Optional) SUL co-deployment scenario

```
//(Optional) Modifying an energy saving policy of intelligent carrier shutdown
MOD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0,
PowerSavingType=INTER_GNB_MULTI_CARR_SD, SulUserNumThld=4, SulUserNumOffset=4;
//(Optional) Modifying the other energy saving policy of intelligent carrier shutdown
MOD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=1,
PowerSavingType=INTER_GNB_MULTI_CARR_SD, SulUserNumThld=4, SulUserNumOffset=4;
```

(Optional) Scenario where the number of UEs in an energy saving policy group needs to be checked for entering or exiting intelligent carrier shutdown

```
//(Optional) Performing the following configurations if the number of UEs in an energy saving policy
group needs to be checked for entering or exiting intelligent carrier shutdown
//For a capacity-layer cell
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;
ADD NRCELLPOLICYGRP: NrCellId=1, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=USER_NUM_CSD_EXIT_SW-1,
GrpUserNumThld=7, GrpUserNumOffset=4;
//For a basic-layer cell
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
ADD NRCELLPOLICYGRP: NrCellId=11, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE;
```

Optimization Command Examples

```
//Changing the length of the penalty timers applied to resource-related handovers to 30s when the
handover success rate decreases due to intelligent carrier shutdown
MOD GNBMOBILITYCOMMPARAM: NecHoPrepFailPunishTimer=30, OptHoPrepFailPunishTimer=30;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- **Intra-base-station multi-carrier scenario**

```
//(Optional) Not checking the number of UEs in an energy saving policy group for entering or exiting
intelligent carrier shutdown
MOD NRCELLPOLICYGRP: NrCellId=0, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=USER_NUM_CSD_EXIT_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;
//(Optional) Turning off the precise shutdown switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=MULTI_CARR_PRECISE_SHDN_SW-0&GRID_MODE_SW-0;
//Disabling intelligent carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingSwitch=INTRA_GNB_MULTI_CARR_SD_SW-0;
//Removing an energy saving policy of intelligent carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
//(Optional) Removing the other energy saving policy of intelligent carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1;
//For high-frequency TDD: Removing an energy saving policy of intelligent carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=2;
```

- **Inter-base-station multi-carrier scenario**

```
//(Optional) Not checking the number of UEs in an energy saving policy group for entering or exiting
intelligent carrier shutdown
MOD NRCELLPOLICYGRP: NrCellId=1, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=USER_NUM_CSD_EXIT_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;
//(Optional) Turning off the precise shutdown switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
PowerSavingExtSwitch=MULTI_CARR_PRECISE_SHDN_SW-0&GRID_MODE_SW-0;
//Disabling intelligent carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
PowerSavingSwitch=INTER_GNB_MULTI_CARR_SD_SW-0;
//Removing an energy saving policy of intelligent carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0;
//(Optional) Removing the other energy saving policy of intelligent carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=1;
//For high-frequency TDD: Removing an energy saving policy of intelligent carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=2;
```


6.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.2.4.2 Activation Verification

As activation verification for intelligent carrier shutdown in intra-base-station multi-carrier scenarios is similar to that in inter-base-station multi-carrier scenarios, the following uses the activation in the former scenarios as an example, which can be verified using either of the following methods:

- Method 1:
 - a. Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Switch** parameter is **Intra-gNodeB Multi-Carrier Shutdown Sw:On** in the command output, intelligent carrier shutdown has been enabled.
 - b. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **Intra-gNodeB Multi-Carrier Shutdown** and **Carrier Shutdown Mode** being **Load-based Carrier Shutdown**.

NOTE

If there are two or more records with **Power Saving Type** being **Intra-gNodeB Multi-Carrier Shutdown** and **Carrier Shutdown Mode** being **Load-based Carrier Shutdown**, intelligent carrier shutdown has been enabled in two or more time periods.

- c. Run the **DSP TIME** command. Check whether the system time is within the periods for this function to take effect.
 - If it is, go to **d**.
 - If it is not, wait until the start time arrives.
 - d. Wait for a certain period of time. Then, run the **DSP NRDUCELL** command. If the **Power Saving State** parameter indicates **Intra-gNodeB Multi-Carrier Shutdown:Enabled**, intelligent carrier shutdown has taken effect.
 - If the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter is set to **MINUTE_LEVEL**, the maximum wait time is equal to the sum of **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** and **NRDUCellPowerSaving.PwrSavingPenaltyTime**.
 - If the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter is set to **SECOND_LEVEL**, the maximum wait time is equal to the sum of **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** and **NRDUCellPowerSaving.PwrSavingPenaltyTime**.
- Method 2:

Run the **DSP NRDUCELLPOWERSAVING** command to query the status of intelligent carrier shutdown and the reason why it did not take effect.

- If the value of **Power Saving Type** is **Intra-gNodeB Multi-Carrier Shutdown** and the value of **Current State** is **Enabled**, intelligent carrier shutdown has taken effect.
- If the value of **Power Saving Type** is **Intra-gNodeB Multi-Carrier Shutdown** and the value of **Current State** is **Disabled**, check the **Current Disabled Cause**, **Latest Disabled Cause**, and **Latest Exit Cause** parameter values to find out the possible reason why intelligent carrier shutdown did not take effect. **Table 6-15** lists possible reasons. You can also check **Latest Exit Time** or **Latest Disabled Time** in the command output for the time the latest exit or the latest application failure occurred, respectively. As the displayed value of the **Current Disabled Cause** parameter is updated every minute, the value displayed in the command output may be the result less than 1 minute ago.

Latest Exit Time, **Latest Exit Cause**, **Latest Disabled Time**, and **Latest Disabled Cause** are used to query the time and reasons for the latest exit and the latest application failure within the specified period. The query may be performed outside the specified period. For example, if the specified period is from 00:00:00 to 06:00:00, **Latest Exit Cause** or **Latest Disabled Cause** indicates the reason for the latest exit or the latest application failure that occurred within the period from 00:00:00 to 06:00:00. This reason, together with the time the latest exit or application failure occurred, can be queried even by an operation performed within the time from 06:00:00 to 23:59:59.

Table 6-15 Possible reasons why intelligent carrier shutdown did not take effect

Reason	Description
Insufficient License	No license for this function is available.
Switch Turned Off	None

Reason	Description
Neighboring Cell Not Configured	- No inter-frequency neighboring cells are configured for the current serving cell. - Inter-frequency neighboring cells are configured for the current serving cell, but there is no neighboring cell for which Power Saving Cell Flag is set to COVERAGE. - In multi-operator sharing scenarios: - The PLMN IDs of the non-reserved operators of the capacity-layer cell are not a subset of the PLMN IDs of the non-reserved operators of any basic-layer cell. - The PLMN IDs of the reserved operators of the capacity-layer cell are not a subset of the PLMN IDs of the reserved and non-reserved operators of any basic-layer cell.
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable for other reasons.
Triggering Time Not Reached	The current time is not within the range specified for intelligent carrier shutdown.
Neighboring Cell Unavailable	- A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is unavailable. - A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is a multi-TRP cell such as a hyper cell or combined cell, and some TRPs of this cell are unavailable. - A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is an NSA cell.
Xn Interface Unavailable	In inter-base-station multi-carrier scenarios, the Xn interface is in the abnormal state.

Reason	Description
Impacted Feature Taken Effect	None
Downlink PRB Threshold Not Reached	None
Uplink PRB Threshold Not Reached	None
UE Number Threshold Not Reached	None
Policy Group UE Number Exceeding Threshold	None
UE Being Transferred or Transfer Failed	• The current serving cell meets the load conditions for carrier shutdown, and UEs are being transferred. • The current serving cell meets the load conditions for carrier shutdown, but some UEs fail to be transferred from the cell.
Triggering Decision Period Not Met	The current time is within the effective period specified for intelligent carrier shutdown and is not within the first detection period (lasting 1 minute or 5 seconds). However, it is within the evaluation period for triggering intelligent carrier shutdown (specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod or NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter).
In Penalty State	The cell is in the penalty state after it exits the intelligent carrier shutdown state or UEs fail to be transferred out of the cell. (The causes of UE transfer failures include UE capability mismatch and missing configuration of neighboring NR frequencies.)
RF Hardware Capability Not Supported	None
Basic Cell PRB Usage Limit Exceeded	None

Reason	Description
Parameter Inconsistency Between Cells	For high-frequency TDD: The parameters in the intelligent carrier shutdown policies for different cells are inconsistent.
Multi-Carrier Not Coordinated	For high-frequency TDD: Some carriers do not meet the conditions for intelligent carrier shutdown to take effect, in which case intelligent carrier shutdown cannot take effect for other carriers that meet the conditions.
Ultra-Long Holdover in Progress	None
Others	- The current time is within the effective period specified for intelligent carrier shutdown as well as the first detection period (lasting 1 minute or 5 seconds). - In the current period, the conditions for this feature to take effect are met, but some conditions for LTE and NR intelligent carrier shutdown are not met. - Internal software or other reasons cause the function to fail.

6.2.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after intelligent carrier shutdown is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.

- The duration of cell unavailability due to intelligent carrier shutdown can be obtained using the **N.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access.
- The duration in which the cell stays in the carrier shutdown state due to intelligent carrier shutdown can be obtained using the **N.PowerSaving.CarrierShutdown.Dur** counter on the MAE-Access.

6.3 Fast Carrier Shutdown

6.3.1 Principles

Fast carrier shutdown adds a mechanism for identifying the downlink traffic volume of UEs to the intelligent carrier shutdown function. **This mechanism mitigates the impacts on user experience while achieving energy saving effects. Specifically, when the network is under light or no load and a small amount of downlink traffic is required for UEs, the gNodeB transfers UEs from capacity-layer cells to basic-layer cells and shuts down the capacity-layer carriers; when the network is under heavy load or there are UEs with heavy downlink traffic, the carrier shutdown state stops.**

The following options must be selected to enable fast carrier shutdown:

- Capacity-layer cells:
 - **INTRA_GNB_MULTI_CARR_SD_SW** option (intra-base-station multi-carrier scenarios) or **INTER_GNB_MULTI_CARR_SD_SW** option (inter-base-station multi-carrier scenarios) of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter
 - **FAST_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter
- Basic-layer cells: **FAST_CARR_SHUT_SERV_ASSUR_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter

NOTE

The concepts of basic-layer cells and capacity-layer cells in fast carrier shutdown are the same as those in intelligent carrier shutdown. For details about these concepts, see [Related Concepts](#).

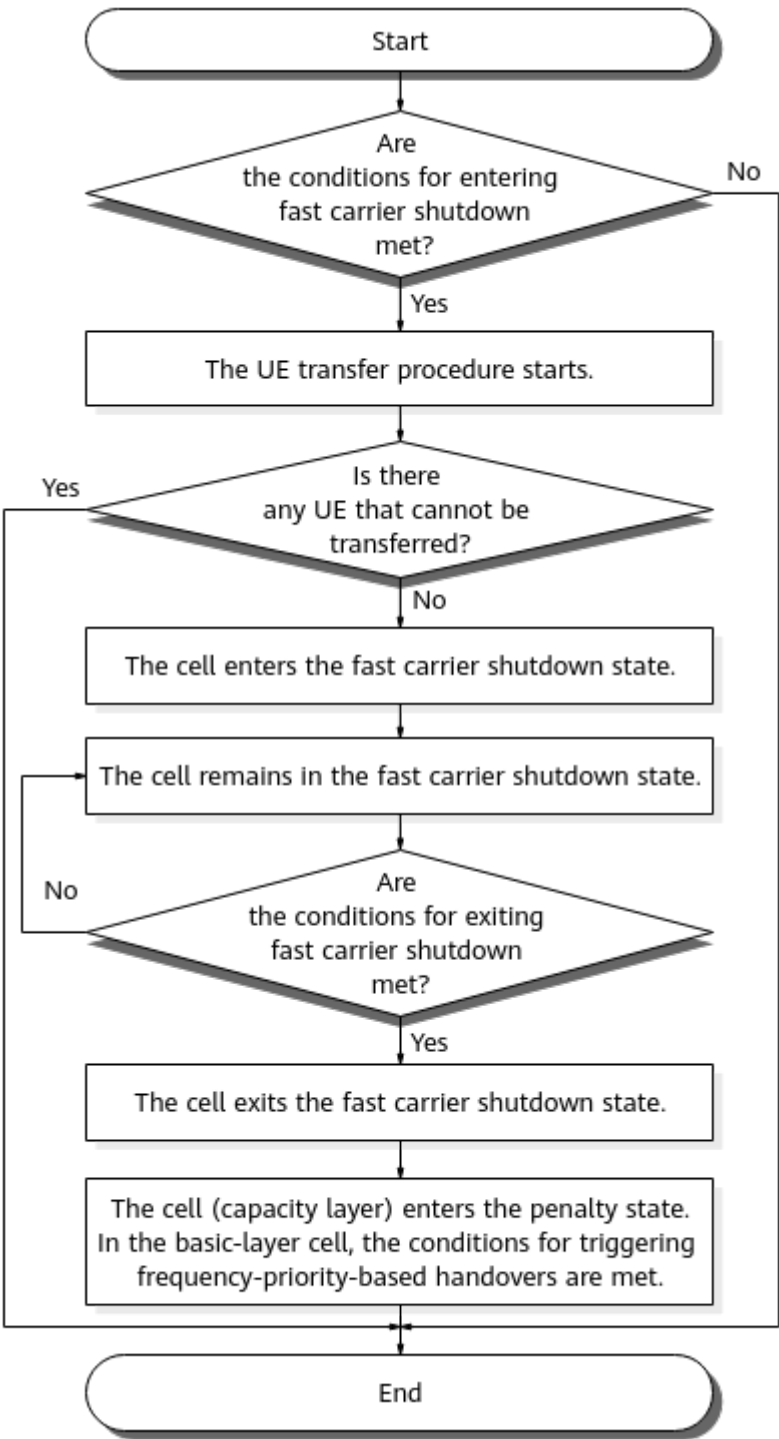
6.3.2 Energy Saving Procedure

This function takes effect every day within the configured fast carrier shutdown period. The time period, energy saving type, and other information under an energy saving policy are specified by the parameters in the **NRDUCellPowerSaving** MO.

- The **NRDUCellPowerSaving.PowerSavingPolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.
- The **NRDUCellPowerSaving.PowerSavingType** parameter specifies the energy saving type indicated by an energy saving policy index. Fast carrier shutdown requires that this parameter be set to **INTRA_GNB_MULTI_CARR_SD** (in intra-base-station multi-carrier scenarios) or **INTER_GNB_MULTI_CARR_SD** (in inter-base-station multi-carrier scenarios).
- The **NRDUCellPowerSaving.CarrierShutdownMode** parameter specifies the carrier shutdown mode indicated by an energy saving policy index. Fast carrier shutdown requires that this parameter be set to **FAST_CARR_SHUTDOWN**.

Figure 6-6 shows the procedure for fast carrier shutdown.

Figure 6-6 Fast carrier shutdown procedure



Conditions for Entering Fast Carrier Shutdown

The base station periodically checks whether a cell meets the entering conditions on load. If any of the conditions is not met, the base station waits for the next carrier shutdown evaluation period. If all the conditions are met, the base station

further checks whether the cell meets the entering conditions for user experience assurance.

The entering conditions on load listed below are the same as the conditions for entering non-precise intelligent carrier shutdown. However, in fast carrier shutdown, the evaluation period can only be set at minute-level granularity. That is, the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter must be set to **MINUTE_LEVEL**, and the evaluation period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter. The entering conditions on load are as follows:

- The capacity-layer cell must meet all of the following conditions:
 - The current time is within the fast carrier shutdown period specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
 - The capacity-layer cell is not in the state of penalty on fast carrier shutdown. For details about the penalty state, see [UE Transfer](#) and [Conditions for Exiting Fast Carrier Shutdown](#).
- The basic-layer cells corresponding to the capacity-layer cell must meet one of the following conditions:
 - All basic-layer cells are available.
 - Some basic-layer cells are available, and all the unavailable cells are in the energy saving state. An example of this case is that an unavailable cell acts as a capacity-layer cell for another basic-layer cell in an energy saving function, such as NR fast carrier shutdown, LTE and NR intelligent carrier shutdown, NR load-based RF module deep dormancy, or LNR load-based RF module dormancy.
- The capacity-layer cell and the corresponding basic-layer cells must meet all the following conditions:
 - The total downlink PRB usage of the capacity-layer cell and basic-layer cells is less than or equal to the value of the **NRDUCellPowerSaving.DlPrbThld** parameter. For the formula used to calculate the total downlink PRB usage of the capacity-layer cell and basic-layer cells, see [Table 6-16](#).
 - The total uplink PRB usage of the capacity-layer cell and basic-layer cells is less than or equal to the value of the **NRDUCellPowerSaving.UlPrbThld** parameter. For the formula used to calculate the total uplink PRB usage of the capacity-layer cell and basic-layer cells, see [Table 6-16](#).
 - The total number of RRC_CONNECTED UEs on each pair of the capacity-layer cell and a basic-layer cell is less than or equal to the value of the **NRDUCellPowerSaving.UserNumThld** parameter.
If multiple basic-layer cells correspond to one capacity-layer cell, each pair of the basic- and capacity-layer cells must meet this condition.
 - For each operator, the operator-specific total number of RRC_CONNECTED UEs on each pair of the capacity-layer cell and a basic-layer cell is less than or equal to the value of the **NRDUCellPowerSaving.OpUserNumThld** parameter for the relevant operator.
If multiple basic-layer cells correspond to one capacity-layer cell, each pair of the basic- and capacity-layer cells must meet this condition.

- In multi-operator sharing scenarios:
 - The non-reserved operators of the capacity-layer cell are a subset of the non-reserved operators of any basic-layer cell.
 - The reserved operators of the capacity-layer cell are a subset of the reserved and non-reserved operators of any basic-layer cell.

For each of the capacity-layer and basic-layer cells, operators are classified into non-reserved operators and reserved operators. A non-reserved operator is an operator with the **NRExternalNCellPlmn.PlmnReservedFlag** parameter set to **FALSE**, and a reserved operator is an operator with the **NRExternalNCellPlmn.PlmnReservedFlag** parameter set to **TRUE**.

- In SUL co-deployment scenarios, the total number of SUL UEs in the capacity-layer and basic-layer cells is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter.
If multiple basic-layer cells correspond to one capacity-layer cell, each pair of the basic- and capacity-layer cells must meet this condition.
- When all of the following conditions are met, the number of UEs in each energy saving policy group is checked against the following rule: Total number of UEs in the energy saving policy groups of capacity-layer and basic-layer cells $\leq$ Threshold specified by **NRCellPsPolicyGrp.GrpUserNumThld**.
 - The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.
 - For the capacity-layer cell, the **NRCellPsPolicyGrp** MO is configured, and the **USER_NUM_CSD_EXIT_SW** option of the **NRCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter corresponding to the **NRCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter is selected. The **NRCellPsPolicyGrp.UserIdentMethod** parameter is set to **SERVICE_TYPE**.

UEs in an energy saving policy group refer to those who have established QCI-specific bearers bound to this **NRCellPsPolicyGrp.PowerSavingPolicyGrpIndex**. QCIs are bound to energy saving policy groups through the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter. The following rules must also be followed for checking the number of UEs in an energy saving policy group:

- If multiple basic-layer cells correspond to one capacity-layer cell, the total number of UEs in energy saving groups of each pair of the basic- and capacity-layer cells must meet the threshold requirements.
- If the capacity-layer cell is configured with multiple energy saving policy groups with the **USER_NUM_CSD_EXIT_SW** option of the **NRCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter selected, the number of UEs in each energy saving policy group must meet the requirements.
- If the capacity-layer cell is configured with **NRCellPsPolicyGrp.PowerSavingPolicyGrpIndex** for which the

USER_NUM_CSD_EXIT_SW option of the **NRCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter is selected, all basic-layer cells corresponding to the capacity-layer cell must be configured. Otherwise, the number of UEs in a power saving policy group does not meet the requirements.

- In inter-base-station multi-carrier scenarios, the Xn interface is in the normal state.

 NOTE

If the capacity-layer cell is a hyper cell or combined cell:

- The total downlink PRB usage and total uplink PRB usage of the capacity-layer cell and basic-layer cells are calculated based on the maximum number of used PRBs among all TRPs. When fast carrier shutdown takes effect, all TRPs are shut down.
- It is recommended that basic-layer cells also be hyper cells or combined cells. Common cells are not recommended as basic-layer cells in this case.

The cell bandwidth, slot configuration, and subcarrier spacing are converted for evaluation of the total uplink or downlink PRB usage of the capacity- and basic-layer cells against the configured uplink or downlink PRB threshold, respectively.

When the preceding entering conditions on load are met, the base station checks against the entering conditions for user experience assurance within a period of 5s as follows:

- If the downlink PRB usage of the capacity-layer cell and that of each basic-layer cell are less than the PRB usage threshold for fast evaluation on carrier shutdown (specified by the **NRDUCellMobility.CarrShutdFastEvalPrbThld** parameter), and none of these cells serves UEs with heavy downlink traffic, a UE transfer procedure starts in the capacity-layer cell.

For the formulas used to calculate the downlink PRB usage of the capacity-layer cell and that of each basic-layer cell, see [Table 6-16](#).

- If the downlink PRB usage of the capacity-layer cell or that of any basic-layer cell is greater than or equal to the PRB usage threshold for fast evaluation on carrier shutdown, whether a UE transfer procedure starts depends on the setting of the **NRDUCellPowerSaving.CarrShutdServiceAssurLevel** parameter.

The value **LEVEL0** indicates the lowest level of service assurance, at which the probability of entering the carrier shutdown state is the highest. The value **LEVEL4** indicates the highest level of service assurance, at which services are almost not affected. A larger value of this parameter results in smaller impact on user experience and fewer energy saving gains.

Table 6-16 Formulas for calculating PRB statistics

PRB Usage	Formula
Total downlink PRB usage of the capacity-layer cell and basic-layer cells	(Number of PRBs used by PDSCH DRBs of the capacity-layer cell (N.PR.B.DL.DrbUsed.Avg – N.PR.B.DL.ExtendUsed.Avg) + Total number of used downlink PRBs of the basic-layer cells (N.PR.B.DL.Used.Avg – N.PR.B.DL.ExtendUsed.Avg)) ÷ Total number of available downlink PRBs of the basic-layer cells (N.PR.B.DL.Actual.Avail.Avg)
Total uplink PRB usage of the capacity-layer cell and basic-layer cells	(Number of used uplink PRBs of the capacity-layer cell (N.PR.B.PUSCH.Used.Avg) + Total number of used uplink PRBs of the basic-layer cells (N.PR.B.PUSCH.Used.Avg)) ÷ Total number of available uplink PRBs of the basic-layer cells (N.PR.B.UL.PUSCH.Actual.Avail.Avg)
Downlink PRB usage of the capacity-layer cell	Number of PRBs used by PDSCH DRBs of the capacity-layer cell (N.PR.B.DL.DrbUsed.Avg – N.PR.B.DL.ExtendUsed.Avg) ÷ Number of available downlink PRBs of the capacity-layer cell (N.PR.B.DL.Actual.Avail.Avg)
Downlink PRB usage of each basic-layer cell	Number of PRBs used by PDSCH DRBs of a basic-layer cell (N.PR.B.DL.DrbUsed.Avg – N.PR.B.DL.ExtendUsed.Avg) ÷ Number of available downlink PRBs of the basic-layer cell (N.PR.B.DL.Actual.Avail.Avg)
Note: The preceding formulas can be used to estimate the PRB usages. The actual PRB usages are automatically calculated by the base station and differ to some extent from the estimated values.	

UE Transfer

UE transfer from the capacity-layer cell varies as follows:

- In SA networking, SA UEs are transferred as follows:
 - Non-CA UEs
 - UEs not performing voice services are transferred to basic-layer cells through inter-frequency handovers or to LTE cells through inter-RAT

handovers. If the handovers fail, the UEs are transferred to inter-frequency basic-layer cells or LTE cells through blind redirections.

- UEs performing voice services are transferred to basic-layer cells through inter-frequency handovers or to LTE cells through inter-RAT handovers.
- CA UEs
 - For CA UEs that treat the capacity-layer cell as their SCell, the SCell is removed.
 - For CA UEs that treat the capacity-layer cell as their PCell, they are transferred in the same way as non-CA UEs.

If UEs cannot be transferred out of the capacity-layer cell 13s after the UE transfer procedure starts (for example, they do not support the operating band of the target basic-layer cell or the handover fails), the fast carrier shutdown procedure stops and cannot be triggered for the capacity-layer cell again within a period specified by the

NRDUCellPowerSaving.PwrSavingPenaltyTime parameter, in which the capacity-layer cell is in the penalty state.

In SA multi-base-station scenarios, new UE access to the capacity-layer NR cell is prohibited during UE transfer. If a UE attempts to be handed over from another cell to this cell, the handover fails. Upon the failure, a penalty is imposed on all UEs in the serving cell of that UE. No handovers of these UEs to the capacity-layer NR cell are allowed within a period that is specified by the **gNBMobilityCommParam.NecHoPrepFailPunishTimer** or **gNBMobilityCommParam.OptHoPrepFailPunishTimer** parameter.

- In NSA networking, NSA UE connections on the NR side are released as follows:
 - If fast carrier shutdown is to take effect in the PSCell of a UE, all connections of the UE on the NR side are released.
 - If fast carrier shutdown is to take effect in an SCell of a UE, its connection to this SCell on the NR side is released but its other connections on the NR side are retained.

In NSA networking, the **PWR_SAVE_SCENARIO_NR_DELAY_SW** option of the **ENodebAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter must be selected. This is because shutting down the NR capacity-layer cell for energy saving may cause SCG releases. When this happens, with this option selected, the eNodeB does not send an SCG addition request to the NR capacity-layer cell for its served UEs within the period specified by **CellUSParaCfg.NsUeNrDelayTimer**.

NOTE

If the RFSP-based handover policy conflicts with the handover policy related to fast carrier shutdown, only the former takes effect.

Entering Fast Carrier Shutdown

Once all UEs are transferred out of the capacity-layer cell, no RRC_CONNECTED UEs remain in the cell, and the cell enters the fast carrier shutdown state. If the PA (of an RF module) serving the shut-down carrier is not serving any other operating carrier, the PA will also be shut down.

Conditions for Exiting Fast Carrier Shutdown

After the capacity-layer cell enters the fast carrier shutdown state, the base station periodically evaluates the conditions for exiting this state. The capacity-layer cell automatically exits the fast carrier shutdown state and cannot enter this state again within a period specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter if any of the conditions listed below is met. (The cell is in the penalty state during this period.) If none of these conditions is met, the cell remains in the fast carrier shutdown state.

- Exiting conditions on load: The base station checks whether any of the following conditions is met throughout a period specified by the **NRDUCellPowerSaving.ExitEvaluationPeriod** parameter. These conditions are the same as the conditions for exiting non-precise intelligent carrier shutdown.
 - The current time is not within the fast carrier shutdown period specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
 - The total downlink PRB usage of the basic-layer cells is greater than the sum of **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset**.
If the sum of **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset** is greater than 100 for a basic-layer cell, the cell cannot exit the fast carrier shutdown state.
 - The total uplink PRB usage of the basic-layer cells is greater than the sum of **NRDUCellPowerSaving.UlPrbThld** and **NRDUCellPowerSaving.NCellUlPrbOffset**.
If the sum of **NRDUCellPowerSaving.UlPrbThld** and **NRDUCellPowerSaving.NCellUlPrbOffset** is greater than 100 for a basic-layer cell, the cell cannot exit the fast carrier shutdown state.
 - The number of RRC_CONNECTED UEs in a basic-layer cell is greater than the sum of **NRDUCellPowerSaving.UserNumThld** and **NRDUCellPowerSaving.UserNumOffset**.
If multiple basic-layer cells correspond to one capacity-layer cell, only one basic-layer cell needs to meet this condition.
 - The operator-specific number of RRC_CONNECTED UEs in a basic-layer cell is greater than the sum of **NRDUCellPowerSaving.OpUserNumThld** and **NRDUCellPowerSaving.OpUserNumOffset**.
If multiple basic-layer cells correspond to one capacity-layer cell, only one basic-layer cell needs to meet this condition.
 - In multi-operator sharing scenarios:
 - The non-reserved operators of the capacity-layer cell are not a subset of the non-reserved operators of any basic-layer cell.
 - The reserved operators of the capacity-layer cell are not a subset of the reserved and non-reserved operators of any basic-layer cell.
 - In SUL co-deployment scenarios, the number of SUL UEs in a basic-layer cell is greater than the sum of the

NRDUCellPowerSaving.SulUserNumThld and **NRDUCellPowerSaving.SulUserNumOffset** parameter values.

If multiple basic-layer cells correspond to one capacity-layer cell, only one basic-layer cell needs to meet this condition.

- When all of the following conditions are met, the number of UEs in each energy saving policy group is checked against the following rule: Number of UEs in a power saving policy group of a basic-layer cell > Threshold specified by **NRCellPsPolicyGrp.GrpUserNumThld** + Offset specified by **NRCellPsPolicyGrp.GrpUserNumOffset**.
 - The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.
 - For the capacity-layer cell, the **NRCellPsPolicyGrp** MO is configured, and the **USER_NUM_CSD_EXIT_SW** option of the **NRCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter corresponding to the **NRCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter is selected. The **NRCellPsPolicyGrp.UserIdentMethod** parameter is set to **SERVICE_TYPE**.

UEs in an energy saving policy group refer to those who have established QCI-specific bearers bound to this

NRCellPsPolicyGrp.PowerSavingPolicyGrpIndex. QCIs are bound to energy saving policy groups through the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter. The following rules must also be followed for checking the number of UEs in an energy saving policy group:

- If multiple basic-layer cells correspond to one capacity-layer cell, the number of UEs in the energy saving policy group of only one basic-layer cell needs to meet the requirements.
- If the capacity-layer cell is configured with multiple energy saving policy groups with the **USER_NUM_CSD_EXIT_SW** option of the **NRCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter selected, the number of UEs in one of the energy saving policy groups needs to meet the requirements.
- In inter-base-station multi-carrier scenarios, the Xn interface is in the abnormal state.
- Any of the basic-layer cells becomes unavailable for a reason other than fast carrier shutdown, LTE and NR intelligent carrier shutdown, and NR load-based RF module deep dormancy.
- The downlink PRB usage of each basic-layer cell is less than the PRB usage threshold for fast evaluation on carrier shutdown (specified by the **NRDUCellMobility.CarrShutdFastEvalPrbThld** parameter), and UEs with heavy downlink traffic are newly admitted to any of the basic-layer cells.
- Exiting conditions for user experience assurance: The base station checks whether the following conditions are met throughout a period of 5s:
 - If the downlink PRB usage of each basic-layer cell is less than the PRB usage threshold for fast evaluation on carrier shutdown (specified by the **NRDUCellMobility.CarrShutdFastEvalPrbThld** parameter) and any basic-layer cell serves UEs with heavy downlink traffic, the capacity-layer

- cell exits the fast carrier shutdown state. After that, a frequency-priority-based handover is triggered. For details about the triggering, see [Triggering Frequency-Priority-based Handovers](#).
- If the downlink PRB usage of any basic-layer cell is greater than or equal to the PRB usage threshold for fast evaluation on carrier shutdown, whether the capacity-layer cell exits the fast carrier shutdown state depends on the setting of the **NRDUCellPowerSaving.CarrShutdServiceAssurLevel** parameter, which specifies the level of service assurance. The higher the level, the smaller the impact on user experience and the lower the energy saving gains.
 - When the fast carrier shutdown load event evaluation function (controlled by the **CARR_SHDN_LOAD_EVENT_EVAL_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter) is enabled, the downlink PRB usage of a basic-layer cell is evaluated per second. If the downlink PRB usage of any basic-layer cell is greater than the threshold specified by the **NRDUCellMobility.CarrShdnEventEvalPrbThld** parameter in the current period but does not exceed this threshold in the previous period, the capacity-layer cell exits the fast carrier shutdown state. The fast carrier shutdown load event evaluation function takes effect only when the **NRDUCellMobility.CarrShdnEventEvalPrbThld** parameter is set to a value other than 255. A smaller value of this parameter results in a smaller impact on user experience and fewer energy saving gains.

NOTE

- When a basic-layer cell becomes unavailable in inter-base-station scenarios, there may be a minute-level delay for the wake-up of the capacity-layer cell. This is because the messages need to be frequently transmitted over the Xn interface to indicate the energy saving status and capacity-layer cell wake-up notifications, leading to possible flow control.
- If a basic-layer cell is a multi-TRP cell such as a hyper cell or combined cell and some of its TRP carriers become unavailable, the capacity-layer cell also automatically exits the fast carrier shutdown state.
- The penalty is not imposed if the capacity-layer cell exits the fast carrier shutdown state because the function switch is turned off, the cell is deactivated and then reactivated, or the base station software version is upgraded. After the capacity-layer cell is deactivated and then reactivated or the base station software version upgrade is complete, the base station evaluates whether the cell enters the fast carrier shutdown state again based on related entering conditions.
- The capacity-layer cell may exit the fast carrier shutdown state because the exit conditions are met during evaluation of basic-layer neighboring cell addition for the capacity-layer cell.
- With the fast carrier shutdown load event evaluation function, if a basic-layer cell is a hyper cell or combined cell and any of its TRP meets the downlink PRB usage requirement, the capacity-layer cell exits the carrier shutdown state.

Triggering Frequency-Priority-based Handovers

After the capacity-layer cell exits the fast carrier shutdown state, a procedure is immediately triggered for each UE with heavy downlink traffic in the basic-layer cells to evaluate whether a frequency-priority-based handover can be performed, as long as this type of handover is enabled in the basic-layer cells. This enables such UEs to be transferred back to the capacity-layer cell promptly.

The time-to-trigger in this procedure is specified by the **NRCCellMobilityConfig.CarrShutUserMeasTimeToTrig** parameter. For details

about frequency-priority-based handover, see the section on frequency-priority-based inter-frequency handover in *SA Mobility Management in Connected Mode* and the section on frequency-priority-based inter-frequency PSCell change in *NSA Mobility Management*.

Frequency-priority-based handover will not be triggered in a basic-layer cell with this type of handover disabled.

6.3.3 Network Analysis

6.3.3.1 Benefits

Fast carrier shutdown reduces energy consumption when the network is under light load. Based on intelligent carrier shutdown, this function adds a downlink traffic volume identification mechanism whereby exit from carrier shutdown is triggered in the presence of UEs with heavy downlink traffic. As such, this function is recommended when intelligent carrier shutdown is not enabled. Impact on user experience may be mitigated but energy saving gains may decrease if switches for fast carrier shutdown are turned on (by selecting **FAST_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter for a capacity-layer cell and the **FAST_CARR_SHUT_SERV_ASSUR_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter for a basic-layer cell) when intelligent carrier shutdown has been enabled (by selecting the **INTRA_GNB_MULTI_CARR_SD_SW** or **INTER_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter for a capacity-layer cell). The energy saving gains are limited in the following scenarios:

- When multiple carriers share the same PA of an RF module, the PA cannot be shut down if an NR carrier is shut down while the other carriers are still working. Energy saving gains decrease in this case.
- To prevent possible hardware faults caused by an excessively large temperature differential, this function may not take effect or may be terminated when the temperature differential of RF modules is excessively large. Energy saving gains may decrease in this case.

6.3.3.2 Impacts

The impacts between this function and other functions are described in [6.3.4.2.3 Function Impacts](#).

Fast carrier shutdown affects certain performance indicators. As whether this function takes effect depends on UE distribution and downlink traffic volume of UEs, changes in these factors will lead to changes in the impacts on performance indicators. The impacts are as follows:

- When fast carrier shutdown takes effect, the values of counters related to inter-frequency handovers and redirections change.
- In SA networking, if a UE attempts to be handed over to a capacity-layer cell whose carrier is to be shut down when a UE transfer process (which lasts 30s at most) is ongoing for this cell, the source cell does not initiate the handover. In this case, the values of counters related to outgoing and incoming handover attempts increase in the source cell and capacity-layer cell, respectively. However, the values of counters related to handover executions

and successful handovers do not increase. As Huawei-defined KPIs **Inter-Frequency Handover Out Success Rate (CU)** and **Intra-RAT Handover In Success Rate (CU)** are calculated using the number of outgoing/incoming handover executions as the denominator, these KPIs are not affected in this scenario. However, if the user-defined KPIs are calculated using the number of outgoing/incoming handover attempts as the denominator, these KPIs may be slightly affected in this scenario.

- In LTE and NR NSA DC networking, the eNodeB may send an SCG addition request to a capacity-layer cell whose carrier is to be shut down when a UE transfer process is ongoing for this cell. SCG addition will fail at the first attempt. As a result, the SCG addition success rate (equal to **L.NsaDc.SgNB.Add.Succ/L.NsaDc.SgNB.Add.Att**) decreases.
- In SA multi-base-station scenarios, new UE access to a capacity-layer NR cell to be shut down is prohibited during UE transfer in the cell. If UEs attempt to be handed over from other cells to this cell, the handover preparations fail. As a result, the handover preparation success rate decreases. Typically, the amount of the decrease is about 0% to 4% during the effective period of fast carrier shutdown. The smaller the number of SA UEs, the greater the change.

Handover preparation success rate =

$$\frac{(\text{N.HO.IntraFreq.Ng.IntergNB.ExecSuccOut} + \text{N.HO.InterFreq.Ng.IntergNB.ExecSuccOut} + \text{N.HO.IntraFreq.Xn.IntergNB.ExecSuccOut} + \text{N.HO.InterFreq.Xn.IntergNB.ExecSuccOut} + \text{N.HO.IntraFreq.IntragNB.ExecSuccOut} + \text{N.HO.InterFreq.IntragNB.ExecSuccOut})}{(\text{N.HO.IntraFreq.Ng.IntergNB.PrepAttOut} + \text{N.HO.InterFreq.Ng.IntergNB.PrepAttOut} + \text{N.HO.IntraFreq.Xn.IntergNB.PrepAttOut} + \text{N.HO.InterFreq.Xn.IntergNB.PrepAttOut} + \text{N.HO.IntraFreq.IntragNB.ExecAttOut} + \text{N.HO.InterFreq.IntragNB.ExecAttOut})} \times 100\%$$

- During carrier shutdown for a cell, the number of inter-frequency handover attempts (indicated by **N.HO.InterFreq.IntragNB.PrepAttOut**, **N.HO.InterFreq.Xn.IntergNB.PrepAttOut**, and **N.HO.InterFreq.FreqPri.PrepAttOut**) and the number of incoming handover rejections (indicated by **N.HO.InterFreq.FailOut.PrepFailure**) increase.
- If a cell is shut down but its adjacent cells are still working, the adjacent cells will cover larger areas, which may lead to larger areas under weak coverage. In this case, the **Intra-Frequency Handover Out Success Rate (CU)** and **Inter-Frequency Handover Out Success Rate (CU)** slightly decrease, and the **Service Call Drop Rate (CU)** and **Call Drop Rate (CU, VoNR)** slightly increase.
- During carrier shutdown for an NR cell, if UEs fail to be handed over to target cells, the values of counters related to the number of intra-NR handover failures (such as **N.HO.FailIn.HOCancel**, **N.HO.FailIn.AdmitFail**, **N.HO.FailIn.AdmitFail.TNL**, **N.HO.FailIn.AdmitFail.UserLic**, and **N.HO.FailIn.AdmitFail.Load**) slightly increase.
- UEs that do not support the operating frequencies of basic-layer cells will camp on LTE cells. As a result, the number of 5G-served UEs (indicated by **N.User.RRConn.Avg**) and 5G traffic volume (indicated by **N.ThpVol.DL**) slightly decrease.

- As basic-layer cells have poorer air interface performance than capacity-layer cells, when fast carrier shutdown takes effect, the overall **User Downlink Average Throughput (DU)** in capacity-layer cells and basic-layer cells decreases, the first packet delay changes, and the delay at the RLC layer increases.
 - First packet delay = $\frac{N.Traffic.DL.RlcFirstPktDelay.Time}{N.Qos.DL.RlcFirstPktDelay.Num}$
 - Delay at the RLC layer = $\frac{N.QoS.DL.PktDelayAirInterface.Time}{N.QoS.DL.PktDelayAirInterface.Num} + \frac{N.QoS.DL.PktDelaygNBDU.Time}{N.QoS.DL.PktDelaygNBDU.Num}$
- When fast carrier shutdown takes effect, UE distribution changes, which has the following impacts:
 - The basic layer serves more UEs and has heavier traffic. For a basic-layer cell, the values of the following indicators increase: **Average CPU Load** of the main control board, **Cell Downlink Average Throughput (DU)**, **Cell Uplink Average Throughput (DU)**, **Downlink Resource Block Utilizing Rate (DU)**, and **Uplink Resource Block Utilizing Rate (DU)**.
 - The capacity layer serves fewer UEs and has lighter traffic. For a capacity-layer cell, the values of the following indicators decrease: **Average CPU Load** of the main control board, **Cell Downlink Average Throughput (DU)**, **Cell Uplink Average Throughput (DU)**, **Downlink Resource Block Utilizing Rate (DU)**, and **Uplink Resource Block Utilizing Rate (DU)**.
 - The uplink RBLER and uplink retransmission rate may increase.

$$\text{Uplink RBLER} = \frac{(N.UL.SCH.HalfPiBPSK.ErrTB.Rbler + N.UL.SCH.QPSK.ErrTB.Rbler + N.UL.SCH.16QAM.ErrTB.Rbler + N.UL.SCH.64QAM.ErrTB.Rbler)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)} \times 100\%$$

$$\text{Uplink retransmission rate} = \frac{(N.UL.SCH.HalfPiBPSK.TB.Retrans + N.UL.SCH.QPSK.TB.Retrans + N.UL.SCH.16QAM.TB.Retrans + N.UL.SCH.64QAM.TB.Retrans + N.UL.SCH.256QAM.TB.Retrans)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)} \times 100\%$$
- Assuming that the neighboring base stations are in the carrier shutdown state, cell-edge UEs in weak coverage areas may access the cells served by the local base station. This affects KPIs related to the aspects such as access and service drops of the local base station. For example, the value of the **N.NsaDc.DRB.AbnormRel** counter may increase.

6.3.4 Requirements

6.3.4.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-081250	Fast Carrier Shutdown	NR0S00FSSD00	per Cell

In addition, the following licenses are required.

- 3900 and 5900 series base stations

Feature ID	Feature Name	Model	Sales Unit
FOFD-050203	Intelligent Carrier Shutdown	NR0S00ICSC00	per Cell

- DBS3900 LampSite and DBS5900 LampSite

Feature ID	Feature Name	Model	Sales Unit
FOFD-050203	Intelligent Carrier Shutdown	NR0SSHUTDW00	per Carrier per pRRU

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.3.4.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.4.2.1 Prerequisite Functions

None

6.3.4.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The period specified for fast carrier shutdown in a cell cannot overlap that specified for non-precise intelligent carrier shutdown in the same cell.
Low-frequency TDD	Precise shutdown mode	MULTI_CARR_PRECISE_SHDN_SW option of the NRDUCellAI goSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Fast carrier shutdown and precise intelligent carrier shutdown cannot be both enabled.
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The period specified for fast carrier shutdown in a cell cannot overlap that specified for timing carrier shutdown in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.
Low-frequency TDD	Multi-carrier joint carrier shutdown	MULTI_CARR_JOINT_CARR_SHDN_SW option of the NRDUCellAI goSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i> <i>Multi-RAT Coordinated Energy Saving</i>	Multi-carrier joint carrier shutdown does not work with fast carrier shutdown.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-frequency deep carrier shutdown	MULTI_FREQ_DEEP_CARR_SHDN_SW option of the NRDUCellAIgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Multi-frequency deep carrier shutdown does not work with fast carrier shutdown.
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAIgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Fast carrier shutdown cannot take effect when the low latency and high reliability function is enabled.
Low-frequency TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Fast carrier shutdown cannot be enabled if the primary/backup function in intra-frequency dual-network redundancy networking has been enabled.
Low-frequency TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Fast carrier shutdown cannot be enabled if dual-BBU backup in dual-network redundancy networking has been enabled.
Low-frequency TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	Fast carrier shutdown is incompatible with the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic MIMO layer number decrease	DYNAMIC_MIMO_LAYER_DEC_SW option of the gNodeBAlgo.DedicatedNetworkFuncSw parameter	None	In 8T8R scenarios where the NRDUCell.NrDualNetworkMode parameter is set to NORMAL_CELL , dynamic MIMO layer number decrease does not work with fast carrier shutdown.

6.3.4.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CarAlgoSwitch parameter	<i>Carrier Aggregation</i>	After fast carrier shutdown takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CarAlgoSwitch parameter	<i>Carrier Aggregation</i>	After fast carrier shutdown takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.
Low-frequency TDD	Inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CarAlgoSwitch parameter	<i>CA Experience Improvement</i>	After fast carrier shutdown takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Symbol power saving	NRDUCellAIgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	When fast carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of symbol power saving is not affected.
Low-frequency TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When fast carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, deep symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of deep symbol power saving is not affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When fast carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, power saving scheduling does not take effect. • If the RF module is serving other NR carriers, the status of power saving scheduling is not affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When fast carrier shutdown takes effect: • If the local cell is a capacity-layer cell, RF channel intelligent shutdown cannot take effect in the scenarios listed in Table 5-6. • If the local cell is a basic-layer cell and RF channel intelligent shutdown takes effect in the scenarios listed in Table 5-6, the related capacity-layer cells may exit the fast carrier shutdown state.
Low-frequency TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both fast carrier shutdown and energy saving based on flexible frequency-domain scheduling take effect, the latter function does not deliver energy saving gains.
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	TTI-level channel shutdown does not take effect when fast carrier shutdown is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low-power RF channel dynamic shutdown	LOW_POWER_RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Low-power RF channel dynamic shutdown does not take effect when fast carrier shutdown is in effect.
Low-frequency TDD	Intelligent power control energy saving	SMART_POWER_CTRL_ENERGY_SAVE_SWITCH option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both fast carrier shutdown and intelligent power control energy saving take effect, the latter function does not deliver energy saving gains.
Low-frequency TDD	Initial transmission power reduction	INIT_TRANSMISSION_POWER_REDUCE_SW option of the NRDUCellPowerSavingAlgorithmPowerSavingAlgorithmSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both initial transmission power reduction and fast carrier shutdown take effect, the former function does not deliver any gains.
Low-frequency TDD	Retransmission power reduction	RETRANSMISSION_POWER_REDUCE_SW option of the NRDUCellPowerSavingAlgorithmPowerSavingAlgorithmSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both retransmission power reduction and fast carrier shutdown take effect, the former function does not deliver any gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR coordinated symbol power saving	MULTI_RAT_SYMBOL_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When fast carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, LTE and NR coordinated symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of LTE and NR coordinated symbol power saving is not affected.
Low-frequency TDD	LTE and NR bundled scheduling	LNR_BUNDLED_SCHEDULING_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When fast carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, LTE and NR bundled scheduling does not take effect. • If the RF module is serving other NR carriers, the status of LTE and NR bundled scheduling is not affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Fast carrier shutdown does not take effect when RF module deep dormancy is in effect.
Low-frequency TDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Fast carrier shutdown does not take effect when RF module super dormancy is in effect.
Low-frequency TDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When both fast carrier shutdown and LTE and NR intelligent carrier shutdown are enabled for a cell, this cell enters the carrier shutdown state only if the entering conditions for both functions are met.
Low-frequency TDD	Cell load simulation	STR_NRDUCCELLSIMULATELOAD	None	Cell load simulation does not take effect when it is enabled together with fast carrier shutdown.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Automatic removal of neighboring cells	NR_NR_ANR_DEL_SW option of the NRCellAlgoSwitch . <i>AnrSwitch</i> parameter	<i>ANR</i>	When automatic removal of neighboring cells in ANR takes effect, fast carrier shutdown does not take effect if the neighbor relationships with the basic-layer cells are automatically removed for capacity-layer cells.
Low-frequency TDD	CMAS	gNBSibConfig . <i>SibType</i> set to SIB_TYPE_8	<i>CMAS</i>	When an NR carrier is shut down, the corresponding NR cell is blocked and CMAS warning notifications cannot be properly sent in this cell. However, UEs can still receive CMAS warning notifications from other carriers.
Low-frequency TDD	Super uplink with SUL and NR FDD co-deployment	FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch . <i>SuperUplinkSwitch</i> parameter	<i>Super Uplink</i>	After an NR FDD carrier is shut down, SUL scheduling cannot be performed for super uplink UEs.
Low-frequency TDD	UL and DL Decoupling	NRDUCellAlgoSwitch . <i>ULDLDecouplingSwitch</i>	<i>UL and DL Decoupling</i>	After an NR FDD carrier is shut down, SUL scheduling cannot be performed for SUL UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-cell cable sequence detection	AntPortOptDetPolicy parameter in the STR ANTENNAP ORTOPTDET command set to INTER_CELL_DETECT	<i>MIMO (TDD)</i>	If fast carrier shutdown takes effect during inter-cell cable sequence detection, the detection results will show that the detection fails in the corresponding cells.
Low-frequency TDD	Basic VoNR functions	VONR_SW option of the NRCellAlgoSwitch . <i>VonrSwitch</i> parameter	<i>VoNR</i>	Fast carrier shutdown does not take effect when there are VoNR UEs that cannot be handed over from the capacity-layer cell during a UE transfer procedure.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Virtual Grid-based Multi-Frequency Coordination	• RSRP prediction model – VG_MODEL_ALLOW_BUILD_FLAG option of the NRCellFreqRelation. <i>Aggregation Attribute</i> parameter – NR_IF_HO_MEAS_QTY_PRED_FUNCTION_SWITCH option of the NRCellSmartMultiCarrierAlgorithmSwitch parameter • Downlink intra-frequency spectral efficiency prediction model – DL_INTERF_S	<i>Virtual Grid-based Multi-Frequency Coordination</i>	After fast carrier shutdown takes effect, the building of virtual grid models will be suspended and existing virtual grid models will be brought offline and cannot be used.

RAT	Function Name	Function Switch	Reference	Description
		E_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarrier.ModelBuildingSwitch parameter - DL_FREQUENCY_SPECT_EFFECT_PRED_SW option of the NRCellSmartMultiCarrier.NrMultiCarrierAlgorithmSwitch parameter • Intra-frequency SRS beam prediction model INTRA_FREQUENCY_SRS_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarrier.Model		

RAT	Function Name	Function Switch	Reference	Description
		Buildingswitch parameter		
Low-frequency TDD	Channel calibration	Macro base station: None LampSite base station: INTER_RRU_CHN_CALIB_SW option of the NRDUCellAlgoSwitch.ChnCalibSwitch parameter	None	If channel calibration is being performed when fast carrier shutdown takes effect, channel calibration fails and will not be started again. You can run the DSP NRDUCELLCHNCALIB command to check whether channel calibration is successful.
Low-frequency TDD	RB allocation for downlink experience optimization	NRDUCellDLISchRes.DLExpOptRbIncreasePct	<i>Downlink Scheduling</i>	When both RB allocation for downlink experience optimization and fast carrier shutdown take effect, an exit from fast carrier shutdown may be performed earlier than expected.
Low-frequency TDD	Cell-level MRO	MRO_SCENARIO_IDENT_SW option of the gNBMrScenarioIdentifySwitch parameter and the INTRA_FREQ_MRO_SW option of the NRCellMr.MroSwitch parameter	<i>MRO</i>	When both fast carrier shutdown and intra-frequency MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-level MRO	UE_MRO_SWITCH option of the NRCellMro.MroSwitch parameter	<i>MRO</i>	When both fast carrier shutdown and UE-level MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO may be affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAIgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	(TDD) The impacts are as follows: • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResS hrCfg.DynPwr ShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of fast carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResS hrCfg.DynPwr ShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control

RAT	Function Name	Function Switch	Reference	Description
				information changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SWITCH option of the NRDUCellAdaptivePowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	A carrier exits a power adaptive scheduling group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	• LTE: LTE_NR_DYN_POWER_SHARING_SW option of the CellDynamicPowerSharing.DynPowerSharingSwitch parameter • NR: LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSbrCfg.DynPwrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of fast carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResSbrCfg.DynPwrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter

RAT	Function Name	Function Switch	Reference	Description
				adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	L/NR Inter-band Deep Coordination	• LTE: LNR_INTE R_BAND_DEEP_COORD_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter • NR: LNR_INTE R_BAND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPowerShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPowerShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of fast carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPowerShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter

RAT	Function Name	Function Switch	Reference	Description
				adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.
Low-frequency TDD	Inter-RAT redirection back to the HPLMN based on roaming UE identification	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	When fast carrier shutdown is enabled, only blind redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 5G network.
Low-frequency TDD	Clock out-of-synchronization detection	CLK_DETECT_SW option of the gNodeBParam.ClkOutofSyncDetSwitch parameter	<i>Synchronization</i>	When fast carrier shutdown is in effect, any instance of clock out-of-synchronization may fail to be detected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Ultra-long holdover for sites with abnormal clock sources	CLK_OTASY NC_PROC_SW option of the TASM.MSCLKDIAGPROC ESSSW parameter	<i>Synchronization</i>	Assume that the Ultra-Long Hold Switch is turned on but the Ultra-Long Hold Enhancement Switch is turned off. If sequence detection is performed on a to-be-detected base station during the period when fast carrier shutdown takes effect for all cells of the base station, the sequence detection fails. In this case, a possible clock fault may cause the base station to be unable to enter the ultra-long holdover state, or to exit the ultra-long holdover state if sequence detection cannot be performed for a long time due to energy conservation. If the Ultra-Long Hold Enhancement Switch is turned on, fast carrier shutdown does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	Assume that voice-capability-based inter-frequency handover back to the HPLMN is enabled on the gNodeB for the home operator of a VoNR-incapable roaming UE, the INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected for the cell serving the UE, handover evaluation for fast carrier shutdown is ongoing, and the related timer has not expired. - When the gNBFreqPriorityGroup.RoamingUeHplmnInterFHoFlag parameter is set to TRUE for the target cell's operating frequency in the frequency group bound to the roaming UE, the following mechanism applies: - If the roaming UE supports

RAT	Function Name	Function Switch	Reference	Description
				handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - If the roaming UE does not support handovers, inter-frequency handover for fast carrier shutdown is not performed for the roaming UE. • When the gNBFreqPriorityGroup.RoamingUeHplmnInterFHoFlag parameter is set to FALSE for the target cell's operating frequency in the frequency group bound to the roaming UE, inter-frequency handover for fast carrier shutdown is not performed for the roaming UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUelidentSw	None	Fast carrier shutdown fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect: - Any option of the gNodeBParam.StUelidentSw parameter is selected. - The gNBAsAppParam.AppDomainName parameter is set. - The gNBAsAppParam.AppIpv4 parameter is set. - The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue: - Select the SPEED_TURBO_PWR_SAVING_SW option of the
Low-frequency TDD	gNodeB acceleration service application	gNBAsAppParam.AppDomainName gNBAsAppParam.AppIpv4 gNBAsAppParam.AppIpv6	None	

RAT	Function Name	Function Switch	Reference	Description
				gNBPwrSavingAlgo.<i>PwrSavin</i>gAlgoSwitch parameter. Fast carrier shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBPwrSavingAlgo.<i>SpeedT</i>urboPwrSavStartTime and gNBPwrSavingAlgo.<i>SpeedT</i>urboPwrSavEndTime parameters, increasing the energy saving gains. • Select the STATIC_ST_PWR_SAVING_SW option of the gNBPwrSavingAlgo.<i>PwrSavin</i>gAlgoSwitch parameter. Fast carrier shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by

RAT	Function Name	Function Switch	Reference	Description
				the gNBPwrSavin gAlgo.StaticST PwrSavStartTi me and gNBPwrSavin gAlgo.StaticST PwrSavEndTi me parameters, increasing the energy saving gains.

6.3.4.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

BBUs

The BBU5900C, BookBBU5901a, and BookBBU5901 cannot be used in inter-base-station scenarios. No restriction is imposed on the other BBUs.

Boards

- Main control boards: Only the UMPT series boards support this function.
- Baseband processing units: Only the UBBP series boards support this function.
- The universal baseband radio unit (UBRU) does not support fast carrier shutdown.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.3.4.4 Cells

- This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**.
- In NR TDD, the **NRDUCell.SupplementaryCellInd** parameter cannot be set to **INTRA_FREQ_ASSISTANT_CELL**.
- In NR TDD, the **NRDUCell.DeployPosition** parameter cannot be set to **DEPLOY_IAB**.
- This function requires that the **NRDUCell.FrequencyBand** parameter not be set to **N257**, **N258**, **N260**, or **N261** for the cells.
- In intra-base-station multi-carrier scenarios (where the **INTRA_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter is selected), if the networking option is NSA for capacity-layer cells, it must be either NSA or SA_NSA for basic-layer cells; if the networking option is SA or SA_NSA for capacity-layer cells, it must be SA or SA_NSA for basic-layer cells. In multi-operator sharing scenarios, the networking option must be SA or SA_NSA for capacity-layer cells.
- In inter-base-station multi-carrier scenarios (where the **INTER_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter is selected), the networking option must be SA or SA_NSA for both capacity-layer cells and basic-layer cells.

6.3.4.5 Others

None

6.3.5 Operation and Maintenance

6.3.5.1 Data Configuration

6.3.5.1.1 Data Preparation

Table 6-17 describes the parameters used for function activation. Function optimization configurations are not involved. This section does not describe parameters related to cell establishment.

Table 6-17 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	When fast carrier shutdown is required: • In intra-base-station multi-carrier scenarios, select the INTRA_GNB_MULTI_CARR_SD_SW option of this parameter for capacity-layer cells. • In inter-base-station multi-carrier scenarios, select the INTER_GNB_MULTI_CARR_SD_SW option of this parameter for capacity-layer cells.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	When fast carrier shutdown is required, select the following options: • Capacity-layer cells: FAST_CARRIER_SHUTDOWN_SW • Basic-layer cells: FAST_CARR_SHUT_SERV_ASUR_SW
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	When fast carrier shutdown is required: • In intra-base-station multi-carrier scenarios, set this parameter to INTRA_GNB_MULTI_CARR_SD for capacity-layer cells. • In inter-base-station multi-carrier scenarios, set this parameter to INTER_GNB_MULTI_CARR_SD for capacity-layer cells.
Carrier Shutdown Mode	NRDUCellPowerSaving.CarrierShutdownMode	When fast carrier shutdown is required, set this parameter to FAST_CARR_SHUTDOWN for capacity-layer cells.

Parameter Name	Parameter ID	Setting Notes
Carrier Shutdown Service Assurance Level	NRDUCellPowerSaving.CarrShutdServiceAssurLevel	The value LEVEL4 is recommended if the impacts of fast carrier shutdown on service experience need to be minimized. The value LEVEL0 is recommended if the gains of fast carrier shutdown need to be maximized. A larger value of this parameter results in a smaller impact on user experience, a shorter duration of carrier shutdown, and fewer energy saving gains.
Carrier Shutdown Fast Eval PRB Threshold	NRDUCellMobility.CarrShutdFastEvalPrbThld	You are advised to use the recommended value. Set this parameter for both capacity-layer cells and basic-layer cells.
Carrier Shutdown User Measure Time-To-Trig	NRCCellMobilityConfig.CarrShutUserMeasTimeToTrig	You are advised to use the recommended value. Set this parameter for both capacity-layer cells and basic-layer cells.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	Set this parameter to MINUTE_LEVEL when fast carrier shutdown is required.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	The recommended value is 5. Set this parameter to 5 or a larger value when fast carrier shutdown is required.
Exit Evaluation Period	NRDUCellPowerSaving.ExitEvaluationPeriod	The recommended value is 5. Set this parameter to 5 or a larger value when fast carrier shutdown is required.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: - For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. - For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. - The period specified for fast carrier shutdown in a cell cannot overlap the period specified for the function with the NRDUCellPowerSaving.PowerSavingType parameter set to TIMING_CARRIER_SHUTDOWN for the same cell.

Parameter Name	Parameter ID	Setting Notes
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan and the following requirements: • For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. • For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. • The period specified for fast carrier shutdown in a cell cannot overlap the period specified for the function with the NRDUCellPowerSaving.PowerSavingType parameter set to TIMING_CARRIER_SHUTDOWN for the same cell.

Parameter Name	Parameter ID	Setting Notes
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	You are advised to use the recommended value. It is recommended that the sum of NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset be less than or equal to 100. If the sum exceeds 100, the cell cannot exit the fast carrier shutdown state.
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	You are advised to use the recommended value. It is recommended that the sum of NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset be less than or equal to 100. If the sum exceeds 100, the cell cannot exit the fast carrier shutdown state.
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	You are advised to use the recommended value. It is recommended that the sum of NRDUCellPowerSaving.UlPrbThld and NRDUCellPowerSaving.NCellUlPrbOffset be less than or equal to 100. If the sum exceeds 100, the cell cannot exit the fast carrier shutdown state.
Neighboring Cell Uplink PRB Offset	NRDUCellPowerSaving.NCellUlPrbOffset	You are advised to use the recommended value. It is recommended that the sum of NRDUCellPowerSaving.UlPrbThld and NRDUCellPowerSaving.NCellUlPrbOffset be less than or equal to 100. If the sum exceeds 100, the cell cannot exit the fast carrier shutdown state.

Parameter Name	Parameter ID	Setting Notes
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	You are advised to use the recommended value. Fast carrier shutdown requires that the NRDUCellPowerSaving.UserNumThld parameter be set to a value less than or equal to 150 .
User Number Offset	NRDUCellPowerSaving.UserNumOffset	You are advised to use the recommended value.
Operator-specific User Number Threshold	NRDUCellPowerSaving.OpUserNumThld	You are advised to use the recommended value. If this parameter is set to 65535 , this parameter and the NRDUCellPowerSaving.OpUserNumOffset parameter are invalid.
Operator-specific User Number Offset	NRDUCellPowerSaving.OpUserNumOffset	You are advised to use the recommended value.

Parameter Name	Parameter ID	Setting Notes
SUL User Number Threshold	NRDUCellPowerSaving.SulUserNumThld	You are advised to use the recommended value. • This parameter is invalid when it is set to 65535. • If the value of NRDUCellPowerSaving.SulUserNumThld is not 65535, the sum of NRDUCellPowerSaving.SulUserNumThld and NRDUCellPowerSaving.SulUserNumOffset must be less than 65535, and the value of NRDUCellPowerSaving.SulUserNumThld must be less than that of NRDUCellPowerSaving.UserNumThld. • When the FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is deselected and there is a record with NRDUCellPowerSaving.PowerSavingType set to INTRA_GNB_MULTI_CARR_S D or INTER_GNB_MULTI_CARR_S D_SW for the cell, the NRDUCellPowerSaving.SulUserNumThld parameter must be set to 65535 for this record.

Parameter Name	Parameter ID	Setting Notes
SUL User Number Offset	NRDUCellPowerSaving.SulUserNumOffset	You are advised to use the recommended value. - This parameter is invalid when it is set to 65535. - When the value of NRDUCellPowerSaving.SulUserNumThld is not 65535, the sum of NRDUCellPowerSaving.SulUserNumThld and NRDUCellPowerSaving.SulUserNumOffset must be less than 65535.
Power Saving Penalty Duration	NRDUCellPowerSaving.PwrSavingPenaltyTime	The value 1 is recommended for fast carrier shutdown.
Power Saving Cell Flag	NRCellRelation.PowerSavingCellFlag	This parameter indicates the flag of fast carrier shutdown for a neighboring cell. Set this parameter based on the network plan. When fast carrier shutdown is enabled, this parameter can be set to COVERAGE for at most 16 neighboring cells and to CAPACITY also for at most 16 neighboring cells.
No Removal Flag	NRCellRelation.NoRmvFlag	When automatic removal of neighboring cells in ANR is enabled (with the NR_NR_ANR_DEL_SW option of the NRCellAlgoSwitch.AnrSwitch parameter selected), it is recommended that the NRCellRelation.NoRmvFlag parameter be set to FORBID_ANR_RMV for the bidirectional neighbor relationships between capacity-layer and basic-layer cells.
PLMN Reserved Flag	NRExternalNCellPlmn.PlmnReservedFlag	In multi-operator sharing scenarios, set this parameter for capacity-layer cells based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Multi Networking Option Opt Switch	EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw	Select the PWR_SAVE_SCENARIO_NR_DELAY_SW option of this LTE parameter for fast carrier shutdown in NSA networking.
NS UE NR Delay Timer	CellUSParaCfg.NsUeNrDelayTimer	The value 13 is recommended for fast carrier shutdown in NSA networking.
Necessary HO Prepare Fail Punish Timer	gNBMobilityCommParam.NecHoPrepFailPunishTimer	The value 15 is recommended on the neighboring base station side for fast carrier shutdown to take effect in SA multi-base-station scenarios.
Optimized HO Prepare Fail Punish Timer	gNBMobilityCommParam.OptHoPrepFailPunishTimer	The value 15 is recommended on the neighboring base station side for fast carrier shutdown to take effect in SA multi-base-station scenarios.
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	If fast carrier shutdown load event evaluation is required, select the CARR_SHDN_LOAD_EVENT_EVAL_SW option for capacity-layer cells.
Fast Carrier Shutdown Event Evaluation PRB Thld	NRDUCellMobility.CarrShdnEventEvalPrbThld	This parameter needs to be configured only for basic-layer cells. If fast carrier shutdown load event evaluation is required, the recommended value is NRDUCellPowerSaving.DlPrbThld plus NRDUCellPowerSaving.DlPrbOffset . An inappropriate setting may cause the failure to quickly exit fast carrier shutdown or frequent entry into and exit from fast carrier shutdown.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	Select the SERV_TYPE_DYN_EXIT_PWR_SAVE_SW option if the number of UEs in an energy saving policy group needs to be checked for entering or exiting intelligent carrier shutdown.

Parameter Name	Parameter ID	Setting Notes
Power Saving Policy Group Index	gNBQciBearer.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan.
Power Saving Policy Group Index	NRCellPsPolicyGrp.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan. The value of this parameter must be the same as that of the gNBQciBearer.PowerSavingPolicyGrpIndex parameter for services to be guaranteed. Before setting this parameter, ensure that the QCI for which the gNBQciBearer.PowerSavingPolicyGrpIndex parameter is set has been configured.
User Identification Method	NRCellPsPolicyGrp.UserIdentificationMethod	Set this parameter to SERVICE_TYPE .
Service Type Dynamic Exit Power Saving Switch	NRCellPsPolicyGrp.ServTypeDynamicExitPowerSavingSw	Select the USER_NUM_CSD_EXIT_SW option if the number of UEs in an energy saving policy group needs to be checked for entering or exiting intelligent carrier shutdown.
PS Policy Group User Number Threshold	NRCellPsPolicyGrp.GrpUserNumberThld	Use the recommended value.
PS Policy Group User Number Offset	NRCellPsPolicyGrp.GrpUserNumberOffset	Use the recommended value.

6.3.5.1.2 Using MML Commands

Before using MML commands, refer to [6.3.4 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- **Intra-base-station multi-carrier scenario**

Assume that a cell with **NrCellId** being **0** is a capacity-layer cell and another cell with **NrCellId** being **10** is a basic-layer cell in an intra-base-station multi-carrier scenario. The **PowerSavingCellFlag** parameter in the **NRCellRelation** MO must be set in both capacity- and basic-layer cells for fast carrier shutdown. For details about how to configure intra-base-station neighboring cells, see *SA Mobility Management in Connected Mode*.

For details about activating frequency-priority-based handover in basic-layer cells, see *SA Mobility Management in Connected Mode*. The activation methods also apply to NSA networking.

```
//Enabling fast carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingSwitch=INTRA_GNB_MULTI_CARR_SD_SW-1,
PowerSavingExtSwitch=FAST_CARRIER_SHUTDOWN_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=10,
PowerSavingExtSwitch=FAST_CARR_SHUT_SERV_ASSUR_SW-1;
//Setting an energy saving policy of fast carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, StartTime=00&00&00, StopTime=06&00&00,
DIPrbThld=25, DIPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
OpUserNumOffset=65535, SulUserNumThld=65535, SulUserNumOffset=65535,
PwrSavingPenaltyTime=1, UlPrbThld=25, NCellUlPrbOffset=25, EvalPeriodGranularity=MINUTE_LEVEL,
ShutdownTrigJudgePeriod=5, ExitEvaluationPeriod=5,
CarrierShutdownMode=FAST_CARR_SHUTDOWN, CarrShutdServiceAssurLevel=LEVEL4;
//((Optional) Setting another energy saving policy of fast carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, StartTime=12&00&00, StopTime=14&00&00,
DIPrbThld=25, DIPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
OpUserNumOffset=65535, SulUserNumThld=65535, SulUserNumOffset=65535,
PwrSavingPenaltyTime=1, UlPrbThld=25, NCellUlPrbOffset=25, EvalPeriodGranularity=MINUTE_LEVEL,
ShutdownTrigJudgePeriod=5, ExitEvaluationPeriod=5,
CarrierShutdownMode=FAST_CARR_SHUTDOWN, CarrShutdServiceAssurLevel=LEVEL4;
//Setting CarrShutdFastEvalPrbThld and CarrShutUserMeasTimeToTrig
MOD NRDUCELLMOBILITY: NrDuCellId=0, CarrShutdFastEvalPrbThld=1;
MOD NRDUCELLMOBILITY: NrDuCellId=10, CarrShutdFastEvalPrbThld=1;
MOD NRCELLMOBILITYCONFIG: NrCellId=0, CarrShutUserMeasTimeToTrig=320MS;
MOD NRCELLMOBILITYCONFIG: NrCellId=10, CarrShutUserMeasTimeToTrig=320MS;
//Setting PowerSavingCellFlag on the local base station, and (recommended) setting NoRmvFlag to
FORBID_ANR_RMV for the bidirectional neighbor relationships between capacity-layer and basic-layer
cells when automatic removal of neighboring cells in ANR takes effect
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=10,
PowerSavingCellFlag=COVERAGE, NoRmvFlag=FORBID_ANR_RMV;
MOD NRCELLRELATION: NrCellId=10, Mcc="302", Mnc="220", gNBId=0, CellId=0,
PowerSavingCellFlag=CAPACITY, NoRmvFlag=FORBID_ANR_RMV;
//((Optional) Selecting the PWR_SAVE_SCENARIO_NR_DELAY_SW option in NSA networking or NSA
and SA hybrid networking
MOD ENODEBALGOEXTSWITCH:
MultiNetworkingOptionOptSw=PWR_SAVE_SCENARIO_NR_DELAY_SW-1;
MOD CELLUSPARACFG: LocalCellId=0, NsUeNrDelayTimer=13;
//Setting NecHoPrepFailPunishTimer and OptHoPrepFailPunishTimer for neighboring NR base stations
configured with intra-frequency or inter-frequency neighboring cells in SA or NSA and SA hybrid
networking
MOD GNBMOBILITYCOMMPARAM: NecHoPrepFailPunishTimer=15, OptHoPrepFailPunishTimer=15;
```

```

// (Optional) Enabling fast carrier shutdown load event evaluation
MOD NRDUCELLPWRSERVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=CARR_SHDN_LOAD_EVENT_EVAL_SW-1;
MOD NRDUCELLMOBILITY: NrDuCellId=10, CarrShdnEventEvalPrbThld=50;

```

(Optional) Multi-operator sharing scenario

```

// (Optional) Modifying an energy saving policy of fast carrier shutdown
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, OpUserNumThld=12, OpUserNumOffset=12;
// (Optional) Modifying the other energy saving policy of fast carrier shutdown
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, OpUserNumThld=12, OpUserNumOffset=12;

```

(Optional) SUL co-deployment scenario

```

// (Optional) Modifying an energy saving policy of intelligent carrier shutdown
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, SulUserNumThld=4, SulUserNumOffset=4;
// (Optional) Modifying the other energy saving policy of intelligent carrier shutdown
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1,
PowerSavingType=INTRA_GNB_MULTI_CARR_SD, SulUserNumThld=4, SulUserNumOffset=4;

```

(Optional) Scenario where the number of UEs in an energy saving policy group needs to be checked for entering or exiting fast carrier shutdown

```

// (Optional) Performing the following configurations if the number of UEs in an energy saving policy group needs to be checked for entering or exiting fast carrier shutdown
// For a capacity-layer cell
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;
ADD NRCELLPOLICYGRP: NrCellId=0, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=USER_NUM_CSD_EXIT_SW-1,
GrpUserNumThld=7, GrpUserNumOffset=4;
// For a basic-layer cell
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
ADD NRCELLPOLICYGRP: NrCellId=10, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE;

```

- **Inter-base-station multi-carrier scenario**

Assume that, in an inter-base-station multi-carrier scenario, a cell with **NrCellId** being 1 is a capacity-layer cell served by a base station with **gNBId** being 0 and another cell with **NrCellId** being 11 is a basic-layer cell served by a base station with **gNBId** being 1. The **PowerSavingCellFlag** parameter in the **NRCellRelation** MO must be set in both capacity- and basic-layer cells for fast carrier shutdown. For details about how to configure inter-base-station neighboring cells, see *SA Mobility Management in Connected Mode*.

For details about activating frequency-priority-based handover in basic-layer cells, see *SA Mobility Management in Connected Mode*. The activation methods also apply to NSA networking.

```

// Enabling fast carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
PowerSavingSwitch=INTER_GNB_MULTI_CARR_SD_SW-1,
PowerSavingExtSwitch=FAST_CARRIER_SHUTDOWN_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=11,
PowerSavingExtSwitch=FAST_CARR_SHUT_SERV_ASSUR_SW-1;
// Setting an energy saving policy of fast carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0,
PowerSavingType=INTER_GNB_MULTI_CARR_SD, StartTime=00&00&00, StopTime=06&00&00,
DIPrbThld=25, DIPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
OpUserNumOffset=65535, SulUserNumThld=65535, SulUserNumOffset=65535,
PwrSavingPenaltyTime=1, UIPrbThld=25, NCellUIPrbOffset=25, EvalPeriodGranularity=MINUTE_LEVEL,
ShutdownTrigJudgePeriod=5, ExitEvaluationPeriod=5,
CarrierShutdownMode=FAST_CARR_SHUTDOWN, CarrShutdServiceAssurLevel=LEVEL4;
// (Optional) Setting another energy saving policy of fast carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=1,
PowerSavingType=INTER_GNB_MULTI_CARR_SD, StartTime=12&00&00, StopTime=14&00&00,

```

DlPrbThld=25, DlPrbOffset=25, UserNumThld=20, UserNumOffset=20, OpUserNumThld=65535,
 OpUserNumOffset=65535, SulUserNumThld=65535, SulUserNumOffset=65535,
 PwrSavingPenaltyTime=1, UlPrbThld=25, NCellUlPrbOffset=25, EvalPeriodGranularity=MINUTE_LEVEL,
 ShutdownTrigJudgePeriod=5, ExitEvaluationPeriod=5,
 CarrierShutdownMode=FAST_CARR_SHUTDOWN, CarrShutdServiceAssurLevel=LEVEL4;
 //Setting CarrShutdFastEvalPrbThld and CarrShutUserMeasTimeToTrig
 MOD NRDUCELLMOBILITY: NrDuCellId=1, CarrShutdFastEvalPrbThld=1;
 MOD NRDUCELLMOBILITY: NrDuCellId=11, CarrShutdFastEvalPrbThld=1;
 MOD NRCELLMOBILITYCONFIG: NrCellId=1, CarrShutUserMeasTimeToTrig=320MS;
 MOD NRCELLMOBILITYCONFIG: NrCellId=11, CarrShutUserMeasTimeToTrig=320MS;
 //Setting PowerSavingCellFlag on the local base station, and (recommended) setting NoRmvFlag to
 FORBID_ANR_RMV for the neighbor relationships with the basic-layer cell when automatic removal of
 neighboring cells in ANR takes effect
 MOD NRCELLRELATION: NrCellId=1, Mcc="302", Mnc="220", gNBId=1, CellId=11,
 PowerSavingCellFlag=COVERAGE, NoRmvFlag=FORBID_ANR_RMV;
 //Setting PowerSavingCellFlag on the peer base station, and (recommended) setting NoRmvFlag to
 FORBID_ANR_RMV for the neighbor relationships with the capacity-layer cell when automatic
 removal of neighboring cells in ANR takes effect
 MOD NRCELLRELATION: NrCellId=11, Mcc="302", Mnc="220", gNBId=0, CellId=1,
 PowerSavingCellFlag=CAPACITY, NoRmvFlag=FORBID_ANR_RMV;
 //(Optional) Selecting the PWR_SAVE_SCENARIO_NR_DELAY_SW option in NSA networking or NSA
 and SA hybrid networking
 MOD ENODEBALGOEXTSWITCH:
 MultiNetworkingOptionOptSw=PWR_SAVE_SCENARIO_NR_DELAY_SW-1;
 MOD CELLUSPARACFG: LocalCellId=0, NsUeNrDelayTimer=13;
 //Setting NecHoPrepFailPunishTimer and OptHoPrepFailPunishTimer for neighboring NR base stations
 configured with intra-frequency or inter-frequency neighboring cells in SA or NSA and SA hybrid
 networking
 MOD GNBMOBILITYCOMMPARAM: NecHoPrepFailPunishTimer=15, OptHoPrepFailPunishTimer=15;
 //(Optional) Enabling fast carrier shutdown load event evaluation
 MOD NRDUCELLPWRSAVEALGO: NrDuCellId=1,
 PowerSavingAlgoSwitch=CARR_SHDN_LOAD_EVENT_EVAL_SW-1;
 MOD NRDUCELLMOBILITY: NrDuCellId=11, CarrShdnEventEvalPrbThld=50;

(Optional) Multi-operator sharing scenario

//Setting PlmnReservedFlag of the basic-layer cell based on the network plan at the base station
 serving the capacity-layer cell
 MOD NREXTERNALNCELLPLMN: Mcc="302", Mnc="220", gNBId=1, CellId=11, SharedMcc="302",
 SharedMnc="610", PlmnReservedFlag=FALSE;
 //(Optional) Modifying an energy saving policy of fast carrier shutdown
 MOD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0,
 PowerSavingType=INTER_GNB_MULTI_CARR_SD, OpUserNumThld=12, OpUserNumOffset=12;
 //(Optional) Modifying the other energy saving policy of fast carrier shutdown
 MOD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=1,
 PowerSavingType=INTER_GNB_MULTI_CARR_SD, OpUserNumThld=12, OpUserNumOffset=12;

(Optional) SUL co-deployment scenario

//(Optional) Modifying an energy saving policy of intelligent carrier shutdown
 MOD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0,
 PowerSavingType=INTER_GNB_MULTI_CARR_SD, SulUserNumThld=4, SulUserNumOffset=4;
 //(Optional) Modifying the other energy saving policy of intelligent carrier shutdown
 MOD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=1,
 PowerSavingType=INTER_GNB_MULTI_CARR_SD, SulUserNumThld=4, SulUserNumOffset=4;

(Optional) Scenario where the number of UEs in an energy saving policy group needs to be checked for entering or exiting fast carrier shutdown

//(Optional) Performing the following configurations if the number of UEs in an energy saving policy
 group needs to be checked for entering or exiting fast carrier shutdown
 //For a capacity-layer cell
 MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
 MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
 PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;
 ADD NRCELLPOLICYGRP: NrCellId=1, PowerSavingPolicyGrpIndex=0,
 UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=USER_NUM_CSD_EXIT_SW-1,
 GrpUserNumThld=7, GrpUserNumOffset=4;
 //For a basic-layer cell
 MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
 ADD NRCELLPOLICYGRP: NrCellId=11, PowerSavingPolicyGrpIndex=0,
 UserIdentMethod=SERVICE_TYPE;

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- **Intra-base-station multi-carrier scenario**

```
//(Optional) Not checking the number of UEs in an energy saving policy group for entering or exiting fast carrier shutdown
MOD NRCELLPOLICYGRP: NrCellId=0, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=USER_NUM_CSD_EXIT_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;
//Disabling fast carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingSwitch=INTRA_GNB_MULTI_CARR_SD_SW-0,
PowerSavingExtSwitch=FAST_CARRIER_SHUTDOWN_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=10,
PowerSavingExtSwitch=FAST_CARR_SHUT_SERV_ASSUR_SW-0;
//Removing an energy saving policy of fast carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
//(Optional) Removing the other energy saving policy of fast carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1;
//(Optional) Deselecting the PWR_SAVE_SCENARIO_NR_DELAY_SW option in NSA networking or NSA and SA hybrid networking
MOD ENODEBALGOEXTSWITCH:
MultiNetworkingOptionOptSw=PWR_SAVE_SCENARIO_NR_DELAY_SW-0;
//(Optional) Disabling fast carrier shutdown load event evaluation
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=CARR_SHDN_LOAD_EVENT_EVAL_SW-0;
MOD NRDUCELLMOBILITY: NrDuCellId=10, CarrShdnEventEvalPrbThld=255;
```

- **Inter-base-station multi-carrier scenario**

```
//(Optional) Not checking the number of UEs in an energy saving policy group for entering or exiting fast carrier shutdown
MOD NRCELLPOLICYGRP: NrCellId=1, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=USER_NUM_CSD_EXIT_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;
//Disabling fast carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
PowerSavingSwitch=INTER_GNB_MULTI_CARR_SD_SW-0,
PowerSavingExtSwitch=FAST_CARRIER_SHUTDOWN_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=11,
PowerSavingExtSwitch=FAST_CARR_SHUT_SERV_ASSUR_SW-0;
//Removing an energy saving policy of fast carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0;
//(Optional) Removing the other energy saving policy of fast carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=1;
//(Optional) Deselecting the PWR_SAVE_SCENARIO_NR_DELAY_SW option in NSA networking or NSA and SA hybrid networking
MOD ENODEBALGOEXTSWITCH:
MultiNetworkingOptionOptSw=PWR_SAVE_SCENARIO_NR_DELAY_SW-0;
//(Optional) Disabling fast carrier shutdown load event evaluation
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=1,
PowerSavingAlgoSwitch=CARR_SHDN_LOAD_EVENT_EVAL_SW-0;
MOD NRDUCELLMOBILITY: NrDuCellId=11, CarrShdnEventEvalPrbThld=255;
```

6.3.5.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.3.5.2 Activation Verification

As activation verification for fast carrier shutdown in intra-base-station multi-carrier scenarios is similar to that in inter-base-station multi-carrier scenarios, the following uses the activation in the former scenarios as an example, which can be verified using any of the following methods:

- Method 1:
 - a. Run the **LST NRDUCELLALGOSWITCH** command for both the capacity-layer and basic-layer cells. If the following information is displayed, fast carrier shutdown has been enabled.
 - Capacity-layer cell: value of **Power Saving Extension Switch** being **Fast Carrier Shutdown Switch:On**
 - Basic-layer cell: value of **Power Saving Extension Switch** being **Fast Carr Shutdown Service Assurance Sw:On**
 - b. Run the **LST NRDUCELLPOWERSAVING** command for the capacity-layer cell. Verify that there is at least one record with **Power Saving Type** being **Intra-gNodeB Multi-Carrier Shutdown** and **Carrier Shutdown Mode** being **Fast Carrier Shutdown**.

 NOTE

If there are two or more records with **Power Saving Type** being **Intra-gNodeB Multi-Carrier Shutdown** and **Carrier Shutdown Mode** being **Fast Carrier Shutdown**, fast carrier shutdown has been enabled in two or more time periods.

- c. Run the **DSP TIME** command for the capacity-layer cell. Check whether the system time is within a fast carrier shutdown period.
 - If it is, go to **d**.
 - If it is not, wait until the start time of fast carrier shutdown arrives.
 - d. Wait for a period (at most equal to the sum of **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** and **NRDUCellPowerSaving.PwrSavingPenaltyTime**), and then run the **DSP NRDUCELL** command for the capacity-layer cell. If the value of **Power Saving State** is **Intra-gNodeB Multi-Carrier Shutdown:Enabled** in the command output, fast carrier shutdown has taken effect.
- Method 2:

Run the **DSP NRDUCELLPOWERSAVING** command for the capacity-layer cell to query the status of fast carrier shutdown.

 - If the value of **Power Saving Type** is **Intra-gNodeB Multi-Carrier Shutdown** and the value of **Current State** is **Enabled**, fast carrier shutdown has taken effect.
 - If the value of **Power Saving Type** is **Intra-gNodeB Multi-Carrier Shutdown** and the value of **Current State** is **Disabled**, check the **Current Disabled Cause**, **Latest Disabled Cause**, and **Latest Exit Cause** parameter values to find out the possible reason why fast carrier shutdown did not take effect. **Table 6-18** lists possible reasons. You can also check **Latest Exit Time** and **Latest Disabled Time** in the command output for the time the latest exit or the latest failure to take effect occurred.

Latest Exit Time, Latest Exit Cause, Latest Disabled Time, and Latest Disabled Cause are used to query the time and reasons for the latest exit and the latest application failure within the specified period. The query may be performed outside the specified period. For example, if the specified period is from 00:00:00 to 06:00:00, **Latest Exit Cause** or **Latest Disabled Cause** indicates the reason for the latest exit or the latest application failure that occurred within the period from 00:00:00 to 06:00:00. This reason, together with the time the latest exit or application failure occurred, can be queried even by an operation performed within the time from 06:00:00 to 23:59:59.

Table 6-18 Possible reasons why fast carrier shutdown did not take effect

Reason	Description
Insufficient License	No license for this function is available.
Switch Turned Off	None
Neighboring Cell Not Configured	- No inter-frequency neighboring cells are configured for the current serving cell. - Inter-frequency neighboring cells are configured for the current serving cell, but there is no neighboring cell for which Power Saving Cell Flag is set to COVERAGE. - In multi-operator sharing scenarios: - The PLMN IDs of the non-reserved operators of the capacity-layer cell are not a subset of the PLMN IDs of the non-reserved operators of any basic-layer cell. - The PLMN IDs of the reserved operators of the capacity-layer cell are not a subset of the PLMN IDs of the reserved and non-reserved operators of any basic-layer cell.
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable for other reasons.
Triggering Time Not Reached	The current time is not within the range specified for fast carrier shutdown.

Reason	Description
Neighboring Cell Unavailable	• A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is unavailable. • A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is a multi-TRP cell such as a hyper cell or combined cell, and some TRPs of this cell are unavailable. • A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is an NSA cell. • The FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter is deselected for the neighboring cell with Power Saving Cell Flag set to COVERAGE.
Xn Interface Unavailable	In inter-base-station multi-carrier scenarios, the Xn interface is in the abnormal state.
Impacted Feature Taken Effect	None
Downlink PRB Threshold Not Reached	None
Uplink PRB Threshold Not Reached	None
UE Number Threshold Not Reached	None
Policy Group UE Number Exceeding Threshold	None
UE Being Transferred or Transfer Failed	• The current serving cell meets the load conditions for carrier shutdown, and UEs are being transferred. • The current serving cell meets the load conditions for carrier shutdown, but some UEs fail to be transferred from the cell.

Reason	Description
Triggering Decision Period Not Met	The current time is within the effective period specified for fast carrier shutdown and is not within the first detection period. However, it is within the evaluation period for triggering fast carrier shutdown (specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter).
In Penalty State	The cell is in the penalty state after it exits the fast carrier shutdown state or UEs fail to be transferred out of the cell. (The causes of UE transfer failures include UE capability mismatch and missing configuration of neighboring NR frequencies.)
RF Hardware Capability Not Supported	None
Basic Cell PRB Usage Limit Exceeded	None

Reason	Description
Fast Carrier Shutdown Condition Not Met	• The downlink PRB usage of each basic-layer cell is less than the PRB usage threshold for fast evaluation on carrier shutdown, but one of the basic-layer cells serves UEs with heavy downlink traffic. • The downlink PRB usage of each basic-layer cell is greater than or equal to the PRB usage threshold for fast evaluation on carrier shutdown, and exit from fast carrier shutdown is triggered based on the Carrier Shutdown Service Assurance Level parameter value. • With the fast carrier shutdown load event evaluation function enabled, if the downlink PRB usage of any basic-layer cell is greater than the PRB usage threshold for fast carrier shutdown event evaluation in the current period but does not exceed this threshold in the previous period, the capacity-layer cell exits the fast carrier shutdown state.
Ultra-Long Holdover in Progress	None
Others	• The current time is within the periods specified for fast carrier shutdown to take effect as well as the first detection period. • Internal software or other reasons cause the function to fail.

- Method 3
If the value of the **N.PowerSaving.FastCarrierShutdown.Dur** counter is not 0, fast carrier shutdown has taken effect in the cell.

6.3.5.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTSBoard** on the MAE-Access. On the live network, the one-week base station energy consumption data obtained before and after fast carrier shutdown is enabled can be compared to assess the energy

saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.

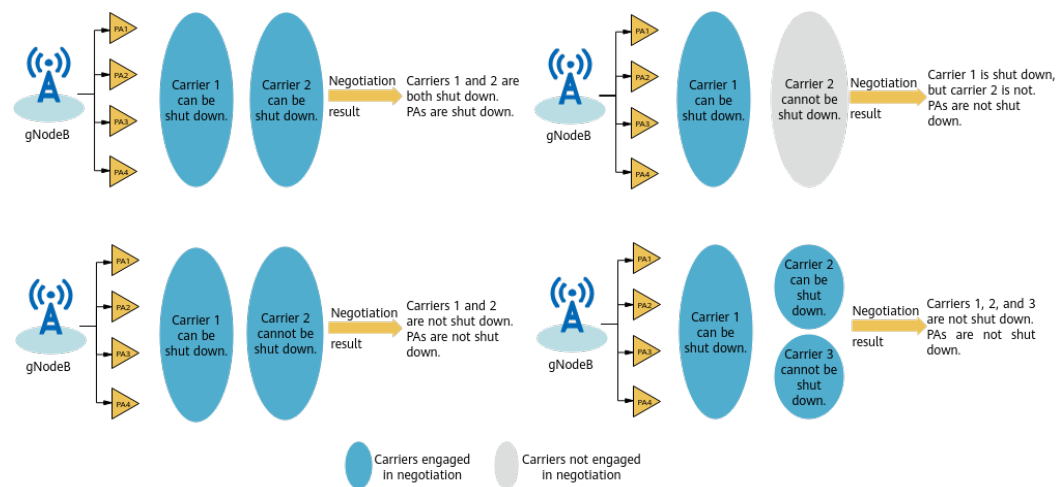
- The duration in which the cell stays in the carrier shutdown state due to fast carrier shutdown can be obtained using the **N.PowerSaving.FastCarrierShutdown.Dur** counter on the MAE-Access.
- The duration of cell unavailability due to fast carrier shutdown can be obtained using the **N.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access.

6.4 Multi-Carrier Joint Carrier Shutdown

6.4.1 Principles

When a power amplifier (PA) in an RF module serves more than one carrier, carrier shutdown is evaluated on a per carrier basis. A carrier is shut down when it meets the triggering conditions for a specific carrier shutdown function. The PA is shut down only when all carriers served by the PA are shut down. In case the PA keeps working because some of the carriers are not shut down, energy saving gains are relatively low. Against this backdrop, multi-carrier joint carrier shutdown is introduced. [Figure 6-7](#) shows how this function works.

Figure 6-7 Multi-carrier joint carrier shutdown



Multi-carrier joint carrier shutdown is controlled by the **MULTI_CARR_JOINT_CARR_SHDN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter.

- All cells with this option selected participate in multi-carrier negotiation. If the carriers of these cells all meet the triggering conditions for a carrier shutdown function and the number of channels configured for the carriers is equal to the negotiated number of channels that can be shut down, all these carriers are shut down. Otherwise, none of these carriers is shut down. Again, the PA is shut down only when all carriers served by the PA are shut down. It is recommended that this option be selected for all cells on the carriers sharing a PA.

- Any cell with this option deselected does not participate in multi-carrier negotiation.

LTE cells, if any, must be enabled with the LTE function multi-carrier joint carrier shutdown so that they can participate in the multi-carrier negotiation. For details about this LTE function, see *Energy Conservation and Emission Reduction* in the eRAN feature documentation.

The following carrier shutdown functions are involved in the negotiation:

- Smart carrier shutdown: See [6.2 Intelligent Carrier Shutdown](#). With multi-carrier joint carrier shutdown enabled, multi-carrier negotiation takes some time, and therefore the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter for intelligent carrier shutdown must be greater than or equal to 5.
- Timing carrier shutdown: See [6.1 Timing Carrier Shutdown](#).
- LTE and NR intelligent carrier shutdown: See *Multi-RAT Coordinated Energy Saving*.
- Intelligent power-off of carriers in the same coverage: See *Energy Conservation and Emission Reduction* in the eRAN feature documentation.
- Cell Switch-off Based on Automatic Co-coverage Identification: See *Energy Conservation and Emission Reduction* in the eRAN feature documentation.
- Multi-carrier coordinated energy saving: See *Energy Conservation and Emission Reduction* in the eRAN feature documentation.

6.4.2 Network Analysis

6.4.2.1 Benefits

Multi-carrier joint carrier shutdown maximizes energy saving gains. However, in case any carrier no longer meets the conditions for the carrier shutdown function involved in the multi-carrier negotiation to take effect, all the carriers exit that function. In this case, the energy saving gains may decrease.

6.4.2.2 Impacts

The impacts between this function and other functions are described in [6.4.3.2.3 Function Impacts](#).

Multi-carrier joint carrier shutdown alters the UE distribution and traffic volume, which in turn results in changes in the performance indicators related to inter-frequency handovers and redirections.

If multi-carrier joint carrier shutdown is enabled for a cell in the energy saving state, the cell exits this state before multi-carrier joint carrier shutdown is evaluated in a coordinated manner.

Assuming that the neighboring base stations are in the carrier shutdown state, cell-edge UEs in weak coverage areas may access the cells served by the local base station. This affects KPIs related to the aspects such as access and service drops of the local base station. For example, the value of the **N.NsaDc.DRB.AbnormRel** counter may increase.

6.4.3 Requirements

6.4.3.1 Licenses

None

6.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Multi-carrier joint carrier shutdown requires one of the following functions: - Intelligent carrier shutdown^a - Timing carrier shutdown
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	
a: Intelligent carrier shutdown can be enabled only when one of the following conditions is met: - The NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to MINUTE_LEVEL. - The NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL, and the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter is set to a value greater than or equal to 6.				

6.4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	Energy Conservation and Emission Reduction	If the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL and the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter is set to a value less than 6 for intelligent carrier shutdown, multi-carrier joint carrier shutdown must be disabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option (inter-base-station multi-carrier scenarios) of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SW option	<i>Energy Conservation and Emission Reduction</i>	Multi-carrier joint carrier shutdown does not work with fast carrier shutdown.

RAT	Function Name	Function Switch	Reference	Description
		of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected Basic-layer cells: FAST_CARRIER_SHUTDOWN_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected		
Low-frequency TDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	If the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL and the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter is set to a value less than 6 for LTE and NR intelligent carrier shutdown, multi-carrier joint carrier shutdown must be disabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Multi-carrier joint carrier shutdown does not work with low power consumption mode.
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAIgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	Multi-carrier joint carrier shutdown does not work with distributed massive MIMO.

6.4.3.2.3 Function Impacts

None

6.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.4.3.4 Others

None

6.4.4 Operation and Maintenance

6.4.4.1 Data Configuration

6.4.4.1.1 Data Preparation

Table 6-19 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-19 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	It is recommended that the MULTI_CARR_JOINT_CARR_SHDN_SW option be selected for all cells on the carriers sharing a PA.

6.4.4.1.2 Using MML Commands

Before using MML commands, refer to **6.4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling multi-carrier joint carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=MULTI_CARR_JOINT_CARR_SHDN_SW-1;
```

Deactivation Command Examples

```
//Disabling multi-carrier joint carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=MULTI_CARR_JOINT_CARR_SHDN_SW-0;
```

6.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.4.2 Activation Verification

- Step 1** Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Extension Switch** parameter is **Multi-Carrier Joint Carrier Shutdown Switch:On** in the command output, multi-carrier joint carrier shutdown has been enabled.
- Step 2** Verify the carrier shutdown function involved in the multi-carrier negotiation. For details, see the descriptions of activation verification for that function.
- Step 3** You can run the **DSP NRDUCELLPOWERSAVING** command to query the status of the carrier shutdown function involved in multi-carrier negotiation and the reason why that function did not take effect. If the value of **Current State** is **Disabled** for that function, check the **Current Disabled Cause**, **Latest Exit Cause**, and **Latest Disabled Cause** parameter values to find out the possible reason why that function did not take effect.
- In the case that the carrier shutdown function involved in the negotiation between carriers A and B did not take effect for either of the carriers because carrier A (not B) did not meet the triggering conditions for that function, the **Current Disabled Cause** and **Latest Disabled Cause** parameter values are **Multi-Carrier Not Coordinated** for carrier B.
 - In the case that carriers A and B both exit that carrier shutdown function because carrier A no longer meets the conditions for that function to take effect, the **Current Disabled Cause**, **Latest Exit Cause**, and **Latest Disabled Cause** parameter values are **In Penalty State** for carrier B.

----End

6.4.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.
On the live network, the one-week base station energy consumption data obtained before and after multi-carrier joint carrier shutdown is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.
- The duration of cell unavailability due to multi-carrier joint carrier shutdown can be obtained using the **N.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access.
- The duration in which the cell stays in the carrier shutdown state due to multi-carrier joint carrier shutdown can be obtained using the **N.PowerSaving.CarrierShutdown.Dur** counter on the MAE-Access.

6.5 Multi-Frequency Deep Carrier Shutdown (Low-Frequency TDD)

6.5.1 Principles

6.5.1.1 Overview

Multi-frequency deep carrier shutdown works on the basis of intelligent carrier shutdown (see [6.2 Intelligent Carrier Shutdown](#)). It further shuts down the digital processors relevant to the capacity-layer carriers that have been shut down, increasing energy saving gains.

After intelligent carrier shutdown takes effect, the following processing logic applies on the gNodeB:

- When the load of capacity- and basic-layer cells further meets the conditions for entering multi-frequency deep carrier shutdown, multi-frequency deep carrier shutdown takes effect. In this case, the relevant digital processors are shut down.
- When the load of the capacity- and basic-layer cells increases, multi-frequency deep carrier shutdown stops.

Multi-frequency deep carrier shutdown is controlled by the **MULTI_FREQ_DEEP_CARR_SHDN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter.

For details about basic-layer cells and capacity-layer cells, see [Related Concepts](#).

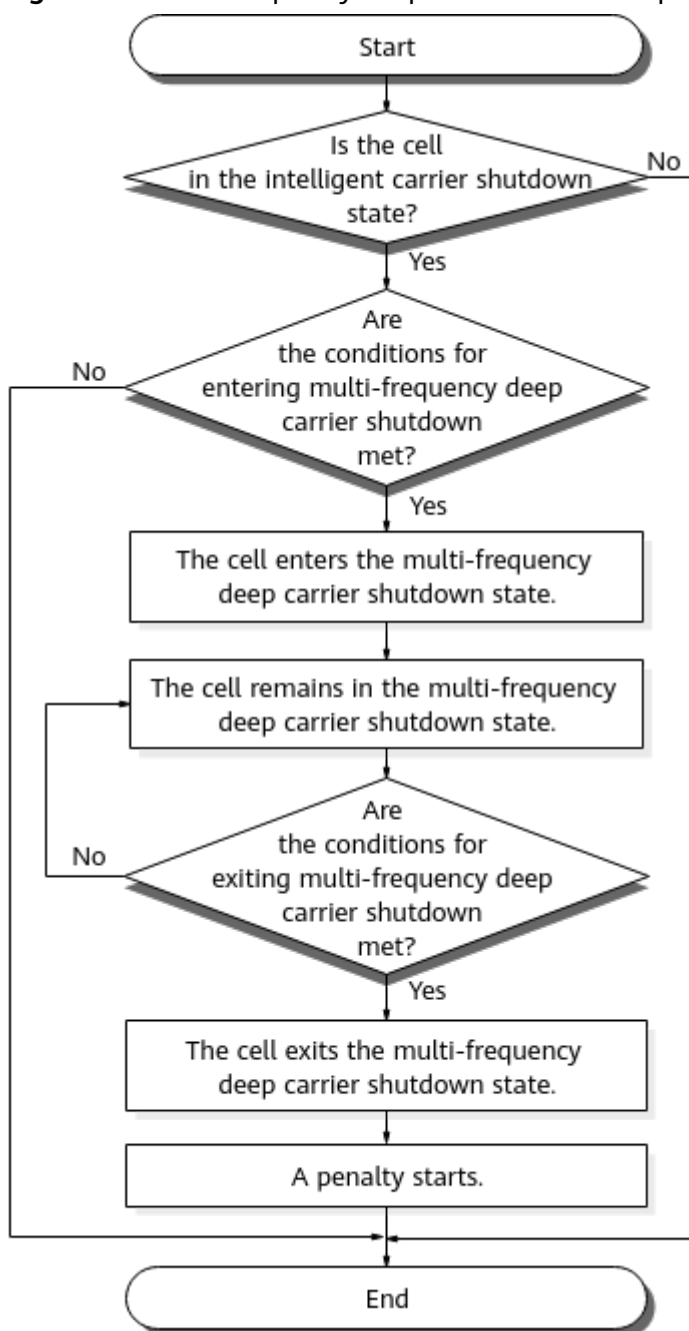
6.5.1.2 Energy Saving Procedure

This function takes effect every day within the configured multi-frequency deep carrier shutdown period. The time period, energy saving type, and other information under an energy saving policy are specified by the parameters in the **NRDUCellPowerSaving** MO.

- The **NRDUCellPowerSaving.PowerSavingPolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.
- The **NRDUCellPowerSaving.PowerSavingType** parameter specifies the energy saving type indicated by an energy saving policy index. Multi-frequency deep carrier shutdown requires that this parameter be set to **MULTI_FREQ_DEEP_CARR_SHUTDOWN**, regardless of whether the frequencies belong to the same base station or not.

[Figure 6-8](#) shows the procedure for multi-frequency deep carrier shutdown.

Figure 6-8 Multi-frequency deep carrier shutdown procedure



Conditions for Entering Multi-Frequency Deep Carrier Shutdown

The base station periodically evaluates whether a cell meets the conditions for entering the multi-frequency deep carrier shutdown state listed below. The evaluation period (specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod**) can only be set at minute-level granularity. That is, the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter must be set to **MINUTE_LEVEL**.

If all of the conditions are met, a capacity-layer cell enters the procedure for multi-frequency deep carrier shutdown. If any of the conditions is not met, the base station waits for the next evaluation period.

- The capacity-layer cell must meet all of the following conditions:
 - The current time is within the multi-frequency deep carrier shutdown period specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
 - The capacity-layer cell is in the intelligent carrier shutdown state.
 - The capacity-layer cell is not in the state of penalty on multi-frequency deep carrier shutdown. For details about the penalty state, see [Conditions for Exiting Multi-Frequency Deep Carrier Shutdown](#).
- The basic-layer cells corresponding to the capacity-layer cell must meet one of the following conditions:
 - All basic-layer cells are available.
 - Some basic-layer cells are available, and all the unavailable cells are in the energy saving state. An example of this case is that an unavailable cell acts as a capacity-layer cell for another basic-layer cell in an energy saving function, such as NR intelligent carrier shutdown, LTE and NR intelligent carrier shutdown, NR load-based RF module deep dormancy, or LNR load-based RF module dormancy.
- The capacity-layer cell and the corresponding basic-layer cells must meet all the following conditions:
 - The total downlink PRB usage of the capacity-layer cell and basic-layer cells is less than or equal to the value of the **NRDUCellPowerSaving.DlPrbThld** parameter. For the formula used to calculate the total downlink PRB usage of the capacity-layer cell and basic-layer cells, see [Table 6-20](#).
 - The total uplink PRB usage of the capacity-layer cell and basic-layer cells is less than or equal to the value of the **NRDUCellPowerSaving.UlPrbThld** parameter. For the formula used to calculate the total uplink PRB usage of the capacity-layer cell and basic-layer cells, see [Table 6-20](#).
 - The total number of RRC_CONNECTED UEs on each pair of the capacity-layer cell and a basic-layer cell is less than or equal to the value of the **NRDUCellPowerSaving.UserNumThld** parameter.
If multiple basic-layer cells correspond to one capacity-layer cell, each pair of the basic- and capacity-layer cells must meet this condition.
 - In multi-operator sharing scenarios:
 - The non-reserved operators of the capacity-layer cell are a subset of the non-reserved operators of any basic-layer cell.
 - The reserved operators of the capacity-layer cell are a subset of the reserved and non-reserved operators of any basic-layer cell.
 For each of the capacity-layer and basic-layer cells, operators are classified into non-reserved operators and reserved operators. A non-reserved operator is an operator with the **NRExternalNCellPlmn.PlmnReservedFlag** parameter set to **FALSE**, and a reserved operator is an operator with the **NRExternalNCellPlmn.PlmnReservedFlag** parameter set to **TRUE**.
 - In inter-base-station multi-carrier scenarios, the Xn interface is in the normal state.

 NOTE

- If the capacity-layer cell is a hyper cell or combined cell:
- The total downlink PRB usage and total uplink PRB usage of the capacity-layer cell and basic-layer cells are calculated based on the maximum number of used PRBs among all TRPs.
 - It is recommended that basic-layer cells also be hyper cells or combined cells. Common cells are not recommended as basic-layer cells in this case.
- The cell bandwidth, slot configuration, and subcarrier spacing are converted for evaluation of the total uplink or downlink PRB usage of the capacity- and basic-layer cells against the configured uplink or downlink PRB threshold, respectively.

Table 6-20 Formulas for calculating various PRB statistics

PRB Usage	Mode	Formula	Description
Total downlink PRB usage of the capacity-layer cell and basic-layer cells	Non-precise shutdown ^a	(Number of PRBs used by PDSCH DRBs of the capacity-layer cell + Total number of used downlink PRBs of the basic-layer cells)/Total number of available downlink PRBs of the basic-layer cells	The base station automatically obtains the number of PRBs used by PDSCH DRBs of the capacity-layer cell, the number of used downlink PRBs of each basic-layer cell, and the number of available downlink PRBs of each basic-layer cell. It then calculates the total downlink PRB usage of the capacity-layer cell and basic-layer cells.
	Precise shutdown ^a	(Number of used downlink PRBs of the capacity-layer cell + Overlap rate of each neighboring cell x Number of used downlink PRBs of each basic-layer cell)/(Number of available downlink PRBs of the capacity-layer cell + Overlap rate of each neighboring cell x Number of available downlink PRBs of each respective basic-layer cell)	The overlap rate of each neighboring cell is specified by the NRCellRelation.OverlapRate parameter at the capacity-layer cell side.

PRB Usage	Mode	Formula	Description
Total uplink PRB usage of the capacity-layer cell and basic-layer cells	Non-precise shutdown	(Number of used uplink PRBs of the capacity-layer cell + Total number of used uplink PRBs of the basic-layer cells)/Total number of available uplink PRBs of the basic-layer cells	None
	Precise shutdown	(Number of used uplink PRBs of the capacity-layer cell + Overlap rate of each neighboring cell x Number of used uplink PRBs of each basic-layer cell)/ (Number of available uplink PRBs of the capacity-layer cell + Overlap rate of each neighboring cell x Number of available uplink PRBs of each respective basic-layer cell)	The overlap rate of each neighboring cell is specified by the NRCellRelation.OverlapRate parameter at the capacity-layer cell side.
a: For details about non-precise and precise shutdown modes, see 6.2.1.1 Overview .			

Entering Multi-Frequency Deep Carrier Shutdown

If a capacity-layer cell in the intelligent carrier shutdown state meets the conditions for entering multi-frequency deep carrier shutdown, the cell enters the multi-frequency deep carrier shutdown state. The digital processor relevant to the capacity-layer carrier that has been shut down is also shut down when it is no longer serving any working carrier.

Conditions for Exiting Multi-Frequency Deep Carrier Shutdown

After the capacity-layer cell enters the multi-frequency deep carrier shutdown state, the base station periodically evaluates the conditions for exiting this state. The evaluation period (specified by the **NRDUCellPowerSaving.ExitEvaluationPeriod**) can only be set at minute-level granularity. That is, the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter must be set to **MINUTE_LEVEL**.

The capacity-layer cell automatically exits the multi-frequency deep carrier shutdown state and cannot enter this state again within a period specified by the

NRDUCellPowerSaving.PwrSavingPenaltyTime parameter if any of the conditions listed below is met. (The cell is in the penalty state during this period.) If none of these conditions is met, the cell remains in the multi-frequency deep carrier shutdown state.

- The current time is not within the multi-frequency deep carrier shutdown period specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
- The total downlink PRB usage of the basic-layer cells is greater than the sum of **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset**.
- The total uplink PRB usage of the basic-layer cells is greater than the sum of **NRDUCellPowerSaving.UlPrbThld** and **NRDUCellPowerSaving.NCellUlPrbOffset**.
- The number of RRC_CONNECTED UEs in a basic-layer cell is greater than the sum of **NRDUCellPowerSaving.UserNumThld** and **NRDUCellPowerSaving.UserNumOffset**.
If multiple basic-layer cells correspond to one capacity-layer cell, only one basic-layer cell needs to meet this condition.
- In multi-operator sharing scenarios:
 - The non-reserved operators of the capacity-layer cell are not a subset of the non-reserved operators of any basic-layer cell.
 - The reserved operators of the capacity-layer cell are not a subset of the reserved and non-reserved operators of any basic-layer cell.
- In inter-base-station multi-carrier scenarios, the Xn interface is in the abnormal state.
- Any of the basic-layer cells becomes unavailable for a reason other than intelligent carrier shutdown, LTE and NR intelligent carrier shutdown, and NR load-based RF module deep dormancy.

NOTE

- When a basic-layer cell becomes unavailable in inter-base-station scenarios, there may be a minute-level delay for the wake-up of the capacity-layer cell. This is because the messages need to be frequently transmitted over the Xn interface to indicate the energy saving status and capacity-layer cell wake-up notifications, leading to possible flow control.
- If a basic-layer cell is a multi-TRP cell such as a hyper cell or combined cell and some of its TRP carriers become unavailable, the capacity-layer cell also automatically exits the multi-frequency deep carrier shutdown state.
- The penalty is not imposed if the capacity-layer cell exits the multi-frequency deep carrier shutdown state because the function switch is turned off, the cell is deactivated and then reactivated, or the base station software version is upgraded. After the capacity-layer cell is deactivated and then reactivated or the base station software version upgrade is complete, the base station evaluates whether the cell enters the multi-frequency deep carrier shutdown state again based on related entering conditions.
- The capacity-layer cell may exit the multi-frequency deep carrier shutdown state because the exit conditions are met during evaluation of basic-layer neighboring cell addition for the capacity-layer cell.

6.5.2 Network Analysis

6.5.2.1 Benefits

On a multi-frequency co-coverage network that is lightly loaded or unloaded, multi-frequency deep carrier shutdown further reduces power consumption on the basis of intelligent carrier shutdown. However, the energy saving gains are limited in the following scenarios:

- When multiple carriers share the same digital processor in an RF module and an NR carrier enters the multi-frequency deep carrier shutdown state, the digital processor cannot be shut down if other carriers are still working. Energy saving gains decrease in this case.
- To prevent possible hardware faults caused by an excessively large temperature differential, this function may not take effect or may be terminated when the temperature differential of RF modules is excessively large. Energy saving gains may decrease in this case.

6.5.2.2 Impacts

The impacts between this function and other functions are described in [6.5.3.2.3 Function Impacts](#).

As digital processors are shut down for multi-frequency deep carrier shutdown, it takes a longer time to exit this shutdown state than to exit the intelligent carrier shutdown state. Therefore, the **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** of basic-layer cells may decrease.

Assuming that the neighboring base stations are in the carrier shutdown state, cell-edge UEs in weak coverage areas may access the cells served by the local base station. This affects KPIs related to the aspects such as access and service drops of the local base station. For example, the value of the **N.NsaDc.DRB.AbnormRel** counter may increase.

6.5.3 Requirements

6.5.3.1 Licenses

None

6.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Multi-frequency deep carrier shutdown can be enabled only if intelligent carrier shutdown has been enabled.

6.5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: - INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter - FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter - Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	Energy Conservation and Emission Reduction	Multi-frequency deep carrier shutdown does not work with fast carrier shutdown.

6.5.3.2.3 Function Impacts

None

6.5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Only AAU5368md supports this function.

6.5.3.4 Others

None

6.5.4 Operation and Maintenance

6.5.4.1 Data Configuration

6.5.4.1.1 Data Preparation

Table 6-21 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-21 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	To enable multi-frequency deep carrier shutdown, select the MULTI_FREQ_DEEP_CARR_SHDN_SW option.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	When multi-frequency deep carrier shutdown is enabled, set this parameter to MULTI_FREQ_DEEP_CARR_SHUTDOWN .

Parameter Name	Parameter ID	Setting Notes
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	When multi-frequency deep carrier shutdown is enabled, this parameter must be set to MINUTE_LEVEL .
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	Use the recommended value.
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: • For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. • For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.

Parameter Name	Parameter ID	Setting Notes
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan and the following requirements: - For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. - For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	Use the recommended value.
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	Use the recommended value.
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	Use the recommended value.
Neighboring Cell Uplink PRB Offset	NRDUCellPowerSaving.NCellUlPrbOffset	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	Use the recommended value.
User Number Offset	NRDUCellPowerSaving.UserNumOffset	Use the recommended value.
Exit Evaluation Period	NRDUCellPowerSaving.ExitEvaluationPeriod	Use the recommended value.
Power Saving Penalty Duration	NRDUCellPowerSaving.PwrSavingPenaltyTime	Use the recommended value.
Power Saving Cell Flag	NRCellRelation.PowerSavingCellFlag	Set this parameter based on the network plan.
No Removal Flag	NRCellRelation.NoRmvFlag	When automatic removal of neighboring cells in ANR is enabled (with the NR_NR_ANR_DEL_SW option of the NRCellAlgoSwitch.AnrSwitch parameter selected), it is recommended that the NRCellRelation.NoRmvFlag parameter be set to FORBID_ANR_RMV for the bidirectional neighbor relationships between capacity-layer and basic-layer cells.
PLMN Reserved Flag	NRExternalNCellPlmn.PlmnReservedFlag	Set this parameter based on the network plan.

6.5.4.1.2 Using MML Commands

Before using MML commands, refer to [6.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Assume that a cell with **NrCellId** being 0 is a capacity-layer cell and another cell with **NrCellId** being 10 is a basic-layer cell. The **PowerSavingCellFlag** parameter in the **NRCellRelation** MO must be set in both capacity- and basic-layer cells for multi-frequency deep carrier shutdown. For details about how to configure neighboring cells, see *SA Mobility Management in Connected Mode*.

```
//Enabling multi-frequency deep carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=MULTI_FREQ_DEEP_CARR_SHDN_SW-1;
//Setting an energy saving policy of multi-frequency deep carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=MULTI_FREQ_DEEP_CARR_SHUTDOWN, StartTime=00&00&00, StopTime=06&00&00,
DlPrbThld=10, DlPrbOffset=10, UserNumThld=20, UserNumOffset=20, PwrSavingPenaltyTime=15,
UlPrbThld=10, NCellUlPrbOffset=10, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1,
ExitEvaluationPeriod=1;
//Optional Setting another energy saving policy of multi-frequency deep carrier shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1,
PowerSavingType=MULTI_FREQ_DEEP_CARR_SHUTDOWN, StartTime=12&00&00, StopTime=14&00&00,
DlPrbThld=10, DlPrbOffset=10, UserNumThld=20, UserNumOffset=20, PwrSavingPenaltyTime=15,
UlPrbThld=10, NCellUlPrbOffset=10, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1,
ExitEvaluationPeriod=1;
//Setting PowerSavingCellFlag on the local base station, and (recommended) setting NoRmvFlag to
FORBID_ANR_RMV for the bidirectional neighbor relationships between capacity-layer and basic-layer cells
when automatic removal of neighboring cells in ANR takes effect
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=10,
PowerSavingCellFlag=COVERAGE, NoRmvFlag=FORBID_ANR_RMV;
MOD NRCELLRELATION: NrCellId=10, Mcc="302", Mnc="220", gNBId=0, CellId=0,
PowerSavingCellFlag=CAPACITY, NoRmvFlag=FORBID_ANR_RMV;
//Optional Setting the PlmnReservedFlag parameter in the NR external-cell PLMN list for the capacity-
layer cell based on the network plan in multi-operator sharing scenarios
MOD NREXTERNALNCELLPLMN: Mcc="302", Mnc="220", gNBId=1, CellId=10, SharedMcc="302",
SharedMnc="610", PlmnReservedFlag=FALSE;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling multi-frequency deep carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=MULTI_FREQ_DEEP_CARR_SHDN_SW-0;
//Removing an energy saving policy of multi-frequency deep carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
//Optional Removing the other energy saving policy of multi-frequency deep carrier shutdown
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1;
```

6.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.5.4.2 Activation Verification

After activating multi-frequency deep carrier shutdown, use either of the following methods to check whether this function has taken effect:

- Method 1:
 - a. Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Extension Switch** parameter is **Multi-Frequency Deep Carrier Shutdown Sw:On** in the command output, multi-frequency deep carrier shutdown has been enabled.
 - b. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **Multi-Frequency Deep Carrier Shutdown**.

NOTE

If there are two or more records with **Power Saving Type** being **Multi-Frequency Deep Carrier Shutdown**, multi-frequency deep carrier shutdown has been enabled in two or more time periods.

- c. Run the **DSP TIME** command. Check whether the system time is within the periods for this function to take effect.
 - If it is, go to **d**.
 - If it is not, wait until the start time arrives.
 - d. Wait for a certain period of time. Then, run the **DSP NRDUCELL** command. If the **Power Saving State** parameter indicates **Multi-Frequency Deep Carrier Shutdown:Enabled**, multi-frequency deep carrier shutdown has taken effect.

Maximum wait time =
NRDUCellPowerSaving.ShutdownTrigJudgePeriod +
NRDUCellPowerSaving.PwrSavingPenaltyTime
- Method 2:

Run the **DSP NRDUCELLPOWERSAVING** command to query the status of multi-frequency deep carrier shutdown and the reason why it did not take effect.

 - If the value of **Power Saving Type** is **Multi-Frequency Deep Carrier Shutdown** and the value of **Current State** is **Enabled**, multi-frequency deep carrier shutdown has taken effect.
 - If the value of **Power Saving Type** is **Multi-Frequency Deep Carrier Shutdown** and the value of **Current State** is **Disabled**, check the **Current Disabled Cause** parameter value to find out the possible reason why multi-frequency deep carrier shutdown did not take effect. [Table 6-22](#) lists possible reasons.

Table 6-22 Possible reasons why multi-frequency deep carrier shutdown did not take effect

Reason	Description
Switch Turned Off	None

Reason	Description
Neighboring Cell Not Configured	In multi-operator sharing scenarios: • The PLMN IDs of the non-reserved operators of the capacity-layer cell are not a subset of the PLMN IDs of the non-reserved operators of any basic-layer cell. • The PLMN IDs of the reserved operators of the capacity-layer cell are not a subset of the PLMN IDs of the reserved and non-reserved operators of any basic-layer cell.
Triggering Time Not Reached	The current time is not within the effective period of multi-frequency deep carrier shutdown.
Neighboring Cell Unavailable	• A neighboring cell for which NRCellRelation.PowerSavingCellFlag is set to COVERAGE is unavailable. • A neighboring cell for which NRCellRelation.PowerSavingCellFlag is set to COVERAGE is a multi-TRP cell such as a hyper cell or combined cell, and some TRPs of this cell are unavailable. • A neighboring cell for which NRCellRelation.PowerSavingCellFlag is set to COVERAGE is an NSA cell.
Xn Interface Unavailable	In inter-base-station multi-carrier scenarios, the Xn interface is in the abnormal state.
Impacted Feature Taken Effect	None
Prerequisite Feature Not Taken Effect	None
Downlink PRB Threshold Not Reached	None
Uplink PRB Threshold Not Reached	None
UE Number Threshold Not Reached	None

Reason	Description
Triggering Decision Period Not Met	The current time is within the effective period of multi-frequency deep carrier shutdown and is not within the first detection period (lasting 1 minute). However, it is within the evaluation period for triggering multi-frequency deep carrier shutdown (specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter).
In Penalty State	The cell is in the penalty state after it exits the multi-frequency deep carrier shutdown state.
RF Hardware Capability Not Supported	• The hardware is incompatible. • The hardware is not reliable enough, for example, when multi-frequency deep carrier shutdown has taken effect in other frequency bands.
Others	• The current time is within the effective period of multi-frequency deep carrier shutdown as well as the first detection period (lasting 1 minute). • Internal software or other reasons cause the function to fail.

6.5.4.3 Network Monitoring

Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after multi-frequency deep carrier shutdown is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.

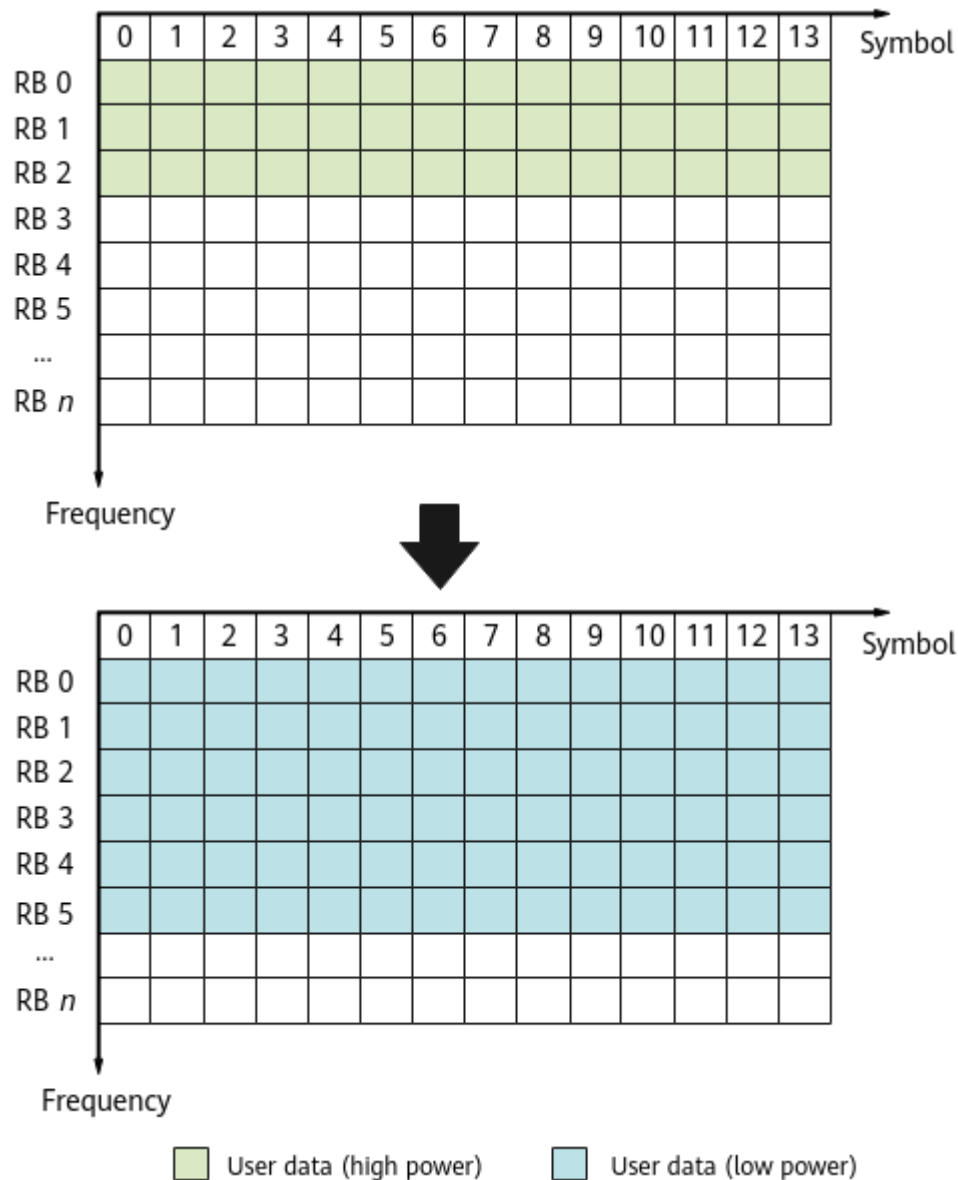
7 Device-Level Energy Saving (Power Domain)

7.1 Energy Saving Based on Flexible Frequency-Domain Scheduling

7.1.1 Principles

Energy saving based on flexible frequency-domain scheduling expands available frequency-domain resources for target UEs when there are idle frequency-domain resources on the network, and it dynamically lowers the transmit power of the cell. This reduces the overall energy consumption of the base station. [Figure 7-1](#) shows how this function works.

Figure 7-1 Working principles for energy saving based on flexible frequency-domain scheduling



Energy saving based on flexible frequency-domain scheduling can be enabled by selecting the **FLEX_FREQ_SCH_ENERGY_SAVING_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter. [Table 7-1](#) describes the parameters in the **NRDUCellPowerSaving** MO that specify the time period, load threshold, and other information under an energy saving policy of energy saving based on flexible frequency-domain scheduling.

Table 7-1 Parameters related to a policy of energy saving based on flexible frequency-domain scheduling

Parameter Name	Parameter ID	Description
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	This parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	This parameter specifies the energy saving type indicated by an energy saving policy index. Energy saving based on flexible frequency-domain scheduling requires that this parameter be set to FLEX_FREQ_SCH_ENERGY_SAVING .

Parameter Name	Parameter ID	Description
Start Time	NRDUCellPowerSaving.StartTime	These parameters specify the effective period for energy saving based on flexible frequency-domain scheduling, as well as the evaluation period and conditions for a cell to enter and exit the state of energy saving based on flexible frequency-domain scheduling. Within the effective period specified by NRDUCellPowerSaving.StartTime and NRDUCellPowerSaving.StopTime for energy saving based on flexible frequency-domain scheduling in a cell, the base station periodically determines whether the cell enters the state of energy saving based on flexible frequency-domain scheduling. The period is specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter. • If the downlink PRB usage of a cell or TRP^c is not greater than the value of NRDUCellPowerSaving.DlPrbThld after energy saving based on flexible frequency-domain scheduling is enabled, the cell or TRP automatically enters the state of energy saving based on flexible frequency-domain scheduling^a. • If the downlink PRB usage of a cell or TRP^c is greater than the sum of the NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset parameter values or the current time is not within the period during which energy saving based on flexible frequency-domain scheduling can take effect, the cell or TRP automatically exits the state of energy saving based on flexible frequency-domain scheduling^b.
Stop Time	NRDUCellPowerSaving.StopTime	
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	

Parameter Name	Parameter ID	Description
Flexible Freq Scheduling Mode	NRDUCellPowerSaving.FlexFreqSchMode	This parameter specifies the mode of energy saving based on flexible frequency-domain scheduling. • If this parameter is set to BASIC, energy saving is performed based on the actual scheduling conditions. • If this parameter is set to ESTIMATE, energy saving is performed based on an estimation algorithm. • If this parameter is set to EXTEND_RB, energy saving is performed based on an extension algorithm, with more frequency-domain resources used for scheduling.

Parameter Name	Parameter ID	Description
		a: A cell or TRP enters the state of energy saving based on flexible frequency-domain scheduling. • A common cell automatically enters the state of energy saving based on flexible frequency-domain scheduling if its downlink PRB usage is less than or equal to the value of NRDUCellPowerSaving.DlPrbThld. • A TRP of a hyper cell or combined cell automatically enters the state of energy saving based on flexible frequency-domain scheduling if its downlink PRB usage is less than or equal to the value of NRDUCellPowerSaving.DlPrbThld. b: A cell or TRP exits the state of energy saving based on flexible frequency-domain scheduling. • A common cell automatically exits the state of energy saving based on flexible frequency-domain scheduling if its downlink PRB usage is greater than the sum of the NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset parameter values. • A TRP of a hyper cell or combined cell automatically exits the state of energy saving based on flexible frequency-domain scheduling if its downlink PRB usage is greater than the sum of the NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset parameter values. The hyper cell or combined cell exits this state only when all TRPs exit this state. • A combined cell in joint transmission mode automatically exits the state of energy saving based on flexible frequency-domain scheduling if the downlink PRB usage of any TRP of this cell is greater than the sum of the NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset parameter values. c: The following formulas are used to calculate the downlink PRB usage of a cell or TRP: • Downlink PRB usage of a common cell = (Average number of PRBs used by downlink DRB services on the PDSCH – Average number of extended PRBs used by the PDSCH)/Average number of available downlink PRBs • Downlink PRB usage of a TRP of a hyper cell or combined cell = (Average number of PRBs used by downlink DRB services on the PDSCH in the TRP – Average number of downlink extended PRBs used in the TRP)/Average number of available downlink PRBs in the TRP

NOTE

The following UEs are not target UEs for energy saving based on flexible frequency-domain scheduling:

- UEs involved in coordinated multipoint transmission/reception (CoMP), coordinated scheduling (CS), coordinated beamforming (CBF), multi-user MIMO (MU-MIMO), fixed wireless access (FWA) services with a guaranteed rate configured but not actually in effect, voice over NR (VoNR), Non-Slot, or high-reliability services
- UEs performing delay-sensitive services
- UEs involved in frequency selective scheduling

To ensure performance of UEs with specific service types, energy saving based on flexible frequency-domain scheduling can be configured to not take effect for these UEs. This is known as service-type-based dynamic exit from energy saving based on flexible frequency-domain scheduling. The service types involved are specified by the **gNBQciBearer.ServiceType** parameter.

Service-type-based dynamic exit from energy saving based on flexible frequency-domain scheduling takes effect when all of the following conditions are met:

- Energy saving based on flexible frequency-domain scheduling is enabled.
- The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.
- The **FLEX_FREQ_SCH_UE_RESTR_SW** option of the **NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter is selected in the group identified by the **NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter that is set to the same value as the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter for a specific service type.

7.1.2 Network Analysis

7.1.2.1 Benefits

The gains vary with network loads as follows:

- When the network is under no load, energy saving based on flexible frequency-domain scheduling produces no energy saving gains.
- When the network is under extremely light load, energy saving based on flexible frequency-domain scheduling produces low energy saving gains due to low energy consumption over the air interface for UE scheduling.
- When network is under low or moderate load, energy saving based on flexible frequency-domain scheduling expands available frequency-domain resources for target UEs and dynamically lowers the transmit power of the cell. This reduces the overall energy consumption of the base station. The energy saving gains increase in comparison to the gains achieved when the network is under no or extremely light load. However, if the network load increases, the frequency-domain resources that can be scheduled decrease, thereby decreasing the energy saving gains.

Energy saving gains may also decrease when some of the UEs on the network are not target UEs of energy saving based on flexible frequency-domain scheduling. If none of the UEs are target UEs of this function, there will be no energy saving gains.

Different settings of the **NRDUCellPowerSaving.FlexFreqSchMode** parameter contribute to different gains:

- If this parameter is set to **ESTIMATE**, the power saving gains are higher than those when this parameter is set to **BASIC**.
- If this parameter is set to **EXTEND_RB**, the power saving gains are even higher than those when this parameter is set to **BASIC** or **ESTIMATE**.

Service-type-based dynamic exit from energy saving based on flexible frequency-domain scheduling relieves the impact of energy saving based on flexible

frequency-domain scheduling on specific service types but reduces energy saving gains.

7.1.2.2 Impacts

The impacts between this function and other functions are described in [7.1.3.2.3 Function Impacts](#).

After energy saving based on flexible frequency-domain scheduling takes effect:

- The IBLER will increase.
- The MAC-layer processing delay in the gNodeB may increase. This delay is calculated as follows: **Average Downlink Packet Delay in MAC** = $(N.QoS.DL.PktDelayAirInterface.Time / N.QoS.DL.PktDelayAirInterface.Num) / 1000$.
- In downlink CA scenarios, the traffic volume of downlink CA UEs in SCells may decrease.
- The average MCS index for downlink scheduling decreases.
- In cells exposed to strong interference from neighboring cells, when energy saving based on flexible frequency-domain scheduling takes effect in neighboring cells, interference from neighboring cells decreases.
 - The average CQI may increase.
 - The values of the **N.PRB.DL.DrbUsed.Avg** and **N.PRB.DL.Used.Avg** counters may decrease.
- The values of the following counters related to downlink PRB usage may increase in common cells:
 - **N.PRB.DL.DrbUsed.Avg**
 - **N.PRB.DL.ExtendUsed.Avg**
 - **N.PRB.DL.Used.Avg**
- The values of the following counters related to the downlink PRB usage of TRPs may increase in hyper cells or combined cells:
 - **N.HyperCell.TRP.PRB.DL.Used.Avg**
 - **N.HyperCell.TRP.PRB.DL.ExtendUsed.Avg**
- When the **NRDUCellPowerSaving.FlexFreqSchMode** parameter is set to different values, the user-perceived rate varies as follows:
 - If this parameter is set to **BASIC**, the user-perceived data rate remains almost unchanged.
 - If this parameter is set to **ESTIMATE** or **EXTEND_RB**, the user-perceived rate slightly decreases.

7.1.3 Requirements

7.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-060203	Energy Saving Based on Flexible Frequency-Domain Scheduling	NR0SESBFFS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Service-type-based dynamic exit from energy saving based on flexible frequency-domain scheduling takes effect only when energy saving based on flexible frequency-domain scheduling is enabled.

7.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Primary/ Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Energy saving based on flexible frequency-domain scheduling is incompatible with the primary/ backup function in intra-frequency dual-network redundancy networking.
Low-frequency TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Energy saving based on flexible frequency-domain scheduling is incompatible with dual-BBU backup in dual-network redundancy networking.
Low-frequency TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	Energy saving based on flexible frequency-domain scheduling is incompatible with the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios.

7.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR multi-band power adaptive scheduling	• LTE: LNR_MULTI_B AND_PWR_ADPT_SCH_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter • NR: LNR_MULTI_B AND_PWR_ADPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When LTE and NR multi-band power adaptive scheduling takes effect together with energy saving based on flexible frequency-domain scheduling, the benefits of both functions decrease.
Low-frequency TDD	L/NR Inter-band Deep Coordination	• LTE: LNR_INTER_B AND_DEEP_COORD_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter • NR: LNR_INTER_B AND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	When L/NR Inter-band Deep Coordination takes effect together with energy saving based on flexible frequency-domain scheduling, the benefits of both functions decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When both NR inter-carrier dynamic power sharing and energy saving based on flexible frequency-domain scheduling take effect, the gains of energy saving based on flexible frequency-domain scheduling decrease. As a result, the gains of power sharing decrease.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SWITCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When NR multi-band power adaptive scheduling and energy saving based on flexible frequency-domain scheduling both take effect, the gains provided by both functions may decrease.
Low-frequency TDD	Power aggregation in unoccupied bandwidths	NRDUCellChnPower.MaxPdschConvPwrOffset	<i>Power Control</i>	When power aggregation in unoccupied bandwidths is enabled and energy saving based on flexible frequency-domain scheduling takes effect, power aggregation in unoccupied bandwidths offers lower gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVO ID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When both remote interference adaptive avoidance and energy saving based on flexible frequency-domain scheduling take effect, the gains of energy saving based on flexible frequency-domain scheduling decrease.
Low-frequency TDD	Target IBLER adaptation based on interference from neighboring cells	DL_INTRF_BASED_IBLER_ADAPT_SW option of the NRDUCellDlAmc.DlMcsSelAlgoSw parameter	<i>Downlink Scheduling</i>	When both target IBLER adaptation based on interference from neighboring cells and energy saving based on flexible frequency-domain scheduling take effect, the former function may not deliver any gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink RB estimation optimization	RB_ESTIMATE_OPT_SW option of the NRDUCellPdsch.DLPdschAlgoSwitch parameter	<i>Downlink Scheduling</i>	If energy saving based on flexible frequency-domain scheduling takes effect but downlink RB estimation optimization does not take effect, the user-perceived rate may decrease. It is recommended that downlink RB estimation optimization be enabled when energy saving based on flexible frequency-domain scheduling is in effect.
Low-frequency TDD	RB allocation for downlink experience optimization	NRDUCellDISchRes.DIExpOptRblncreasePct	<i>Downlink Scheduling</i>	When both RB allocation for downlink experience optimization and energy saving based on flexible frequency-domain scheduling take effect, the gains offered by energy saving based on flexible frequency-domain scheduling will decrease and an exit from energy saving based on flexible frequency-domain scheduling may be performed earlier than expected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink type 1 resource allocation for UEs performing small-packet services	SMALL_PKT_TYPE1_SCH_SW option of the NRDUCELLPDSCH.DLPdschAlgoSwitch parameter	<i>Downlink Scheduling</i>	When energy saving based on flexible frequency-domain scheduling takes effect, selecting the SMALL_PKT_TYPE1_SCH_SW option can produce additional energy saving gains. In this case, if the SMALL_PKT_MCS_SELECT_SW option of the NRDUCellPdsch.DLPdschAlgoSwitch parameter is deselected: • The IBLER increases slightly. • The proportion of downlink HARQ retransmissions slightly increases. • The average MCS index for downlink scheduling increases. If the SMALL_PKT_MCS_SELECT_SW option of the NRDUCellPdsch.DLPdschAlgoSwitch parameter is selected: • The IBLER decreases slightly. • The proportion of downlink

RAT	Function Name	Function Switch	Reference	Description
				HARQ retransmissions slightly decreases. The average MCS index for downlink scheduling decreases.
Low-frequency TDD	Interference coordination between public and dedicated networks	HIGH_SPEED_INT RF_AVOID_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	When both interference coordination between public and dedicated networks and energy saving based on flexible frequency-domain scheduling take effect, the latter function does not deliver any energy saving gains in public-network cells within the time window when a train arrives.
Low-frequency TDD	Intelligent scheduling for power saving	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both power saving scheduling (sub-function of intelligent scheduling for power saving) and energy saving based on flexible frequency-domain scheduling take effect, the gains of energy saving based on flexible frequency-domain scheduling decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both timing carrier shutdown and energy saving based on flexible frequency-domain scheduling take effect, the latter function does not deliver energy saving gains.
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTICARRIER_SHUTDOWN_SW or INTER_GNB_MULTICARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both intelligent carrier shutdown and energy saving based on flexible frequency-domain scheduling take effect, the latter function does not deliver energy saving gains.
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both TTI-level channel shutdown and energy saving based on flexible frequency-domain scheduling take effect, the gains of energy saving based on flexible frequency-domain scheduling decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both energy saving based on flexible frequency-domain scheduling and dynamic RF channel shutdown are enabled, energy saving based on flexible frequency-domain scheduling will deliver fewer energy saving gains.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCY_SW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Energy saving based on flexible frequency-domain scheduling does not take effect when RF module deep dormancy is in effect.
Low-frequency TDD	RF module super dormancy	RRU.DORMANCY_SW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	RF module super dormancy takes precedence over energy saving based on flexible frequency-domain scheduling when both functions are enabled. Energy saving based on flexible frequency-domain scheduling does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS W (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over energy saving based on flexible frequency-domain scheduling when both functions are enabled. Energy saving based on flexible frequency-domain scheduling does not take effect when RF module fast deep dormancy is in effect.
Low-frequency TDD	Cell Combination	NRDUCell.NrDualCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	The gains of energy saving based on flexible frequency-domain scheduling decrease in combined cells.
Low-frequency TDD	MetaAAU power reduction for cell-center UEs	NRDUCellChnPower.PwrReductionValue not set to 0	<i>Energy Conservation and Emission Reduction</i>	When MetaAAU power reduction for cell-center UEs and energy saving based on flexible frequency-domain scheduling take effect simultaneously, MetaAAU power reduction for cell-center UEs delivers fewer energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option (inter-base-station multi-carrier scenarios) of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected - Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected	<i>Energy Conservation and Emission Reduction</i>	When both fast carrier shutdown and energy saving based on flexible frequency-domain scheduling take effect, the latter function does not deliver energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUeIdentSw	None	Energy saving based on flexible frequency-domain scheduling fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
Low-frequency TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.AppIpv4 • gNBAsAppParam.AppIpv6	None	• Any option of the gNodeBParam.StUeIdentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.AppIpv4 parameter is set. • The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue: • Select the SPEED_TURBO_PWR_SAVING

RAT	Function Name	Function Switch	Reference	Description
				_SW option of the gNB PwrSaving Algo. PwrSavin gAlgoSwitch parameter. Energy saving based on flexible frequency-domain scheduling can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNB PwrSaving Algo. SpeedTur boPwrSavStartTime and gNB PwrSaving Algo. SpeedTur boPwrSavEndTime parameters, increasing the energy saving gains. • Select the STATIC_ST_PWR_SAVING_SW option of the gNB PwrSaving Algo. PwrSavin gAlgoSwitch parameter. Energy saving based on flexible frequency-domain scheduling can take effect for

RAT	Function Name	Function Switch	Reference	Description
				cells served by the base station in some of the preceding scenarios within the period specified by the gNBPwrSaving Algo.StaticSTPwrSavStartTime and gNBPwrSaving Algo.StaticSTPwrSavEndTime parameters, increasing the energy saving gains.

7.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands, except those for which the **Description** parameter value includes "V3" in the command output of **DSP BRDMFRINFO**, support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.1.3.4 Cells

This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**. This function can be configured but does not take effect in cells with the **NRDUCell.DuplexMode** parameter set to **CELL_SUL**.

In NR TDD, the **NRDUCell.DeployPosition** parameter cannot be set to **DEPLOY_IAB**.

7.1.3.5 Others

None

7.1.4 Operation and Maintenance

7.1.4.1 Data Configuration

7.1.4.1.1 Data Preparation

[Table 7-2](#) and [Table 7-3](#) describe the parameters used for function activation and optimization, respectively. This section does not describe parameters related to cell establishment.

Table 7-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellAlgoSwitch.NrDuCellId	Set this parameter based on the network plan.
NR DU Cell ID	NRDUCellPowerSaving.NrDuCellId	Set this parameter based on the network plan.
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	Select the FLEX_FREQ_SCH_ENERGY_SAVING_SW option of this parameter when energy saving based on flexible frequency-domain scheduling is required.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: • For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. • For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.

Parameter Name	Parameter ID	Setting Notes
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan and the following requirements: • For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. • For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to FLEX_FREQ_SCH_ENERGY_SAVING when energy saving based on flexible frequency-domain scheduling is required.
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	You are advised to use the recommended value.
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	You are advised to use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Flexible Freq Scheduling Mode	NRDUCellPowerSaving.FlexFreqSchMode	You are advised to use the recommended value.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	For energy saving based on flexible frequency-domain scheduling, this parameter can only be set to its default value.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	You are advised to use the recommended value.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	It is recommended that the SERV_TYPE_DYN_EXIT_PWR_SAVE_SW option be selected when there are specific types of services and user experience of these services needs to be ensured.
Power Saving Policy Group Index	gNBQciBearer.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan.
Power Saving Policy Group Index	NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan. The value of this parameter must be the same as that of the gNBQciBearer.PowerSavingPolicyGrpIndex parameter for services to be guaranteed. Before setting this parameter, ensure that the QCI for which the gNBQciBearer.PowerSavingPolicyGrpIndex parameter is set has been configured.
User Identification Method	NRDUCellPsPolicyGrp.UserIdentMethod	Set this parameter to SERVICE_TYPE .

Parameter Name	Parameter ID	Setting Notes
Service Type Dynamic Exit Power Saving Switch	NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw	To enable service-type-based dynamic exit from energy saving based on flexible frequency-domain scheduling, select the FLEX_FREQ_SCH_UE_RESTR_SW option.

Table 7-3 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Downlink PDSCH Algorithm Switch	NRDUCellPdsch.DlPdschAlgoSwitch	When energy saving based on flexible frequency-domain scheduling is enabled, select the RB_ESTIMATE_OPT_SW option of this parameter to enable downlink RB estimation optimization. For details about the impact of downlink RB estimation optimization on network performance, see <i>Downlink Scheduling</i> .

7.1.4.1.2 Using MML Commands

Before using MML commands, refer to [7.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling energy saving based on flexible frequency-domain scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=FLEX_FREQ_SCH_ENERGY_SAVING_SW-1;
```

- **Scenario 1:**
//Setting a policy of energy saving based on flexible frequency-domain scheduling (with FlexFreqSchMode set to BASIC)
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=01&00&00, StopTime=03&00&00, PowerSavingType=FLEX_FREQ_SCH_ENERGY_SAVING, DlPrbThld=40, DlPrbOffset=10, FlexFreqSchMode=BASIC, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5;
- **Scenario 2:**
//(Optional) Setting a policy of energy saving based on flexible frequency-domain scheduling (with FlexFreqSchMode set to ESTIMATE)


```
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1, StartTime=04&00&00,
StopTime=05&00&00, PowerSavingType=FLEX_FREQ_SCH_ENERGY_SAVING, DlPrbThld=40,
DlPrbOffset=10, FlexFreqSchMode=ESTIMATE, EvalPeriodGranularity=MINUTE_LEVEL,
ShutdownTrigJudgePeriod=5;
```

- **Scenario 3:**

```
//(Optional) Setting a policy of energy saving based on flexible frequency-domain scheduling (with
FlexFreqSchMode set to EXTEND_RB)
```

```
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1, StartTime=04&00&00,
StopTime=05&00&00, PowerSavingType=FLEX_FREQ_SCH_ENERGY_SAVING, DlPrbThld=40,
DlPrbOffset=10, FlexFreqSchMode=EXTEND_RB, EvalPeriodGranularity=MINUTE_LEVEL,
ShutdownTrigJudgePeriod=5;
```

```
//(Optional) Enabling service-type-based dynamic exit from energy saving based on flexible frequency-
domain scheduling
```

```
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
```

```
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
```

```
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;
```

```
ADD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,
```

```
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=FLEX_FREQ_SCH_UE_RESTR_SW-1;
```

Optimization Command Examples

```
//Enabling downlink RB estimation optimization
```

```
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPdschAlgoSwitch=RB_ESTIMATE_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//(Optional) Disabling service-type-based dynamic exit from energy saving based on flexible frequency-
domain scheduling
```

```
MOD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=FLEX_FREQ_SCH_UE_RESTR_SW-0;
```

```
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
```

```
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;
```

```
//Disabling energy saving based on flexible frequency-domain scheduling
```

```
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=FLEX_FREQ_SCH_ENERGY_SAVING_SW-0;
```

```
//Removing a policy of energy saving based on flexible frequency-domain scheduling
```

```
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
```

(Optional) If another policy of energy saving based on flexible frequency-domain scheduling has been configured:

```
//(Optional) Removing the other policy of energy saving based on flexible frequency-domain scheduling
```

```
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1;
```

7.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.1.4.2 Activation Verification

After energy saving based on flexible frequency-domain scheduling is activated, use any of the following methods to check whether the function has taken effect:

- Method 1:

- a. Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Switch** parameter is **Flexible Freq Sch for Energy Saving Sw:On** in the command output, energy saving based on flexible frequency-domain scheduling has been enabled.
- b. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **Flexible Freq Sch for Energy Saving**.

 NOTE

If there are two or more records with **Power Saving Type** being **Flexible Freq Sch for Energy Saving**, energy saving based on flexible frequency-domain scheduling has been enabled in two or more time periods.

- c. Run the **DSP TIME** command. Check whether the system time is within an effective period of energy saving based on flexible frequency-domain scheduling.
 - If it is, go to **d**.
 - If it is not, wait until the start time of energy saving based on flexible frequency-domain scheduling arrives.
 - d. In the case of a common cell, check whether the downlink PRB usage of the cell is less than or equal to the downlink PRB usage threshold. In the case of a hyper cell or combined cell, check whether the downlink PRB usage of any TRP of the cell is less than or equal to the downlink PRB usage threshold. The **NRDUCellPowerSaving.DlPrbThld** parameter specifies the downlink PRB usage threshold for starting energy saving based on flexible frequency-domain scheduling.
 - If it is, go to **e**.
 - If it is not, wait until the downlink PRB usage of the common cell or that of a TRP of the hyper cell or combined cell is less than or equal to the downlink PRB usage threshold for starting energy saving based on flexible frequency-domain scheduling.
 - e. Wait for 0–20 minutes. Then, run the **DSP NRDUCELL** command. Verify that the value of **Power Saving State** is **Flexible Freq Sch for Energy Saving:Enabled** in the command output, indicating that energy saving based on flexible frequency-domain scheduling has taken effect.
- Method 2:
Run the **DSP NRDUCELLPOWERSAVING** command to query the status of energy saving based on flexible frequency-domain scheduling.
 - If the value of **Power Saving Type** is **Flexible Freq Sch for Energy Saving** and the value of **Current State** is **Enabled** in the command output, energy saving based on flexible frequency-domain scheduling has taken effect.
 - If the value of **Power Saving Type** is **Flexible Freq Sch for Energy Saving** and the value of **Current State** is **Disabled** in the command output, check the **Current Disabled Cause** value to determine the reason why energy saving based on flexible frequency-domain scheduling has not taken effect. **Table 7-4** lists the possible reasons.

Table 7-4 Possible reasons why energy saving based on flexible frequency-domain scheduling has not taken effect

Reason	Description
Insufficient License	No license for this function is available.
Switch Turned Off	None
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable for other reasons.
Triggering Time Not Reached	The current time is not within the period specified for energy saving based on flexible frequency-domain scheduling.
Impacted Feature Taken Effect	- Service-type-based dynamic exit from energy saving based on flexible frequency-domain scheduling has taken effect. - Any other impacted feature has taken effect.
Downlink PRB Threshold Not Reached	None
In Penalty State	A penalty is being imposed upon an exit from the state of energy saving based on flexible frequency-domain scheduling.
RF Hardware Capability Not Supported	None
Ineffective for All TRPs	Energy saving based on flexible frequency-domain scheduling does not take effect for any TRP.
TRP Unavailable	A TRP of a hyper cell or combined cell is blocked or deactivated, or is unavailable due to other reasons.
Others	- Though it is currently within the range of time in which energy saving based on flexible frequency-domain scheduling can take effect, this function is in its triggering evaluation period, which is specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter. - Internal software or other reasons cause the function to fail.

7.1.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station),

VS.EnergyCons.BTS.Adding.NR (for an entire base station), or
VS.EnergyCons.BTSBoard (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after energy saving based on flexible frequency-domain scheduling is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.

- The increase in PRB utilization due to energy saving based on flexible frequency-domain scheduling can be observed using the **N.PRB.DL.ExtendUsed.Avg** and **N.HyperCell.TRPRB.DL.ExtendUsed.Avg** counters.

7.2 MetaAAU Power Reduction for Cell-Center UEs (Low-Frequency TDD)

7.2.1 Principles

This function reduces the PDSCH power for certain cell-center UEs in cells served by MetaAAUs, so that the base station consumes less energy. The power is reduced by an amount specified by the **NRDUCellChnPwr.PwrReductionValue** parameter. A target UE for this function must meet one of the following conditions on the downlink MCS index:

- If the UE is engaged in transmission using rank 2, its downlink MCS index must be greater than or equal to the threshold specified by **NRDUCellChnPwr.LowRankPwrReduceMcsThld**.
- If the UE is engaged in transmission using rank 3 or higher, its downlink MCS index must be greater than or equal to the threshold specified by **NRDUCellChnPwr.HighRankPwrReduceMcsThld**.

MetaAAU power reduction for cell-center UEs can take effect only if the **NRDUCellChnPwr.PwrReductionValue** parameter is set to a value other than 0.

7.2.2 Network Analysis

7.2.2.1 Benefits

This function takes effect only for cell-center UEs whose downlink MCS index reaches the corresponding threshold in cells served by MetaAAUs. The benefits vary with the threshold:

- A higher MCS index threshold (specified by **NRDUCellChnPwr.LowRankPwrReduceMcsThld** or **NRDUCellChnPwr.HighRankPwrReduceMcsThld**) results in fewer UEs for which the power is reduced, smaller decreases in the downlink user-perceived rates, and fewer energy saving gains.
- A lower MCS index threshold results in more UEs for which the power is reduced, larger decreases in the downlink user-perceived rates, and greater energy saving gains.

7.2.2.2 Impacts

The impacts between this function and other functions are described in [7.2.3.2.3 Function Impacts](#).

This function reduces the PDSCH power for UEs. As a result, the downlink user-perceived rates may slightly decrease for UEs qualified for this function to take effect. In addition, the average MCS index decreases, the IBLER increases, and the downlink PRB usage increases in a cell.

- A higher MCS index threshold (specified by **NRDUCellChnPwr.LowRankPwrReduceMcsThld** or **NRDUCellChnPwr.HighRankPwrReduceMcsThld**) results in smaller impacts on the preceding indicators.
- A lower MCS index threshold results in greater impacts on the preceding indicators.

7.2.3 Requirements

7.2.3.1 Licenses

None

7.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.2.3.2.1 Prerequisite Functions

None

7.2.3.2.2 Mutually Exclusive Functions

None

7.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When MetaAAU power reduction for cell-center UEs and RF channel intelligent shutdown take effect simultaneously, MetaAAU power reduction for cell-center UEs delivers fewer energy saving gains.
Low-frequency TDD	Low-power RF channel dynamic shutdown	LOW_POWER_RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When MetaAAU power reduction for cell-center UEs and low-power RF channel dynamic shutdown take effect simultaneously, MetaAAU power reduction for cell-center UEs delivers fewer energy saving gains.
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When MetaAAU power reduction for cell-center UEs and TTI-level channel shutdown take effect simultaneously, MetaAAU power reduction for cell-center UEs delivers fewer energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When MetaAAU power reduction for cell-center UEs and energy saving based on flexible frequency-domain scheduling take effect simultaneously, MetaAAU power reduction for cell-center UEs delivers fewer energy saving gains.

7.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Only MetaAAUs support this function.

7.2.3.4 Cells

This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**.

7.2.3.5 Others

None

7.2.4 Operation and Maintenance

7.2.4.1 Data Configuration

7.2.4.1.1 Data Preparation

[Table 7-5](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-5 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Power Reduction Value	NRDUCellChnPwr.PwrReductionValue	Set this parameter based on the network plan.
Low Rank Power Reduction MCS Threshold	NRDUCellChnPwr.LowRankPwrReduceMcsThld	Set this parameter based on the network plan.
High Rank Power Reduction MCS Threshold	NRDUCellChnPwr.HighRankPwrReduceMcsThld	Set this parameter based on the network plan.

7.2.4.1.2 Using MML Commands

Before using MML commands, refer to [7.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling MetaAAU power reduction for cell-center UEs
MOD NRDUCELLCHNPWR: NrDuCellId=0, PwrReductionValue=1;
//Setting the MCS index thresholds
MOD NRDUCELLCHNPWR: NrDuCellId=0, LowRankPwrReduceMcsThld=14,
HighRankPwrReduceMcsThld=15;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.


```
//Disabling MetaAAU power reduction for cell-center UEs  
MOD NRDUCELLCHNPWR: NrDuCellId=0, PwrReductionValue=0;
```

7.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.2.4.2 Activation Verification

Run the **LST NRDUCellCHNPWR** command to check whether the value of **Power Reduction Value** is a non-zero value. If it is, this function has been enabled.

Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after MetaAAU power reduction for cell-center UEs is enabled can be compared to assess the energy saving gains of this function.

7.2.4.3 Network Monitoring

Monitor the network gains and impacts of this function by comparing the performance indicators listed in [7.2.2 Network Analysis](#) before and after MetaAAU power reduction for cell-center UEs is enabled.

7.3 AMC-based Precise Power Control Energy Saving

7.3.1 Principles

AMC-based Precise Power Control Energy Saving is introduced in light- and moderate-load scenarios. It enables refined power allocation for initial transmission scheduling and retransmission scheduling by using the initial transmission power reduction function and retransmission power reduction function. The two functions reduce the redundant power for PDSCH data transmission during downlink initial MCS index selection and downlink retransmission, thereby achieving energy saving for base stations.

The initial transmission power reduction function and retransmission power reduction function are controlled by the **INIT_TRANS_POWER_REDUCE_SW** and **RETRANS_POWER_REDUCE_SW** options of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter, respectively. [Table 7-6](#) describes the parameters in the **NRDUCellPowerSaving** MO that specify the effective period, load threshold, and other information under an energy saving policy of initial transmission power reduction or retransmission power reduction.

Table 7-6 Parameters for AMC-based Precise Power Control Energy Saving

Parameter Name	Parameter ID	Description
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	This parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	This parameter specifies the energy saving type indicated by an energy saving policy index. For initial transmission power reduction or retransmission power reduction in AMC-based Precise Power Control Energy Saving, set this parameter to INIT_TRANS_POWER_REDUCE or RETRANS_POWER_REDUCE .
Start Time	NRDUCellPowerSaving.StartTime	These parameters specify the effective period for initial transmission power reduction or retransmission power reduction, the evaluation period and conditions for a cell to enter and exit the related function state, and the power protection margin for executing power reduction. These parameters can be configured separately for initial transmission power reduction and retransmission power reduction.
Stop Time	NRDUCellPowerSaving.StopTime	
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	
Power Protection Margin	NRDUCellPowerSaving.PowerProtectionMargin	

Initial transmission power reduction and retransmission power reduction do not take effect for the following non-target UEs:

- VoNR/ViNR/MCPTT/Cloud-VR/Game (with 5QI 3) UEs
- UEs running low-latency and high-reliability services
- Network slice UEs
- Non-Slot UEs
- 64QAM UEs with low bit rates

Entering the State of Initial Transmission Power Reduction or Retransmission Power Reduction

Within the time period specified by **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** for initial transmission power reduction or retransmission power reduction, the base station periodically checks the downlink PRB usage of a cell or TRP, with the periodicity being specified by **NRDUCellPowerSaving.ShutdownTrigJudgePeriod**.

- A common cell automatically enters the state of initial transmission power reduction or retransmission power reduction if its downlink PRB usage is less than or equal to the value of **NRDUCellPowerSaving.DlPrbThld**.
- A TRP of a hyper cell or combined cell (except a combined cell in joint transmission mode) automatically enters the state of initial transmission power reduction or retransmission power reduction if its downlink PRB usage is less than or equal to the value of **NRDUCellPowerSaving.DlPrbThld**.
- Each TRP of a combined cell in joint transmission mode can automatically enter the state of initial transmission power reduction or retransmission power reduction only when the downlink PRB usage of all TRPs of the cell is less than or equal to the value of **NRDUCellPowerSaving.DlPrbThld**.

The downlink PRB usage of a cell or TRP is calculated as follows:

- Downlink PRB usage of a common cell = ((Average number of PRBs used by PDSCH DRBs (**N.PR.B.DL.DrbUsed.Avg**) – Average number of extended PRBs used on the PDSCH (**N.PR.B.DL.ExtendUsed.Avg**))/Average number of actually available downlink PRBs (**N.PR.B.DL.Actual.Avail.Avg**)
- Downlink PRB usage of a TRP of a hyper cell or combined cell = [(Average number of PRBs used by downlink DRB services on the PDSCH in the TRP (**N.HyperCell.TRP.PR.B.DL.Used.Avg**) – Average number of downlink extended PRBs used in the TRP (**N.HyperCell.TRP.PR.B.DL.ExtendUsed.Avg**)]/Average number of available downlink PRBs in the TRP (**N.HyperCell.TRP.PR.B.DL.Avail.Avg**)

Executing Power Reduction

After a cell enters the state of initial transmission power reduction or retransmission power reduction, the base station calculates per TTI the power value that can be reduced based on the channel quality, and this value minus the power protection margin is the actual power reduction value. Power reduction can be performed only when the power value that can be reduced exceeds the power protection margin. The **NRDUCellPowerSaving.PowerProtectionMargin** parameter specifies the power protection margin.

Exiting the State of Initial Transmission Power Reduction or Retransmission Power Reduction

After a cell enters the initial transmission power reduction or retransmission power reduction state, the base station periodically checks whether any of the following conditions is met, with the periodicity being **NRDUCellPowerSaving.ShutdownTrigJudgePeriod**. If this is true, the cell or TRP automatically exits the initial transmission power reduction or retransmission power reduction state.

- The current time is not within the time period specified by **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime**.
- The downlink PRB usage of the cell or TRP meets the exit conditions. For details about how to calculate the downlink PRB usage of a cell or TRP, see [Entering the State of Initial Transmission Power Reduction or Retransmission Power Reduction](#).
 - A common cell exits the state of initial transmission power reduction or retransmission power reduction if its downlink PRB usage is greater than the sum of the **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset** parameter values.
 - A TRP of a hyper cell or combined cell (except a combined cell in joint transmission mode) exits the state of initial transmission power reduction or retransmission power reduction if its downlink PRB usage is greater than the sum of the **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset** parameter values. A hyper cell or combined cell exits this state only when all TRPs exit this state.
 - A combined cell in joint transmission mode exits the state of initial transmission power reduction or retransmission power reduction if the downlink PRB usage of any TRP of this cell is greater than the sum of the **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset** parameter values.

Initial transmission power reduction and retransmission power reduction can also be configured to not take effect for specific service types.

- Service-type-based dynamic exit from initial transmission power reduction
To ensure performance of UEs with specific service types, initial transmission power reduction can be configured to not take effect for these UEs. The service types involved are specified by the **gNBQciBearer.ServiceType** parameter. Service-type-based dynamic exit from initial transmission power reduction takes effect when all of the following conditions are met:
 - Initial transmission power reduction is enabled.
 - The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.
 - The **INIT_TRANS_PWR_RD_EXIT_SW** option of the **NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter is selected in the group identified by the **NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter that is set to the same value as the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter for a specific service type.
- Service-type-based dynamic exit from retransmission power reduction
To ensure performance of UEs with specific service types, retransmission power reduction can be configured to not take effect for these UEs. The service types involved are specified by the **gNBQciBearer.ServiceType** parameter. Service-type-based dynamic exit from retransmission power reduction takes effect when all of the following conditions are met:
 - Retransmission power reduction is enabled.
 - The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.
 - The **RETRANS_PWR_RD_EXIT_SW** option of the **NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter is

selected in the group identified by the **NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter that is set to the same value as the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter for a specific service type.

7.3.2 Network Analysis

7.3.2.1 Benefits

When the network is under low or moderate load, initial transmission power reduction and retransmission power reduction dynamically lower the transmit power of cells. This reduces the overall energy consumption of the base station. In other load scenarios, the energy saving gains are limited.

Energy saving gains may also decrease when some of the UEs on the network are not target UEs of initial transmission power reduction and retransmission power reduction. If none of the UEs are target UEs of the two functions, there will be no energy saving gains.

Service-type-based dynamic exit from initial transmission power reduction can guarantee the performance of UEs of specific service types but reduces energy saving gains.

Service-type-based dynamic exit from retransmission power reduction can guarantee the performance of UEs of specific service types but reduces energy saving gains.

7.3.2.2 Impacts

The impacts between this function and other functions are described in [7.3.3.2.3 Function Impacts](#).

After initial transmission power reduction and retransmission power reduction take effect:

- The initial block error rate (IBLER) and residual block error rate (RBLER) increase.
- The MAC-layer processing delay in the gNodeB may increase. This delay is calculated as follows: **Average Downlink Packet Delay in MAC** = $(N.QoS.DL.PktDelayAirInterface.Time / N.QoS.DL.PktDelayAirInterface.Num) / 1000$.
- The average MCS index for downlink scheduling decreases.
- The downlink user-perceived rate decreases.
- The number of retransmissions and the retransmission rate increase.
- In cells exposed to strong interference from neighboring cells, when initial transmission power reduction and retransmission power reduction take effect in neighboring cells, interference from neighboring cells decreases.
 - The average CQI may increase. $Average\ CQI = ((0 \times N.ChMeas.CQI.SingleCW.0 + 1 \times N.ChMeas.CQI.SingleCW.1 + 2 \times N.ChMeas.CQI.SingleCW.2 + 3 \times N.ChMeas.CQI.SingleCW.3 + \dots + 13 \times N.ChMeas.CQI.SingleCW.13 + 14 \times N.ChMeas.CQI.SingleCW.14 + 15 \times N.ChMeas.CQI.SingleCW.15) / (N.ChMeas.CQI.SingleCW.0 +$

$$N.ChMeas.CQI.SingleCW.1 + N.ChMeas.CQI.SingleCW.2 + N.ChMeas.CQI.SingleCW.3 + \dots + N.ChMeas.CQI.SingleCW.13 + N.ChMeas.CQI.SingleCW.14 + N.ChMeas.CQI.SingleCW.15))$$

- The values of the **N.PRB.DL.DrbUsed.Avg** and **N.PRB.DL.Used.Avg** counters may decrease.
- The values of the following counters related to downlink PRB usage may increase in common cells:
 - **N.PRB.DL.DrbUsed.Avg**
 - **N.PRB.DL.ExtendUsed.Avg**
 - **N.PRB.DL.Used.Avg**
- The values of the following counters related to the downlink PRB usage of TRPs may increase in hyper cells or combined cells:
 - **N.HyperCell.TRP.PRB.DL.Used.Avg**
 - **N.HyperCell.TRP.PRB.DL.ExtendUsed.Avg**

7.3.3 Requirements

7.3.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-100252	AMC-based Precise Power Control Energy Saving	NR0SPCPWMG00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Initial transmission power reduction	INIT_TRANS_POWER_REDUCE_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Service-type-based dynamic exit from initial transmission power reduction can take effect only when initial transmission power reduction is enabled.
Low-frequency TDD	Retransmission power reduction	RETRANS_POWER_REDUCE_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Service-type-based dynamic exit from retransmission power reduction can take effect only when retransmission power reduction is enabled.

7.3.3.2.2 Mutually Exclusive Functions

None

7.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both initial transmission power reduction and timing carrier shutdown take effect, the former function does not deliver any gains. When both retransmission power reduction and timing carrier shutdown take effect, the former function does not deliver any gains.
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both initial transmission power reduction and intelligent carrier shutdown take effect, the former function does not deliver any gains. When both retransmission power reduction and intelligent carrier shutdown take effect, the former function does not deliver any gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS W (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When both initial transmission power reduction and RF module fast deep dormancy take effect, the former function does not deliver any gains. When both retransmission power reduction and RF module fast deep dormancy take effect, the former function does not deliver any gains.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCY SW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Initial transmission power reduction does not take effect when RF module deep dormancy is in effect. Retransmission power reduction does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module super dormancy	RRU.DORMANCY SW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Initial transmission power reduction does not take effect when RF module super dormancy is in effect. Retransmission power reduction does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option (inter-base-station multi-carrier scenarios) of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected - Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected	<i>Energy Conservation and Emission Reduction</i>	When both initial transmission power reduction and fast carrier shutdown take effect, the former function does not deliver any gains. When both retransmission power reduction and fast carrier shutdown take effect, the former function does not deliver any gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUeIdentSw	None	Initial transmission power reduction and retransmission power reduction fail to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
Low-frequency TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.AppIpv4 • gNBAsAppParam.AppIpv6	None	• Any option of the gNodeBParam.StUeIdentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.AppIpv4 parameter is set. • The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue:

RAT	Function Name	Function Switch	Reference	Description
				• Select the SPEED_TURBO_PWR_SAVING_SW option of the gNBPwrSavingAlgo.PwrSavin<i>gAlgoSwitch</i> parameter. Initial transmission power reduction and retransmission power reduction can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBPwrSavingAlgo.SpeedTurboPwrSavStartTime and gNBPwrSavingAlgo.SpeedTurboPwrSavEndTime parameters, increasing the energy saving gains. • Select the STATIC_ST_PWR_SAVING_SW option of the gNBPwrSavingAlgo.PwrSavin<i>gAlgoSwitch</i> parameter. Initial transmission power

RAT	Function Name	Function Switch	Reference	Description
				reduction and retransmission power reduction can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNB PwrSaving Algo.StaticSTP wrSavStartTime and gNB PwrSaving Algo.StaticSTP wrSavEndTime parameters, increasing the energy saving gains.

7.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

TDD: All RF modules can be used.

7.3.3.4 Cells

This function requires that the **NRDUCell.FrequencyBand** parameter be set to **N1, N2, N3, N5, N7, N8, N12, N13, N20, N25, N28, N38, N40, N41, N66, N71, N75, N77, N78, or N79.**

7.3.3.5 Others

None

7.3.4 Operation and Maintenance

7.3.4.1 Data Configuration

7.3.4.1.1 Data Preparation

Table 7-7 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-7 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellAlgoSwitch.NrDuCellId	Set this parameter based on the network plan.
NR DU Cell ID	NRDUCellPowerSaving.NrDuCellId	Set this parameter based on the network plan.
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	Select the INIT_TRANS_POWER_REDUCE_SW option of this parameter if initial transmission power reduction is required.
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	Select the RETRANS_POWER_REDUCE_SW option of this parameter if retransmission power reduction is required.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: - For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. - For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Stop Time	NRDUCellPowerSaving.StopTime	
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to INIT_TRANS_POWER_REDUCE if initial transmission power reduction is required.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to RETRANS_POWER_REDUCE if retransmission power reduction is required.

Parameter Name	Parameter ID	Setting Notes
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	You are advised to use the recommended value. In cell combination scenarios, it is good practice to set this parameter to a small value because all TRPs share RBs of all cells.
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	You are advised to use the recommended value.
Power Protection Margin	NRDUCellPowerSaving.PowerProtectionMargin	The default value is recommended.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	It is recommended that the SERV_TYPE_DYN_EXIT_PWR_SAVE_SW option be selected when there are specific types of services and user experience of these services needs to be ensured.
Power Saving Policy Group Index	gNBQciBearer.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan.
Power Saving Policy Group Index	NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan. The value of this parameter must be the same as that of the gNBQciBearer.PowerSavingPolicyGrpIndex parameter for services to be guaranteed. Before setting this parameter, ensure that the QCI for which the gNBQciBearer.PowerSavingPolicyGrpIndex parameter is set has been configured.
User Identification Method	NRDUCellPsPolicyGrp.UserIdentMethod	Set this parameter to SERVICE_TYPE .

Parameter Name	Parameter ID	Setting Notes
Service Type Dynamic Exit Power Saving Switch	NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw	Select the INIT_TRANS_PWR_RD_EXIT_SW option if service-type-based dynamic exit from initial transmission power reduction is required. Select the RETRANS_PWR_RD_EXIT_SW option if service-type-based dynamic exit from retransmission power reduction is required.

7.3.4.1.2 Using MML Commands

Before using MML commands, refer to [7.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling initial transmission power reduction
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=INIT_TRANS_POWER_REDUCE_SW-1;
//Configuring an energy saving policy for initial transmission power reduction
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1, StartTime=00&00&00,
StopTime=23&59&59, PowerSavingType=INIT_TRANS_POWER_REDUCE, DlPrbThld=50, DlPrbOffset=10,
PowerProtectionMargin=0;
//((Optional) Enabling service-type-based dynamic exit from initial transmission power reduction
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;
ADD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=1,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=INIT_TRANS_PWR_RD_EXIT_SW-1;

//Enabling retransmission power reduction
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0, PowerSavingAlgoSwitch=RETRANS_POWER_REDUCE_SW-1;
//Configuring an energy saving policy for retransmission power reduction
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=2, StartTime=00&00&00,
StopTime=23&59&59, PowerSavingType=RETRANS_POWER_REDUCE, DlPrbThld=50, DlPrbOffset=15,
PowerProtectionMargin=0;
//((Optional) Enabling service-type-based dynamic exit from retransmission power reduction
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=2;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;
ADD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=2,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=RETRANS_PWR_RD_EXIT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//(Optional) Disabling service-type-based dynamic exit from initial transmission power reduction
MOD NRDUCELLPSPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=1,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=INIT_TRANS_PWR_RD_EXIT_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;
//Disabling initial transmission power reduction
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=INIT_TRANS_POWER_REDUCE_SW-0;
//Deleting the energy saving policy for initial transmission power reduction
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1;

//(Optional) Disabling service-type-based dynamic exit from retransmission power reduction
MOD NRDUCELLPSPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=2,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=RETRANS_PWR_RD_EXIT_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;
//Disabling retransmission power reduction
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0, PowerSavingAlgoSwitch=RETRANS_POWER_REDUCE_SW-0;
//Deleting the energy saving policy for retransmission power reduction
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=2;
```

7.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.3.4.2 Activation Verification

After AMC-based precise power control energy saving is activated, use any of the following methods to check whether the function has taken effect:

- Method 1:
 - a. Run the **LST NRDUCELLPWRSAVEALGO** command. If the value of the **Power Saving Algorithm Switch** parameter is **Initial Transmission Power Reduction Sw:On** in the command output, initial transmission power reduction has been enabled. If the value of the **Power Saving Algorithm Switch** parameter is **Retransmission Power Reduction Sw:On** in the command output, retransmission power reduction has been enabled.
 - b. For initial transmission power reduction, run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **Initial Transmission Power Reduction**. For retransmission power reduction, run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **Retransmission Power Reduction**.

 NOTE

If there are two or more records with **Power Saving Type** being **Initial Transmission Power Reduction**, initial transmission power reduction has been enabled in two or more time periods.

If there are two or more records with **Power Saving Type** being **Retransmission Power Reduction**, retransmission power reduction has been enabled in two or more time periods.

- c. Run the **DSP TIME** command. Check whether the system time is within an effective period of initial transmission power reduction or retransmission power reduction.
 - If it is, go to **d**.
 - If it is not, wait until the start time arrives.
- d. In the case of a common cell, check whether the downlink PRB usage of the cell is less than or equal to the downlink PRB usage threshold. In the case of a hyper cell or combined cell, check whether the downlink PRB usage of any TRP of the cell is less than or equal to the downlink PRB usage threshold. The **NRDUCellPowerSaving.DlPrbThld** parameter specifies the downlink PRB usage threshold for starting initial transmission power reduction or retransmission power reduction.
 - If it is, go to **e**.
 - If it is not, wait until the downlink PRB usage of the common cell or that of a TRP of the hyper cell or combined cell is less than or equal to the downlink PRB usage threshold for starting initial transmission power reduction or retransmission power reduction.
- e. Wait for 0 to 20 minutes. Then, run the **DSP NRDUCELL** command.
 - Verify that the value of **Power Saving State** is **Initial Transmission Power Reduction:Enabled** in the command output and that the value of the **N.PowerSaving.PrecisePwrMgmt.Dur** counter on the MAE-Access is not 0, indicating that initial transmission power reduction has taken effect.
 - Verify that the value of **Power Saving State** is **Retransmission Power Reduction:Enabled** in the command output and that the value of the **N.PowerSaving.PrecisePwrMgmt.Dur** counter on the MAE-Access is not 0, indicating that retransmission power reduction has taken effect.
- Method 2:

You can run the **DSP NRDUCELLPOWERSAVING** command to query the status of initial transmission power reduction or retransmission power reduction.

 - If the value of **Power Saving Type** is **Initial Transmission Power Reduction** and the value of **Current State** is **Enabled** in the command output, and the value of the **N.PowerSaving.PrecisePwrMgmt.Dur** counter on the MAE-Access is not 0, then initial transmission power reduction has taken effect.
 - If the value of **Power Saving Type** is **Initial Transmission Power Reduction** and the value of **Current State** is **Disabled**, check the **Current**

Disabled Cause parameter value to determine the reason why initial transmission power reduction did not take effect. [Table 7-8](#) lists possible reasons.

- If the value of **Power Saving Type** is **Retransmission Power Reduction** and the value of **Current State** is **Enabled** in the command output, and the value of the **N.PowerSaving.PrecisePwrMgmt.Dur** counter on the MAE-Access is not 0, then retransmission power reduction has taken effect.
- If the value of **Power Saving Type** is **Retransmission Power Reduction** and the value of **Current State** is **Disabled**, check the **Current Disabled Cause** parameter value to determine the reason why retransmission power reduction did not take effect. [Table 7-8](#) lists possible reasons.

Table 7-8 Possible reasons why initial transmission power reduction or retransmission power reduction did not take effect

Reason	Description
Insufficient License	No license for this function is available.
Switch Turned Off	None
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable for other reasons.
Triggering Time Not Reached	The current time is not within the range specified for initial transmission power reduction or retransmission power reduction.
Triggering Decision Period Not Met	The current time is within the range specified for initial transmission power reduction or retransmission power reduction. However, it is within the evaluation period for triggering initial transmission power reduction or retransmission power reduction (specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter).
Impacted Feature Taken Effect	Any other impacted feature has taken effect.
Downlink PRB Threshold Not Reached	None
In Penalty State	A penalty is being imposed on the power saving function due to other functions.
Others	Internal software or other reasons cause the function to fail.

7.3.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station),

VS.EnergyCons.BTS.Adding.NR (for an entire base station), or
VS.EnergyCons.BTSBoard (for RF modules) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after AMC-based precise power control energy saving is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases after this function is enabled.

- The effective duration of AMC-based precise power control energy saving can be obtained by observing the **N.PowerSaving.PrecisePwrMgmt.Dur** counter.

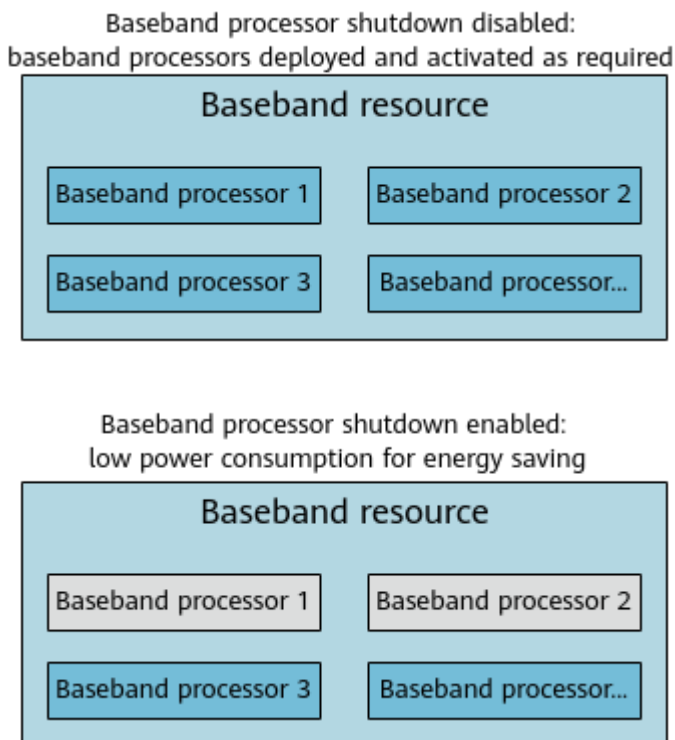
8 Site Level Energy Saving

8.1 Baseband Processor Shutdown

8.1.1 Principles

A baseband resource typically consists of multiple baseband processors, as shown in [Figure 8-1](#). The baseband processors can be deployed and activated as required. When baseband processors are not in use by services, the baseband processor shutdown function enables them to enter the low power consumption state for energy saving. When services change and require some baseband processors in the low power consumption state to be activated, these baseband processors exit this state.

Figure 8-1 Baseband processor shutdown



Baseband resources are deployed on baseband processing units (BBPs) and AAUs.

Baseband processor shutdown now can work at the BBP level (adaptive baseband shutdown and power saving first) and base station level (baseband shutdown for energy saving).

- Adaptive baseband shutdown: When the **BBP.CELLDEPLOY** parameter is set to **ADAPT_BB_SHUTDOWN**, BBP-level baseband processor shutdown is supported. On a BBP, baseband processors carrying no services can enter the low power consumption state. It is recommended that this function be enabled together with RF module deep dormancy or RF module super dormancy.
- Power saving first: When the **BBP.CELLDEPLOY** parameter is set to **POWERSAVING**, base-station-level baseband processor shutdown is supported. Compared to adaptive baseband shutdown, this function prioritizes power saving by deploying services centrally on fewer BBPs. This further reduces power consumption, but may affect cell performance. It is recommended that this function be enabled together with RF module deep dormancy or RF module super dormancy.
- Baseband shutdown for energy saving: When the **BB_SHUTDOWN_POWERSAVING_SW** option of the **gNodeBParam.BaseStationDeepDormancySw** parameter is selected, base-station-level baseband processor shutdown is supported. On BBPs and AAUs, baseband processors carrying no services can enter the low power consumption state.

Baseband processors carrying no services refer to the those where no cell is deployed or where cells are deployed but the cells enter the RF module deep dormancy or super dormancy state.

When baseband processor shutdown is enabled at both the BBP and base station level, the BBP-level function takes effect preferentially.

8.1.2 Network Analysis

8.1.2.1 Benefits

The energy saving gains are dependent on the duration of baseband processor shutdown.

In addition, as the ambient temperature has a great impact on the power consumption of baseband processors, the ambient temperature affects the gains of this function.

8.1.2.2 Impacts

The impacts between this function and other functions are described in [8.1.3.2.3 Function Impacts](#).

During baseband processor shutdown, activation or establishment of cells causes some baseband processors to exit the low power consumption state, and therefore the activation or establishment duration is prolonged. The duration typically lasts for 2 to 5 minutes.

After power saving first is enabled, resources are deployed based on power saving optimization, leading to changes in user-plane performance indicators such as the number of UEs, user-perceived rate, and traffic volume, as well as slight increase of the CPU usage of BBPs.

8.1.3 Requirements

8.1.3.1 Licenses

None

8.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.1.3.2.1 Prerequisite Functions

- When there are no activated cells on the baseband processing resources, baseband processor shutdown (baseband adaptive shutdown, power saving first, and baseband shutdown for energy saving) does not depend on other functions.
- When there are activated cells on the baseband processing resources, baseband processor shutdown (baseband adaptive shutdown, power saving

first, and baseband shutdown for energy saving) depends on the functions described in [Table 8-1](#).

Table 8-1 Prerequisite functions for baseband processor shutdown

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCYS <i>W</i> set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>
Low-frequency TDD	RF module super dormancy	RRU.DORMANCYS <i>W</i> set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>
Low-frequency TDD	pRRU deep dormancy	RRU.DORMANCYS <i>W</i> set to ON	<i>Multi-RAT Coordinated Energy Saving</i>

8.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Secondary Baseband Equipment ID	NRDUCellTrp.SecondaryBasebandEqmId	None	When NRDUCellTrp.SecondaryBasebandEqmId parameter is set to a value other than 255, baseband shutdown for energy saving must be disabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Secondary Baseband Equipment ID	NRDUCellTrp. SecondaryBasebandEqmId	None	When the NRDUCellTrp. SecondaryBasebandEqmId parameter is set to a value other than 255, neither power saving first nor baseband adaptive shutdown can be enabled for the BBPs contained in the baseband equipment corresponding to the NRDUCellTrp. BasebandEqmId and NRDUCellTrp. SecondaryBasebandEqmId parameters.
Low-frequency TDD	Primary/ Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Baseband shutdown for energy saving does not work with the primary/ backup function in intra-frequency dual-network redundancy networking.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Baseband shutdown for energy saving does not work with dual-BBU backup in dual-network redundancy networking.
Low-frequency TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	Baseband shutdown for energy saving does not work with the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios.

8.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	NRDUCell.NrDualCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	If carrier shutdown takes effect for the TRPs of a hyper cell, the baseband processors that work with the TRPs cannot be shut down. If deep dormancy takes effect for the TRPs, the baseband processors can be shut down.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDualCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	If carrier shutdown takes effect for the TRPs of a combined cell, the baseband processors that work with the TRPs cannot be shut down. If deep dormancy takes effect for the TRPs, the baseband processors can be shut down.
Low-frequency TDD	Adaptive BBU deep dormancy	• BBP_DEEP_DORMANCY_SWITCH option of the gNBPowerSavingAlgo.DormancySwitch parameter selected • BBP.PWRSVGSW (5G gNodeB, LTE eNodeB) set to BBP_DEEP_DORMANCY_ADAPT	<i>Multi-RAT Coordinated Energy Saving</i>	When adaptive BBU deep dormancy is in effect, adaptive baseband shutdown, power saving first, and baseband shutdown for energy saving do not take effect.
Low-frequency TDD	Inter-Board Baseband Resource Redundancy	• RRUCHAIN.TT • RRUCHAIN.BM	<i>Base Station Equipment Reliability</i>	Adaptive baseband shutdown, power saving first, and baseband shutdown for energy saving do not take effect for LampSite distributed massive MIMO cells with Inter-Board Baseband Resource Redundancy enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	When the primary/backup function in intra-frequency dual-network redundancy networking is in effect, adaptive baseband shutdown and power saving first do not take effect.
Low-frequency TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	When dual-BBU backup in dual-network redundancy networking is in effect, adaptive baseband shutdown and power saving first do not take effect.
Low-frequency TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	When the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios is in effect, adaptive baseband shutdown and power saving first do not take effect.

8.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and all NR-capable baseband processing units except the UBBPg1 support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*. However, if a BBP is configured to work in multiple RATs (for example, LTE TDD and NR), the baseband processors deployed for all these RATs will not be shut down.

Only resources of the UBBPg3/UBBPh3a/UBBPh3b can be deployed for the power saving first function.

RF Modules

When baseband processors are deployed on BBPs, all NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

When baseband processors are deployed on AAUs, all NR-capable AAUs, except the AAU5685md, that comply with the eCPRI protocol, work in low frequency bands, and are configured with NR carriers support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

However, if an RF module is configured to work in multiple RATs (for example, LTE TDD and NR), the baseband processors deployed for all these RATs will not be shut down.

8.1.3.4 Others

None

8.1.4 Operation and Maintenance

8.1.4.1 Data Configuration

8.1.4.1.1 Data Preparation

Table 8-2 describes the parameters used for function activation. Function optimization configurations are not involved.

Table 8-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	BBP.CN	Set this parameter based on the network plan.
Subrack No.	BBP.SRN	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Slot No.	BBP.SN	Set this parameter based on the network plan.
Cell Deployment Policy	BBP.CELLDEPLOY	Set this parameter to ADAPT_BBP_SHUTDOWN when adaptive baseband shutdown is required. The value POWERSAVING is recommended when energy saving is preferred.
Base Station Deep Dormancy Switch	gNodeBParam.BaseStationDeepDormancySw	Select the BB_SHUTDOWN_POWERSAVING_SW option of this parameter when baseband shutdown for energy saving is required.

8.1.4.1.2 Using MML Commands

Before using MML commands, refer to [8.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTE

Before enabling baseband processor shutdown, ensure that no ALM-26819 Data Configuration Exceeding Licensed Limit is reported. If such an alarm is reported, the cell may fail to be set up due to license insufficiency after baseband processors exit the shutdown state.

Activation Command Examples

BBP-level baseband processor shutdown

- Adaptive baseband shutdown
//Setting CELLDEPLOY to ADAPT_BBP_SHUTDOWN for the BBP
MOD BBP: CN=0, SRN=0, SN=4, CELLDEPLOY=ADAPT_BBP_SHUTDOWN;
- Power saving first
//Setting CELLDEPLOY to POWERSAVING for the BBP
MOD BBP: CN=0, SRN=0, SN=4, CELLDEPLOY=POWERSAVING;

Base-station-level baseband processor shutdown (baseband shutdown for energy saving)


```
//Turning on BB_SHUTDOWN_POWERSAVING_SW  
MOD GNODEBPARAM: BaseStationDeepDormancySw=BB_SHUTDOWN_POWERSAVING_SW-1;
```

Optimization Command Examples

N/A

Deactivation Command Examples

BBP-level baseband processor shutdown

- Adaptive baseband shutdown
//Setting CELLDEPLOY to NORMAL
MOD BBP: CN=0, SRN=0, SN=4, CELLDEPLOY=NORMAL;
- Power saving first
//Setting CELLDEPLOY to NORMAL for the BBP, deactivating the cell, and then activating the cell again
MOD BBP: CN=0, SRN=0, SN=4, CELLDEPLOY=NORMAL;
DEA NRCELL: NrCellId=0;
ACT NRCELL: NrCellId=0;

Base-station-level baseband processor shutdown (baseband shutdown for energy saving)

```
//Turning off BB_SHUTDOWN_POWERSAVING_SW  
MOD GNODEBPARAM: BaseStationDeepDormancySw=BB_SHUTDOWN_POWERSAVING_SW-0;
```

8.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.1.4.2 Activation Verification

You can check whether baseband processor shutdown has been enabled as follows:

- Run the **LST BBP** command. If the value of the **Cell Deployment Policy** parameter is **ADAPT_BBP_SHUTDOWN**, adaptive baseband shutdown has been enabled.
- Run the **LST BBP** command. If the value of the **Cell Deployment Policy** parameter is **POWERSAVING**, power saving first has been enabled.
- Run the **LST GNODEBPARAM** command. If the value of the **Base Station Deep Dormancy Switch** parameter is **Baseband Shutdown Energy Saving Switch:On** in the command output, baseband shutdown for energy saving has been enabled.

Run the **DSP BRDDORMANCY** command and check the value of the **Baseband Shutdown State** parameter to determine whether baseband processor shutdown has taken effect.

- If RF module deep dormancy has taken effect, the value of the **Baseband Shutdown State** parameter is **Enable** regardless of whether baseband processor shutdown takes effect.
- If RF module deep dormancy has not taken effect:

- If baseband processor shutdown has taken effect, the value of the **Baseband Shutdown State** parameter is **Enable**.
- If baseband processor shutdown has not taken effect, the value of the **Baseband Shutdown State** parameter is **Disable**.

If baseband processor shutdown is enabled but RF module deep dormancy is not, the value of the **Baseband Shutdown State** parameter being **Enable** indicates that baseband processor shutdown has taken effect.

8.1.4.3 Network Monitoring

Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), **VS.EnergyCons.BTSBoard** (for RF modules), or **VS.EnergyCons.BU** (for BBUs) on the MAE-Access.

On the live network, the one-week base station energy consumption data obtained before and after baseband processor shutdown is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases slightly after this function is enabled.

9 Network Level Energy Saving

9.1 Multi-RF-Module Coordinated Energy Saving

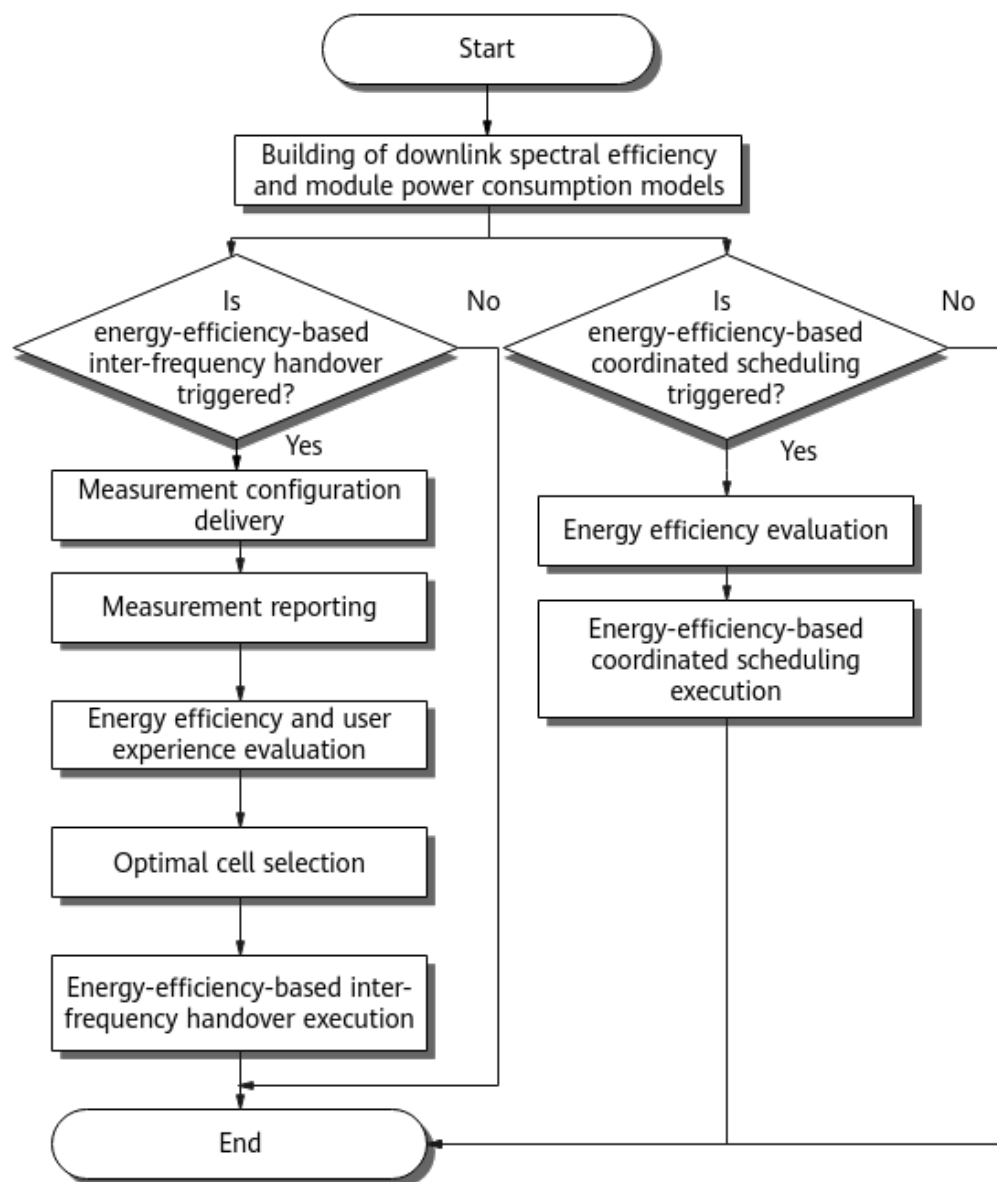
9.1.1 Principles

Multi-RF-module coordinated energy saving includes the following functions:

- Energy-efficiency-based inter-frequency handover
The gNodeB hands over UEs to the optimal inter-frequency cell based on factors such as cell energy efficiency and user experience to achieve network energy saving. This function is controlled by the **ENERGY_EFFICIENCY_BASED_HO** option of the **NRCellAlgoSwitch**.*InterFreqHoSwitch* parameter.
- Energy-efficiency-based coordinated scheduling
During downlink CA for a UE, the gNodeB transmits data on the carriers with higher energy efficiency to the UE for network energy saving. This function is controlled by the **ENERGY_EFF_BASED_COLLAB_SCH_SW** option of the **NRCellAlgoSwitch**.*MultiFreqAlgoSwitch* parameter.

Figure 9-1 shows the feature principles. This feature requires collaboration of the Mobile Intelligent Engine (MIE). The MIE calculates energy saving factors based on reported historical data including the PRB load and user-perceived rate. It then sends the factors to the base station. The base station can perform the preceding energy saving functions only if the factors are valid. For details about the feature principles and operation guide on the MIE side, contact Huawei engineers.

Figure 9-1 Multi-RF-module coordinated energy saving



9.1.1.1 Downlink Spectral Efficiency and Module Power Consumption Model Building

Energy efficiency is the amount of data that a base station can provide per unit of energy consumption. In this feature, energy efficiency is calculated by dividing the downlink spectral efficiency by the module power consumption load factor k .

- The downlink spectral efficiency is provided by virtual grid models for downlink spectral efficiency after the models are successfully built and launched.

For details about the building and use of virtual grid models for downlink spectral efficiency, see *Virtual Grid-based Multi-Frequency Coordination*.

- The module power consumption load factor k is provided by module power consumption models after the models are built either normally or intelligently.

Normal building of module power consumption models is controlled by the **PWR_EFF_MODEL_BUILD_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter. Intelligent building of module power consumption models is controlled by the **PWR_EFF_MODEL_INTEL_BUILD_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter. When the normal and intelligent building functions are both enabled, the intelligent building function takes effect preferentially. This function uses the neural network to build models of the relationships between RF module power consumption and cell load.

In TDD, the calculation of k varies as follows:

- For a dual-band 32T32R RF module: The normal building function calculates a value of k , while the intelligent building function calculates a value of k for each frequency band.
- For a non-dual-band 32T32R RF module: The normal and intelligent building functions each calculate a value of k .

With either the normal building function or the intelligent building function enabled, the base station automatically collects loads of the cells that it serves and the power consumption of the corresponding RF modules (for collected information, see [Collected Information](#)). The collected information is used to build models of the relationships between RF module power consumption and cell load (for the purposes and steps to build the models, see [Module Power Consumption Model Building](#)).

Collected Information

- Average number of used downlink PRBs for an individual NR cell:
N.PR.B.DL.Used.Avg
- Number of available downlink PRBs for an individual NR cell:
N.PR.B.DL.Avail.Avg
- Average number of downlink PRBs used for an individual LTE cell (only in LTE-NR multimode scenarios): **L.ChMeas.PR.B.DL.Used.Avg**
- Number of available downlink PRBs for an individual LTE cell (only in LTE-NR multimode scenarios): **L.ChMeas.PR.B.DL.Avail**
- Power consumption of the RF modules serving an NR cell:
VS.EnergyCons.BTSBoard
- Sample collection time

Module Power Consumption Model Building

The module power consumption load factor k at different times and under different loads can be obtained by using module power consumption models. This factor reflects the module power consumption changes with the PRB load. A larger value of the factor indicates higher power consumption for the same PRB load. The detailed model building process is as follows:

1. With the normal building function, the base station collects samples of two to three days for each RF module. With the intelligent building function, the base station collects samples of four to five days for each RF module. The specific number of days is subject to the collection speed.

2. After the sample size meets the model building requirements, models of the relationships between power consumption and load are started to build.
3. The module power consumption models are checked every day after they are successfully built.
 - After the model accuracy is verified to meet the requirements, the models can be successfully put into use.
 - If the accuracy of a model does not meet the requirements, this model is marked as unavailable. Energy-efficiency-based inter-frequency handover and energy-efficiency-based coordinated scheduling do not take effect for the corresponding cells. Another model building process is then started for this RF module.

 NOTE

Module power consumption models cannot be built (either normally or intelligently) for high-frequency cells, hyper cells, combined cells, LampSite cells, SUL cells, or Distributed MIMO cells.

When the network is almost under no load or the PRB load fluctuates slightly in the whole day, module power consumption models may fail to be normally or intelligently built because the PRB load is approximately constant.

9.1.1.2 Energy-Efficiency-based Inter-Frequency Handover

UEs are handed over to the cell with optimal energy efficiency by the gNodeB based on factors such as cell energy efficiency and user experience to achieve network energy saving. This function is controlled by the **ENERGY_EFFICIENCY_BASED_HO** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter.

If this option is selected, energy-efficiency-based inter-frequency handover is enabled. Such a handover is triggered for a UE that meets any of the following conditions:

- The UE has initially accessed the cell.
- The UE has been handed over to the cell based on coverage or voice quality.
- The UE has accessed the cell through RRC reestablishment.

The procedure for an energy-efficiency-based inter-frequency handover includes the phases described in [9.1.1.2.1 Measurement Configuration Delivery](#), [9.1.1.2.2 Measurement Reporting](#), [9.1.1.2.3 Energy Efficiency and User Experience Evaluation](#), [9.1.1.2.4 Optimal Cell Selection](#), and [9.1.1.2.5 Energy-Efficiency-based Inter-Frequency Handover Execution](#).

To ensure performance of UEs with specific service types, the energy-efficiency-based inter-frequency handover function can be configured to not take effect for these UEs. This is known as service-type-based dynamic exit from energy-efficiency-based inter-frequency handover. The service types involved are specified by the **gNBQciBearer.ServiceType** parameter.

Service-type-based dynamic exit from energy-efficiency-based inter-frequency handover takes effect when all of the following conditions are met:

- Energy-efficiency-based inter-frequency handover is enabled.
- The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.

- The **ENERGY_EFF_BASED_HO_EXIT_SW** option of the **NRCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter is selected in the group identified by the **NRCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter that is set to the same value as the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter for a specific service type.

9.1.1.2.1 Measurement Configuration Delivery

If the conditions for triggering an energy-efficiency-based inter-frequency handover are met, the gNodeB determines whether to deliver measurement configurations to the UE based on the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter for QCLs of services running on the UE.

- If this option is deselected for all QCLs of services running on the UE, energy-efficiency-based inter-frequency handover takes effect for the UE. In this case, the gNodeB delivers A5 measurement configurations (threshold 1 and threshold 2 for event A5 being -43 dBm and -140 dBm, respectively) related to the handover.
- If this option is selected for any QCL of services running on the UE, energy-efficiency-based inter-frequency handover does not take effect for the UE. In this case, the gNodeB does not deliver measurement configurations related to an energy-efficiency-based handover to the UE.

A measurement configuration message related to an energy-efficiency-based inter-frequency handover contains a maximum of four intra-RAT neighboring frequencies. If the number of configured intra-RAT neighboring frequencies exceeds four, the measurement configurations are delivered using multiple messages. Then, a candidate cell is selected from the serving cell and cells on all these neighboring frequencies for energy efficiency and user experience evaluation.

- If the gNodeB receives the measurement reports related to all intra-RAT neighboring frequencies in a message within x seconds after the message is delivered, it immediately removes the A5 measurement configurations in this message.
 - If the number of configured intra-RAT neighboring frequencies is less than or equal to four, the energy efficiency and user experience evaluation starts.
 - If the number of configured intra-RAT neighboring frequencies is greater than four, other messages containing A5 measurement configurations are delivered until the configurations related to all the intra-RAT neighboring frequencies are sent to the UE. Then, the energy efficiency and user experience evaluation starts.
- If the gNodeB does not receive the measurement reports related to all intra-RAT neighboring frequencies in a message within x seconds after the message is delivered, it also removes the A5 measurement configurations in this message. Then:
 - If the number of configured intra-RAT neighboring frequencies is less than or equal to four, the energy efficiency and user experience evaluation starts.

- If the number of configured intra-RAT neighboring frequencies is greater than four, other messages containing A5 measurement configurations are delivered until the configurations related to all the intra-RAT neighboring frequencies are sent to the UE. Then, the energy efficiency and user experience evaluation starts.

 NOTE

- x is determined by the number of frequencies delivered for inter-frequency measurements. Its value is 1.2s in the case of one neighboring frequency, 1.5s in the case of two neighboring frequencies, and 2s in the case of three or four neighboring frequencies.
- During gap-assisted inter-frequency measurements, the gaps may conflict with the time-domain positions of NR resources such as resources for sending scheduling requests (SRs) on the physical uplink control channel (PUCCH). As a result, KPIs such as the service drop rate, RRC connection reestablishment success rate, and contention-based random access success rate are affected. To avoid the conflicts between measurement gaps and NR resources, non-overlapping between gap and NR resources can be enabled. This function is controlled by the **GAP_AVOID_NR_RES_SW** option of the **gNBMobilityCommParam.MeasAlgoSwitch** parameter. For details about this function, see *SA Mobility Management in Connected Mode*.

9.1.1.2.2 Measurement Reporting

The UE performs A5 measurements for an energy-efficiency-based handover based on the event measurement configurations and reports the measurement result.

9.1.1.2.3 Energy Efficiency and User Experience Evaluation

Based on the inter-frequency event A5 measurement reports, the gNodeB selects the serving cell and the top-RSRP cell with RSRP greater than the RSRP threshold on each of the reported neighboring frequencies as candidate cells and performs energy efficiency evaluation and user experience evaluation as described in [Energy Efficiency Evaluation](#) and [User Experience Evaluation](#). The RSRP threshold is specified as follows:

- If the **NRCeIIInterFHoMeaGrp.MultiFreqSmtSelA5RsrpThld2** parameter is set to a value other than 255, the threshold is specified by the **NRCeIIInterFHoMeaGrp.MultiFreqSmtSelA5RsrpThld2** parameter.
- If the **NRCeIIInterFHoMeaGrp.MultiFreqSmtSelA5RsrpThld2** parameter is set to 255, the threshold is specified by the **NRCeIIInterFHoMeaGrp.IntrfInterFreqHoRsrpThld** parameter.

Energy Efficiency Evaluation

The gNodeB obtains the following information for evaluation of the energy efficiency of the candidate cells:

- UE-level downlink spectral efficiency: The gNodeB inputs the SSB measurement information of the candidate cells into the virtual grid models for downlink spectral efficiency and obtains the UE-level downlink spectral efficiency in the cells. For details about virtual grid model building, see *Virtual Grid-based Multi-Frequency Coordination*.
- Power consumption load factor k : The gNodeB inputs the PRB load and time of NR carriers and LTE carriers (only when LTE and NR share the same RF

module) of the candidate cells into the module power consumption models and obtains k of the modules under the current load and time.

The gNodeB divides the UE-level downlink spectral efficiency by the factor k to obtain the evaluated UE-level energy efficiency in the candidate cells.

User Experience Evaluation

The gNodeB obtains the following information for evaluation of user experience of the candidate cells in terms of downlink air interface capability:

- UE-level downlink spectral efficiency: The gNodeB inputs the SSB measurement information of the candidate cells into the virtual grid models for downlink spectral efficiency and obtains the UE-level downlink spectral efficiency in the cells. For details about virtual grid model building, see *Virtual Grid-based Multi-Frequency Coordination*.
- Equivalent downlink cell bandwidth: This bandwidth is calculated based on factors such as the downlink bandwidths of the candidate cells, slot configuration, and UE-supported downlink bandwidth.
- Downlink load factor: This factor is calculated based on the number of UEs to be scheduled in the downlink per second.

The gNodeB evaluates the downlink air interface capability which involves the UE-level downlink spectral efficiency, equivalent downlink cell bandwidth, and downlink load factor to determine user experience of the candidate cells.

NOTE

The gNodeB evaluates user experience based on the measurement reports from UEs. The evaluation may be affected by factors such as measurement report accuracy, gNodeB configuration differences, and UE compatibility differences. Therefore, the average downlink UE throughputs after energy-efficiency-based inter-frequency handovers may not be as expected.

9.1.1.2.4 Optimal Cell Selection

The optimal cell is selected for the UE based on the results of energy efficiency evaluation and user experience evaluation. The principles are as follows:

- Energy efficiency evaluation: A cell with higher energy efficiency has a higher priority. Hysteresis of 5% is always applied to comparison of the energy efficiency of any inter-frequency candidate cell with that of the serving cell. An inter-frequency candidate cell is considered to have higher energy efficiency than the serving cell only when its energy efficiency is 5% higher than that of the serving cell.
- User experience evaluation: The evaluated user experience of the candidate cell exceeds 1.4 times the value of **NRCellMobilityConfig.EnergyEffInterFHoDlRateThd**.
- The results of energy efficiency evaluation and user experience evaluation meet the requirements of energy saving factors delivered by the MIE.

If multiple candidate cells meet the preceding requirements, these cells are sorted in descending order based on the results of energy efficiency evaluation. The cell with the highest result is the optimal cell.

9.1.1.2.5 Energy-Efficiency-based Inter-Frequency Handover Execution

If the optimal cell is not the serving cell, the base station directly initiates a handover to the optimal cell. Otherwise, the procedure ends.

9.1.1.3 Energy-Efficiency-based Coordinated Scheduling

During downlink CA for a UE, the gNodeB transmits data on the carriers with higher energy efficiency to the UE for network energy saving. This function is controlled by the **ENERGY_EFF_BASED_COLLAB_SCH_SW** option of the **NRCellAlgoSwitch.MultiFreqAlgoSwitch** parameter.

If this option is selected, energy-efficiency-based coordinated scheduling is enabled. The working procedure is as follows:

1. Triggering

When the base station transmits downlink data over multiple carriers to a CA UE, it initiates energy-efficiency-based coordinated scheduling.

2. Energy efficiency evaluation

The gNodeB obtains the following information to evaluate the energy efficiency of the downlink carriers for the CA UE:

- Real-time downlink spectral efficiency of the UE on these carriers
- Power consumption load factor k obtained using the power consumption model for the RF module serving each carrier

The gNodeB divides the downlink spectral efficiency by the factor k to obtain the UE energy efficiency evaluation value for each carrier.

3. Execution of energy-efficiency-based coordinated scheduling

If the energy saving factors delivered by the MIE are valid, the gNodeB transmits more downlink data on high-energy-efficiency carriers based on the energy efficiency of each carrier evaluated in step 2.

To ensure performance of UEs with specific service types, energy-efficiency-based coordinated scheduling can be configured to not take effect for these UEs. This is known as service-type-based dynamic exit from energy-efficiency-based coordinated scheduling. The service types involved are specified by the **gNBQciBearer.ServiceType** parameter.

Service-type-based dynamic exit from energy-efficiency-based coordinated scheduling takes effect when all of the following conditions are met:

- Energy-efficiency-based coordinated scheduling is enabled.
- The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.
- The **ENERGY_EFF_COLLAB_SCH_EXIT_SW** option of the **NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter is selected in the group identified by the **NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter that is set to the same value as the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter for a specific service type.

9.1.2 Network Analysis

9.1.2.1 Benefits

For dual-band TDD 32T32R RF modules, the module power consumption load factor k generated by intelligently built module power consumption models is more accurate than that generated by normally built ones.

With energy-efficiency-based inter-frequency handover and energy-efficiency-based coordinated scheduling enabled, the energy consumption of the base station decreases given that the traffic volume remains unchanged after the base station selects the carriers with optimal energy efficiency for UEs. This function does not produce energy saving gains to an unloaded network but produces low energy saving gains to a lightly-loaded network.

- Energy-efficiency-based inter-frequency handover:
 - Energy saving gains may decrease when there are some non-target UEs for energy-efficiency-based inter-frequency handovers. There are no energy saving gains if all UEs on the network are non-target UEs. Non-target UEs refer to those not meeting the three conditions described in [9.1.1.2 Energy-Efficiency-based Inter-Frequency Handover](#) or those running services with QCLs for which energy-efficiency-based inter-frequency handover is not allowed.
 - If other types of handover-related features (such as Frequency-Priority-based Inter-Frequency Handover, Service Based Inter-frequency Handover, Operator-specific-Priority-based Inter-Frequency Handover, and Experience-based Smart Carrier Selection) are enabled on the network, the energy saving gains of this feature decrease or even no gains are obtained.
 - Service-type-based dynamic exit from energy-efficiency-based inter-frequency handover relieves the impact of energy-efficiency-based inter-frequency handover on specific service types but reduces energy saving gains.
- Energy-efficiency-based coordinated scheduling:
 - The fewer the UEs that support downlink CA on the network, the fewer the energy saving gains offered by energy-efficiency-based coordinated scheduling. The lower the probability of SCell activation for CA UEs and the shorter the duration in which the UEs stay in the SCell-activated state, the fewer the energy saving gains offered by energy-efficiency-based coordinated scheduling. For the factors that affect SCell activation for CA UEs, see *Carrier Aggregation*.
 - Service-type-based dynamic exit from energy-efficiency-based coordinated scheduling relieves the impact of energy-efficiency-based coordinated scheduling on specific service types but reduces energy saving gains.

9.1.2.2 Impacts

The impacts between this function and other functions are described in [9.1.3.2.3 Function Impacts](#).

Energy-efficiency-based inter-frequency handover has the following impacts on the network:

- Inter-band UE distribution and the PRB load of each cell change. The number of signaling messages over the air interface and the number of handovers

increase. For a cell with an increasing number of UEs, the CPU usage of its serving BBP increases.

- After this function is enabled, data collection, model building, and energy efficiency and user experience evaluation are performed. As a result, the CPU usage of the main control board increases.
- After UEs are handed over to the cells with optimal energy efficiency, the **User Downlink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)** may decrease.

Energy-efficiency-based coordinated scheduling has the following impacts on the network:

- After this function is enabled, data collection, model building, and energy efficiency and user experience evaluation are performed. As a result, the CPU usage of the main control board increases.
- The **User Downlink Average Throughput (DU)** may decrease.

9.1.3 Requirements

9.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-071904	Intelligent Multi-RF-Module Coordinated Energy Saving	NR0SMRMCES00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

9.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Building of downlink intra-frequency spectral efficiency prediction models	DL_INTRA_FREQ_SPCT_EFF_PRED_SW option of the NRCellSmartMultiCarr.NrMultiCarrierAlgoSwitch parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	Energy-efficiency-based inter-frequency handover can be enabled only if at least one of the following options is selected: the DL_INTRA_FREQ_SPCT_EFF_PRED_SW option of the NRCellSmartMultiCarr.NrMultiCarrierAlgoSwitch parameter and the DL_FREQ_SE_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarr.ModelBuildingSwitch parameter.
Low-frequency TDD	DL Freq Spct Eff Mdl Build Allowed	DL_FREQ_SE_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarr.ModelBuildingSwitch parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Energy-efficiency-based coordinated scheduling can be enabled only when fast CA is enabled together with downlink intra-band CA or downlink intra-FR inter-band CA.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast inter-gNodeB CA	INTER_GNB_ULTRALOW_L AT_SPLT_SW option of the NRDUCellCar rMgmt.CaEn hancedAlgoE xtSwitch parameter	<i>CA Experience Improvement</i>	
Low-frequency TDD	Normal building of module power consumption models	PWR_EFF_M ODEL_BUILD_ SW option of the NRDUCellAlg oSwitch.Pow erSavingSwit ch parameter	<i>Energy Conservation and Emission Reduction</i>	When energy-efficiency-based inter-frequency handover and energy-efficiency-based coordinated scheduling are enabled, at least one of normal and intelligent building of module power consumption models must be enabled.
Low-frequency TDD	Intelligent building of module power consumption models	PWR_EFF_M ODEL_INTEL_ BUILD_SW option of the NRDUCellAlg oSwitch.Pow erSavingExtS witch parameter	<i>Energy Conservation and Emission Reduction</i>	
Low-frequency TDD	Energy-efficiency-based inter-frequency handover	ENERGY_EFFI CIENCY_BASE D_HO option of the NRCe llAlgoS witch.InterFr eqHoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Service-type-based dynamic exit from energy-efficiency-based inter-frequency handover takes effect only if energy-efficiency-based inter-frequency handover has been enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Energy-efficiency-based coordinated scheduling	ENERGY_EFF_BASED_COLLAB_SCH_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Service-type-based dynamic exit from energy-efficiency-based coordinated scheduling takes effect only if energy-efficiency-based coordinated scheduling has been enabled.

9.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	The following functions cannot be enabled for cells with distributed massive MIMO already enabled: normal building of module power consumption models, intelligent building of module power consumption models, energy-efficiency-based inter-frequency handover, and energy-efficiency-based coordinated scheduling.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Energy-efficiency-based inter-frequency handover and energy-efficiency-based coordinated scheduling do not work in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Energy-efficiency-based inter-frequency handover and energy-efficiency-based coordinated scheduling do not work in combined cells.
Low-frequency TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	CA Performance Boost Based on Multi-dimensional Optimization cannot be enabled when energy-efficiency-based coordinated scheduling has been enabled.

9.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Frequency-priority-based inter-frequency handover	FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	It is not recommended that frequency-priority-based inter-frequency handover and energy-efficiency-based inter-frequency handover be both enabled. If they are both enabled, energy saving gains will decrease.
Low-frequency TDD	UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and NRCellMlb.MlbTriggerMode set to CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	Assume that the INTER_FREQ_HO_COLLABORATION_SW option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected and that both UE-number-based connected mode MLB and energy-efficiency-based inter-frequency handover are enabled. When an energy-efficiency-based inter-frequency handover is triggered, cells in the MLB state will not be selected as the target cell. This prevents ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	INTER_FREQ_CONNECTE D_MLB_SW option of the NRCellAlgoS witch.MlbAl goSwitch parameter, and NRCellMlb. MlbTrigger Mode set to SPEC_EFF_B ASED_USER_ NUM	<i>Mobility Load Balancing in Connected Mode</i>	Assume that the INTER_FREQ_HO _COLLABORATIO N_SW option of the NRCellAlgoSwitc h.InterFreqHoSwi tch parameter is selected and that both spectral-efficiency-based connected mode MLB and energy-efficiency-based inter-frequency handover are enabled. If an inter-frequency handover is triggered based on energy efficiency, cells in the MLB state will not be selected as the target cell. This prevents ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	5QI-specific UE-number-based connected mode MLB	INTER_FREQ_CONNECTE D_MLB_SW option of the NRCellAlgoS witch.MlbAl goSwitch parameter, and NRCellMlb. MlbTrigger Mode set to SPECIFIED_5 QI_USER_NU M	<i>Mobility Load Balancing in Connected Mode</i>	Assume that the INTER_FREQ_HO _COLLABORATIO N_SW option of the NRCellAlgoSwitc h.InterFreqHoSwi tch parameter is selected and that both 5QI-specific UE-number-based connected mode MLB and energy-efficiency-based inter-frequency handover are enabled. If an inter-frequency handover is triggered based on energy efficiency, cells in the MLB state will not be selected as the target cell. This prevents ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Radio-resource-based connected mode MLB	INTER_FREQ_CONNECTE D_MLB_SW option of the NRCellAlgoS witch.MlbAl goSwitch parameter and NRCellMlb. MlbTrigger Mode set to RADIO_RES OURCE	<i>Mobility Load Balancing in Connected Mode</i>	Assume that the INTER_FREQ_HO _COLLABORATIO N_SW option of the NRCellAlgoSwitc h.InterFreqHoSwi tch parameter is selected and that both radio-resource-based connected mode MLB and energy-efficiency-based inter-frequency handover are enabled. If an inter-frequency handover is triggered based on energy efficiency, cells in the MLB state will not be selected as the target cell. This prevents ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell-level MRO	MRO_SCENARIO_IDENT_SW option of the gNB-MRO-ScenarioIdentifySwitch parameter and the INTRA_FREQ_MRO_SW option of the NRCell-MRO-MroSwitch parameter	<i>MRO</i>	When both multi-RF-module coordinated energy saving and intra-frequency MRO are enabled, event A3 reporting for multi-RF-module coordinated energy saving is affected. As a result, the downlink experience evaluation of the serving cell may not be as expected, which may cause the average downlink UE throughput to decrease.
Low-frequency TDD	UE-level MRO	UE_MRO_SWITCH option of the NRCell-MRO-MroSwitch parameter	<i>MRO</i>	When both multi-RF-module coordinated energy saving and UE-level MRO are enabled, event A3 reporting will be affected if multi-RF-module coordinated energy saving takes effect for ping-pong UEs. As a result, the downlink experience evaluation of the serving cell may not be as expected, which may cause the average downlink UE throughput to decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SEL_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	It is not recommended that experience-based smart carrier selection and energy-efficiency-based inter-frequency handover be both enabled. If they are both enabled, experience gains and energy saving gains will decrease.
Low-frequency TDD	Inter-carrier frequency selective scheduling	CROSS_CARRIER_FREQ_SEL_SCH_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	When inter-carrier frequency selective scheduling and energy-efficiency-based coordinated scheduling are both enabled, inter-carrier frequency selective scheduling gains and energy saving gains both decrease.
Low-frequency TDD	Multi-frequency convergent scheduling	MULTI_FREQ_CONVERGENT_SCH_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	When both energy-efficiency-based coordinated scheduling and multi-frequency convergent scheduling are enabled, the gains of multi-frequency convergent scheduling decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunctionSw parameter	<i>RedCap</i>	Energy-efficiency-based inter-frequency handover does not take effect for RedCap UEs.
Low-frequency TDD	CA-UE-specific frequency-priority-based inter-frequency handover	CA_FREQ_PRIORITY_BASED_INTER_FREQ_HO_SWITCH option of the NRCellAlgorithmSwitch.InterFreqHoSwitch parameter	<i>CA Experience Improvement</i>	It is not recommended that CA-UE-specific frequency-priority-based inter-frequency handover and energy-efficiency-based inter-frequency handover be both enabled. If they are both enabled, energy saving gains will decrease.

9.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

The main control board must be a UMPTha, UMPThb, UMPTg, UMPTga, or UMPTe (NR-only).

All NR-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

9.1.3.4 Networking

The networking requirements are as follows:

- Two- or three-frequency networking is used.
- Cell type:
 - Normal and intelligent building of module power consumption models cannot be enabled for SUL cells, combined cells, hyper cells, or LampSite cells.
 - Energy-efficiency-based inter-frequency handover cannot be enabled for SUL cells, combined cells, hyper cells, or LampSite cells.
 - Energy-efficiency-based coordinated scheduling cannot be enabled for SUL cells, combined cells, hyper cells, or LampSite cells.
- Operating band: The **NRDUCell.FrequencyBand** parameter cannot be set to **N257**, **N258**, **N260**, or **N261**.

9.1.3.5 Others

When NR and LTE share RF modules, it is recommended that the LTE side be upgraded to eRAN19.1 or later. Without the upgrade, the gains of multi-RF-module coordinated energy saving may be affected.

This function requires the Customer Policy-based On-Demand Energy Saving feature to be enabled on the MIE.

The total number of module power consumption models, either normally or intelligently built, must be less than or equal to 30 in the following scenarios:

- The main control board is a UMPTg.
- The main control board is a UMPTga and the **NODE.WM** parameter is set to **NON-CONCURRENT**.
- The main control board is a UMPTe and the **NODE.WM** parameter is set to **NON-CONCURRENT**.

The total number of module power consumption models, either normally or intelligently built, must be less than or equal to 15 when the main control board is a UMPTga and the **NODE.WM** parameter is set to **CONCURRENT**.

9.1.4 Operation and Maintenance

9.1.4.1 Data Configuration

9.1.4.1.1 Data Preparation

This function requires the Customer Policy-based On-Demand Energy Saving feature to be enabled on the MIE only. For details, contact Huawei engineers.

Table 9-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 9-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Energy Eff-based Inter-Freq HO DL Rate Thld	NRCellMobilityConfig.EnergyEffInterFHoDLRateThd	Use the recommended value. - Cells with more than eight TX channels: value (in the unit of Mbit/s) of the downlink cell bandwidth - Cells with eight or fewer TX channels: rounded number of 1/2 multiplied by the value (in the unit of Mbit/s) of the downlink cell bandwidth For example: Set this parameter to 100 for a 64T cell with a bandwidth of 100 MHz. Set this parameter to 10 for a 4T cell with a bandwidth of 20 MHz.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	It is recommended that the SERV_TYPE_DYN_EXIT_PWR_SAVE_SW option be selected when there are specific types of services and user experience of these services needs to be ensured.
Power Saving Policy Group Index	gNBQciBearer.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan.
Power Saving Policy Group Index	NRCellPsPolicyGrp.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan. The value of this parameter must be the same as that of the gNBQciBearer.PowerSavingPolicyGrpIndex parameter for services to be guaranteed. Before setting this parameter, ensure that the QCI for which the gNBQciBearer.PowerSavingPolicyGrpIndex parameter is set has been configured.

Parameter Name	Parameter ID	Setting Notes
User Identification Method	NRCellPsPolicyGrp.<i>UserIdentMethod</i>	Set this parameter to SERVICE_TYPE .
Service Type Dynamic Exit Power Saving Switch	NRCellPsPolicyGrp.<i>ServTypeDynExitPwrSavingSw</i>	To enable service-type-based dynamic exit from energy-efficiency-based inter-frequency handover, select the ENERGY_EFF_BASED_HO_EXIT_SW option.
Power Saving Policy Group Index	NRDUCellPsPolicyGrp.<i>PowerSavingPolicyGrpIndex</i>	Set this parameter based on the network plan. The value of this parameter must be the same as that of the gNBQciBearer.<i>PowerSavingPolicyGrpIndex</i> parameter for services to be guaranteed. Before setting this parameter, ensure that the QCI for which the gNBQciBearer.<i>PowerSavingPolicyGrpIndex</i> parameter is set has been configured.
User Identification Method	NRDUCellPsPolicyGrp.<i>UserIdentMethod</i>	Set this parameter to SERVICE_TYPE .
Service Type Dynamic Exit Power Saving Switch	NRDUCellPsPolicyGrp.<i>ServTypeDynExitPwrSavingSw</i>	To enable service-type-based dynamic exit from energy-efficiency-based coordinated scheduling, select the ENERGY_EFF_COLLAB_SCH_EXIT_SW option.

NOTE

On a multi-frequency network, these options must be selected for all cells on all frequencies. Otherwise, the function may not take effect properly.

9.1.4.1.2 Using MML Commands

Before using MML commands, refer to [9.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Activating this function requires the Customer Policy-based On-Demand Energy Saving feature to be enabled on the MIE only. For details, contact Huawei engineers.

```
/(Optional) Changing the setting of EnergyEffInterFHoDlRateThd
MOD NRCELLMOBILITYCONFIG: NrCellId=0, EnergyEffInterFHoDlRateThd=5;
/(Optional) Enabling service-type-based dynamic exit from energy-efficiency-based inter-frequency
handover
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;
ADD NRCELLPOLICYGRP: NrCellId=0, PowerSavingPolicyGrpIndex=0, UserIdentMethod=SERVICE_TYPE,
ServTypeDynExitPwrSavingSw=ENERGY_EFF_BASED_HO_EXIT_SW-1;
/(Optional) Enabling service-type-based dynamic exit from energy-efficiency-based coordinated scheduling
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;
ADD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=ENERGY_EFF_COLLAB_SCH_EXIT_SW-1;
```

Deactivation Command Examples

Deactivating this function requires the Customer Policy-based On-Demand Energy Saving feature to be disabled on the MIE only. For details, contact Huawei engineers.

```
/(Optional) Disabling service-type-based dynamic exit from energy-efficiency-based coordinated scheduling
MOD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=ENERGY_EFF_COLLAB_SCH_EXIT_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;
/(Optional) Disabling service-type-based dynamic exit from energy-efficiency-based inter-frequency
handover
MOD NRCELLPOLICYGRP: NrCellId=0, PowerSavingPolicyGrpIndex=0, UserIdentMethod=SERVICE_TYPE,
ServTypeDynExitPwrSavingSw=ENERGY_EFF_BASED_HO_EXIT_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;
```

9.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.1.4.2 Activation Verification

If any of the counters listed in [Table 9-2](#) returns a non-zero value, energy-efficiency-based inter-frequency handover has taken effect.

Table 9-2 Counters related to energy-efficiency-based inter-frequency handover

Counter ID	Counter Name
1911837898	N.HO.InterFreq.EnergyEff.PrepAttOut
1911837900	N.HO.InterFreq.EnergyEff.ExecAttOut
1911837899	N.HO.InterFreq.EnergyEff.ExecSuccOut
1911837897	N.NsaDc.InterFreq.EnergyEff.PSCell.Change.Att
1911837896	N.NsaDc.InterFreq.EnergyEff.PSCell.Change.Succ

The status of module power consumption models can be queried by running the **DSP NRDUCELLPWREFFMODEL** command.

9.1.4.3 Network Monitoring

Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, you can compare the total energy consumption data of base stations in the area within one week before and after multi-RF-module coordinated energy saving is enabled to evaluate its energy saving gains. It is expected that the total energy consumption of base stations in the area decreases after this function is enabled.

9.2 Intelligent Power Control Energy Saving

9.2.1 Principles

Intra-frequency interference between cells is common in urban NR networks because of small inter-site distances. **When some cells in a cluster are heavily loaded and suffer from severe interference from neighboring cells, inter-cell coordination can be used to reduce the PDSCH and DMRS transmit power for UEs in the neighboring cells. This not only lowers the interference to the heavy-load cells, but also reduces the energy consumption of the RF modules serving the neighboring cells.**

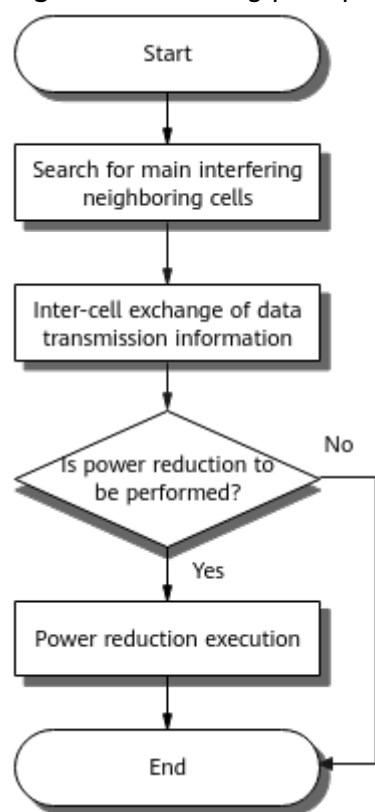
Intelligent power control energy saving is controlled by the **SMART_PWR_CTRL_ENERGY_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter. [Table 9-3](#) describes the parameters in the **NRDUCellPowerSaving** MO that specify the effective period and parameters related to function triggering under a policy of intelligent power control energy saving.

Table 9-3 Parameters related to a policy of intelligent power control energy saving

Parameter Name	Parameter ID	Description
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	This parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	This parameter specifies the energy saving type indicated by an energy saving policy index. Intelligent power control energy saving requires that this parameter be set to SMART_PWR_CTRL_ENERGY_SAVE .
Start Time	NRDUCellPowerSaving.StartTime	These parameters specify the effective period for intelligent power control energy saving and the conditions for entering or exiting this function.
Stop Time	NRDUCellPowerSaving.StopTime	
Smart Power Control NCell Load Threshold	NRDUCellPowerSaving.SmartPwrCtrlNCellLoadThld	
Smart Power Control Maximum PDSCH Offset	NRDUCellPowerSaving.SmartPwrCtrlMaxPd-schOffset	
Smart Power Control MCS Threshold	NRDUCellPowerSaving.SmartPwrCtrlMcsThld	

Figure 9-2 shows the procedure for intelligent power control energy saving.

Figure 9-2 Working principles for intelligent power control energy saving



Search for Main Interfering Neighboring Cells

The base station searches for interfering neighboring cells based on intra-frequency neighboring cell measurement, which is controlled by the **INTRA_FREQ_NCELL_STAT_SW** option of the **NRDUCellCollabServ.NeighborNrCellMgmtSw** parameter. With this option selected, the gNodeB delivers event A3 measurement configurations and collects event A3 measurement reports from UEs to obtain information about intra-frequency neighboring cells. The A3 threshold is specified by the **NRCellComp.IntraFreqNCellStatA3Offset** parameter.

A collected A3 measurement report is valid if the reported measurement result of RSRP in a neighboring cell is greater than the value of **NRDUCellChnPwr.SmartPwrCtrlNCellRsrpThld**.

The neighboring cells are sorted in descending order based on the proportion of valid measurement reports. Cells with relatively high proportions of valid measurement reports are selected as main interfering neighboring cells.

Inter-Cell Exchange of Data Transmission Information

The serving cell calculates data transmission information such as the downlink PRB load and periodically sends the latest calculation result to the main interfering neighboring cells. If an interfering neighboring cell and the serving cell are served by different base stations, data transmission information is transmitted over the eXn interface. It is recommended that this feature be enabled when the one-way delay over the eXn interface is less than 4 ms to ensure that data transmission information arrives at the receiver in a timely manner.

Power Reduction Evaluation

Each interfering neighboring cell receives the data transmission information from surrounding cells and then calculates its overall load based on the information.

If the overall load of an interfering neighboring cell is higher than the value of **NRDUCellPowerSaving.SmartPwrCtrlNCellLoadThld** and the impact of power reduction is limited, the procedure goes to **Power Reduction Execution**. Otherwise, the procedure ends.

However, if the MCS index for the UE is less than the value of the **NRDUCellPowerSaving.SmartPwrCtrlMcsThld** parameter, power reduction will not be performed.

Power Reduction Execution

The PDSCH and DMRS transmit power for UEs in the interfering neighboring cell is reduced by a maximum of **NRDUCellPowerSaving.SmartPwrCtrlMaxPdschOffset**.

Service-Type-based Dynamic Exit from Intelligent Power Control Energy Saving

To ensure performance of UEs with specific service types, intelligent power control energy saving can be configured to not take effect for these UEs. This is known as service-type-based dynamic exit from intelligent power control energy saving. The service types involved are specified by the **gNBQciBearer.ServiceType** parameter.

Service-type-based dynamic exit from intelligent power control energy saving takes effect when all of the following conditions are met:

- Intelligent power control energy saving is enabled.
- The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.
- The **SMT_PWR_CTRL_NRG_SAVE_EXIT_SW** option of the **NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter is selected in the group identified by the **NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter that is set to the same value as the **gNBQciBearer.PowerSavingPolicyGrpIndex** parameter for a specific service type.

9.2.2 Network Analysis

9.2.2.1 Benefits

After the base station reduces the transmit power for UEs, the energy consumption of the base station decreases given that the traffic volume remains unchanged. This function does not produce energy saving gains to an unloaded network but produces low energy saving gains to a lightly-loaded network.

Service-type-based dynamic exit from intelligent power control energy saving relieves the impact of intelligent power control energy saving on specific service types but reduces energy saving gains.

9.2.2.2 Impacts

The impacts between this function and other functions are described in [9.2.3.2.3 Function Impacts](#).

Intelligent power control energy saving has the following impacts on the network:

- The CQI may slightly change because interference from neighboring cells decreases.
- The **User Downlink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)** may decrease, causing a slight increase in loads indicated by the **N.PRB.DL.DrbUsed.Avg** and **N.PRB.DL.Used.Avg** counters.
- After the data channel power decreases, the overall interference on the network changes. As a result, some KPIs may slightly change.
- The network load, including the CPU load of boards and uplink load, may slightly change due to the increase in the number of measurement reports.

9.2.3 Requirements

9.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-071903	Intelligent Power Control Energy Saving	NR0S0PSBPC00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

9.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-frequency neighboring cell measurement	INTRA_FREQ_NCELL_STAT_SW option of the NRDUCellCollabServ.NeighborNrCellMgmtSw parameter	None	Intelligent power control energy saving requires that intra-frequency neighboring cell measurement be enabled.
Low-frequency TDD	Intelligent power control energy saving	SMART_PWR_CTL_ENERGY_SAVE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Service-type-based dynamic exit from intelligent power control energy saving takes effect only if intelligent power control energy saving has been enabled.

9.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	TDD: <i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	(TDD) If distributed massive MIMO is enabled for a cell, intelligent power control energy saving cannot be enabled for the cell.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDualCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Intelligent power control energy saving does not work in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination/Joint Transmission Cell Combination	NRDUCell.NrDualCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Intelligent power control energy saving does not work in combined cells.
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Intelligent power control energy saving does not work in high-speed cells.
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The period specified for timing carrier shutdown in a cell cannot overlap that specified for intelligent power control energy saving in the same cell. You are advised to preferentially plan the period during which timing carrier shutdown takes effect.

9.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CoMpSwitch parameter	<i>CoMP</i>	Power reduction does not take effect for DL CoMP UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Basic VoNR functions	VONR_SW option of the NRCellAlgoSwitch . VonrSwitch parameter	<i>VoNR</i>	Power reduction does not take effect for VoNR UEs.
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch . HighReliabilitySwitch parameter	<i>URLLC</i>	Power reduction does not take effect for UEs running low-latency and high-reliability services.
Low-frequency TDD	URLLC for Vehicle Communications	HYBRID_SERVISE_SWITCH option of the NRDUCellAlgoSwitch . HighReliabilitySwitch parameter	<i>URLLC for Vehicle Communications</i>	Intelligent power control energy saving does not take effect for vehicle communication UEs.
Low-frequency TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW or INTER_GNB_DL_CS_CBF_SW option of the NRDUCellAlgoSwitch . CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	Intelligent power control energy saving does not take effect for CS or CBF UEs.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SLEEP	<i>Multi-RAT Coordinated Energy Saving</i>	Intelligent power control energy saving does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	RF module super dormancy takes precedence over intelligent power control energy saving when both functions are enabled. Intelligent power control energy saving does not take effect when RF module super dormancy is in effect.
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over intelligent power control energy saving when both functions are enabled. Intelligent power control energy saving does not take effect when RF module fast deep dormancy is in effect.
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellA1goSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both intelligent carrier shutdown and intelligent power control energy saving take effect, the latter function does not deliver energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option (inter-base-station multi-carrier scenarios) of the NRDU CellAI goSwitch.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SW option	<i>Energy Conservation and Emission Reduction</i>	When both fast carrier shutdown and intelligent power control energy saving take effect, the latter function does not deliver energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
		of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected Basic-layer cells: FAST_CAR_R_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected		

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUelidentSw	None	Intelligent power control energy saving fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
Low-frequency TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.AppIpv4 • gNBAsAppParam.AppIpv6	None	• Any option of the gNodeBParam.StUelidentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.AppIpv4 parameter is set. • The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue: • Select the SPEED_TURBO_PWR_SAVING_SW option of

RAT	Function Name	Function Switch	Reference	Description
				the gNBWPwrSavin gAlgo.PwrSavi ngAlgoSwitch parameter. Intelligent power control energy saving can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBWPwrSavin gAlgo.SpeedT urboPwrSavSt artTime and gNBWPwrSavin gAlgo.SpeedT urboPwrSavEn dTime parameters, increasing the energy saving gains. • Select the STATIC_ST_PW R_SAVING_SW option of the gNBWPwrSavin gAlgo.PwrSavi ngAlgoSwitch parameter. Intelligent power control energy saving can take effect for cells served by the base station in some of the preceding scenarios

RAT	Function Name	Function Switch	Reference	Description
				within the period specified by the gNBPwrSavin gAlgo.StaticST PwrSavStartTime and gNBPwrSavin gAlgo.StaticST PwrSavEndTime parameters, increasing the energy saving gains.
Low-frequency TDD	UE selection for neighboring cell measurement	NCELL_MEAS_UE_SEL_SW option of the gNBQciBearer.PwrSavin gAlgoSwitch parameter	None	If the NCELL_MEAS_UE_SEL_SW option of the gNBQciBearer.PwrSavin gAlgoSwitch parameter is deselected for any QCI of the services running on a UE, the UE cannot perform intra-frequency measurements. This affects intelligent power control energy saving in searching for main interfering neighboring cells.
Low-frequency TDD	FWA	gNBQciBearer.ServiceType set to FWA_INTERNET , FWA_VIDEO , or FWA_PRIVATE_LINE	FWA	Intelligent power control energy saving does not take effect for FWA users.

9.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

The main control board must be a UMPTe, UMPTg, UMPTga, UMPTgb, or UMPTba.

All NR-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

9.2.3.4 Networking

- In densely populated urban areas, the average load must be greater than 20%. It is recommended that this feature be enabled for all sites in these areas.
- Xn and eXn interfaces must be configured between base stations.
- The one-way delay over the eXn interface must be less than 4 ms.

9.2.3.5 Cells

This function requires that the duplex mode of cells be NR TDD. That is, the **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**.

(TDD) This function requires that the **NRDUCell.FrequencyBand** parameter not be set to **N257**, **N258**, **N260**, or **N261** for the cells.

9.2.3.6 Others

None

9.2.4 Operation and Maintenance

9.2.4.1 Data Configuration

9.2.4.1.1 Data Preparation

[Table 9-4](#) describes the parameters used for function activation.

Table 9-4 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellAlgoSwitch.NrDuCellId	Set this parameter based on the network plan.
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	Select the SMART_PWR_CTRL_ENERGY_SAVE_SW option when intelligent power control energy saving is required.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	It is recommended that this function be enabled all day long. Set these parameters based on the network plan. Their settings must meet the following requirements: - For a single policy of an energy saving function, the interval between the start time and stop time must be longer than 20 minutes. If the stop time is earlier than or the same as the start time, it is assumed that the stop time is a time on the next day. For example: If the start time is set to 08:00 and the stop time is set to 07:00, the energy saving function can take effect within the period from 08:00 to 07:00 the next day. - For different policies of an energy saving function, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving function needs to be enabled after policy A of this function has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Stop Time	NRDUCellPowerSaving.StopTime	
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to SMART_PWR_CTRL_ENERGY_SAVE if intelligent power control energy saving is required.
Intra-Freq Neighbor Cell Statistics A3 Offset	NRCCellComp.IntraFreqNCellStatA3Offset	Retain the default value.

Parameter Name	Parameter ID	Setting Notes
Smart Power Control NCell RSRP Threshold	NRDUCellChnPwr.SmartPwrCtrlNCellRsrpThld	Use the recommended value.
Smart Power Control NCell Load Threshold	NRDUCellPowerSaving.SmartPwrCtrlNCellLoadThld	Use the recommended value.
Smart Power Control Maximum PDSCH Offset	NRDUCellPowerSaving.SmartPwrCtrlMaxPd-schOffset	Use the recommended value.
Smart Power Control MCS Threshold	NRDUCellPowerSaving.SmartPwrCtrlMcsThld	Use the recommended value.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	It is recommended that the SERV_TYPE_DYN_EXIT_PWR_SAVE_SW option be selected when there are specific types of services and user experience of these services needs to be ensured.
Power Saving Policy Group Index	gNBQciBearer.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Power Saving Policy Group Index	NRDUCellPsPolicyGrp.PowerSavingPolicyGrpIndex	Set this parameter based on the network plan. The value of this parameter must be the same as that of the gNBQciBearer.PowerSavingPolicyGrpIndex parameter for services to be guaranteed. Before setting this parameter, ensure that the QCI for which the gNBQciBearer.PowerSavingPolicyGrpIndex parameter is set has been configured.
User Identification Method	NRDUCellPsPolicyGrp.UserIdentMethod	Set this parameter to SERVICE_TYPE .
Service Type Dynamic Exit Power Saving Switch	NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw	To enable service-type-based dynamic exit from intelligent power control energy saving, select the SMT_PWR_CTRL_NRG_SAVE_EXIT_SW option.

9.2.4.1.2 Using MML Commands

Before using MML commands, refer to [9.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling intelligent power control energy saving
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=SMART_PWR_CTRL_ENERGY_SAVE_SW-1;
//Setting an energy saving policy of intelligent power control energy saving
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=SMART_PWR_CTRL_ENERGY_SAVE, StartTime=00&00&00, StopTime=23&59&59,
SmartPwrCtrlMaxPdschOffset=1, SmartPwrCtrlNCellLoadThld=90, SmartPwrCtrlMcsThld=OFF;
//Setting the SmartPwrCtrlNCellRsrpThld parameter
MOD NRDUCELLCHNPWR: NrDuCellId=0, SmartPwrCtrlNCellRsrpThld=-115;
//((Optional) Setting the IntraFreqNCellStatA3Offset parameter
MOD NRCELLCOMP: NrCellId=0, IntraFreqNCellStatA3Offset=-30;
//((Optional) Enabling service-type-based dynamic exit from intelligent power control energy saving
```

```
MOD GNBQCIBEARER: Qci=201, PowerSavingPolicyGrpIndex=0;  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-1;  
ADD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,  
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=SMT_PWR_CTRL_NRG_SAVE_EXIT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
///(Optional) Disabling service-type-based dynamic exit from intelligent power control energy saving  
MOD NRDUCELLPOLICYGRP: NrDuCellId=0, PowerSavingPolicyGrpIndex=0,  
UserIdentMethod=SERVICE_TYPE, ServTypeDynExitPwrSavingSw=SMT_PWR_CTRL_NRG_SAVE_EXIT_SW-0;  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=SERV_TYPE_DYN_EXIT_PWR_SAVE_SW-0;  
//Disabling intelligent power control energy saving  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=  
SMART_PWR_CTRL_ENERGY_SAVE_SW-0;  
//Removing an energy saving policy of intelligent power control energy saving  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
```

9.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.2.4.2 Activation Verification

After intelligent power control energy saving is activated, use either of the following methods to check whether the function has taken effect:

- Method 1:
 - a. Run the **LST NRDUCELLALGOSWITCH** command. If the value of **Power Saving Switch** is **Intelligent Power Control Energy Saving:On**, intelligent power control energy saving has been enabled.
 - b. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one entry with **Power Saving Type** being **Intelligent Power Control Energy Saving**.

NOTE

- If there are two or more entries with **Power Saving Type** being **Intelligent Power Control Energy Saving**, intelligent power control energy saving has been enabled in two or more time periods.
 - c. Run the **DSP TIME** command. Check whether the system time is within a period for intelligent power control energy saving to take effect.
 - If it is, go to step **d**.
 - If it is not, wait until the start time of intelligent power control energy saving arrives.
 - d. After 30 to 50 minutes, run the **DSP NRDUCELL** command. Verify that the value of **Power Saving State** is

SMART_PWR_CTRL_ENERGY_SAVE:Enabled, which indicates that intelligent power control energy saving has taken effect.

- Method 2:
Run the **DSP NRDUCELLPOWERSAVING** command to query the status of intelligent power control energy saving.
 - If the value of **Power Saving Type** is **Intelligent Power Control Energy Saving** and the value of **Current State** is **Intelligent Power Control Enabled**, intelligent power control energy saving has taken effect.
 - If the value of **Power Saving Type** is **Intelligent Power Control Energy Saving** and the value of **Current State** is **Intelligent Power Control Disabled**, check the **Current Disabled Cause** parameter value to determine the reason why intelligent power control energy saving did not take effect. [Table 9-5](#) lists the possible reasons.

Table 9-5 Possible reasons why intelligent power control energy saving does not take effect

Reason	Description
Insufficient License	No license for this function is available.
Switch Turned Off	None
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable for other reasons.
Triggering Time Not Reached	The current time is not within the time range specified for intelligent power control energy saving to take effect.
Neighboring Cell Information Unreachable	There is no eXn interface available between the serving cell and the main interfering cell, or no interfering neighboring cell is obtained.
Others	Internal software or other reasons cause the function to fail.

NOTE

As external conditions change (for example, the eXn status changes), the feature working status also changes. There is a delay up to 15 minutes for the feature working status to be updated.

9.2.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.

On the live network, you can compare the total energy consumption data of base stations in the area within one week before and after intelligent power control energy saving is enabled to evaluate its energy saving gains. It is

expected that the total energy consumption of base stations in the area decreases after this feature is enabled.

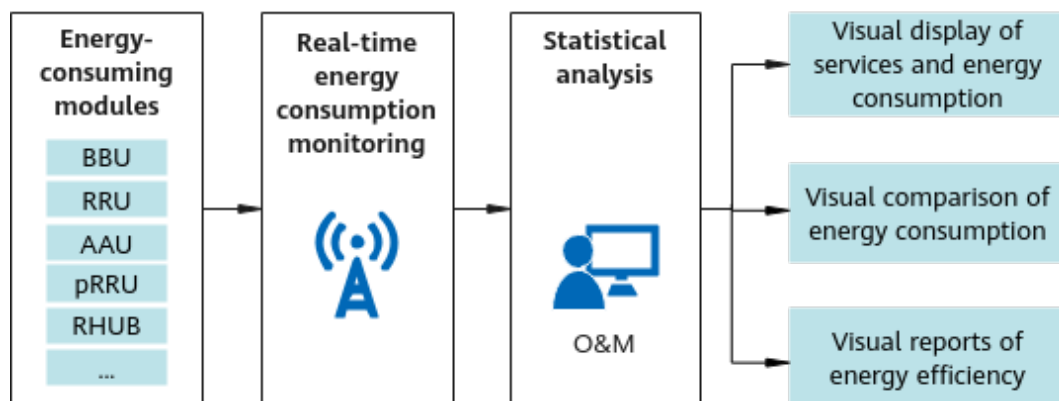
- The duration of intelligent power control energy saving can be observed using the counter **N.PowerSaving.SmartPwrCtrl.Dur** (for a cell) on the MAE-Access.

10 Energy Consumption Monitoring

10.1 Principles

Energy consumption monitoring enables a base station to periodically measure its energy consumption and report the statistics to the MAE-Access, making it easy for operators to monitor the base station's energy consumption. The MAE-Access can perform further statistical analysis on the energy consumption data, as shown in [Figure 10-1](#). For related descriptions of the MAE-Access, see the corresponding section in the MAE-Access online help: **RAN Management > SON Management > Base Station Energy Conservation Management**.

Figure 10-1 Working principles for energy consumption monitoring



The energy consumption of the BBU and RF modules, as well as that of the entire base station, can be monitored.

NOTE

This section describes only the energy consumption measurement method on the base station side.

Energy Consumption Statistics of the Entire Base Station

The energy consumption statistics of the entire base station can be obtained through cumulative or direct measurements. Cumulative measurement is the most common, whereas direct measurement applies only to the base stations configured with the PMU. Neither method requires parameter configuration or engineering deployment.

- Cumulative measurement

The base station measures the energy consumption of modules, such as the BBU, RF modules, and fans, as scheduled based on actual configurations. All the values add up to the total energy consumption to be reported to the MAE-Access.

- The modules with digital power supply in the base station automatically report their energy consumption to the BBU.
- For macro base stations, RRUs or AAUs report their energy consumption, including that of the remote electrical tilt devices, to the BBU on a scheduled basis.

For LampSite base stations, pRRUs do not support energy consumption measurement but RHUBs and RRUs do. If pRRUs are used, the RHUBs measure the energy consumption of pRRUs, add it up with their own energy consumption, and report the total value to the BBU on a scheduled basis. If RRUs are used, the RRUs and RHUBs separately measure and report their own energy consumption to the BBU on a scheduled basis. The BBU then reports the total energy consumption of the RRUs, RHUBs, and the BBU to the MAE-Access.

- For electromechanical devices (such as the EMU, FMU, TCU, PMU, CCU, MEB, PSU, and BDU) configured in a base station, their energy consumption data is obtained by the BBU on a scheduled basis.

The cumulative measurement values of the NR energy consumption are indicated by the **VS.EnergyCons.BTS.Adding.NR** and **VS.EnergyCons.BTS.Adding.Service.NR** counters, which are reported to the MAE-Access.

- For an NR-only base station, these counters indicate the energy consumption of the entire base station.
- For an NR-capable multimode macro base station, these counters indicate the energy consumption of the NR part of the base station.

NOTE

Because a multimode base station serves multiple RATs, the **VS.EnergyCons.BTS.Adding.NR** and **VS.EnergyCons.BTS.Adding.Service.NR** counters do not reflect the exact NR energy consumption and therefore should be used for reference only. In addition, these counters do not apply to NR-capable multimode LampSite base stations. For details about the energy consumption measurement of such base stations, see [Energy Consumption Statistics of the BBU and RF Modules](#).

Each counter value equals the sum of NR energy consumption of all physical devices. The NR energy consumption is calculated in different ways for different counters:

- For the **VS.EnergyCons.BTS.Adding.NR** counter: NR energy consumption of a single physical device = (Energy consumption of the physical device/

Number of main control boards that manage this physical device)/
Number of RATs on the local end that use this physical device. In this formula, "Number of RATs on the local end that use this physical device" refers to the number of RATs served by the main control board that works in NR and manages the physical device.

For example:

In G+LNR separate-MPT scenarios where G and LNR share the same RRU:
NR RRU energy consumption = (RRU energy consumption/2)/2 = 1/4 RRU energy consumption

In GL+NR separate-MPT scenarios where GL and NR share the same RRU:
NR RRU energy consumption = (RRU energy consumption/2)/1 = 1/2 RRU energy consumption

- For the **VS.EnergyCons.BTS.Adding.Service.NR** counter, the NR energy consumption of a single physical device is calculated as follows:

NR energy consumption of a BBU = Energy consumption of the BBU x
Number of NR physical cells/Total number of physical cells of all RATs in the BBU

NR energy consumption of an RF module = Energy consumption of the RF module x Average air-interface transmit power for NR/Sum of average transmit power values for all RATs of the RF module

For example:

Assume that a BBU serves 1 GSM physical cell, 2 UMTS physical cells, 3 NB-IoT physical cells, 4 LTE (excluding NB-IoT) physical cells, and 5 NR physical cells. Then, the NR energy consumption of the BBU is equal to (Energy consumption of the BBU x 5/15).

Assume that there are GSM, UMTS, NB-IoT, LTE (excluding NB-IoT), and NR carriers on an RF module, and the average transmit power values for these RATs of the RF module in a measurement period are 20 W, 15 W, 5 W, 30 W, and 25 W, respectively. Then, the NR energy consumption of the RF module is equal to (Energy consumption of the RF module x 25/95).

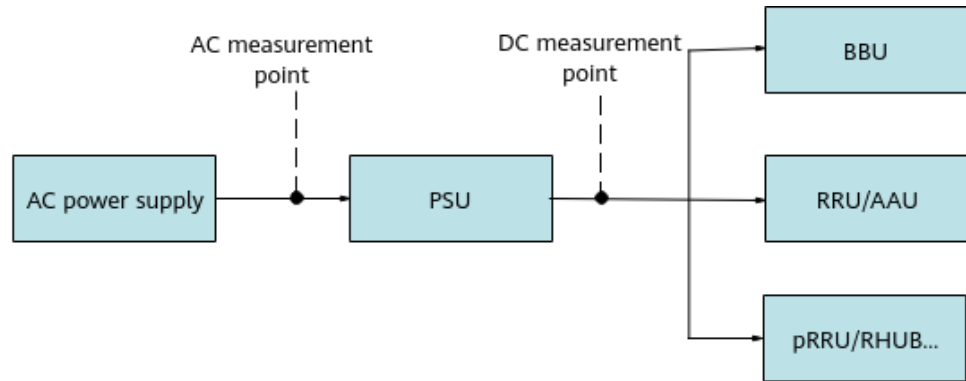
- Direct measurement

The energy consumption of a base station configured with a PMU can also be measured based on the PSUs. The measured energy consumption is the total value of all devices (including but not limited to the BBU, RF units, and electromechanical devices) powered by the PSUs. The base station obtains the total energy consumption from the PMU and reports the value to the MAE-Access.

The base station can report either AC or DC energy consumption (as shown in [Figure 10-2](#)), depending on the setting of the **ENERGYCON.MP** parameter in the **SET ECMP** command.

- If the **ENERGYCON.MP** parameter is set to **AC**, the total AC energy consumption before PSUs perform AC-to-DC conversion is reported.
- If the **ENERGYCON.MP** parameter is set to **DC**, the total DC energy consumption after PSUs perform AC-to-DC conversion is reported.

Figure 10-2 Direct measurement



The **VS.EnergyCons.BTS.Measuring.NR** counter on the MAE-Access indicates the energy consumption of the NR base station calculated based on the current and voltage of the power supply system.

- If **ENERGYCON.MP** is set to **AC**:
 - For an NR-only base station, this counter indicates the input power of PSUs calculated by the power supply system.
 - For a multimode base station supporting NR, this counter indicates partial input power of PSUs. For example, in dual-mode scenarios, the value of this counter is half of the total input power of PSUs.
- If **ENERGYCON.MP** is set to **DC**:
 - For an NR-only base station, this counter indicates the output power of PSUs calculated by the power supply system.
 - For a multimode base station supporting NR, this counter indicates partial output power of PSUs. For example, in dual-mode scenarios, the value of this counter is half of the total output power of PSUs.

Energy Consumption Statistics of the BBU and RF Modules

The base station can also measure the energy consumption of the BBU and RF modules and report the statistics to the MAE-Access.

- The **VS.EnergyCons.BU** counter indicates the energy consumption of all boards in a BBU.
- The **VS.EnergyCons.BTSBoard** counter indicates the energy consumption of RF modules.

NOTE

For LampSite base stations, the RHUB measures pRRU energy consumption and reports the total result to the BBU. This is because pRRUs are not capable of measuring energy consumption. In such cases, the **VS.EnergyCons.BTSBoard** counter value adds up the energy consumption of the RHUB and that of the RHUB-powered pRRUs.

The counters measuring the energy consumption of the BBU and RF modules do not distinguish between RATs. The energy consumption of a multimode base station supporting NR is the total energy consumption of the BBU and RF modules in this base station. In a separate-MPT multimode base station, the energy

consumption of common devices is separately reported by each main control board. Therefore, when adding up the values of the reported energy consumption counters, note the following:

- For the BBU, only one counter needs to be added up among all reported counters with the same cabinet number and subrack number.
- For RF modules, only one counter needs to be added up among all reported counters with the same cabinet number, subrack number, and slot number.

In a separate-MPT multimode base station, the sampling period and reporting period of the counters may not be synchronized between different main control boards. Therefore, a deviation of one sampling period (5 minutes) may exist.

10.2 Network Analysis

10.2.1 Benefits

Although energy consumption monitoring does not provide energy saving gains, it provides a straightforward method of monitoring the base station energy consumption. You can use this function to evaluate the energy saving gains achieved after an energy-saving function is enabled.

10.2.2 Impacts

The impacts between this function and other functions are described in [10.3.2.3 Function Impacts](#).

This function has no impact on the network.

10.3 Requirements

10.3.1 Licenses

None

10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.3.2.1 Prerequisite Functions

None

10.3.2.2 Mutually Exclusive Functions

None

10.3.2.3 Function Impacts

None

10.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and power supply boards support this function. For details, see the related BBU technical specifications and hardware description in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

10.3.4 Others

- The base station is correctly connected to the MAE, the MAE-Access supports energy consumption monitoring, and the PowerSavingStatistic module is enabled.
- **Table 10-1** describes the PMU requirements of direct measurement of energy consumption for the entire base station. For details about the PMU, see *Power Supply Management*.

Table 10-1 PMU requirements of direct measurement

Power Supply System Type	PMU Model	PMU Software Version
ETP	PMU 11A	No requirements
	PMU 11B	No requirements
	APM5930 (CCUC)	No requirements

Power Supply System Type	PMU Model	PMU Software Version
	OPM200 (PMU 16A)	No requirements
	CCUB	Not supported
	PMU 15A	Not supported
	OPM50M	Not supported
	OPM30M	Not supported
	OPM30A	Not supported
APM30	PMU 01B	No requirements
	PMU 11A	No requirements
	PMU 11B	No requirements
	CCUB	No requirements
	CCUC	No requirements
	EPMU01	Not supported
EPS4890	PMU 01B	No requirements
	PMU 11A	No requirements
	PMU 11B	No requirements
	EPMU01	Not supported
EPS4815	PMU 11A	No requirements
	PMU 11B	No requirements
	EPMU03	Not supported
EPU	None	Not supported
SC48200	None	Not supported

10.4 Operation and Maintenance

10.4.1 Data Configuration

Energy consumption data can be observed on the MAE-Access without any configuration on the base station. If the energy consumption data of the base station needs to be further summarized and analyzed by the MAE-Access, enable the PowerSavingStatistic component by following instructions in "Managing Components" in *OSS Management of MAE Product Documentation(EulerOS, TaiShan)*.

10.4.2 Activation Verification

The energy consumption of the base station can be observed by subscribing to counters listed in [Table 10-2](#) on the MAE-Access.

Table 10-2 Counters for measuring energy consumption of the NR base station

Counter ID	Counter Name	Description
1593835804	VS.EnergyCons.BTS.Adding.NR	This counter provides the NR-specific energy consumption of the base station by adding up the energy consumption of all physical devices working in NR (cumulative measurement).
1593835961	VS.EnergyCons.BTS.Adding.Service.NR	This counter provides the NR-specific energy consumption of the base station by adding up the rated energy consumption of all boards of the base station (cumulative measurement).
1593835805	VS.EnergyCons.BTS.Measuring.NR	This counter provides the NR-specific energy consumption of the base station by using the energy consumption measured by the power system (direct measurement).
1593835835	VS.EnergyCons.BU	This counter measures the energy consumption of the BBU.
1593835836	VS.EnergyCons.BTSBoard	This counter measures the energy consumption of RF modules.

10.4.3 Network Monitoring

None

11 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

12 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

13 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

14 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *License Control Item Lists*
- *FWA*
- *Hyper Cell*
- *Cell Combination*
- *LTE and NR Power Allocation*
- *CMAS*
- *Remote Interference Management (Low-Frequency TDD)*
- *Super Uplink*
- *UL and DL Decoupling*
- *Synchronization*
- *MIMO (TDD)*
- *CoMP*
- *URLLC*
- *Network Slicing*
- *Downlink Scheduling*
- *Extended Cell Range*
- *Modulation Schemes*
- *Channel Management*
- *Carrier Aggregation*
- *ANR*
- *SA Mobility Management in Connected Mode*
- *Power Supply Management*
- *UE Power Saving*
- *High Speed Mobility*
- *VoNR*
- *Virtual Grid-based Multi-Frequency Coordination*
- *Interference Avoidance*

- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *Indoor Coverage*
- *Beam Management*
- *Fusion Cell (Low-Frequency TDD)*
- *Multi-Frequency Smart Selection*
- *Multi-Frequency Smart Aggregation*
- *Mobility Load Balancing in Connected Mode*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *MRO*
- *Co-MM (Low-Frequency TDD)*
- *CA Experience Improvement*
- *RedCap*
- *Green Site*
- *NR Inter-Carrier Dynamic Power Allocation*
- *Standards Compliance*
- *Positioning*
- *NSA Mobility Management*
- *License Management*
- *URLLC*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *5G Networking and Signaling*
- *Base Station Equipment Reliability*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *Emergency Communications Solution (Low-Frequency TDD)*
- *Power Control*
- *URLLC for Vehicle Communications*
- *3D Networking Experience Improvement*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *FWA*
- *mmWave Super Coverage (High-Frequency TDD)*
- *NB-IoT Enhancements (FDD) in eRAN Feature Documentation*
- *OSS Management in MAE Product Documentation (EulerOS, TaiShan)*
- *PSU in 3900 & 5900 Series Base Station Product Documentation*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*